# EE6301
# Concept-first Beginner Guide
# 知识框架与零基础中英文讲解

8 chapter maps + 57 core lessons + 52 supplements
8周知识框架 + 57项核心讲解 + 52项补充
48 figures + 74 formula cards
48张图 + 74组公式 · 2026-10-14

Body text / 正文：11 pt
English first, Chinese immediately below. / 英文在前，中文紧随。

2026-10-14 · Course notes, sample explanations and original practice
课件知识、样题参考解析与原创练习

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# Start here / 从这里开始

**EN:** This edition explains all 25 main cards and 32 additional topics from the reviewed guide. Each lesson uses three steps: build the idea, follow the mechanism, then work through an example. Examples and analogies are teaching aids. Source anchors connect the explanations to the supplied slides and transcripts.

**中文：** 本版为复核版的25张主卡和32条补充知识全部配上讲解。每项按三步展开：建立直觉、跟着机制理解、再看例子。例子与类比用于帮助学习；来源定位对应你提供的课件与字幕。

**EN:** Scope and format: Weeks 1–7 plus Week 8 Parts One and Two, physical PDF p5–29; 25 single-answer MCQs, 60 minutes, closed book. Read in Chinese first when needed, then match the English terms. The 25-question mock remains the reviewed practice paper, with 17 concepts and 8 short calculations.

**中文：** 范围和形式：Week1–7加Week8前两章，实际PDF p5–29；25道单选，60分钟，闭卷。需要时先读中文理解，再对应英文。25题模拟沿用已复核卷，含17道概念和8道短计算。

**EN:** Lesson IDs C01–C25 match the previous main cards; B01–B32 match the additional concepts. They are now arranged in a learning sequence. Integer LOD is taught beside calibration, and grating resolution beside spectroscopy, even though those topics were revisited in later lectures.

**中文：** C01–C25对应原主卡，B01–B32对应原补充概念。本版按学习顺序排列：整数LOD紧接校准，光栅分辨紧接光谱；它们虽在后续课堂重讲，但放在相关基础旁更容易理解。

**EN:** 2026-10-12 coverage update: the eight original PDFs have been read again page by page. All 57 existing lessons and their recall statements remain. 52 short S lessons now explain missing distinctions and conditions beside the relevant topic, with 39 original slide figures and bilingual reading guidance. Study the whole chapter, not just the statement that gives a sample answer.

**中文：** 2026-10-12补漏更新：重新逐页核对八份原PDF，保留57项原讲解及其重点摘要；在对应知识章旁增加52个S补充小节，讲清缺漏的区别和条件，并加入39张原课件图及中英文读图说明。复习整章知识，不只记得到样题答案的那一句。

**EN:** For each concept, ask what it measures, how the signal is generated, which conditions the statement assumes, and how it differs from similar methods. A true option also needs a reason. Lower-priority derivations are kept at the level needed to recognize their concepts; the scope still ends before Week 8 Human–Machine Interface.

**中文：** 每个概念都问：它测什么、信号怎样产生、这句话依赖什么条件、与相似方法有什么区别。正确选项也要能说出理由。较低优先级的推导保留到能辨认概念的层次；范围仍止于Week8 Human–Machine Interface之前。

**EN:** Visual update: 48 figures now sit beside their concepts: 39 original slide crops and 9 teaching diagrams, including 32 newly added pictures. Read the bilingual guidance below each figure. In the HTML, use Zoom to inspect labels, then Actual size if necessary. The classmate PDF informs the visual layout; the reviewed course scope and content remain the basis.

**中文：** 图文更新：共48张图紧跟对应知识点，其中39张原课件截图、9张教学示意，本次新增32张。每张新图下方都有中英文读图说明。HTML可点“放大”检查标签，再选“原始大小”。同学PDF用于参考图文排版；知识内容和考试范围仍以已复核的课程资料为依据。

**EN:** Reading colours: blue builds the concept, purple follows the mechanism, green explains examples or recall, and amber marks wording and conditions. P1/P2/P3 separately indicate review priority. Original slide figures retain their own colours; interpret their axes, legends and arrows using the nearby reading guide.

**中文：** 阅读颜色：蓝色建立概念，紫色追踪机制，绿色看例子与记忆要点，橙黄色提示题目措辞和条件。P1/P2/P3另行表示复习优先级。原课件截图保留原有配色，需结合旁边读图说明判断坐标、图例和箭头的含义。

**EN:** 2026-10-13 reading order: begin each week with its concept map, then read the key points at the start of each lesson before following the explanation and example. The existing topics and figures remain. Comparison with the supplied classmate notes adds short missing explanations and clarifies conditions; the reference is study material, while the slides and classroom evidence determine course scope.

**中文：** 2026-10-13阅读顺序：每周先看知识框架，再读每节开头的分点必会知识，随后理解机制与例子。原有知识项和配图保留。对照同学资料后，补上少量缺漏说明并讲清适用条件；参考资料用于学习，课程范围仍依据原课件和课堂证据。

From a biological target to a readable result / 从生物目标到读数




Recognition chooses the target; transduction produces the signal.
识别负责选择目标，换能负责形成信号。

Biosensor workflow / 生物传感流程

# Study priorities and route / 重点标记与复习路线

**EN:** P1: master first and explain each statement; P2: important foundations that connect or qualify P1; P3: supporting details to consult after the first pass. These levels are study recommendations, not probabilities or a guarantee of the real questions.

**中文：** P1：先掌握，能解释陈述；P2：重要基础，用来连接P1及理解条件；P3：辅助细节，第一轮后按需查。等级是复习建议，不是出题概率或真题保证。

**EN:** Evidence tags are separate: Quiz stated, teacher emphasis, repeated explanation, worked example, sample quiz and final-only emphasis. “Important” alone is not a Quiz promise. Example tags identify practice in the slides/class; they do not invent a separate homework assignment.

**中文：** 依据标签分别为：Quiz明确提及、课堂强调、反复讲解、例题练习、样卷涉及、仅期末强调。“重要”本身不等于Quiz保证。例题标签指课件或课堂练习，不凭空认定为独立课后作业。

**EN:** A mixed lesson has a specific review target: focus on that part first. A sample question marks all its relevant option concepts. Its extra derivations or device details can still be P2/P3. Open “Evidence” below a heading in the HTML to see the review target, transcript references and the lecturer’s short quote when available.

**中文：** 混合小节有具体学习抓手，先学该部分；样卷标记覆盖相关选项的概念，额外推导或器件细节仍可为P2/P3。HTML中展开标题下的“依据”，可看学习抓手、字幕位置及可核对的老师短引语。

**P1: 44 • P2: 53 • P3: 12**

| First-pass study group / 第一轮学习组 | P1 locations / P1位置 | What to do / 要会什么 |
|---|---|---|
| Measurement and LOD / 测量性能与检测限 | C02, C03, B03 | Distinguish slope, selectivity and LOD; use the linear range and integer-count rule. / 区分斜率、选择性与LOD；会选线性区、计算及整数计数处理。 |
| Photon energy and absorption / 光子与吸收 | C04, S105 | Match wavelength, energy and photon count; calculate absorbance, transmission and ε. / 对应波长、能量、光子数；会算吸光度、透过率与ε。 |
| Fluorescence / 荧光 | C05, C06, S110, S111 | Explain intensity versus color, source/filter roles and Φ, τ, radiative rate. / 解释强度与颜色、光源及滤光片作用；区分Φ、τ与辐射速率。 |
| Raman and scattering / 拉曼与散射 | C07, B08 | Distinguish elastic/inelastic signals and Stokes/anti-Stokes; read $cm^{-1}$. / 区分弹性/非弹性与Stokes/anti-Stokes；会读$cm^{-1}$及位移。 |

| First-pass study group / 第一轮学习组 | P1 locations / P1位置 | What to do / 要会什么 |
|---|---|---|
| Grating resolution / 光栅分辨 | C08 | Use R=mN and distinguish groove density, illuminated width and diffraction order. / 会用R=mN，区分刻线密度、照明宽度与衍射级次。 |
| Label-free, fibre and resonance / 无标记、光纤与共振 | C09, C10, C11, B10, S115, S116, S117 | Recognize cavities, SPR and Raman; explain TIR, evanescent sensing and spatial arrangements. / 辨认共振腔、SPR与Raman；解释全反射、倏逝场及光纤传感布置。 |
| ART and LSPR / 抗原快检与LSPR | C12 | Separate antibody recognition from gold-particle color generation and check the control line. / 区分抗体识别与金颗粒显色，并会判断控制线。 |
| Microscopy and resolution / 显微成像与分辨 | C13, C14, B14, B15, B17 | Compare NA rather than magnification alone; distinguish confocal and two-photon mechanisms. / 比较NA而非只看倍率；区分共聚焦与双光子机制。 |
| OCT and light detectors / OCT与光探测器 | C15, C16, S404 | Explain reflectivity-based OCT and CCD/CMOS/PMT roles; separate depth and lateral resolution. / 理解OCT反射读出与CCD/CMOS/PMT；分清轴向、横向分辨。 |
| Ultrasound and photoacoustics / 超声与光声 | B18, S501 | Trace transmit/receive and piezoelectric conversion; distinguish echo depth from a photoacoustic source path. / 会追踪发射接收与压电转换；区分回波深度与光声源传播路径。 |
| Electrochemical readouts / 电化学读出 | C18, C19, S603 | Match voltage/current/impedance readouts, electrode roles and GOx reaction-current relations. / 对应电位/电流/阻抗模式、电极分工与GOx反应—电流关系。 |
| Bioelectricity / 生物电基础 | C20, C21, B27, S707 | Know sources of biopotentials, resting/spike phases and basic charge/voltage/current units. / 理解生物电来源、静息及尖峰阶段，记基础电荷、电压、电流单位。 |
| ECG, EEG and watches / 心电、脑电与手表 | C22, S705, S706 | Identify wave sources and time intervals; estimate frequency and separate watch PPG from EEG/EMG. / 辨认波形来源与时间；估频率，区分手表PPG与EEG/EMG。 |

| First-pass study group / 第一轮学习组 | P1 locations / P1位置 | What to do / 要会什么 |
|---|---|---|
| Patch clamp and MEA / 膜片钳与多电极阵列 | C23, S801 | Recognize pipette versus array, intracellular/extracellular signals and throughput differences. / 辨认吸管与阵列、胞内与胞外，以及通量差别。 |
| Amplification and filters / 放大与滤波 | C24, C25 | Separate gain, input/output impedance and common-mode rejection; recognize four filter curves. / 区分增益、输入输出阻抗与共模抑制；会认四种滤波曲线。 |

**EN:** Suggested order: read the prerequisites you need, complete the P1 groups, use P2 to explain unfamiliar options, then take the 25-question mock in English. Return to P3 when a concept or a diagram still blocks understanding. Keep the full guide for checking; a P3 label does not mean a whole field is excluded.

**中文：** 建议顺序：先补自己缺的先修基础，完成P1学习组；用P2解释不熟悉的选项，再用纯英文做25题。概念或图仍不懂时回查P3。完整资料仍供查阅；P3不等于整个领域已被排除。

# Foundations before the course concepts / 先补足理解所需的基础

## F01 • What connects these eight weeks? / 这八周到底在学什么?

**EN:** The course repeatedly asks one question: how can an invisible biological event become a number we can read? A molecule binds, a cell fires, or light interacts with tissue. An instrument changes that event into a measurable signal. Analysis then connects the signal to the quantity we want.

**中文：** 这门课反复讨论一个问题：看不见的生物事件，怎样变成能读出的数字？比如分子发生结合、细胞发出电活动、光与组织相互作用。仪器把事件变成可测信号，再用分析把信号与想知道的量联系起来。

**EN:** Think in four questions: What is the target? What physical event changes? What does the detector measure? How do we interpret the result? For a glucose meter, the target is glucose, a chemical reaction occurs, the electrode measures current, and calibration turns that current into a glucose estimate.

**中文：** 每个方法都问四件事：目标是什么？哪种物理事件发生变化？探测器实际测什么？怎样解释结果？血糖仪的目标是葡萄糖，发生的是化学反应，电极实际测电流，校准关系再把电流转换为葡萄糖估计值。

**EN:** Read the foundations first. Then read each topic’ s three paragraphs: basic idea, mechanism, and example. Use Chinese mode when a concept is new; use bilingual mode to match terms; finally use English-only mode to practise recognition. The older 32-page guide is useful after understanding, for quick revision.

**中文：** 先读本基础章，再读各主题的三段：基本意思、机制和例子。第一次学可切换中文模式，随后用中英对照对应术语，最后用纯英文练识别。理解之后，原32页版本适合做快速复习。

## F02 • Atoms, molecules, electrons and ions / 原子、分子、电子和离子

**EN:** An atom is a small building block of matter. Atoms joined together form a molecule: $H_2O$ is a water molecule, while $H_2O_2$ is hydrogen peroxide, a different substance. An electron carries negative electric charge. An ion is an atom or group of atoms with net charge, such as $Na^+$ or $K^+$.

**中文：** Atom原子是物质的基本构成单位之一。多个原子结合形成molecule分子：$H_2O$是水，$H_2O_2$是过氧化氢，是不同物质。Electron电子带负电；ion离子是具有净电荷的原子或原子团，例如$Na^+$、$K^+$。

**EN:** A chemical reaction rearranges substances. An enzyme is a biological catalyst: it speeds a particular reaction without being consumed in the overall catalytic cycle. An antibody is a recognition protein that can bind a matching antigen. Binding and reaction are different events; a sensor can use either to obtain information.

**中文：** Chemical reaction化学反应改变物质的组成。Enzyme酶是生物催化剂，能加快特定反应，在整个催化循环中不被消耗。Antibody抗体是能结合相匹配antigen抗原的识别蛋白。Binding结合与reaction反应不是同一事件，传感器可利用其中任何一种获取信息。

**EN:** Concentration means amount per volume. A mole is an amount containing about $6.022\times10^{23}$ entities; it is not a unit of mass. A solution at 1 mM contains 0.001 mol per litre. The same concentration can correspond to different masses for different molecules because their molar masses differ.

**中文：** Concentration浓度表示单位体积中的量。Mole摩尔是物质的量单位，每摩尔约含$6.022\times10^{23}$个实体，不是质量单位。1 mM溶液含0.001 mol/L。不同分子的摩尔质量不同，因此相同摩尔浓度不一定对应相同质量。

## F03 • Units without panic / 不慌张地看单位

**EN:** Prefixes change scale: milli (m)=$10^{-3}$, micro (μ)=$10^{-6}$, nano (n)=$10^{-9}$, pico (p)=$10^{-12}$, kilo (k)=$10^{3}$ and mega (M)=$10^{6}$. Therefore 1 mV=0.001 V, 1 μA=0.000001 A, 1 kΩ=1000 Ω, 1 MΩ=$10^{6}$ Ω and 1 pF=$10^{-12}$ F. Always convert a value together with its unit; changing only the number changes the physical meaning.

**中文：** 单位前缀改变数量级：milli（m，毫）=$10^{-3}$，micro（μ，微）=$10^{-6}$，nano（n，纳）=$10^{-9}$，pico（p，皮）=$10^{-12}$，kilo（k，千）=$10^{3}$，mega（M，兆）=$10^{6}$。所以1 mV=0.001 V，1 μA=0.000001 A，1 kΩ=1000 Ω，1 MΩ=$10^{6}$ Ω，1 pF=$10^{-12}$ F。数值与单位必须一起转换，只改数字会改变物理意义。

**EN:** Watch the context of m: in 2 m it means metres; in mV it means milli; in y=mC+b it is a symbol for slope. Likewise C can mean concentration, capacitance, or the unit coulomb. MΩ uses the mega prefix, but 1 M for concentration means 1 mol/L. A symbol’ s meaning comes from its definition, not just its letter.

**中文：** 注意m的上下文：2 m中的m是米，mV中的m是毫，y=mC+b中的m是斜率符号。同样，C可能表示浓度、电容或库仑单位。MΩ中的M是兆前缀，而浓度1 M表示1 mol/L。字母含义要看定义，不能只凭字母判断。

**EN:** Useful conversions: 1 cm=$10^{7}$ nm; 1 s=$10^{3}$ ms=$10^{6}$ μs. In an echo-depth question, 40 μs is $40\times10^{-6}$ s. In a wavelength question, 600 nm is $6\times10^{-7}$ m. Write the conversion before substitution to avoid thousand-fold or million-fold errors.

**中文：** 常用换算：1 cm=$10^{7}$ nm；1 s=$10^{3}$ ms=$10^{6}$ μs。回波题的40 μs应写成$40\times10^{-6}$ s；波长题的600 nm应写成$6\times10^{-7}$ m。先写换算再代入，避免相差千倍或百万倍。

## F04 • Signal, noise and calibration / 信号、噪声与校准

**EN:** A signal is the measured change carrying information, such as current, wavelength or voltage. Noise is unwanted variation in that measurement. A blank sample contains no target but can still give a nonzero, fluctuating reading. Therefore “I see some output” does not prove that the target is present.

**中文：** Signal信号是携带信息的测量变化，例如电流、波长或电压。Noise噪声是测量中的无用波动。Blank空白样品不含目标，但读数仍可能不为零、也可能波动。所以“有输出”不能直接证明“有目标”。

**EN:** Calibration means measuring known target amounts to learn the response. A straight calibration line y=mC+b says: output y equals slope m times concentration C, plus baseline b. The slope tells how much output changes per unit concentration. The standard deviation σ summarizes how widely repeated blank readings fluctuate.

**中文：** Calibration校准是先测已知目标量，建立响应关系。直线y=mC+b表示：输出y=斜率m×浓度C+基线b。Slope斜率说明每单位浓度带来多少输出变化。Standard deviation标准差σ概括重复空白读数的波动程度。

**EN:** Imagine a signal increase of 1 unit. It is easy to see if blank readings fluctuate by only 0.1 unit, but difficult if they fluctuate by 5 units. This is the intuition behind detection limits. A steep response helps only when noise is considered too; this comparison is not about whether a device can identify the correct molecule.

**中文：** 想象信号增加1单位：空白只波动0.1单位时，很容易看出来；空白波动5单位时，就很难判断。这就是检测限的直观含义。响应斜率大有帮助，但必须一起考虑噪声；这件事与设备能否认对分子是两个不同问题。

## F05 • Light, wavelength, frequency and spectra / 光、波长、频率与光谱

**EN:** Light is electromagnetic radiation. Wavelength λ is the distance between neighbouring wave crests. Frequency f is the number of oscillations per second, measured in hertz (Hz). In vacuum, c=λf, where c is light speed. A longer wavelength therefore means a lower frequency.

**中文：** 光是电磁辐射。Wavelength波长λ是相邻波峰之间的距离；frequency频率f是每秒振动次数，单位hertz赫兹（Hz）。真空中c=λf，c为光速。因此波长越长，频率越低。

**EN:** A photon is a small packet, or quantum, of light energy. Its energy is E=hf=hc/λ. A longer-wavelength photon has less energy, even if a beam contains many such photons and has high total power. Power is energy per second: 1 watt (W)=1 joule (J)/s. Photon energy and total beam power are different quantities, just as one coin’s value differs from the value of a whole bag of coins.

**中文：** Photon光子是一小份光能，即光量子，能量E=hf=hc/λ。长波长光子的单个能量较低，即使光束含很多光子、总功率很大也如此。Power功率是每秒的能量：1 watt（W，瓦）=1 joule（J，焦耳）/s。单个光子能量与整束光的功率不同，类似一枚硬币面值与整袋硬币总值不同。

**EN:** A spectrum plots measured optical strength against wavelength, frequency or wavenumber. A peak shows where a signal is strong; a dip can show where transmission is reduced. Check the horizontal axis before comparing “left” and “right”: moving right in wavelength means lower photon energy, but moving right in frequency means higher energy.

**中文：** Spectrum光谱将测得的光强画在波长、频率或波数坐标上。Peak峰表示信号强的位置；dip凹谷可能表示透射减少的位置。比较左右前先读横轴：在波长轴向右意味着单光子能量更低，在频率轴向右却意味着能量更高。

## F06 • Voltage, current, resistance and impedance / 电压、电流、电阻与阻抗

**EN:** Voltage is an electric potential difference: energy difference per unit charge, measured in volts. Current is charge passing per second, measured in amperes. In a water analogy, voltage resembles a pressure difference and current resembles flow rate. The analogy explains their roles; electrical charge is not literally water.

**中文：** Voltage电压是电势差，即单位电荷对应的能量差，单位伏特。Current电流是每秒通过的电荷量，单位安培。用水流类比时，电压像压力差，电流像流量。类比帮助理解作用，但电荷并不是真正的水。

**EN:** Resistance R limits current: for an ideal resistor, V=IR. A capacitor stores separated charge, with Q=CV; capacitance C is measured in farads (F). An inductor stores energy in a magnetic field. DC current keeps its direction, although its magnitude may vary; AC current reverses direction, usually periodically. Their AC opposition is described by impedance Z, which includes size and phase, not just resistance.

**中文：** Resistance电阻R限制电流；理想电阻满足V=IR。Capacitor电容器储存分离电荷，Q=CV；capacitance电容C的单位是farad（F，法拉）。Inductor电感器把能量储存在磁场中。DC直流保持方向，大小可以变化；AC交流方向交替反转，通常周期变化。交流中的阻碍用impedance阻抗Z描述，包含大小与相位，不只是电阻。

**EN:** A voltmeter measures between two points; one voltage value needs a reference. A measuring instrument can draw current and change the source it is measuring: this is loading. High input impedance draws less current and helps preserve a weak biological voltage. Amplifier gain is a separate ratio, output voltage divided by input voltage.

**中文：** 电压表测两个点之间的差；一个电压值需要参照。测量仪器可能取走电流、改变被测信号，这叫loading加载。高输入阻抗取电流较少，有助于保留微弱生物电压。Amplifier gain放大器增益则是另一个比值：输出电压除以输入电压。

## F07 • Resolution, magnification, contrast and pixels / 分辨率、放大倍率、对比度与像素

**EN:** Resolution is the ability to distinguish nearby features. Magnification makes an image look bigger. Enlarging a blurry photograph produces a bigger blur; it does not recover missing detail. In microscopy, a smaller minimum separable distance means better spatial resolution. In spectroscopy, a smaller separable wavelength gap means better spectral resolution.

**中文：** Resolution分辨率是区分邻近细节的能力；magnification放大倍率是把图像变大。放大模糊照片只会得到更大的模糊，不会找回丢失细节。显微中最小可分辨距离越小，空间分辨越好；光谱中最小可分辨波长间隔越小，光谱分辨越好。

**EN:** Contrast is the difference between an object and its surroundings. You may have sufficient resolution but still fail to see a feature with almost no contrast. A pixel is one sampled position in a digital image; bit depth controls how many intensity levels can be stored. Pixel count, bit depth, contrast and optical resolution describe different things.

**中文：** Contrast对比度是目标与周围的差别。即使分辨能力够，目标与背景几乎一样时也可能看不见。Pixel像素是数字图像中的一个采样位置；bit depth位深决定能存多少强度等级。像素数、位深、对比度和光学分辨率是不同概念。

**EN:** When a question says “higher resolution”, translate it into “can separate closer features”. When it says “larger resolution distance”, it may mean a worse separation limit. Always identify the quantity being compared. A label such as 100× describes magnification; NA describes the objective’s light-collection angle and medium.

**中文：** 题目说higher resolution时，理解为“能区分更近的细节”。若说larger resolution distance，则可能是最小间距更大、分辨更差。必须看比较的物理量。100×表示倍率；NA描述物镜的收光角和介质。

## F08 • How to read three common graphs / 三种常见图怎么读

**EN:** For a calibration graph, identify x and y, then choose two points within the linear region. Slope is vertical change divided by horizontal change, not simply the height of one point. If output starts to saturate, the last points are not suitable for the straight-line slope used in the LOD calculation.

**中文：** 校准图先认x、y，再在线性区选两个点。斜率是纵向变化除以横向变化，不是某一个点的高度。若输出开始饱和，末端点就不适合计算LOD所用的直线斜率。

**EN:** For a resonance spectrum, distinguish peak position, peak height and linewidth. A peak moving sideways is a wavelength or frequency shift; a peak getting taller is an intensity change. Full width at half maximum (FWHM) is the horizontal width measured at half the peak height above baseline. These changes carry different information.

**中文：** 共振光谱要区分峰位置、峰高和线宽。峰横向移动是波长/频率位移；峰变高是强度变化。FWHM半高全宽是在高出基线的峰高一半处测得的横向宽度。这些变化携带不同信息。

**EN:** For a time trace, one cycle goes from one repeating feature to the next equivalent feature. Period T is time per cycle; f=1/T. Eight cycles in 0.2 s give 40 Hz. Counting eight peaks from the first to the eighth gives only seven peak-to-peak intervals, so use the actual interval rather than blindly dividing by eight.

**中文：** 时间波形中，一个周期从某个特征到下一个相同特征。Period周期T是每轮时间，f=1/T。0.2 s内八个完整周期对应40 Hz。若是从第一个峰到第八个峰，则只有七个峰间隔；要按实际完整间隔计算，不能机械除以八。

# Week 1 learning block: Biosensors and measurement quality / 第1学习组：生物传感器与测量质量

## Week 1 concept map / 第1周知识框架

**EN:** Chapter logic: A biosensor recognizes a target, converts its event into a signal, and uses calibration to judge identity and amount.

**中文：** 本章主线：生物传感器先识别目标，再把事件变成信号，最后用校准判断是什么、多少及能否检出。

| English term 中文名词 | Meaning and connection 含义与知识联系 |
|---|---|
| **Analyte • bioreceptor • transducer 分析物•生物受体•换能器** | Target, recognition and signal conversion are separate roles. 被测目标、识别与信号转换是三个不同角色。 → C01 |
| **Biorecognition 生物识别** | Antibodies bind; enzymes catalyse; complementary nucleic acids hybridize. 抗体结合，酶催化，互补核酸杂交。 → B01 |
| **Optical • electrochemical • mechanical 光学•电化学•机械** | Classify the transducer by the measured physical change. 按测到的物理变化判断换能类别。 → B01 |
| **Sensitivity • selectivity 灵敏度•选择性** | Sensitivity is calibration slope; selectivity concerns interferents. 灵敏度是校准斜率，选择性针对干扰物。 → C02 |
| **Limit of detection, LOD 检测限** | LOD=3σ/\|m\| combines blank noise with linear calibration slope. LOD=3σ/\|m\|同时考虑空白噪声与线性校准斜率。 → C02 |
| **Linearity • response time • reproducibility 线性度•响应时间•再现性** | These describe range, speed and consistency; they do not alone establish accuracy. 分别描述范围、速度和一致性，不能单独证明准确。 → B02 |
| **POCT • microfluidics 即时检测•微流控** | POCT describes test location; microfluidics organizes small-volume sample handling. POCT描述检测地点，微流控组织少量液体处理。 → B03 |
| **Minimum integer count 最小整数数量** | Round upward only when a minimum integer count must reach a continuous threshold. 仅当最小整数数量必须达到连续阈值时，向上取整。 → C03 |

**EN:** Formula links for this chapter: FX101 Calibration sensitivity and concentration LOD, FX102 Hall voltage.

**中文：** 本章公式定位：FX101 校准灵敏度与浓度检出限；FX102 霍尔电压。

## C01 • [P2] Sensor, transducer and biosensor / 传感器、换能器与生物传感器

Priority / 复习标记：P2 Important foundations/重要基础 · Repeated explanation/反复讲解

**EN: Key points**

- A sensor responds to a physical or chemical input.
- A transducer converts a recognition event or another input into a measurable signal.
- A biosensor combines a biological recognition element with a transducer;
- the recognition element gives specificity, while the transducer produces the readout.

**中文：必会知识点**

- Sensor响应物理或化学输入；
- transducer把识别事件或其他输入转为可测信号。
- Biosensor由生物识别元件与换能器组成：识别元件赋予特异性，换能器产生读出信号。

**EN:** Start from zero: A sensor helps us measure something that our eyes cannot reliably count, such as glucose in a drop of blood. The analyte is the thing we want to measure. A biosensor uses a biological recognition element, called a bioreceptor, to interact with that target. Recognition alone is not yet a number on a screen.

**中文：** 从零理解：传感器帮助我们测量肉眼无法可靠数清的东西，例如一滴血中的葡萄糖。Analyte（分析物）就是我们想测的目标。Biosensor（生物传感器）用一个生物识别元件，即bioreceptor（生物受体），与目标相互作用。但发生了识别，还不等于屏幕上已经有数值。

**EN:** Follow the mechanism: Think of an antibody as a selective catcher and the transducer as a translator. First, the catcher binds its target. Next, that event changes light, electrical current, or another measurable property. Finally, electronics convert the change into a result. The catcher analogy only describes selective binding; an antibody does not consciously identify a substance.

**中文：** 跟着机制走：可以把antibody（抗体）想成有选择性的捕捉器，把transducer（换能器）想成翻译器。第一步，捕捉器结合目标；第二步，这个事件引起光、电流或其他可测性质的变化；最后，电子系统把变化转换成结果。捕捉器类比只帮助理解选择性结合，抗体并不会有意识地识别物质。

**EN:** Try an example: Teaching example: a glucose sensor uses an enzyme to help react with glucose and an electrode to produce an electrical reading. Glucose is the analyte; the enzyme is the recognition element; the electrode belongs to the transduction system. In a question, ask three separate questions: What is measured? What recognizes it? What produces the signal?

**中文：** 用例子理解：教学例子：葡萄糖传感器用enzyme（酶）帮助葡萄糖发生反应，再用electrode（电极）产生电读数。葡萄糖是分析物，酶是识别元件，电极属于换能系统。做题时分开问三个问题：测的是什么？谁识别它？谁把识别变成信号？

**EN:** Exam wording: Do not confuse the analyte (the target being measured), the bioreceptor (what recognizes it), and the transducer (what converts recognition into a signal).

**中文：** 英文读题抓手：不要混淆analyte（被测目标）、bioreceptor（识别目标的元件）和transducer（把识别转成信号的元件）。

Source / 来源：W1 PDF p13–20; W1 PDF p39–42; W2 PDF p6–12 | T1-0017–0021, T1-0033–0034, T2-0004–0009

## B01 • [P2] Biorecognition and sensor families / 生物识别与传感器类别

Priority / 复习标记：P2 Important foundations/重要基础 · Repeated explanation/反复讲解

**EN: Key points**

• Antibodies bind antigens;
• enzymes catalyse reactions;
• complementary nucleic acids hybridize.
• Recognition supplies selectivity, while the transducer supplies readout.
• Optical sensors read light, electrochemical sensors read electrical changes, and mechanical sensors can read mass-induced displacement or frequency shifts.
• AI interprets measured data;
• it does not replace the physical sensing step.

**中文：必会知识点**

• Antibody抗体结合抗原；
• enzyme酶催化反应；
• 互补核酸通过hybridization杂交识别。
• 识别环节提供选择性，换能器提供读出。
• 光学传感测光，电化学传感测电变化，机械传感可测质量引起的位移或频移。
• AI分析测得的数据，不能替代实际感测。

**EN:** Start from zero: Different targets need different recognition tools. An antibody binds a matching molecular feature; an enzyme helps a particular chemical reaction occur faster; a nucleic-acid strand binds a sufficiently complementary sequence. Catalysis means speeding a reaction without being consumed overall. Hybridization means complementary strands pairing. These words describe recognition processes, not whether the final reading is optical, electrical, or mechanical.

**中文：** 从零理解：不同目标需要不同识别工具。Antibody（抗体）结合匹配的分子特征，enzyme（酶）帮助某类化学反应更快发生，nucleic-acid strand（核酸链）与足够互补的序列结合。Catalysis（催化）指加快反应，而自身不被总体消耗；hybridization（杂交）指互补链配对。这些词描述识别过程，并没有决定最终读数是光学、电学还是机械。

**EN:** Follow the mechanism: After recognition, a transducer needs to turn the event into a measurable change. Optical systems may measure color or light intensity. Electrochemical systems may measure current or potential. Mechanical systems may respond when added molecules change mass or motion. Think of different doorbells announcing the same visitor. The biological recognition is the visitor check; the transducer is the announcement mechanism, so they can be combined in different ways.

**中文：** 跟着机制走：识别之后，换能器需要把这个事件变成可测变化。光学系统可以测颜色或光强，电化学系统可以测电流或电位，机械系统可以响应分子附加后造成的质量或运动变化。可以类比不同门铃都能报告同一位访客：生物识别像核查访客，换能器像发出通知的装置，所以两者可以有不同组合。

**EN:** Try an example: Teaching example: an antibody-coated surface can be read optically or electrically. The antibody alone does not determine the sensor family. Ask what physical quantity is measured. AI can help classify or interpret the measured data, but it still needs data from a physical measurement. In a question, do not treat an analysis algorithm as a replacement for sampling, recognition, and signal generation.

**中文：** 用例子理解：教学例子：涂有抗体的表面可以用光学方法读出，也可以用电学方法读出。仅有抗体并不能确定传感器类别，要看测的物理量是什么。AI可以帮助分类或解释测量数据，但仍需要来自实际测量的数据。做题时，不能把分析算法当成采样、识别和信号产生的替代品。

Source / 来源：W1 PDF p1–3; W1 PDF p21–31; W1 PDF p39–42; W1 PDF p43–44; W1 PDF p45–49; W1 PDF p50–57; W1 PDF p68–79 | T1-0001–0004, T1-0022–0027, T1-0033–0052, T1-0065–0072

## S101 • [P3] Recognize sensor families from their input / 从输入辨认传感器类别

Priority / 复习标记：P3 Supporting reading/辅助阅读 · Supporting concept/辅助理解 · Lower priority/降低优先级

**EN:** A sensor family describes what it senses: magnetic field, chemicals, sound, temperature, electric quantities, mechanical quantities, or light. A magnetic Hall sensor produces a transverse Hall voltage related to magnetic field; with other factors fixed, V_H is proportional to B. A thermal sensor can infer temperature from another changing property, such as resistance. Measuring an electrical output does not automatically make the original stimulus electrical.

**中文：** 传感器类别描述它感受什么：磁场、化学物、声音、温度、电量、机械量或光。磁性Hall sensor（霍尔传感器）产生与磁场有关的横向Hall voltage（霍尔电压）；其他条件固定时，V_H与B成正比。Thermal sensor（温度传感器）可通过电阻等随温度改变的性质间接测温。最后输出电信号，并不表示最初感受的输入一定是电量。

**EN:** SONAR uses sound; RADAR uses electromagnetic radio waves; LiDAR uses laser light. In an ideal reflected time-of-flight measurement, distance is wave speed times round-trip delay divided by two. Pressure is force per area, whereas a force sensor measures force. A gas sensor reports gas composition or amount through an optical or electrical response; it is a biosensor only when it includes the relevant biological recognition element.

**中文：** SONAR（声呐）用声音；RADAR（雷达）用无线电电磁波；LiDAR（激光雷达）用激光。在理想反射飞行时间测量中，距离=波速×往返延时÷2。Pressure（压力）是单位面积上的力，force sensor（力传感器）测的是力。Gas sensor（气体传感器）以光或电响应报告气体成分或含量；有相应生物识别元件时，才能按生物传感器定义归类。

Source / 来源：Week 1 PDF p20–30; T1-0022–0027

## S102 • [P3] Static and dynamic microcantilever sensing / 微悬臂梁的静态与动态传感

Priority / 复习标记：P3 Supporting reading/辅助阅读 · Supporting concept/辅助理解

**EN:** A microcantilever is a tiny beam fixed at one end and free at the other. In static-mode sensing, molecular binding can change surface stress and bend the beam; the readout is deflection. In dynamic-mode sensing, the beam vibrates and a change in attached mass or mechanical properties can shift its resonance frequency. Deflection and frequency are different measured quantities.

**中文：** Microcantilever（微悬臂梁）是一端固定、另一端自由的小梁。Static mode（静态模式）中，分子结合可以改变表面应力，使小梁弯曲，读出deflection（偏转量）。Dynamic mode（动态模式）中，小梁振动，附加质量或机械性质变化可以改变resonance frequency（共振频率）。偏转与频率是两种不同的被测量。

**EN:** Under a simple fixed-stiffness oscillator model, adding mass lowers the resonance frequency. If binding also changes stiffness or stress, the shift needs the appropriate model. The receptor still supplies target recognition: a frequency shift alone does not identify a particular virus, and mechanical sensing is not automatically selective.

**中文：** 在刚度固定的简单振子模型下，增加质量会降低共振频率。如果结合还改变刚度或应力，就需要相应模型判断位移方向。受体仍负责识别目标：仅凭频移，不能认定某一种病毒存在；机械传感也不会自动具有选择性。

Source / 来源：Week 1 PDF p47; T1-0041–0042

## B03 • [P1] POCT, microfluidics and the sensing workflow / 现场检测、微流控与检测流程

Priority / 复习标记：P1 First pass/先掌握 · Sample Q9/样卷Q9 · Repeated explanation/反复讲解 · Supporting concept/辅助理解

**EN: Key points**

- Point-of-care testing (POCT) brings testing near the patient;
- centralized testing sends samples to a laboratory.
- Sampling → recognition → transduction → analysis is the basic workflow.
- Microfluidic biochips manipulate small fluid volumes and support integration and automation;
- wearable devices support repeated or continuous monitoring.
- Portability alone does not guarantee better analytical performance.

**中文：必会知识点**

- Point-of-care testing（POCT，即时/现场检测）把检测带到患者附近；
- centralized testing把样品送到中心实验室。
- 基本流程为采样→识别→换能→分析。
- Microfluidic biochip微流控芯片操控少量液体，便于集成与自动化；
- wearable可穿戴设备支持重复或连续监测。
- 便携本身不保证分析性能更好。

**EN:** Start from zero: Point-of-care testing, POCT, means testing near the person receiving care, such as a bedside test. Centralized testing sends the sample to a laboratory. A microfluidic chip handles liquid in very small channels and can organize several steps on one device. A wearable stays on the body to collect repeated or continuous information. These terms describe location, fluid handling, or monitoring format, rather than one universal detection mechanism.

**中文：** 从零理解：Point-of-care testing（POCT，即时检测）指在接受照护的人附近检测，例如床旁检测；centralized testing（中心化检测）则把样品送到实验室。Microfluidic chip（微流控芯片）在极小通道中处理液体，能把几个步骤安排在一个器件上。Wearable（可穿戴设备）佩戴在身上，收集重复或连续信息。这些词分别描述检测位置、液体处理或监测形式，并不代表某一种统一检测机制。

**EN:** Follow the mechanism: Follow the complete route: collect the sample, bring it to the recognition region, generate a measurable change, and interpret the reading. A chip is like a miniature plumbing system, with channels routing small fluid volumes to different stations. That analogy explains transport, not automatic selectivity. The chip still needs suitable recognition and readout components; making channels smaller does not itself prove a lower detection limit.

**中文：** 跟着机制走：沿完整路线理解：收集样品，把它送到识别区域，产生可测变化，再解释读数。芯片像一个微型管道系统，用通道把少量液体送往不同工作站。这个类比解释运输，并不代表芯片自动拥有选择性。它仍需要合适的识别与读出元件，把通道做小本身不能证明检测限更低。

**EN:** Try an example: Teaching example: a portable cartridge accepts a drop of blood, separates fluid, mixes reagents, and reads a signal. It combines microfluidics with sensing and can support POCT. Portability may reduce travel and waiting, but analytical quality must still be checked. A smartwatch provides ongoing measurements, while a single rapid-test strip usually provides one result.

**中文：** 用例子理解：教学例子：便携检测盒接收一滴血，分离液体、混合试剂，再读取信号。它把微流控与传感结合起来，可以支持POCT。便携可能减少运输和等待，但分析质量仍要检查。智能手表提供持续测量，单次快检试纸通常提供一次结果。

Source / 来源：W1 PDF p32–36; W1 PDF p63–66; W6 p3–12; Sample quiz p2 Q8–9; W4 p35–38, p73–86; W3 p25–32 | T1-0028–0031, T1-0059–0061, T6-0001–0010, T7-0055–0057

## S103 • [P3] Electronic noses and sensing-to-action systems / 电子鼻与从感测到行动的系统

Priority / 复习标记：P3 Supporting reading/辅助阅读 · Supporting concept/辅助理解 · Repeated explanation/反复讲解

**EN:** An electronic nose exposes a group of chemical sensors to volatile compounds and reads their response pattern. A computer extracts features and classifies the pattern, much as a combination of responses distinguishes an aroma. It measures chemical information, not sound or EEG. Electronic readout is convenient for integrating processing and wireless communication such as Bluetooth; this advantage concerns system integration, not a guarantee of greater selectivity.

**中文：** Electronic nose（电子鼻）让一组化学传感器接触volatile compounds（挥发性化合物），读取它们的响应模式。电脑提取特征并分类，依靠一组响应区分气味。它测化学信息，不是声音或EEG。电读出便于集成处理与Bluetooth（蓝牙）等无线通信；这属于系统集成优势，不保证选择性一定更高。

**EN:** A monitoring system records and reports a signal. A closed-loop system also uses the measured signal to decide an action, then measures the result again. The illustrated implantable example links neural recording, processing and stimulation. Recording is sensing; stimulation or drug delivery is an action by an actuator. A wearable describes where a device is worn, while implantable means it is placed inside the body.

**中文：** Monitoring（监测）系统记录并报告信号；closed-loop（闭环）系统还利用测得的信号决定行动，再重新测量行动结果。课件的植入示例连接神经记录、处理和刺激。Recording（记录）是感测，stimulation（刺激）或drug delivery（给药）是actuator（执行器）的行动。Wearable描述佩戴形式，implantable描述器件放在身体内部。

Source / 来源：Week 1 PDF p69–75; T1-0065–0070

## C02 • [P1] Sensitivity, selectivity and limit of detection / 灵敏度、选择性与检测限

Priority / 复习标记：P1 First pass/先掌握 · Quiz stated/Quiz明确提及 · Teacher emphasis/课堂强调 · Sample Q10, Q12/样卷Q10、Q12

**EN: Key points**

- For a linear calibration y=mC+b, sensitivity is the slope m.
- Selectivity means discriminating the target from interfering species.
- Under the course convention, LOD=3σ/|m|, where σ is the standard deviation of the background response.
- Holding the other factor fixed, a larger absolute slope or smaller background noise lowers the LOD.

**中文：必会知识点**

- 线性校准y=mC+b中，灵敏度是斜率m。
- 选择性是区分目标与干扰物的能力。
- 本课程采用LOD=3σ/|m|，σ是背景响应的标准差。
- 在另一因素固定时，斜率绝对值更大或背景噪声更小，才会降低检测限。

**EN:** Start from zero: A calibration curve connects a known target concentration to a measured output. Its slope tells us how much the output changes when concentration increases by one unit; this is sensitivity. Selectivity asks a different question: does another substance also produce a response? The limit of detection, or LOD, asks how small a target amount we can distinguish from background.

**中文：** 从零理解：Calibration curve（校准曲线）把已知的目标浓度与测得的输出联系起来。它的slope（斜率）告诉我们浓度增加一个单位时，输出变化多少，这叫sensitivity（灵敏度）。Selectivity（选择性）问的是另一件事：其他物质会不会也引起响应？Limit of detection，即LOD（检测限），则问多小的目标量还能与背景区分。

**EN:** Follow the mechanism: Even a blank sample can give slightly different readings each time. These unwanted fluctuations are noise. Imagine hearing a quiet voice in a noisy room: a louder voice helps, but so does a quieter room. Similarly, LOD=3σ/|m| combines background variation σ with calibration slope m. A steep slope alone does not prove a low LOD.

**中文：** 跟着机制走：即使是blank sample（空白样品），每次读数也可能稍有不同，这种不需要的波动叫noise（噪声）。想象在嘈杂房间里听很轻的说话声：声音更大有帮助，房间更安静也有帮助。同样，LOD=3σ/|m|同时考虑背景波动σ和校准斜率m。只知道斜率陡，不能证明检测限低。

**EN:** Try an example: Teaching example: let m=10 output units per mM and σ=2 output units. Then LOD=6/10=0.6 mM. If a second sensor has the same slope but twice the noise, its LOD becomes 1.2 mM. For a comparison question, look for both slope and noise. Lower LOD means smaller amounts can be detected; it does not guarantee selectivity.

**中文：** 用例子理解：教学例子：设m=10输出单位/mM，σ=2输出单位，则LOD=6/10=0.6 mM。若另一传感器斜率相同，但噪声加倍，LOD就变为1.2 mM。比较题要同时寻找斜率和噪声。检测限更低，表示能检测更少的量，并不自动保证选择性。

**EN:** Clarify the concept: Matrix effect means that other components of the sample affect the measured response. A calibration slope obtained in pure water need not match the slope in groundwater or another complex sample. Compare sensors and calculate LOD using a calibration and noise estimate appropriate to the sample conditions. A high slope is sensitivity; distinguishing the target from interferents is selectivity.

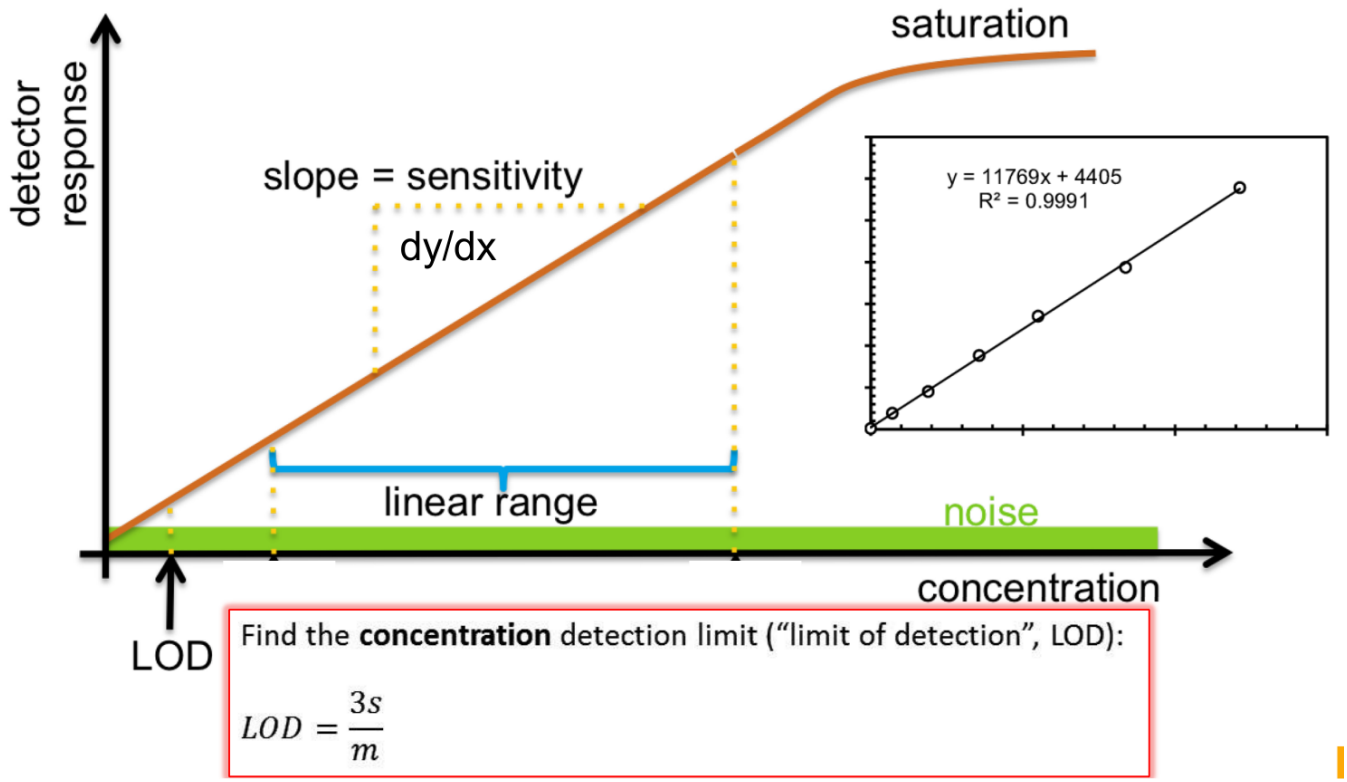
**中文：** 把概念讲清：Matrix effect（基质效应）指样品中其他成分影响测量响应。纯水中的校准斜率不一定适用于地下水或其他复杂样品。比较传感器和计算LOD时，应使用与样品条件匹配的校准及噪声估计。斜率大属于灵敏度；区分目标与干扰物属于选择性。

Source / 来源：Week 1 PDF p64 original image: perchlorate response differs between reagent water and groundwater; T1-0053–0061

**EN:** Exam wording: Use the linear region rather than a saturated endpoint. A low LOD means better detection capability. Diagnostic sensitivity TP/(TP+FN) is a different meaning of sensitivity; do not substitute it for calibration slope.

**中文：** 英文读题抓手：用线性区计算，不能把饱和点纳入斜率。LOD越小，检测能力越好。诊断学sensitivity=TP/(TP+FN)是另一个含义，不能代替校准斜率。

Source / 来源：W1 PDF p58–61; W1 PDF p60–62; W6 p49 | T1-0053–0058, T6-0050–0053




One calibration curve separates sensitivity, linear range, noise and LOD. / 用一张校准曲线区分灵敏度、线性范围、噪声与检出限。

**EN:** Read the figure: Read concentration on the horizontal axis and response on the vertical axis. The orange slope is sensitivity; the blue bracket is the range where one constant slope is useful. The green noise band explains why a small response may be indistinguishable from the blank. With the course criterion LOD = 3σ/m, a steeper slope or smaller blank noise lowers LOD; the saturated section cannot use the same linear calibration.

**中文：** 读图：横轴是浓度，纵轴是响应。橙线斜率表示Sensitivity（灵敏度）；蓝括号内才适合用同一个固定斜率校准。绿色Noise（噪声）带说明：信号太小时可能与空白分不清。按本课LOD = 3σ/m判据，斜率更大或空白噪声更小都会降低LOD；进入Saturation（饱和）段后不能沿用原来的线性校准。

Source / 来源：Week 1 PDF p61; original slide figure / 原课件截图

## B02 • [P2] Performance beyond sensitivity / 灵敏度之外的性能指标

Priority / 复习标记：P2 Important foundations/重要基础 · Supporting concept/辅助理解 · Lower priority/降低优先级

**EN: Key points**

- Linearity describes how closely output follows a straight calibration line.
- Response time describes how quickly output approaches its final value (95% in these slides).
- Reproducibility describes agreement across repeated measurements.
- Selectivity concerns interferents;
- sensitivity is calibration slope;
- LOD combines slope with noise.
- None of these terms alone means overall accuracy.

**中文：必会知识点**

- Linearity线性度描述输出接近直线校准关系的程度；
- response time响应时间描述接近最终值的速度（课件取95%）；
- reproducibility再现性描述重复测量的一致程度。
- Selectivity选择性针对干扰物；
- sensitivity灵敏度是校准斜率；
- LOD兼顾斜率与噪声。
- 任何一个指标都不单独等于整体准确性。

**EN:** Start from zero: A useful sensor needs more than a large output. Linearity asks whether equal concentration increases give approximately equal output increases over a stated range. Response time asks how quickly the reading settles after the target changes. Reproducibility asks how well repeated measurements agree under stated conditions. These describe the shape, speed, and consistency of measurement, so none can be substituted for another.

**中文：** 从零理解：有用的传感器不只需要输出大。Linearity（线性度）问的是：在规定范围内，浓度每次等量增加，输出是否也近似等量增加？Response time（响应时间）问目标改变后读数多久稳定？Reproducibility（再现性）问规定条件下重复测量是否一致？它们分别描述测量关系的形状、速度和一致性，不能互相代替。

**EN:** Follow the mechanism: Think of a bathroom scale. It might repeatedly show the same wrong weight: good consistency, poor accuracy. It might be accurate after a long wait: reasonable accuracy, slow response. For the course's 95% response-time definition, the clock stops when output reaches 95% of its final change, not necessarily when all fluctuation disappears. A biosensor's delays also involve molecular transport and reactions.

**中文：** 跟着机制走：想象一台体重秤：它可能每次都显示同一个错误体重，说明一致性好，但准确性差；也可能等很久后才显示正确值，说明准确性尚可，但响应慢。课程的95%响应时间定义，是输出达到最终变化的95%时停止计时，而不是必须等所有波动完全消失。生物传感器的延迟还涉及分子传输与反应。

**EN:** Try an example: Teaching example: a reading starts at 0 and finally reaches 100 units. The 95% response point is 95 units. If a graph bends toward a plateau, use its straight region for calibration rather than assuming constant sensitivity everywhere. When asked which sensor is better, inspect the task: continuous monitoring may favor speed, while detecting tiny amounts also needs low noise and an appropriate slope.

**中文：** 用例子理解：教学例子：读数从0开始，最终到100单位，95%响应点就是95单位。如果图线弯曲并趋向平台，校准时应使用直线区域，不要假设所有位置的灵敏度都相同。问哪种传感器更好时，要看任务：连续监测可能更重视速度，而检测极少量还需要低噪声和合适的斜率。

Source / 来源：W1 PDF p58–61; W1 PDF p60–62; W1 PDF p63–66 | T1-0053–0061

## C03 • [P1] Integer detection threshold: corrected sample Q10 / 整数检测阈值：修正样题Q10

Priority / 复习标记：P1 First pass/先掌握 · Sample Q10/样卷Q10 · Teacher emphasis/课堂强调 · Worked example/例题练习

**EN: Key points**

- In the sample graph, the linear interval is 20–80 viruses: m=(700−100)/(80−20)=10 pF/virus.
- With σ=14 pF, LOD=4.2 viruses.
- The lecturer explicitly rejects 4 and chooses the minimum integer count that reaches the threshold: 5, option B.

**中文：必会知识点**

- 样题图线性区为20–80个病毒：m=(700−100)/(80−20)=10 pF/virus。
- σ=14 pF时，LOD=4.2个病毒。
- 老师明确否定4，并选择达到阈值的最小整数数量：5，对应B。

**EN:** Start from zero: The sample question measures a number of viruses, so the final answer must be a whole number. However, the LOD formula first gives a continuous threshold. A threshold of 4.2 viruses means the required signal corresponds to more than four viruses under the stated model. Four is below that threshold, even though it is numerically closer than five.

**中文：** 从零理解：样题测的是病毒个数，所以最后答案必须是整数。但是LOD公式先算出来的是连续阈值。4.2个病毒表示：在题目采用的模型下，需要的信号对应超过4个病毒。4虽然在数值上比5更接近4.2，但仍低于检测阈值。

**EN:** Follow the mechanism: Use only the straight part of the graph to calculate slope. Here, output rises from 100 to 700 pF while virus count rises from 20 to 80, giving m=600/60=10 pF per virus. With background standard deviation σ=14 pF, the required output change is 3σ=42 pF. Divide that change by the slope: 42/10=4.2 viruses.

**中文：** 跟着机制走：先只用图中的直线部分求斜率。输出从100增加到700 pF时，病毒数从20增加到80，因此m=600/60=10 pF/个病毒。背景标准差σ=14 pF，需要达到的输出变化为3σ=42 pF。再用这个变化除以斜率，得到42/10=4.2个病毒。

**EN:** Try an example: A helpful analogy is needing at least 4.2 units of money: four whole coins cannot pay the bill; five can. This is rounding upward to meet a minimum, not rounding to the nearest integer. Keep 0.42 μM as 0.42 μM when concentration is continuous. The sample uses the slope as a model near the threshold; experimental validation would require checking that assumption.

**中文：** 用例子理解：可以类比成需要至少4.2单位的钱：4枚整硬币不够，5枚才够。这是为了达到最低要求而向上取整，不是四舍五入到最近整数。如果测的是连续浓度，0.42 μM就保留0.42 μM。样题把该斜率作为阈值附近的模型使用，真实实验还需要检查这个假设是否成立。

**EN:** Exam wording: The previous guide incorrectly used the nearest option, A=4. This edition corrects it to B=5. Round up only when the question asks for a minimum detectable integer count; a concentration LOD such as 0.42 μM stays continuous.

**中文：** 英文读题抓手：旧版错误地选了最近选项A=4，本版改为B=5。只有题目要求“最小可检测整数数量”时才向上取整；0.42 μM这样的浓度检测限仍保留连续数值。

Source / 来源：Sample quiz p2 Q10; W1 p59–62 | T7-0057–0060

# Week 2 learning block: Light and optical signals / 第2学习组：光与光学信号

## Week 2 concept map / 第2周知识框架

**EN:** Chapter logic: Match a light source to the sample, follow how light interacts with it, and distinguish the measured optical quantities.

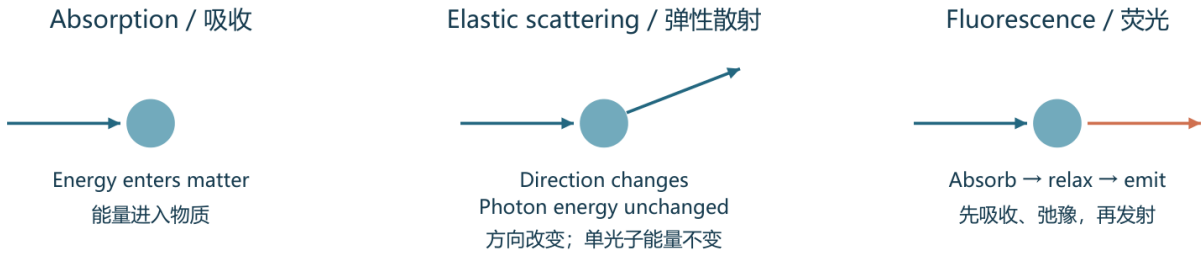
**中文：** 本章主线：先让光源匹配样品，再追踪光怎样作用，最后区分实际测得的光学量。

| English term 中文名词 | Meaning and connection 含义与知识联系 |
|---|---|
| **Ray • wave • photon 光线•波•光子** | Paths, interference and energy exchange call for different light models. 路径、干涉与能量交换使用不同光模型。 → B04 |
| **Photon energy • photon rate 光子能量•光子速率** | $E=hc/\lambda$ describes one photon; P/E gives photons per second. $E=hc/\lambda$描述一个光子，P/E得到每秒光子数。 → S105 |
| **Coherence • noise • power 相干性•噪声•功率** | Select suitable phase stability, noise and power; brighter is not automatically better. 相位稳定、噪声和功率须合适，越亮不一定越好。 → S104 |
| **Absorption • scattering • attenuation 吸收•散射•衰减** | Direct-beam attenuation can contain both absorption and redirection losses. 直达光衰减可同时包含吸收与改变方向造成的损失。 → B05 |
| **Absorbance • transmittance 吸光度•透射率** | $A=-\log_{10}T=\varepsilon CL$; compare at the same wavelength and matching units. $A=-\log_{10}T=\varepsilon CL$；同波长比较，并匹配单位。 → C04 |
| **Pulse oximetry 脉搏血氧测量** | Two wavelengths and the pulsatile component help estimate arterial saturation. 两波长与脉动分量帮助估计动脉血氧饱和度。 → B06 |
| **Fluorescence • filters • Φ • τ 荧光•滤光片•产率•寿命** | Match excitation and emission bands; Φ is emitted/absorbed and $k_r=\Phi/\tau$. 匹配激发/发射波段；Φ=发射/吸收，$k_r=\Phi/\tau$。 → C05 |
| **Raman • Rayleigh • spectral resolution 拉曼•瑞利•光谱分辨** | Raman exchanges vibration energy; Rayleigh is elastic; grating $R=mN$ separates wavelengths. 拉曼交换振动能，瑞利为弹性；光栅$R=mN$描述分开波长的能力。 → C07 |

**EN:** Formula links for this chapter: FX103 Optical wave, frequency, wavelength and phase, FX104 Photon energy, total energy and photon rate, FX105 Photon momentum, FX106 Energy fractions: reflection, absorption and transmission, FX107 Direct-beam attenuation and characteristic distances, FX108 Beer–Lambert: transmission and two logarithm conventions, FX109 Two-wavelength hemoglobin mixture, FX110 Oxygen saturation and total hemoglobin, FX111 Pulse-oximeter example: ratio of ratios, FX112 Raman shift and absolute wavenumber, FX113 Fluorescence lifetime and exponential decay, FX114 Quantum yield, radiative and nonradiative rates, FX115 Molecular fluorescence brightness, FX116 Grating resolving power and illuminated grooves.

**中文：** 本章公式定位：FX103 光波、频率、波长与相位；FX104 单光子能量、总能量与光子率；FX105 光子动量；FX106 反射、吸收与透射能量比例；FX107 直达光衰减与特征距离；FX108 Beer–Lambert：透过率与两种对数约定；FX109 双波长血红蛋白混合关系；FX110 血氧饱和度与总血红蛋白；FX111 血氧仪例题：比值之比；FX112 拉曼位移与绝对波数；FX113 荧光寿命与指数衰减；FX114 量子产率、辐射与非辐射速率；FX115 分子荧光亮度指标；FX116 光栅分辨本领与受照刻线数。

Three different light–sample processes / 三种光与样品作用




Light processes / 光学作用教学示意

## B04 • [P2] Light models, sources and coherence / 光的模型、光源与相干性

Priority / 复习标记：P2 Important foundations/重要基础 · Teacher emphasis/课堂强调

**EN: Key points**

- Rays describe paths, waves explain interference/diffraction, and photons describe energy exchange.
- Lasers commonly offer narrower spectra and greater coherence than LEDs or lamps;
- CW means continuous-wave, while pulsed means time-separated emission.
- Coherence is a stable phase relationship.
- Source power and noise affect usable signal;
- more power is not always better because saturation or photodamage can occur.

**中文：必会知识点**

- Ray光线模型描述路径，wave波模型解释干涉/衍射，photon光子模型描述能量交换。
- 激光通常比LED或灯具具有更窄光谱和更高相干性；
- CW为连续波，pulsed为脉冲发射。
- Coherence相干性指稳定相位关系。
- 光源功率与噪声影响有效信号；
- 功率过大可能饱和或光损伤，并非越大越好。

**EN:** Start from zero: Light can be described in several useful ways. Rays trace where light travels and help explain lenses. Waves have repeating peaks and help explain interference, where waves combine, and diffraction, where light spreads around openings. Photons describe small packets of energy exchanged with matter. These are models for different questions, not three unrelated substances. Choose the model that explains the measured effect.

**中文：** 从零理解：光可以用几种有用模型描述。Ray（光线）追踪传播路径，帮助解释透镜；wave（波）有重复的波峰，帮助解释波相加的interference（干涉）以及光绕开口扩展的diffraction（衍射）；photon（光子）描述与物质交换的一小份能量。这是针对不同问题的模型，不是三种互不相关的物质，要选择能解释被测效应的模型。

**EN:** Follow the mechanism: Coherence means a sufficiently stable relationship between wave phases. Phase tells us where a wave is within its repeating cycle. Compare people stepping together: their steps have a predictable relationship. Light waves have no intention, but a stable phase relationship makes interference predictable. Lasers commonly have narrower wavelength ranges and greater coherence than LEDs or lamps. This is a typical comparison, not a guarantee for every device.

**中文：** 跟着机制走：Coherence（相干性）表示波的相位之间有足够稳定的关系。Phase（相位）告诉我们一个波处于重复周期中的哪个位置。可以类比步伐整齐的人，他们的动作之间存在可预测关系；光波没有意识，但稳定相位关系使干涉更可预测。激光通常比LED或灯具具有更窄的波长范围和更高相干性，但并不是每个器件都绝对如此。

**EN:** Try an example: Teaching example: interference sensing needs suitable coherence, while fluorescence excitation needs an appropriate wavelength and enough power. CW, continuous wave, means sustained emission; pulsed means emission separated in time. More power may increase a useful signal, but can also cause saturation or damage. In questions, separate wavelength, timing, coherence, and power rather than calling all of them simply 'stronger light'.

**中文：** 用例子理解：教学例子：干涉传感需要合适的相干性，荧光激发则需要合适波长和足够功率。CW，即continuous wave（连续波），表示持续发光；pulsed（脉冲）表示发光在时间上分开。功率提高可能增强有用信号，也可能造成饱和或损伤。做题时，把波长、时间方式、相干性和功率分开，不要全部理解成“光更强”。

Source / 来源：W2 PDF p13–22; W2 PDF p21–28 | T2-0010–0027

## S104 • [P2] Wave terminology, coherence and source noise / 波的术语、相干性与光源噪声

Priority / 复习标记：P2 Important foundations/重要基础 · Teacher emphasis/课堂强调

**EN:** For the wave $E(z,t)=E_0\cos(\omega t-kz)$, $E_0$ is field amplitude, $\omega=2\pi f$ is angular frequency, and $k=2\pi/\lambda$ is angular wavenumber. Raman spectra commonly use the different wavenumber convention $\nu_tilde=1/\lambda$. Phase specifies position within an oscillation; polarization describes electric-field orientation, not propagation direction. Keep this k separate from a fluorescence rate constant also called k.

**中文：** 波动表达式$E(z,t)=E_0\cos(\omega t-kz)$中，$E_0$是场振幅，$\omega=2\pi f$是angular frequency（角频率），$k=2\pi/\lambda$是角波数。拉曼图常用另一种波数约定$\nu_tilde=1/\lambda$。Phase（相位）说明振荡位置；polarization（偏振）描述电场方向，不是传播方向。这里的k也不同于同样可能叫k的荧光速率常数。

**EN:** Spatial coherence concerns the phase relationship between different positions across a light field. Temporal coherence concerns how long a predictable phase relationship persists over time; a narrower spectrum commonly gives a longer coherence time. They are independent aspects: the slide shows a superluminescent diode, SLD, with relatively high spatial coherence and low temporal coherence. Coherence is not simply equal brightness everywhere.

**中文：** Spatial coherence（空间相干）关注光场不同空间位置之间的相位关系；temporal coherence（时间相干）关注相位关系随时间能保持多久，较窄光谱通常对应更长相干时间。两者是不同方面：课件中的SLD（superluminescent diode，超辐射发光二极管）具有相对较高的空间相干和较低的时间相干。相干不等于所有位置亮度都相同。

**EN:** Intensity noise changes signal height; wavelength or frequency noise moves its spectral position. The slides distinguish quantum noise, including spontaneous-emission contributions, from technical noise caused by pump/current fluctuations, temperature or vibration. A source should have suitable coherence, sufficiently low noise and adequate power. More power does not by itself correct wavelength drift or prevent sample saturation. RIN stands for relative intensity noise: it characterizes intensity fluctuations relative to the source’ s average optical power, not the mean power alone.

**中文：** Intensity noise（强度噪声）改变信号高度；wavelength/frequency noise（波长/频率噪声）改变光谱位置。课件区分quantum noise（量子噪声，包括自发辐射贡献）与technical noise（技术噪声，由泵浦/电流波动、温度或振动造成）。光源需要合适相干性、足够低噪声和适当功率；增大功率本身不能纠正波长漂移，也不能避免样品饱和。 RIN是relative intensity noise（相对强度噪声），表征相对于光源平均光功率的强度波动，不是平均功率本身。

Source / 来源：Week 2 PDF p17, p23–28; T2-0010–0012, T2-0019–0027

## S105 • [P1] Single-photon energy versus photon rate / 单光子能量与光子数量速率

Priority / 复习标记：P1 First pass/先掌握 · Teacher emphasis/课堂强调 · Worked example/例题练习 · Final emphasis/仅期末强调

**EN:** One photon has energy E=hc/λ. For N photons of the same wavelength, total energy is NE. Power P is energy per second, so photon arrival rate is N_dot=P/E=Pλ/(hc), with units photons/s. Convert watts to J/s and wavelength to metres when using SI h and c. The useful shortcut is E(eV)=1240/λ(nm); 1.24 is the coefficient when wavelength is in micrometres. A photon also has momentum: in vacuum p=h/λ. Longer wavelength gives smaller momentum per photon; transferring photon momentum to matter produces radiation pressure.

**中文：** 单个光子的能量E=hc/λ。同一波长的N个光子，总能量为NE。Power P（功率）是每秒能量，因此光子到达速率N_dot=P/E=Pλ/(hc)，单位photons/s。用SI单位h、c时，把瓦看作J/s，波长换成米。常用简式为E(eV)=1240/λ(nm)；波长用微米时，系数才是1.24。 光子也有momentum（动量），真空中p=h/λ；波长越长，单光子动量越小。光把动量传给物体时可产生radiation pressure（辐射压力）。

**EN:** At 410 nm, E is about $4.84\times10^{-19}$ J; 10 nW therefore corresponds to about $2.07\times10^{10}$ incident photons/s. At fixed power, longer-wavelength photons have less individual energy, so more photons arrive each second. Incident, absorbed and detected photon counts are different: absorption probability, quantum yield and detector efficiency can reduce the count at later stages.

**中文：** 410 nm时，E约为$4.84\times10^{-19}$ J；10 nW因此对应约$2.07\times10^{10}$个入射光子/秒。功率固定时，长波长光子的单个能量较少，所以每秒光子数较多。Incident（入射）、absorbed（吸收）和detected（检测）光子数不同：吸收概率、量子产率和探测效率可能使后续阶段的数量减少。

Source / 来源：Week 2 PDF p19–20, p80–81; T2-0013–0015

**EN:** Clarify the concept: In the particle model, a photon in vacuum also carries momentum p=h/λ. Absorption or reflection can transfer momentum to an object, so light can exert pressure. At the same vacuum wavelength, single-photon energy E=hc/λ and momentum p=h/λ are different quantities; P/E gives photons per second, not momentum.

**中文：** 把概念讲清：粒子模型中，真空中的光子还携带momentum（动量）p=h/λ。吸收或反射可以向物体传递动量，所以光能施加压力。同一真空波长下，单光子能量E=hc/λ与动量p=h/λ是不同量；P/E得到每秒光子数，不是动量。

Source / 来源：Week 2 PDF p19 (explicit Photon momentum and light pressure); T2-0013–0015

## B05 • [P2] Absorption, scattering and penetration / 吸收、散射与穿透

Priority / 复习标记：P2 Important foundations/重要基础 · Repeated explanation/反复讲解 · Lower priority/降低优先级

**EN: Key points**

- Absorption transfers optical energy to matter;
- scattering redirects light.
- Ballistic attenuation can include both: μt=μa+μs.
- Rayleigh and Mie scattering are elastic;
- their regimes depend on particle size relative to wavelength.
- Raman scattering is inelastic.
- Tissue penetration depends on both absorption and scattering;
- the near-infrared window is useful, but not loss-free.

**中文：必会知识点**

- Absorption吸收把光能传给物质；
- scattering散射改变光的传播方向。
- 直达光衰减可同时包含两者：μt=μa+μs。
- Rayleigh瑞利与Mie米氏散射都是弹性散射，其适用情形取决于颗粒相对波长的尺寸；
- Raman拉曼为非弹性。
- 组织穿透取决于吸收与散射；
- 近红外窗口有利于穿透，但并非没有损耗。

**EN:** Start from zero: If less light arrives straight behind tissue, it does not follow that all missing light was absorbed. Absorption transfers light energy to material. Scattering redirects light, so some can leave sideways or backward instead of reaching the straight-through detector. Ballistic light means light that follows the original path without scattering. Its attenuation can include both absorption and scattering: μt=μa+μs.

**中文：** 从零理解：如果组织正后方收到的光减少了，并不表示所有少掉的光都被吸收了。Absorption（吸收）把光能传给材料；scattering（散射）改变方向，因此部分光会向侧面或后方离开，无法到达直透探测器。Ballistic light（直达光）指没有被散射、沿原路径传播的光。其衰减可以同时包括吸收和散射：μt=μa+μs。

**EN:** Follow the mechanism: Imagine a hallway: some people stop inside rooms, while others turn into side passages. Both reduce the number reaching the far end, but for different reasons. Here μa describes absorption loss per distance and μs scattering loss per distance. Rayleigh and Mie are elastic scattering regimes, meaning photon energy is unchanged; particle size relative to wavelength matters. Raman differs because it exchanges energy with molecular vibrations.

**中文：** 跟着机制走：想象一条走廊：有些人停进了房间，另一些人拐进了岔路，两者都会减少到达末端的人数，但原因不同。这里μa描述单位距离的吸收损失，μs描述单位距离的散射损失。瑞利和米氏属于elastic scattering（弹性散射）情形，表示光子能量不变，颗粒尺寸与波长的相对大小很重要。拉曼不同，因为它与分子振动交换能量。

**EN:** Try an example: Teaching example: a cloudy suspension can strongly reduce a narrow transmitted beam even without strong absorption. A tissue's useful imaging depth therefore depends on scattering as well as absorption. A near-infrared window offers favorable losses for some tissues, not zero losses. If a question equates all attenuation with absorption, check whether scattering is being ignored. Scattered photons need not be destroyed.

**中文：** 用例子理解：教学例子：浑浊悬液即使吸收不强，也可能显著削弱一束狭窄的透射光。组织的有效成像深度因此既依赖散射，也依赖吸收。Near-infrared window（近红外窗口）对某些组织具有较有利的损耗，并不代表无损耗。如果题目把所有attenuation（衰减）都等同于吸收，要检查是否忽略了散射。散射光子并不一定被消灭。

Source / 来源：W2 PDF p29–33; W2 PDF p44–46; W2 PDF p55–62; W2 PDF p61–64 | T2-0028–0036, T2-0052–0055, T2-0070–0082

## S106 • [P2] Attenuation coefficients and characteristic distances / 衰减系数与特征距离

Priority / 复习标记：P2 Important foundations/重要基础 · Repeated explanation/反复讲解 · Lower priority/降低优先级

**EN:** For the directly transmitted, unscattered component in a uniform sample, $I(L)=I_0 \exp[-(\mu_a+\mu_s)L]$. The coefficients have units of inverse length, such as $cm^{-1}$; multiplying a coefficient by L gives a dimensionless exponent. Increasing either coefficient or path length reduces this direct component. This model does not describe every multiply scattered photon collected by every detector.

**中文：** 均匀样品中，直接透过、未经散射的光分量满足$I(L)=I_0 \exp[-(\mu_a+\mu_s)L]$。系数单位是长度的倒数，例如$cm^{-1}$；系数乘光程L后，指数无量纲。增大任一系数或光程，都会减少该直达分量。这个模型并不代表每种探测器收集到的全部多次散射光。

**EN:** The slide interprets $1/\mu_a$ and $1/\mu_s$ as characteristic mean distances for absorption and scattering, respectively. For total direct-beam attenuation, $1/\mu_t$ is the distance at which $I/I_0$ falls to 1/e. It is a scale of exponential decay, not a sharp border after which all light disappears. In R+A+T=1, A denotes an absorbed fraction; it is not the logarithmic absorbance A of Beer–Lambert notation.

**中文：** 课件把$1/\mu_a$和$1/\mu_s$分别解释为吸收与散射的特征平均距离。对直达光总衰减，$1/\mu_t$是$I/I_0$降到1/e的距离，这是指数衰减尺度，不是“过了这条边界光全没了”。在R+A+T=1中，A代表吸收比例；它不是Beer–Lambert记号里取对数的absorbance A（吸光度）。

Source / 来源：Week 2 PDF p30–33; T2-0034–0036

## C04 • [P1] Absorbance and Beer–Lambert law / 吸光度与比尔—朗伯定律

Priority / 复习标记：P1 First pass/先掌握 · Sample Q11/样卷Q11 · Worked example/例题练习 · Teacher emphasis/课堂强调

**EN: Key points**

- Absorbance is $A=-\log_{10}(I/I_0)=\varepsilon CL$.
- Transmittance is $T=I/I_0$, so $A=-\log_{10}T$.
- If C is in mol/L and L in cm, ε must be in L $mol^{-1}$ $cm^{-1}$.
- A=1 means 10% transmitted light;
- A=2 means 1%.

**中文：必会知识点**

- 吸光度$A=-\log_{10}(I/I_0)=\varepsilon CL$；
- 透射率$T=I/I_0$，所以$A=-\log_{10}T$。
- 若C用mol/L、L用cm，ε必须用L $mol^{-1}$ $cm^{-1}$。
- A=1代表透过10%，A=2代表透过1%。

**EN:** Start from zero: Shine light through a solution and compare the light before and after it. $I_0$ is the incoming intensity and I is the transmitted intensity. Transmittance $T=I/I_0$ is the fraction that gets through. Absorbance $A=-\log_{10}T$ expresses that loss on a logarithmic scale. Therefore, absorbance is not simply the percentage of light absorbed; the two quantities use different scales.

**中文：** 从零理解：让光穿过一杯溶液，比较进入前和穿出后的光强。$I_0$是入射光强，I是透射光强。Transmittance（透射率）$T=I/I_0$表示有多大比例的光穿过去。Absorbance（吸光度）$A=-\log_{10}T$用对数尺度表示这种损失。因此，吸光度不是直接的吸收百分比，两者使用不同的尺度。

**EN:** Follow the mechanism: Beer–Lambert law says $A=\varepsilon CL$ under its measurement assumptions. C is concentration, L is the distance through the sample, and ε describes how strongly the substance absorbs at the chosen wavelength. More absorbing molecules or a longer path give light more opportunities to be absorbed. Think of walking through a longer crowd, although molecules absorb light rather than physically blocking it like people.

**中文：** 跟着机制走：在相应测量假设下，比尔—朗伯定律为$A=\varepsilon CL$。C是浓度，L是光在样品中经过的距离，ε表示物质在所选波长处的吸收能力。吸收分子更多，或者路径更长，都使光有更多机会被吸收。可以类比走过更长的人群，但分子是在吸收光，并非像人一样直接挡住光。

**EN:** Try an example: Teaching example: T=0.1 gives A=1. Doubling concentration or path length doubles A to 2, so T becomes 0.01, not zero. This multiplication of transmission losses is why the logarithm is useful. Match units: concentration in mol/L and path in cm require ε in L $mol^{-1}$ $cm^{-1}$. Strong scattering or unsuitable concentrations can invalidate the simple absorption model.

**中文：** 用例子理解：教学例子：T=0.1时，A=1。浓度或光程加倍，A就加倍到2，于是T变为0.01，而不是零。透射损失之间这种相乘关系，正是对数方便的原因。单位要匹配：浓度用mol/L、光程用cm时，ε应为L $mol^{-1}$ $cm^{-1}$。强散射或不合适的浓度可能使简单吸收模型不再适用。

**EN:** Clarify the concept: Beer–Lambert uses $A=-\log_{10}T=\varepsilon CL$. The equivalent natural-exponential form is $T=\exp(-\varepsilon' CL)$, with $\varepsilon'=\ln(10)\varepsilon\approx2.303\varepsilon$. The two coefficients are not numerically identical. For the slide example $A_1=0.35$ at $L_1=2$ cm, changing only the path to $L_2=4.5$ cm gives $A_2=0.35\times4.5/2=0.7875\approx0.79$; keep intermediate precision before rounding.

**中文：** 把概念讲清：Beer–Lambert形式为$A=-\log_{10}T=\varepsilon CL$。等价的自然指数形式是$T=\exp(-\varepsilon' CL)$，其中$\varepsilon'=\ln(10)\varepsilon\approx2.303\varepsilon$，两个系数数值不同。课件例子$L_1=2$ cm、$A_1=0.35$，若仅把光程改为$L_2=4.5$ cm，则$A_2=0.35\times4.5/2=0.7875\approx0.79$；中间计算不要提前取整。

Source / 来源：Week 2 PDF p38 and p41–42; numerical check 0.35×4.5/2=0.7875; T2-0041–0051

**EN:** Exam wording: The lecturer names $A=\varepsilon CL$ as a short equation to remember for the closed-book Quiz. Absorbance is dimensionless and is not the same as an absorbed fraction or a current.

**中文：** 英文读题抓手：老师明确举$A=\varepsilon CL$为闭卷Quiz应记忆的短公式。Absorbance无量纲，不等于吸收比例，也不是电流。

Source / 来源：W2 PDF p36–40; W2 PDF p41–43; Sample quiz p2 Q11; Sample quiz p3 Q12; W2 p37–43; W1 p61–62 | T2-0041–0051, T7-0060–0061

## S107 • [P2] Wavelength selection, blanks and absorption spectra / 波长选择、空白与吸收光谱

Priority / 复习标记：P2 Important foundations/重要基础 · Teacher emphasis/课堂强调 · Worked example/例题练习

**EN:** A monochromator selects a narrow wavelength band: an entrance slit admits light, a dispersive element separates wavelengths, and an exit slit selects the band sent through the sample. A detector then measures transmitted light. A grating separates wavelengths; a photodetector measures light; a sample holder supplies the optical path. These components have different jobs.

**中文：** Monochromator（单色器）选择较窄波段：entrance slit（入射狭缝）让光进入，dispersive element（色散元件）分开波长，exit slit（出射狭缝）选出送往样品的波段，再由探测器测透射光。Grating（光栅）分波长，photodetector（光探测器）测光，sample holder（样品池）提供光程，这些元件的功能不同。

**EN:** Compare I and $I_0$ at the same wavelength. A blank/reference containing the solvent and relevant background provides a baseline so their contribution is not mistaken for target absorption; a buffer is not universally transparent at every wavelength. The coefficient ε depends on wavelength. At fixed wavelength and path length, increasing concentration raises absorbance in the valid linear range; different wavelengths can have different ε and peak heights.

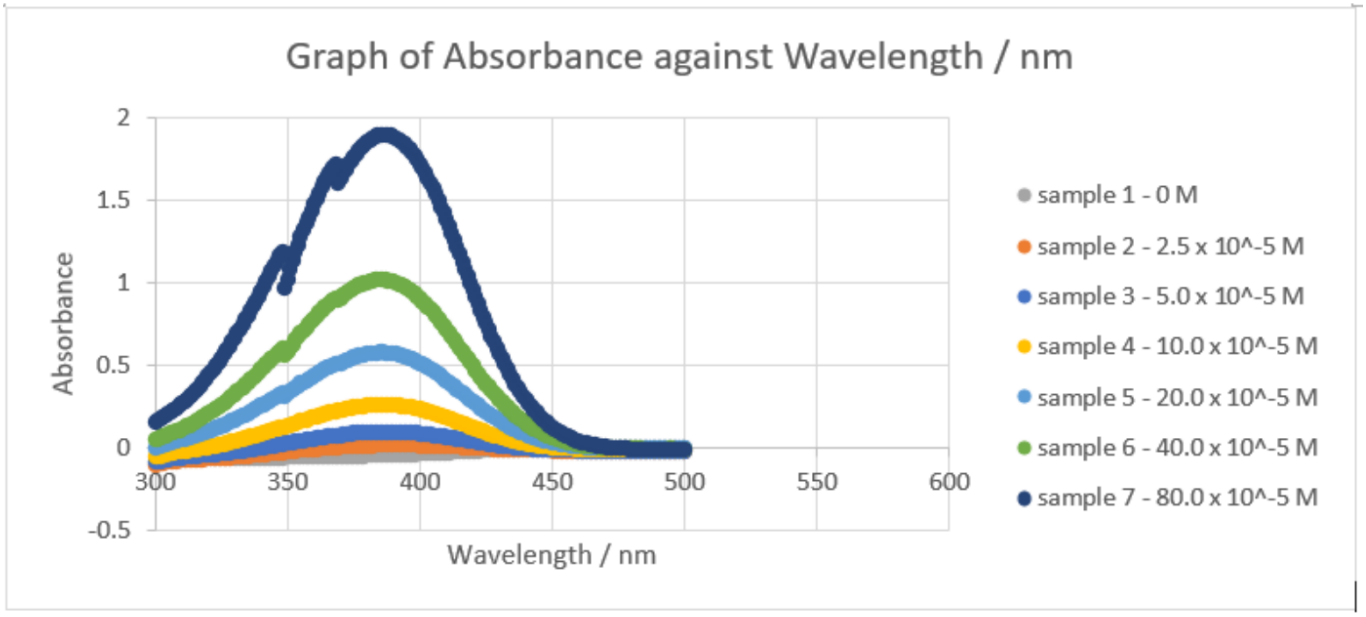
**中文：** I与$I_0$必须在相同波长下比较。含溶剂及相应背景的blank/reference（空白/参照）提供基线，避免把背景贡献误认成目标吸收；不能断言所有buffer（缓冲液）在每个波长都透明。ε随波长变化。波长和光程固定、且处于有效线性范围时，浓度越高吸光度越大；不同波长处的ε和峰高可以不同。

Source / 来源：Week 2 PDF p34–43; T2-0037–0051

**EN:** Clarify the concept: Typical spectroscopy associations in the slide are: X-rays—core-electron transitions; UV–visible—valence-electron transitions; infrared—molecular vibrations; microwaves—molecular rotations; radiofrequency—electron or nuclear spins. Use these to identify the kind of interaction being studied; they are typical associations, not exclusive statements that each band has only one possible interaction.

**中文：** 把概念讲清：课件中的典型光谱对应为：X-rays X射线——内层电子跃迁；UV–visible紫外/可见——价电子跃迁；infrared红外——分子振动；microwaves微波——分子转动；radiofrequency射频——电子或核自旋。可据此辨认研究的相互作用；这些是典型对应，不代表每个波段只有唯一作用。

Source / 来源：Week 2 PDF p35 original spectroscopy diagram; T2-0037–0040 explains different spectroscopy methods




Absorption spectra at different concentrations: compare height at the same wavelength. / 不同浓度的吸收光谱：必须在同一波长处比较峰高。

**EN:** Read the figure: The horizontal axis is wavelength, not concentration; the vertical axis is absorbance A, not transmitted intensity. Near the absorption peak, this sample gives a larger signal than at the spectral tail. For the same analyte, path length and wavelength, Beer–Lambert predicts $A \propto C$ in its valid range. Selecting λmax can improve sensitivity, but interference and usable absorbance range still matter; compare with a matched blank.

**中文：** 读图：横轴是Wavelength（波长），不是浓度；纵轴是Absorbance A（吸光度），不是透射光强。该样品在吸收峰附近的信号比光谱尾部大。同一物质、光程和波长下，在Beer–Lambert定律适用范围内有$A \propto C$。选λmax可提高灵敏度，但仍需考虑干扰和可测吸光度范围，并扣除匹配的Blank（空白）影响。

Source / 来源：Week 2 PDF p39; original slide figure / 原课件截图

## B06 • [P2] Pulse oximetry / 脉搏血氧测量

Priority / 复习标记：P2 Important foundations/重要基础 · Teacher emphasis/课堂强调 · Worked example/例题练习 · Lower priority/降低优先级

**EN: Key points**

- Two wavelengths exploit different absorption by oxyhemoglobin and deoxyhemoglobin;
- the pulsatile component helps isolate the arterial contribution.
- Know oxygenated versus deoxygenated hemoglobin and differential absorption.
- This estimates oxygen saturation, not simply the total amount of blood.
- The long saturation equation is lower priority under the lecturer’ s instruction.

**中文：必会知识点**

- 两个波长利用oxyhemoglobin氧合血红蛋白与deoxyhemoglobin脱氧血红蛋白的吸收差异；
- pulsatile脉动分量帮助分离动脉贡献。
- 要认得氧合/脱氧及差异吸收。
- 估计量是oxygen saturation血氧饱和度，不是简单的总血量。
- 按老师说明，长的饱和度方程降低优先级。

**EN:** Start from zero: Hemoglobin is the oxygen-carrying protein in red blood cells. Oxyhemoglobin carries oxygen; deoxyhemoglobin has released it. Oxygen saturation describes the fraction of available hemoglobin binding sites occupied by oxygen. A pulse oximeter estimates this from light, commonly using red and infrared wavelengths. It is not simply counting blood cells, measuring total blood volume, or reading heart rate alone.

**中文：** 从零理解：Hemoglobin（血红蛋白）是红细胞中携带氧的蛋白质。Oxyhemoglobin（氧合血红蛋白）携带氧，deoxyhemoglobin（脱氧血红蛋白）已经释放氧。Oxygen saturation（血氧饱和度）描述可用血红蛋白结合位点中，有多大比例被氧占据。Pulse oximeter（脉搏血氧仪）通常用红光与红外光估计这个比例，并非简单数血细胞、测总血量或只测心率。

**EN:** Follow the mechanism: The two hemoglobin forms absorb the two wavelengths differently. The device compares their changing optical signals. During each heartbeat, arterial blood volume changes, creating a pulsatile component above a relatively steady background from other tissue and blood. Focusing on this component helps isolate the arterial contribution. Think of noticing a flashing light against a steady lamp; measured time-varying signals help, without removing every background effect.

**中文：** 跟着机制走：两种血红蛋白对两个波长的吸收不同，仪器比较相应的光信号变化。每次心跳时，动脉血量改变，在其他组织和血液形成的相对稳定背景上产生pulsatile component（脉动分量）。关注这一分量，有助于分离动脉的贡献。可以类比在常亮灯旁识别闪烁灯；测得的时间变化信号有帮助，但不能消除所有背景影响。

**EN:** Try an example: Teaching example: if only one wavelength were measured, a weaker signal could reflect several causes, including more blood or different oxygenation. Two wavelengths and the pulse-related component provide more information for the calibrated estimate. When recognizing the method, look for differential hemoglobin absorption plus pulsation. You do not need the long saturation equation to understand why both wavelengths and a time-dependent signal are useful.

**中文：** 用例子理解：教学例子：如果只测一个波长，信号变弱可能来自不同原因，包括血量更多或氧合状态不同。两个波长加上与脉搏相关的分量，为校准后的估计提供更多信息。辨认该方法时，寻找“血红蛋白差异吸收 + 脉动”。不背长的饱和度方程，也能理解为什么两个波长和随时间变化的信号都有用。

Source / 来源：W2 PDF p47–54 | T2-0056–0069

## S108 • [P2] Pulse-oximeter geometry and absorption directions / 血氧仪布置与吸收方向

Priority / 复习标记：P2 Important foundations/重要基础 · Teacher emphasis/课堂强调 · Worked example/例题练习 · Lower priority/降低优先级

**EN:** In transmissive oximetry, source and detector are on opposite sides, as in a finger clip. In reflectance oximetry, they are on the same side and the detector collects returned light, useful for sites such as the wrist. Both exploit wavelength-dependent hemoglobin absorption; reflectance mode does not mean that absorption is irrelevant.

**中文：** Transmissive oximetry（透射式血氧测量）中，光源和探测器在两侧，例如指夹；reflectance oximetry（反射式）中，两者位于同一侧，探测器收集返回光，可用于手腕等位置。两种布置都利用血红蛋白随波长变化的吸收；反射式并不意味着吸收没有作用。

**EN:** For the common red/infrared comparison near 660/940 nm, deoxyhemoglobin absorbs red light more strongly than oxyhemoglobin, while the taught infrared comparison reverses that tendency. The pulse-related components at both wavelengths are compared to estimate arterial saturation. Oxygen saturation is C_ox/(C_ox+C_de); total hemoglobin is C_ox+C_de. Saturation and total concentration are distinct.

**中文：** 对常用约660/940 nm的红光/红外比较，deoxyhemoglobin（脱氧血红蛋白）对红光的吸收强于oxyhemoglobin（氧合血红蛋白），课件中的红外比较趋势相反。比较两个波长与脉搏有关的分量，以估计动脉血氧。血氧饱和度为C_ox/(C_ox+C_de)，总血红蛋白浓度为C_ox+C_de；饱和度与总浓度是不同量。

**EN:** The near-infrared tissue window reflects a favorable balance of absorption and scattering. It does not provide unlimited penetration: scattering blurs spatial detail, and strong water absorption at other infrared wavelengths can increase loss. The long pulse-oximetry equation is lower priority; ratios and logarithms of intensity ratios are dimensionless, even though the slide solution prints nm beside them.

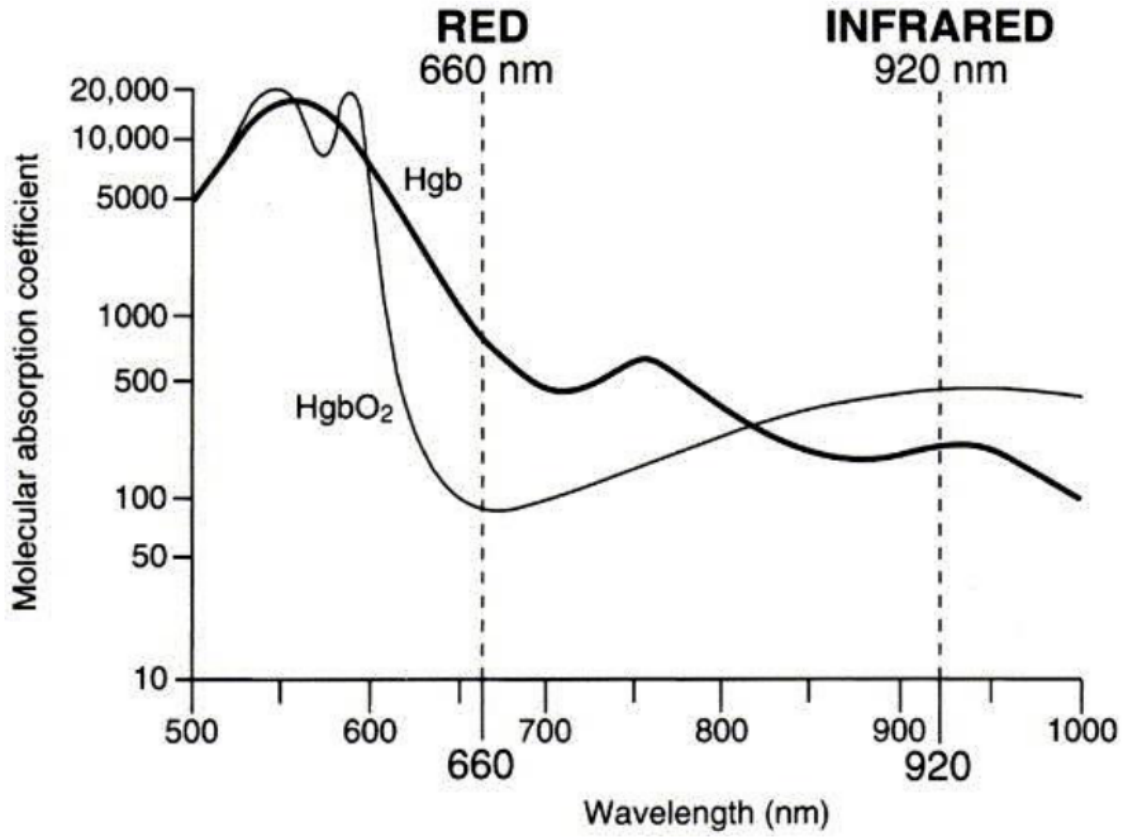
**中文：** Near-infrared tissue window（组织近红外窗口）来自较有利的吸收/散射组合，不意味着无限穿透：散射模糊空间细节，其他红外波段的强水吸收还会增加损耗。长的血氧方程优先级较低；光强比以及光强比的对数均无量纲，不能沿用课件答案旁错误标出的nm。

Source / 来源：Week 2 PDF p44–54; T2-0052–0069

**EN:** Clarify the concept: At an isosbestic wavelength, the two hemoglobin forms have equal extinction coefficients: in the ideal two-component model, absorption then depends on their total concentration rather than their oxygenation ratio.

**中文：** 把概念讲清：Isosbestic wavelength（等吸收点）处两种血红蛋白消光系数相同；理想两组分模型中，该处吸收取决于两者总浓度，而不是氧合比例。

Source / 来源：Week 2 PDF p44–46, p49–52; p46 original hemoglobin absorption curves; T2-0052–0069




Oxyhemoglobin and deoxyhemoglobin absorb red and infrared light differently. / 氧合与脱氧血红蛋白对红光、红外光的吸收不同。

**EN:** Read the figure: At 660 nm, follow the dashed line upward: deoxyhemoglobin (Hgb) absorbs more than oxyhemoglobin ($HgbO_2$). Near the infrared marker, their ordering reverses. Two wavelengths therefore help distinguish their contributions. This original plot marks 920 nm; the typical pulse-oximeter pair described on p52 is 660/940 nm. These curves show absorption coefficients, whereas actual pulse oximetry uses the pulsatile signal and calibration rather than reading saturation directly from one curve.

**中文：** 读图：沿660 nm虚线向上看：脱氧血红蛋白Hgb的吸收高于氧合血红蛋白$HgbO_2$；到红外标线附近，大小关系相反，所以两个波长有助于区分两者贡献。原图标的是920 nm；课件p52所讲典型血氧仪组合是660/940 nm。图展示的是吸收系数；实际Pulse oximetry（脉搏血氧测量）使用脉动信号和校准，不能凭一条曲线直接读出血氧饱和度。

Source / 来源：Week 2 PDF p46; original slide figure / 原课件截图

## C05 • [P1] Fluorescence and Stokes shift / 荧光与斯托克斯位移

Priority / 复习标记：P1 First pass/先掌握 · Sample Q1/样卷Q1 · Repeated explanation/反复讲解 · Worked example/例题练习

**EN: Key points**

• Fluorescence follows absorption and relaxation in an excited state.
• The usual Stokes shift gives a longer emission wavelength and lower photon energy than excitation.
• An excitation filter selects excitation light;
• an emission filter and dichroic mirror help separate fluorescence from the excitation beam.

**中文：必会知识点**

• 荧光经历吸收及激发态弛豫。
• 通常的Stokes shift使发射波长比激发波长更长、光子能量更低。
• Excitation filter选择激发光，emission filter与dichroic mirror帮助分离荧光和激发光。

**EN:** Start from zero: A photon is a small packet of light energy. In fluorescence, a molecule absorbs a photon and temporarily gains energy. It can then release a new photon as it returns to a lower-energy state. Excitation names the incoming light that starts the process; emission names the light produced afterward. They are different stages, rather than the same beam simply bouncing back.

**中文：** 从零理解：Photon（光子）可以理解为一小份光能。在fluorescence（荧光）中，分子吸收一个光子，暂时获得较高能量；随后返回较低能量状态时，可以释放一个新光子。Excitation（激发）指启动过程的入射光，emission（发射）指随后产生的光。这是两个不同阶段，并非同一束光只是反弹回来。

**EN:** Follow the mechanism: Usually, some energy is lost to molecular motion before emission. The emitted photon therefore has less energy. Because photon energy is inversely related to wavelength, less energy means a longer wavelength: the usual Stokes shift. A staircase analogy helps: the molecule first goes up, steps down a little without shining, and then comes down while emitting light. The levels are not literal physical stairs.

**中文：** 跟着机制走：通常，分子在发光之前会把一部分能量传给分子运动。因此，发射光子的能量较低。由于光子能量与波长成反比，能量较低就意味着波长更长，这就是通常的Stokes shift（斯托克斯位移）。可以类比成先上楼，不发光地往下走一小段，再边发光边下楼；但能级并不是真正的物理楼梯。

**EN:** Try an example: Teaching example: blue excitation can produce green emission. An emission filter helps block strong excitation light so the weaker fluorescence can be seen. In an incorrect-statement question, check whether longer wavelength has wrongly been called higher energy. Fluorescence may come from an added dye or naturally fluorescent material. Thus, the fluorescence mechanism itself does not always require an artificial label.

**中文：** 用例子理解：教学例子：蓝光激发可以产生绿光发射。Emission filter（发射滤光片）帮助挡住较强的激发光，让较弱的荧光能被看到。遇到“错误陈述”题，要检查是否把长波长误写成了高能量。荧光既可以来自添加的染料，也可以来自天然会发荧光的材料，所以荧光机制本身不总需要人工标记。

**EN:** Exam wording: If a question asks for the wrong statement, check the direction of wavelength and energy carefully. Two-photon excitation combines two lower-energy photons; it does not mean two individually higher-energy photons.

**中文：** 英文读题抓手：若题目问wrong statement，先核对波长与能量的方向。Two-photon excitation是两个低能光子共同激发，不是两个各自更高能的光子。

Source / 来源：W2 PDF p69–76; W2 PDF p77–79; Sample quiz p1 Q1–3; W2 p73–78; W3 p39–50; W4 p54–61 | T2-0088–0100, T7-0051–0053

## S110 • [P1] Fluorescence microscope: source, filters and dichroic cutoff / 荧光显微镜：光源、滤光片与二向色镜截止

Priority / 复习标记：P1 First pass/先掌握 · Sample Q1/样卷Q1 · Repeated explanation/反复讲解

**EN:** A fluorescence microscope needs a light source capable of exciting the fluorophore: its output must overlap a usable absorption/excitation band and provide sufficient light. This source can be a suitable lamp, LED or laser; it need not always be a laser. A source at a wavelength the dye cannot absorb will not produce the expected fluorescence merely because it is bright.

**中文：** 荧光显微镜需要能激发该fluorophore（荧光团）的光源：输出波段必须与可用的吸收/激发带重合，并提供足够光。合适的灯、LED或laser（激光）均可，不要求一定用激光。染料不能吸收的波长，即使看起来很亮，也不能因此产生预期荧光。

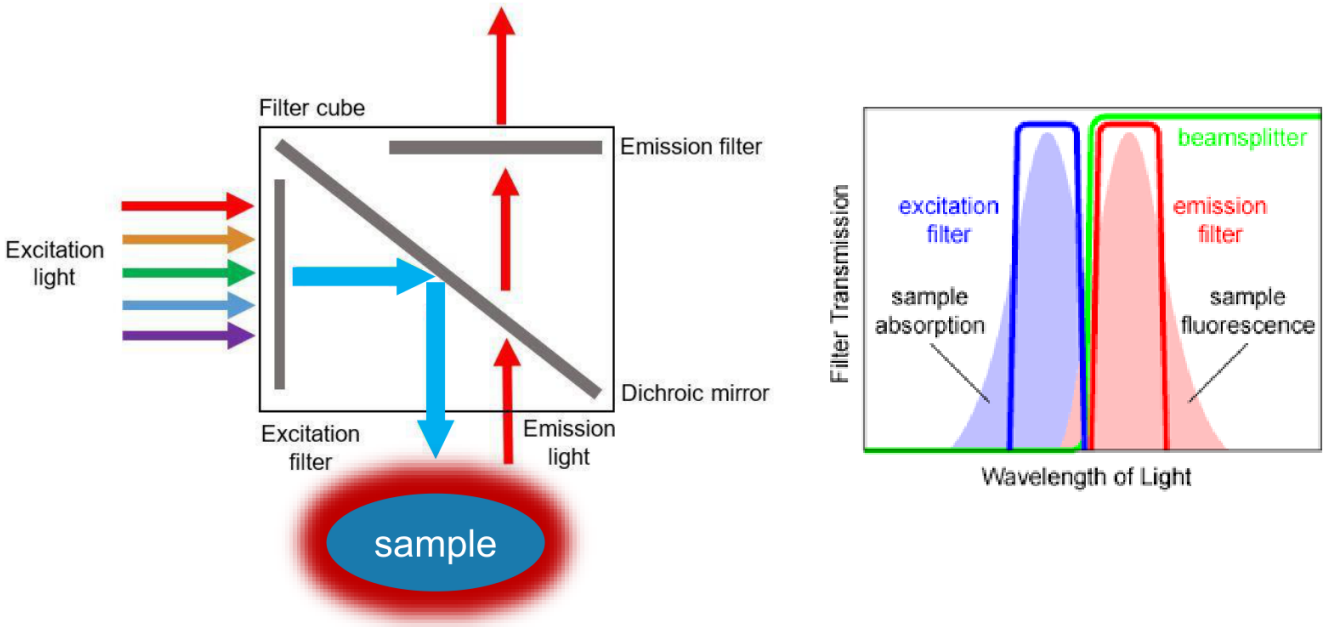
**EN:** In the usual epi-fluorescence filter cube, the excitation filter passes the desired excitation band. The dichroic mirror reflects that shorter-wavelength excitation toward the objective and sample, but transmits the longer-wavelength fluorescence toward the detector. The emission filter then passes the desired emission band and rejects remaining excitation/background. A dichroic mirror sorts wavelengths by reflection versus transmission; it is not just an ordinary mirror reflecting every color equally.

**中文：** 常见epi-fluorescence（落射荧光）滤光组件中，excitation filter（激发滤光片）通过所需激发带；dichroic mirror（二向色镜）把较短波长激发光反射到物镜与样品，却让较长波长荧光透向探测器；emission filter（发射滤光片）再通过所需发射带，抑制残余激发光和背景。二向色镜按波长区分反射与透射，不是对所有颜色都同样反射的普通镜子。

**EN:** For a simple separated-band example with excitation near 480 nm and emission near 530 nm, a suitable cutoff/transition between them allows the mirror to reflect excitation and transmit emission. This is why sample option B is reasonable. Real absorption and emission are bands, so filter choice considers their widths, overlap and transition region, not merely two exact numbers. B and C explain optical design; D is wrong because two-photon excitation uses two individually lower-energy photons, and higher spatial resolution is not automatic.

**中文：** 简单例子：激发约480 nm、发射约530 nm，合适的cutoff/transition（截止/过渡位置）选在两者之间，便可让二向色镜反射激发、透过发射，所以样题B合理。真实吸收/发射是波段，选滤光片时还要看宽度、重叠和过渡区，不能只看两个精确数字。B、C说明光学设计；D错误在于双光子用两个各自能量较低的光子共同激发，而且空间分辨率更高并非自动成立。

Source / 来源：Week 2 PDF p70, p77–78; Week 1 PDF p88; T2-0097–0100; Sample quiz p1 Q1 B/C




Fluorescence filter cube and spectral separation (original lecture slide). / 荧光滤光组件与波段分离（原课件）。

**EN:** Read the figure: Trace source → excitation filter → dichroic reflection → sample, then emission → dichroic transmission → emission filter → detector. On the right, the mirror transition lies between selected excitation and emission bands.

**中文：** 读图：沿光源→激发滤光片→二向色镜反射→样品追光路，再看荧光→二向色镜透射→发射滤光片→探测器；右图的镜面过渡区位于所选激发/发射波段之间。

Source / 来源：Week 2 PDF p77; original slide figure / 原课件截图

## S111 • [P1] Fluorescence readouts: intensity, color, ratio and lifetime / 荧光读出：强度、颜色、比值与寿命

Priority / 复习标记：P1 First pass/先掌握 · Sample Q1/样卷Q1 · Supporting concept/辅助理解

**EN:** Fluorescence intensity means how much emission is detected: a sensor may turn on, turn off, become brighter or become dimmer. Color/wavelength readout instead tracks where the emission spectrum is located or how different color channels are distributed. These can change independently. A greener-looking image does not necessarily contain more total photons, and a brighter signal need not have a shifted emission peak.

**中文：** Fluorescence intensity（荧光强度）指检测到多少发射光：传感器可turn on（亮起）、turn off（熄灭）、变亮或变暗。Color/wavelength（颜色/波长）读出关注发射光谱的位置，或不同颜色通道的分布。两者可独立改变：图像更绿，不一定总光子数更多；信号更亮，也不一定发射峰移动。

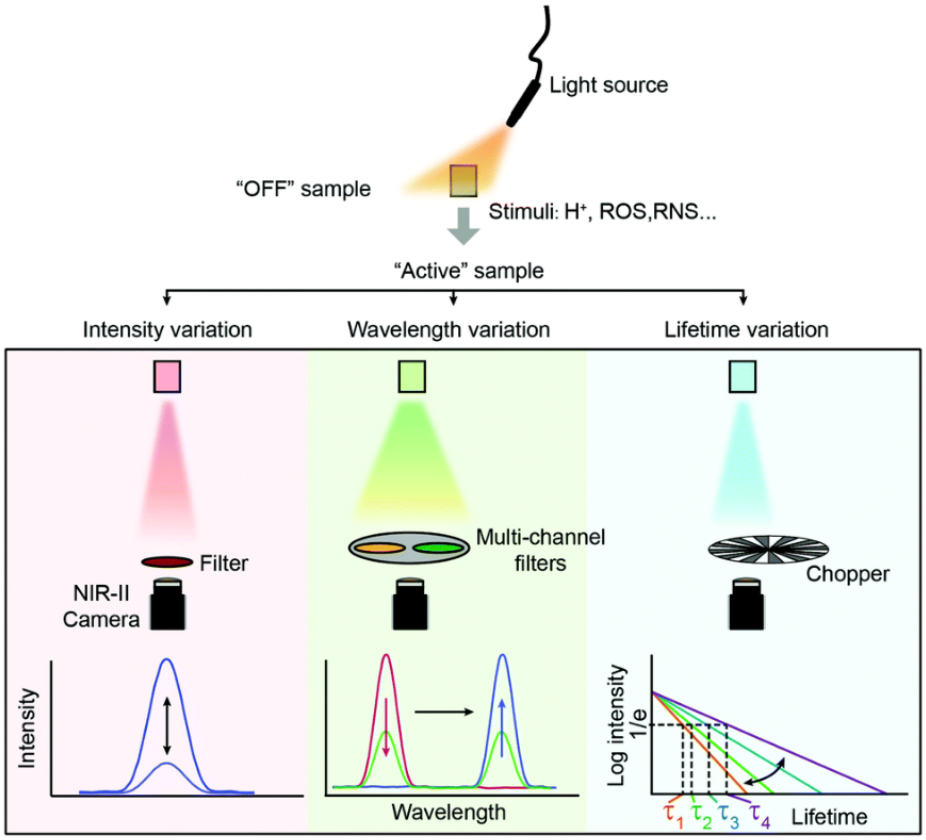
**EN:** A fluorescence array has many sensing spots or channels. Responses across the array can be expressed as intensity changes, color changes, or a pattern of both, depending on the probe design. This explains why sample option A is correct as a statement of possible readouts. It does not mean every fluorescence sensor must change color, or that all spots identify targets without recognition chemistry and calibration.

**中文：** Fluorescence array（荧光阵列）包含多个传感点或通道。根据探针设计，阵列响应可以表现为强度变化、颜色变化，或两者的组合模式，因此样题A作为“可采用的读出方式”是正确的。它不意味着每一种荧光传感器都必须变色，也不意味着无需识别化学和校准就能让所有点识别目标。

**EN:** A ratiometric probe compares two measured channels, for example $I_1/I_2$, instead of using one intensity alone; the ratio can reduce some shared intensity variations when the reference channel behaves suitably. Lifetime readout measures decay after excitation rather than just total brightness. For a spectrum, read the wavelength of the emission peak to identify emitted color; the excitation color is not the emitted color.

**中文：** Ratiometric probe（比率型探针）比较两个测得的通道，例如$I_1/I_2$，而非只用单一强度；参照通道性质合适时，该比值可降低部分共同强度波动。Lifetime readout（寿命读出）测激发之后的衰减，而非只看总亮度。读光谱时，用发射峰波长判断发出的颜色；激发光颜色不等于发射颜色。

Source / 来源：Week 1 PDF p50–51; Week 2 PDF p79, p89–92; Week 3 PDF p8–12; T2-0009, T2-0099–0101; Sample quiz p1 Q1 A




Different fluorescence outputs: intensity, wavelength and lifetime (original lecture slide). / 不同荧光输出：强度、波长与寿命（原课件）。

**EN:** Read the figure: The left diagram compares intensity variation, wavelength variation and lifetime variation. Look for peak height, peak position and decay with time respectively; these are different observables, not three names for one quantity.

**中文：** 读图：左图对比强度、波长、寿命变化：分别观察峰高、峰位置和随时间衰减，是不同观测量，不能只记一种“变亮”。

Source / 来源：Week 2 PDF p91; original slide figure / 原课件截图

## B07 • [P2] Luminescence and state transitions / 发光类别与能级跃迁

Priority / 复习标记：P2 Important foundations/重要基础 · Supporting concept/辅助理解

**EN: Key points**

- Fluorescence usually emits from S1 to S0;
- phosphorescence involves a triplet state and is generally slower.
- Internal conversion is nonradiative relaxation between states of the same spin multiplicity;
- intersystem crossing changes multiplicity.
- Chemiluminescence uses chemical excitation, and bioluminescence is biological chemiluminescence.
- An emission peak need not have the same wavelength as absorption.

**中文：必会知识点**

- Fluorescence荧光通常由S1到S0发射；
- phosphorescence磷光涉及三重态，通常更慢。
- Internal conversion内转换是在相同自旋多重度能态间的非辐射弛豫；
- intersystem crossing系间窜越改变多重度。
- Chemiluminescence化学发光由化学反应激发，bioluminescence生物发光属于生物化学发光。
- 发射峰不必与吸收峰波长相同。

**EN:** Start from zero: Luminescence means light produced through an excited-state process rather than simply because an object is hot. The energy can come from absorbed light, giving photoluminescence, or from a chemical reaction, giving chemiluminescence. Bioluminescence is chemical light production in a biological system. Fluorescence and phosphorescence are forms of photoluminescence distinguished by the states and transitions involved, not just by the visible color.

**中文：** 从零理解：Luminescence（发光）指通过激发态过程产生光，而不是单纯因为物体很热。能量可以来自吸收光，这叫photoluminescence（光致发光）；也可以来自化学反应，这叫chemiluminescence（化学发光）。Bioluminescence（生物发光）是生物系统中的化学发光。荧光与磷光属于光致发光，其区别在涉及的状态和跃迁，并不只是可见颜色不同。

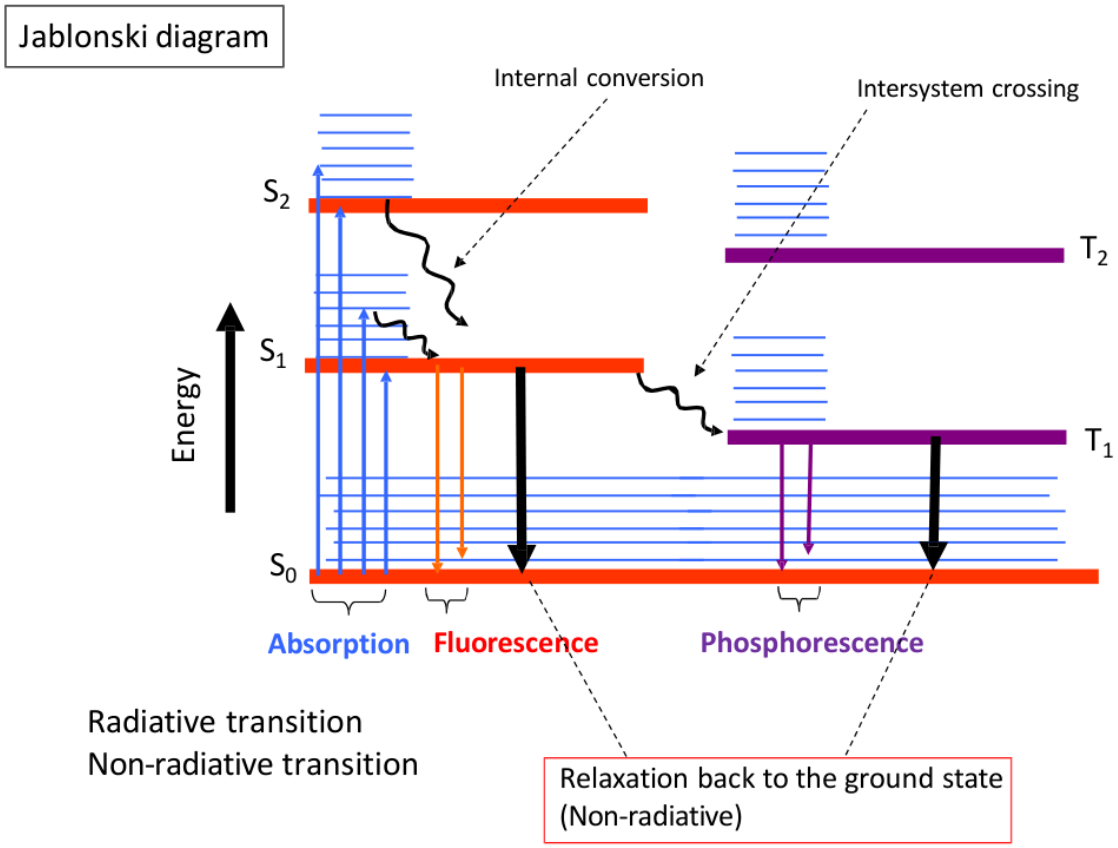
**EN:** Follow the mechanism: An energy-level diagram labels S0 as the ground singlet state, S1 as a higher singlet state, and T1 as a triplet state. Singlet and triplet describe quantum spin arrangements, not the number of emitted photons. Fluorescence commonly returns from S1 to S0. Phosphorescence involves a triplet route that is usually slower. Internal conversion loses energy without light within one multiplicity; intersystem crossing changes multiplicity.

**中文：** 跟着机制走：能级图中，S0是基态单重态，S1是较高的单重态，T1是三重态。Singlet（单重态）和triplet（三重态）描述量子自旋安排，并不是发射了几个光子。荧光通常从S1回到S0，磷光涉及通常更慢的三重态路线。Internal conversion（内转换）在同一多重度内不发光地损失能量，intersystem crossing（系间窜越）则改变多重度。

**EN:** Try an example: Teaching example: glow remaining after excitation stops may suggest a slower phosphorescent route, but identify the actual mechanism rather than assuming every afterglow is phosphorescence. A chemical glow needs chemical energy instead of an excitation lamp. In diagrams, arrows downward do not all mean light emission: some indicate nonradiative relaxation. Follow how energy enters, where it moves, and which step actually releases a photon.

**中文：** 用例子理解：教学例子：停止激发后还持续发光，可能提示较慢的磷光路线，但应看实际机制，不能认定所有余辉都是磷光。化学发光需要化学能，而不是激发灯。在图中，向下箭头不都代表发光，有些表示非辐射弛豫。沿着能量从哪里进入、往哪里转移、哪一步真正释放光子来读图。

Source / 来源：W2 PDF p69–76; W2 PDF p77–79; W2 PDF p82–88 | T2-0088–0100, T2-0105–0107




Jablonski diagram: distinguish absorption, fluorescence, phosphorescence and non-radiative loss. / Jablonski能级图：区分吸收、荧光、磷光与非辐射损失。

**EN:** Read the figure: Blue upward arrows mean absorption; orange downward arrows from $S_1$ to $S_0$ mean fluorescence. Before emission, some excitation energy can be lost without producing a photon, explaining the usual lower-energy, longer-wavelength fluorescence. Intersystem crossing transfers population to a triplet state $T_1$; purple $T_1 \rightarrow S_0$ emission is phosphorescence. Black return paths produce no emitted photon. The vertical axis is energy: an upward arrow is not a longer wavelength.

**中文：** 读图：蓝色向上箭头表示Absorption（吸收）；从$S_1$回到$S_0$的橙色箭头表示Fluorescence（荧光）。发光前可先无光子地损失部分能量，所以普通荧光的光子能量较低、波长较长。Intersystem crossing（系间窜越）把激发态人口转到三重态$T_1$；紫色$T_1 \rightarrow S_0$发光是Phosphorescence（磷光）。黑色返回路径不发射光子。纵轴是能量，不能把向上箭头理解为波长变长。

Source / 来源：Week 2 PDF p74; original slide figure / 原课件截图

## C06 • [P1] Quantum yield, lifetime and radiative rate / 量子产率、寿命与辐射速率

Priority / 复习标记：P1 First pass/先掌握 · Sample Q6/样卷Q6 · Worked example/例题练习 · Teacher emphasis/课堂强调

**EN: Key points**

- Quantum yield Φ is emitted photons divided by absorbed photons.
- With total non-radiative rate $k_{nr}$, $\tau = 1/(k_r + k_{nr})$, $\Phi = k_r\tau$ and $k_r = \Phi/\tau$.
- Fluorescence brightness also depends on absorption, commonly through εΦ in the course examples.

**中文：必会知识点**

- 量子产率Φ是发射光子数除以吸收光子数。
- 非辐射总速率为$k_{nr}$时，$\tau = 1/(k_r + k_{nr})$、$\Phi = k_r\tau$、$k_r = \Phi/\tau$。
- 本课程示例中亮度还受吸收影响，常用εΦ表示。

**EN:** Start from zero: An excited molecule has more than one way to lose energy. It can emit a photon, a radiative route, or lose energy without light, a nonradiative route. Quantum yield Φ is the fraction of absorbed photons that lead to emitted photons. Lifetime τ describes how quickly a population of excited molecules decays; it is not how long the dye remains usable in a bottle.

**中文：** 从零理解：被激发的分子有不止一种释放能量的方法：可以发出光子，这是radiative（辐射）途径；也可以不发光地损失能量，这是nonradiative（非辐射）途径。Quantum yield Φ（量子产率）表示吸收的光子中，有多大比例最终对应发射光子。Lifetime τ（寿命）描述一群激发态分子衰减得多快，并不是染料在瓶子里能保存多久。

**EN:** Follow the mechanism: Imagine two exits from a room: a light-producing exit and a dark exit. Their rates, $k_r$ and $k_{nr}$, add because both empty the excited population. Hence $\tau=1/(k_r+k_{nr})$, and $\Phi=k_r/(k_r+k_{nr})=k_r\tau$. Faster total escape shortens lifetime. The analogy describes competing probabilities; a molecule does not choose an exit deliberately. The rates have units of inverse time, such as $s^{-1}$.

**中文：** 跟着机制走：想象一个房间有两个出口：一个出去会发光，另一个出去不会发光。它们的速率$k_r$与$k_{nr}$相加，因为两条路都会减少激发态分子。所以$\tau=1/(k_r+k_{nr})$，而$\Phi=k_r/(k_r+k_{nr})=k_r\tau$。总离开速率越快，寿命越短。这个类比描述相互竞争的概率，分子不会有意识地选择出口。速率单位是时间的倒数，例如$s^{-1}$。

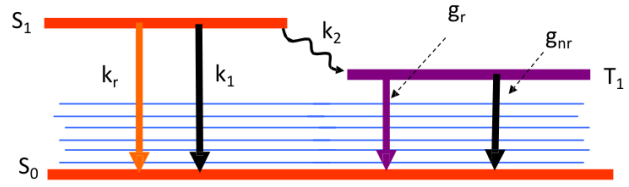
**EN:** Try an example: Teaching example: 100 photons arrive, 20 are absorbed, and 12 are emitted. $\Phi=12/20=0.60$, not 12/100. If $\tau=12$ ns, $k_r=0.60/12=0.05$ $ns^{-1}=5\times10^7$ $s^{-1}$. Lifetime is the time to reach 1/e of the initial population in this model, not the half-life. Brightness also depends on how many photons are absorbed, so yield alone does not determine total brightness.

**中文：** 用例子理解：教学例子：100个光子入射，20个被吸收，12个被发射，$\Phi=12/20=0.60$，而不是12/100。如果$\tau=12$ ns，则$k_r=0.60/12=0.05$ $ns^{-1}=5\times10^7$ $s^{-1}$。这个模型中的寿命，是激发态群体降到初始值1/e所需的时间，不是半衰期。亮度还取决于吸收了多少光子，所以只知道产率还不能确定总亮度。

**EN:** Exam wording: Do not use all incident photons as the denominator unless all are absorbed. Lifetime is an exponential 1/e decay time, not automatically a half-life. Convert ns and s consistently.

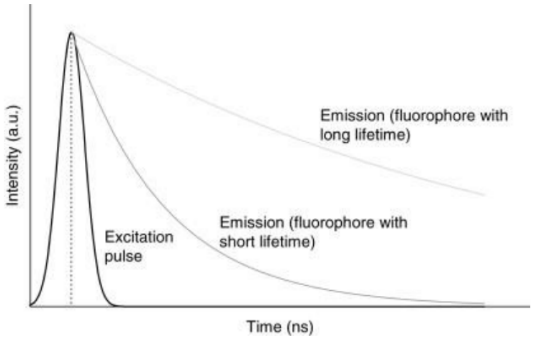
**中文：** 英文读题抓手：除非入射光子全部被吸收，否则分母不能用全部入射光子。Lifetime是指数衰减至1/e的时间，并非直接等于半衰期。ns与s必须统一。

Source / 来源：W2 PDF p82–88; Sample quiz p1 Q6; Sample quiz p2 Q7; W2 p82–88 | T2-0105–0107, T7-0054–0055




Let's look at the population of molecules at $S_1$, [$S_1$]

How many channels does molecule at $S_1$ to leave $S_1$ state?

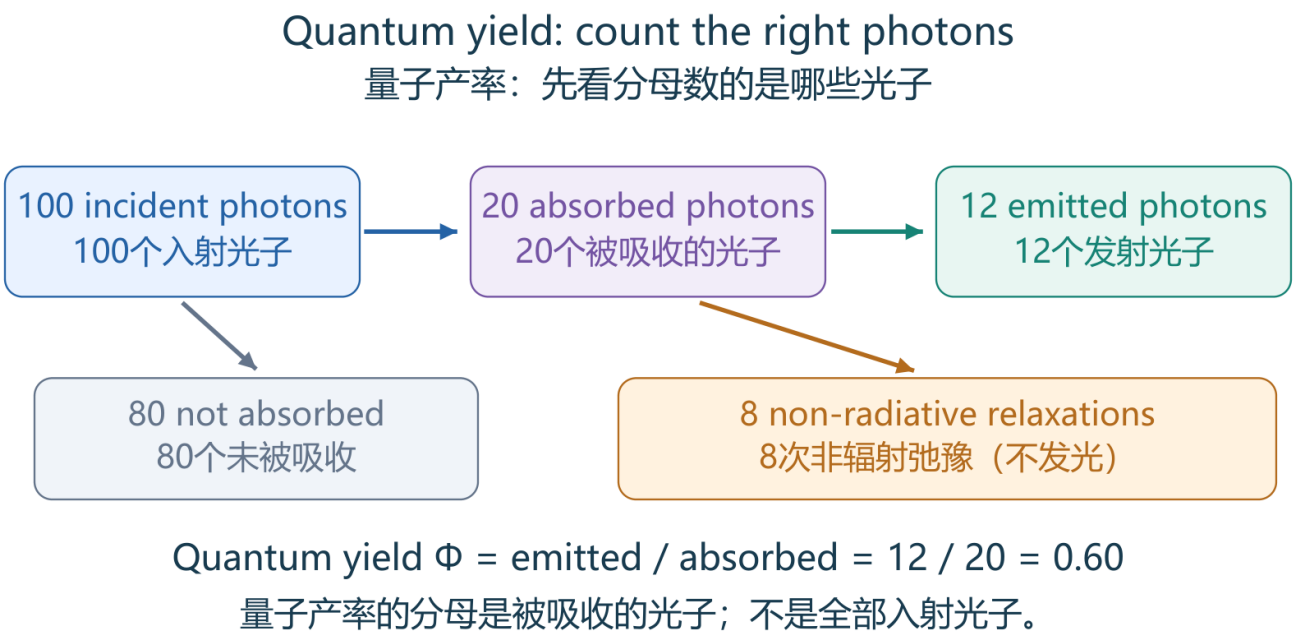



Fluorescence lifetime: competing exit rates on the left, decay after a pulse on the right. / 荧光寿命：左边看激发态离开路径，右边看脉冲后衰减。

**EN:** Read the figure: From $S_1$, the molecule may emit through $k_r$ or leave without fluorescence through $k_1$ and $k_2$; these competing routes together set its fluorescence lifetime. On the right, a short excitation pulse is followed by a fast or slow emission decay. In the single-exponential model, $I(t)=I_0e^{-t}/\tau$ and $\tau=1/(k_r+k_{nr})$, with $k_{nr}=k_1+k_2$ here. Longer τ means slower decay; at τ the intensity is $I_0/e$, not $I_0/2$. A long lifetime alone does not prove a high quantum yield.

**中文：** 读图：分子离开$S_1$时，可以经$k_r$发荧光，也可以经$k_1$、$k_2$不发荧光地离开；这些竞争路径共同决定荧光寿命。右图的短激发脉冲之后，荧光有快、慢两种衰减。在单指数模型下，$I(t)=I_0e^{-t}/\tau$，$\tau=1/(k_r+k_{nr})$，本图$k_{nr}=k_1+k_2$。τ较大表示衰减较慢；t=τ时剩$I_0/e$，不是$I_0/2$。仅凭寿命长，不能断言量子产率高。

Source / 来源：Week 2 PDF p82; original slide figure / 原课件截图




Quantum yield: count the right photons / 量子产率：先看分母数的是哪些光子

**EN:** Read the figure: This is a made-up counting example. Of 100 incoming photons, only 20 are absorbed; 12 of those events produce fluorescence and 8 relax without light. Thus Φ = 12/20 = 0.60. The ratio 12/100 also includes losses before absorption and is not the fluorescence quantum yield.

**中文：** 读图：这是便于理解的假设数字。100个入射光子中只有20个被吸收；其中12次产生荧光，8次不发光而弛豫。因此Φ = 12/20 = 0.60。12/100还混入吸收前的损失，不是Fluorescence quantum yield（荧光量子产率）。

Source / 来源：Week 2 PDF p82–88; original teaching schematic / 自绘教学示意，非实验数据

## S112 • [P2] Environment, quenching and fluorescence brightness / 环境、猝灭与荧光亮度

Priority / 复习标记：P2 Important foundations/重要基础 · Worked example/例题练习 · Sample Q6/样卷Q6

**EN:** A dye’ s fluorescence can depend on solvent and local environment. The slide compares spectra in two solvents, so one dye need not have an identical emission peak in every condition. Concentration-dependent self-quenching is one reason that adding dye does not always increase intensity proportionally. Quenching reduces fluorescence through competing processes; photobleaching refers to light-induced chemical damage or loss of the fluorophore. Both can dim a signal, but they are not the same mechanism. Control concentration, solvent, excitation and detection conditions when comparing samples.

**中文：** 染料荧光可受solvent（溶剂）和局部环境影响。课件比较两种溶剂中的光谱，同一染料不一定总有相同发射峰。与浓度有关的self-quenching（自猝灭）是“加更多染料不一定按比例更亮”的原因之一。Quenching（猝灭）通过竞争过程降低荧光；photobleaching（光漂白）指光引起荧光团化学损伤或损失，两者都可使信号变暗，但不是同一机制。比较样品时，要控制浓度、溶剂、激发和检测条件。

**EN:** In the simple competing-rate model, if k_nr increases while k_r is fixed, $\tau=1/(k_r+k_{nr})$ and $\Phi=k_r/(k_r+k_{nr})$ both decrease. For $\Phi=0.80$ and $\tau=8$ ns, $k_r=0.10$ ns$^{-1}$; the total rate is 0.125 ns$^{-1}$ and $k_{nr}=0.025$ ns$^{-1}$. Quantum yield, radiative rate, total decay rate and measured brightness are related but different quantities. Brightness $\varepsilon\Phi$ applies to a specified excitation wavelength and comparable conditions.

**中文：** 简单竞争速率模型中，若k_r固定而k_nr增大，$\tau=1/(k_r+k_{nr})$与$\Phi=k_r/(k_r+k_{nr})$都会减小。$\Phi=0.80$、$\tau=8$ ns时，$k_r=0.10$ ns$^{-1}$；总速率为0.125 ns$^{-1}$，$k_{nr}=0.025$ ns$^{-1}$。量子产率、辐射速率、总衰减速率和测得亮度互相关联，但并非同一个量。Brightness $\varepsilon\Phi$应对应指定激发波长及可比条件。

Source / 来源：Week 2 PDF p79, p82–86; Week 3 PDF p8; T2-0099–0100, T2-0105–0107

## C07 • [P1] Raman, Rayleigh, Stokes and anti-Stokes / 拉曼、瑞利、斯托克斯与反斯托克斯

Priority / 复习标记：P1 First pass/先掌握 · Sample Q2/样卷Q2 · Repeated explanation/反复讲解 · Teacher emphasis/课堂强调

**EN: Key points**

- Rayleigh scattering is elastic: scattered photon energy is unchanged.
- Raman scattering is inelastic and reports molecular vibrations.
- Stokes Raman light has lower energy and longer wavelength;
- anti-Stokes has higher energy and shorter wavelength.
- Raman shift is usually expressed in cm$^{-1}$.

**中文：必会知识点**

- Rayleigh scattering是弹性散射，光子能量不变。
- Raman scattering是非弹性散射，反映分子振动。
- Stokes Raman能量更低、波长更长；
- anti-Stokes能量更高、波长更短。
- 拉曼位移常用cm$^{-1}$。

**EN:** Start from zero: Scattering means light leaves in a changed direction after interacting with matter. In Rayleigh scattering, the outgoing photon keeps its energy; this is called elastic scattering. In Raman scattering, the photon exchanges energy with a molecular vibration, so its energy changes. Molecules have different vibration patterns, making a Raman spectrum useful as a chemical fingerprint rather than just a brightness measurement.

**中文：** 从零理解：Scattering（散射）表示光与物质作用后，改变了离开的方向。在Rayleigh scattering（瑞利散射）中，出射光子的能量不变，这叫elastic（弹性）散射。在Raman scattering（拉曼散射）中，光子与分子振动交换能量，所以能量发生变化。不同分子有不同振动模式，因此拉曼光谱可以作为化学指纹，而不只是测亮度。

**EN:** Follow the mechanism: For Stokes Raman scattering, the photon gives some energy to a vibration and leaves with lower energy and longer wavelength. For anti-Stokes scattering, a vibration already carrying energy gives some to the photon, which leaves with higher energy and shorter wavelength. Think of an energy trade between light and vibration. This differs from fluorescence, which involves absorption and subsequent emission from an excited electronic state.

**中文：** 跟着机制走：在Stokes Raman（斯托克斯拉曼）中，光子把一部分能量给了振动，离开时能量更低、波长更长。在anti-Stokes（反斯托克斯）中，原本已经带有能量的振动把部分能量给光子，使它离开时能量更高、波长更短。可以理解为光与振动之间的能量交换。这与荧光不同：荧光涉及吸收，以及随后从电子激发态发光。

**EN:** Try an example: Teaching example: use one laser and observe an unchanged-energy peak plus shifted peaks. The unchanged peak is Rayleigh; shifted vibrational peaks are Raman. Surface-enhanced Raman scattering (SERS) uses nearby metal nanostructures to strengthen these Raman signals. Recognize the measured signal: vibrational shifts mean Raman, whereas a changed resonance position means resonance sensing. Plasmonic enhancement does not make SERS an optical-cavity measurement.

**中文：** 用例子理解：教学例子：用一束激光照射，观察到一个能量不变的峰和一些偏移峰。能量不变的峰是瑞利，振动产生的偏移峰是拉曼。Surface-enhanced Raman scattering（SERS，表面增强拉曼散射）利用附近的金属纳米结构增强这些拉曼信号。辨认被测信号：振动位移提示拉曼，共振位置变化提示共振传感。等离激元增强并不把SERS变成光学腔测量。

**EN:** Exam wording: Fluorescence and Raman are different mechanisms. In the sample resonator question, Raman is the clearest non-resonance-sensing option. SERS means surface-enhanced Raman spectroscopy; the metal nanostructure enhances the Raman signal.

**中文：** 英文读题抓手：荧光和拉曼机制不同。样题的共振器辨析中，Raman是最明确的非共振传感选项。SERS是surface-enhanced Raman spectroscopy，金属纳米结构增强拉曼信号。

Source / 来源：W2 PDF p55–62; W2 PDF p65–68; W3 p88–94 | T2-0070–0078, T2-0083–0087, T4-0007–0013

## S109 • [P2] Why Raman is different from fluorescence / 拉曼为什么不同于荧光

Priority / 复习标记：P2 Important foundations/重要基础 · Repeated explanation/反复讲解 · Lower priority/降低优先级

**EN:** Fluorescence absorbs light into a real electronic excited state and then emits after an excited-state interval. The usual Raman diagram instead uses a virtual intermediate state: scattered light exchanges energy with molecular vibration without populating that real fluorescence state. Thus Raman does not require the laser to match an allowed electronic absorption transition in the same way. It is not simply delayed fluorescent light.

**中文：** 荧光先吸收光、进入真实电子激发态，经过激发态停留后再发光。常见拉曼图使用virtual intermediate state（虚中间态）：散射光与分子振动交换能量，而不是占据那个真实荧光激发态。因此，拉曼不需要像荧光那样，让激光匹配允许的电子吸收跃迁；它不是“延迟出来的荧光”。

**EN:** Polarizability describes how an electric field distorts a molecule’ s electron distribution. A vibration is Raman-active when it changes polarizability. Bonding and molecular environment determine vibration frequencies, producing a fingerprint. Raman is normally much weaker than Rayleigh scattering; an instrument must separate the strong unshifted light from weak shifted lines. Rayleigh and Mie are elastic scattering regimes, whereas Stokes and anti-Stokes Raman exchange vibration energy.

**中文：** Polarizability（极化率）描述电场使分子电子分布变形的程度。振动能改变极化率时，该振动具有Raman-active（拉曼活性）。化学键和分子环境决定振动频率，形成指纹。拉曼通常远弱于瑞利，仪器要把很强的无位移光与弱的位移谱线分开。瑞利、米氏是弹性散射情形，斯托克斯及反斯托克斯拉曼则交换振动能量。

Source / 来源：Week 2 PDF p58–62, p94; Week 3 PDF p89–90; T2-0070–0082

## B08 • [P1] Raman shift and spectral axes / 拉曼位移与光谱坐标

Priority / 复习标记：P1 First pass/先掌握 · Teacher emphasis/课堂强调 · Worked example/例题练习 · Final emphasis/仅期末强调

**EN: Key points**

- Wavelength (nm), frequency (Hz) and wavenumber ($cm^{-1}$) are different axes.
- For Stokes scattering, $\Delta\tilde{\nu}=10^{7}(1/\lambda exc-1/\lambda S)$ $cm^{-1}$ when both wavelengths are in nm.
- A 1000 nm excitation and 1080 nm Stokes line give about 741 $cm^{-1}$.
- Anti-Stokes photons have higher energy than excitation;
- Stokes photons have lower energy.

**中文：必会知识点**

- Wavelength波长（nm）、frequency频率（Hz）、wavenumber波数（$cm^{-1}$）是不同坐标。
- 两波长均用nm时，斯托克斯位移$\Delta\tilde{\nu}=10^{7}(1/\lambda exc-1/\lambda S)$，单位$cm^{-1}$。
- 1000 nm激发、1080 nm斯托克斯线对应约741 $cm^{-1}$。
- Anti-Stokes反斯托克斯光子能量高于激发，Stokes斯托克斯光子能量低于激发。

**EN:** Start from zero: A spectrum can use wavelength, frequency, or wavenumber on its horizontal axis. Wavelength is the distance between repeating wave features. Frequency counts oscillations per second. Wavenumber is inverse wavelength; $cm^{-1}$ means reciprocal centimeters, not centimeters. These axes describe related properties, but their numbers cannot be subtracted as if they were the same unit. Raman shift measures a difference in wavenumber from the excitation light.

**中文：** 从零理解：光谱横轴可以用wavelength（波长）、frequency（频率）或wavenumber（波数）。波长是波上重复特征之间的距离；频率是每秒振荡次数；波数是波长的倒数，$cm^{-1}$表示厘米的倒数，并不是厘米。这些坐标描述相关性质，但不能把不同单位的数值当成同一个单位相减。Raman shift（拉曼位移）测的是相对激发光的波数差。

**EN:** Follow the mechanism: Because 1 nm equals $10^{-7}$ cm, a wavelength written in nm has wavenumber $10^7/\lambda$ in $cm^{-1}$. For Stokes scattering, subtract the emitted wavenumber from the excitation wavenumber: $\Delta\tilde{\nu}=10^7(1/\lambda exc-1/\lambda S)$. Here λexc is excitation wavelength and λS the Stokes wavelength, both in nm. The symbol $\tilde{\nu}$ represents wavenumber, not ordinary frequency. Taking reciprocals is essential before subtracting.

**中文：** 跟着机制走：因为1 nm=$10^{-7}$ cm，所以波长λ用nm表示时，波数为$10^7/\lambda$，单位$cm^{-1}$。斯托克斯散射中，用激发波数减去发射波数：$\Delta\tilde{\nu}=10^7(1/\lambda exc-1/\lambda S)$。λexc是激发波长，λS是斯托克斯波长，两者都用nm。$\tilde{\nu}$表示波数，不是普通频率。必须先取倒数，再相减。

**EN:** Try an example: Teaching example: 1000 nm excitation has wavenumber 10000 $cm^{-1}$; 1080 nm emission has about 9259 $cm^{-1}$. The Stokes shift is about 741 $cm^{-1}$, not 80 $cm^{-1}$. The number 80 is the wavelength difference in nm. A longer Stokes wavelength means lower photon energy. For anti-Stokes light the direction reverses; check the question's sign convention if a signed shift is requested.

**中文：** 用例子理解：教学例子：1000 nm激发的波数为10000 $cm^{-1}$，1080 nm发射的波数约为9259 $cm^{-1}$，所以斯托克斯位移约741 $cm^{-1}$，而不是80 $cm^{-1}$。80是以nm表示的波长差。斯托克斯波长更长，表示光子能量更低。反斯托克斯方向相反，如果题目要求带符号的位移，要检查其符号约定。

Source / 来源：W2 PDF p34–35; W2 PDF p55–62; W2 PDF p65–68 | T2-0037–0040, T2-0070–0078, T2-0083–0087

## C08 • [P1] Grating density and spectral resolution / 光栅刻线密度与光谱分辨率

Priority / 复习标记：P1 First pass/先掌握 · Quiz stated/Quiz明确提及 · Sample Q7/样卷Q7 · Worked example/例题练习

**EN: Key points**

• For the same illuminated width and diffraction order, a higher groove density illuminates more grooves N.

• Resolving power R=λ/Δλ=mN increases, and the minimum resolvable wavelength difference Δλ decreases.

• Thus 1200 lines/mm resolves more closely spaced lines than 300 lines/mm under these assumptions.

**中文：必会知识点**

• 在受照宽度与衍射级次相同的条件下，刻线密度更高使受照刻线数N增加。

• 分辨本领R=λ/Δλ=mN增大，最小可分辨波长差Δλ减小。

• 因此在这些假设下，1200 lines/mm比300 lines/mm能分开更接近的谱线。

**EN:** Start from zero: White light contains many wavelengths. A diffraction grating is a surface with many regularly spaced grooves that sends different wavelengths toward different directions. This lets a spectrometer separate colors and measure a spectrum. Spectral resolution asks whether two nearby wavelengths can be seen as two distinct lines. It is different from microscope resolution, which asks whether two nearby objects can be separated in space.

**中文：** 从零理解：白光含有许多波长。Diffraction grating（衍射光栅）表面有大量规律排列的刻线，能把不同波长送向不同方向，因此光谱仪可以分开颜色、测量光谱。Spectral resolution（光谱分辨率）问的是两个接近的波长能否显示为两条不同谱线。它不同于显微镜的分辨率，后者问的是空间中两个接近物体能否分开。

**EN:** Follow the mechanism: Each illuminated groove contributes a wave. Their combined interference makes wavelength-selective bright directions; more contributing grooves can make those features sharper. The ideal relation is R=λ/Δλ=mN: λ is the measured wavelength, Δλ the smallest separable difference, m the diffraction order, and N the illuminated groove count. Grooves outside the light beam do not count. Here m does not mean calibration slope.

**中文：** 跟着机制走：每条受到照明的刻线都贡献一个波，它们相互干涉，在特定方向形成对应波长的亮光。参与的刻线更多，可以使这些特征更尖锐。理想关系为R=λ/Δλ=mN：λ是被测波长，Δλ是最小可分开的波长差，m是衍射级次，N是受照刻线数。光束外的刻线不计入N。这里m不是校准斜率。

**EN:** Try an example: Teaching example: a 5 mm illuminated width covers 1500 grooves at 300 lines/mm, but 6000 grooves at 1200 lines/mm. With the same order, the second grating has four times the ideal resolving power and one-quarter the minimum Δλ. Thus larger R and smaller Δλ both mean better separation. Real spectrometers also depend on slit width and detector sampling, so preserve the comparison assumptions.

**中文：** 用例子理解：教学例子：受照宽度为5 mm时，300 lines/mm覆盖1500条刻线，1200 lines/mm覆盖6000条。在级次相同的条件下，第二块光栅的理想分辨本领是4倍，最小Δλ是四分之一。因此，R更大和Δλ更小都表示分得更清。真实光谱仪还受到狭缝宽度与探测器采样的影响，所以要保留比较条件。

**EN:** Exam wording: The lecturer explicitly revisits this comparison as a Quiz item. Higher resolving power and smaller Δλ both describe better resolution. The 300/1200 example is anchored to the Week 5 oral review, not an invented Week 2 slide page.

**中文：** 英文读题抓手：老师明确说会在Quiz再问该比较。R更大与Δλ更小都表示分辨率更好。300/1200示例依据Week5口头回顾，不虚标为Week2某页原图。

Source / 来源：W2 p35 (spectrometer structure background; no 300/1200 comparison on supplied slides); Sample quiz p1 Q6; Sample quiz p2 Q7; W2 p82–88 | T5-0061–0063, T7-0054–0055

# Week 3 learning block: Labels, fibres and resonance / 第3学习组：标记、光纤与共振

## Week 3 concept map / 第3周知识框架

**EN:** Chapter logic: Follow how unlabelled binding changes a guided or resonant optical response, while separating recognition from the readout.

**中文：** 本章主线：追踪无报告标签的结合怎样改变导光或共振响应，并始终区分识别与读出。

| English term 中文名词 | Meaning and connection 含义与知识联系 |
|---|---|
| **Labelled • label-free 有标记•无标记** | An added reporter defines labelled sensing; label-free does not mean no light or no receptor. 添加报告物是有标记；无标记不等于不用光或不用受体。 → C09 |
| **Quantum dots • flow cytometry 量子点•流式细胞术** | Dot size can tune color within a material family; flow cytometry reads individual passing events. 同材料系列量子点尺寸可调颜色；流式逐个读取经过的事件。 → B09 |
| **Total internal reflection • fibre NA 全内反射•光纤NA** | TIR needs high-to-low index and an angle above critical; acceptance and internal angles differ. TIR要求高到低折射率及超过临界角；接受角不同于内部角。 → S114 |
| **Evanescent field 倏逝场** | A rapidly decaying external field lets nearby molecules affect a guided optical signal. 外侧迅速衰减的场使附近分子影响导光信号。 → S115 |
| **Modes • sensing arrangement 模式•传感布置** | Mode count is a field property; point, multiplexed and distributed describe measurement positions. 模式数是场性质；点式、复用、分布式描述测量位置。 → S116 |
| **Optical resonators • SPR • LSPR 光学共振器•SPR•LSPR** | Identify circulating/standing optical patterns versus metal collective-electron resonance. 辨认循环/驻波光场与金属集体电子共振。 → C11 |
| **Sλ • FWHM • Q • FOM 位移灵敏度•线宽•Q•FOM** | Compare shift with width in consistent units; index sensitivity alone does not establish selectivity. 位移和线宽要统一单位；折射率灵敏度本身不证明选择性。 → B11 |
| **ART • SERS 抗原快检•表面增强拉曼** | ART uses antibody capture and gold-particle color; SERS enhances molecular vibration fingerprints. ART以抗体捕获并用金颗粒显色；SERS增强分子振动指纹。 → C12 |

**EN:** Formula links for this chapter: FX117 Label-free spectral sensitivity and index LOD, FX118 Snell refraction and critical angle, FX119 Fibre numerical aperture and acceptance half-angle, FX120 Evanescent intensity and field-amplitude decay, FX121 Evanescent penetration-depth convention (support), FX122 WGM resonance and small fractional changes, FX123 Two-mirror cavity resonance (support), FX124 FWHM, Q and unit-consistent FOM, FX125 Weak Raman signals and reported SERS enhancement.

**中文：** 本章公式定位：FX117 无标记光谱灵敏度与折射率检出限；FX118 斯涅尔折射与临界角；FX119 光纤数值孔径与接受半角；FX120 倏逝强度与场幅衰减；FX121 倏逝穿透深度约定（辅助）；FX122 WGM 共振与小幅相对变化；FX123 双镜腔共振（辅助）；FX124 半高宽、Q 与单位一致的 FOM；FX125 微弱拉曼信号与所报 SERS 增强量级。

## C09 • [P1] Labelled versus label-free optical sensing / 有标记与无标记光学传感

Priority / 复习标记：P1 First pass/先掌握 · Sample Q3/样卷Q3 · Teacher emphasis/课堂强调 · Repeated explanation/反复讲解

**EN: Key points**

- Labelled sensing attaches a reporter such as a fluorescent dye.
- Label-free sensing detects an intrinsic change such as absorption, refractive index, or resonance shift without adding that reporter.
- In the sample classification, fluorescence intensity is treated as a labelled method.

**中文：必会知识点**

- 有标记传感添加荧光染料等报告物。
- 无标记传感不添加这种报告物，而测量吸收、折射率或共振位移等内在变化。
- 样题分类把fluorescence intensity视为有标记方法。

**EN:** Start from zero: Many biological targets do not produce an easily visible signal by themselves. A label is an added reporter, such as a fluorescent dye attached to an antibody. It helps reveal where the target is or how much is present. Labelled sensing reads that reporter. Label-free sensing instead reads a property change caused by the target without adding that kind of reporter.

**中文：** 从零理解：许多生物目标本身不会产生容易看见的信号。Label（标记）是一种额外添加的报告物，例如连接在抗体上的荧光染料，帮助显示目标在哪里或有多少。Labelled sensing（有标记传感）读取这个报告物；label-free sensing（无标记传感）则不添加这种报告物，而读取目标引起的性质变化。

**EN:** Follow the mechanism: Imagine locating a guest by a colored badge versus noticing their weight on a floor sensor. The badge is a reporter; the weight is an intrinsic effect. In an optical label-free sensor, surface binding can change refractive index, meaning how light propagates through the local material. A light source measures that change. Thus label-free does not mean light-free, antibody-free, or surface-free.

**中文：** 跟着机制走：可以类比为靠彩色胸牌找到客人，与靠地板传感器感受到客人的重量。胸牌相当于报告物，重量则是客人本身造成的作用。在光学无标记传感器中，表面结合可以改变refractive index（折射率），即光在局部材料中传播的性质。光源负责测量这个变化，所以label-free并不等于不用光、不用抗体或不用表面。

**EN:** Try an example: Teaching example: adding a fluorescent antibody and measuring its glow is labelled sensing. Measuring a resonance shift after unlabelled molecules bind can be label-free sensing. First identify whether a separate reporter was added, then identify the measured signal. Some substances naturally fluoresce, called autofluorescence, so do not claim all fluorescence is labelled. The sample classification concerns its taught dye-based fluorescence example.

**中文：** 用例子理解：教学例子：添加荧光抗体并测量它的发光，属于有标记传感；未标记分子结合后测共振位移，可以属于无标记传感。先判断是否添加了独立报告物，再判断测的信号。有些物质天然会发荧光，叫autofluorescence（自发荧光），因此不能说所有荧光都属于有标记。样题分类针对的是课程所讲的染料荧光例子。

**EN:** Exam wording: Label-free does not mean no illumination, no biological recognition, or no sensor surface. Autofluorescence exists, so the course classification should not be turned into an absolute claim about every fluorescence measurement.

**中文：** 英文读题抓手：Label-free不等于不照光、不需生物识别或不需传感界面。自发荧光确实存在，因此不能把课程分类扩大为所有荧光测量的绝对规律。

Source / 来源：W3 PDF p13–18 | T3-0016–0020

## S117 • [P1] Label-free changes and the signal actually measured / 无标记变化与真正测得的信号

Priority / 复习标记：P1 First pass/先掌握 · Sample Q3/样卷Q3 · Teacher emphasis/课堂强调

**EN:** Label-free optical sensing may detect direct absorption, refractive-index change, or a shift of resonance wavelength/frequency caused by the target. The index change is a property change; wavelength/frequency shift is one possible instrument readout of that change. At fixed medium convention, a shift toward longer wavelength corresponds to lower frequency. A sensor can also monitor intensity at a selected point on the shifting response curve.

**中文：** 无标记光学传感可检测目标的直接吸收、折射率变化，或由目标引起的共振波长/频率移动。折射率变化是性质改变，波长/频率位移是仪器对该改变的一种读出。介质约定固定时，波长变长对应频率变低。传感器也可在响应曲线选定位置监测强度，观察曲线移动。

**EN:** In sample Q3, B absorption, C refractive index, D wavelength and E frequency can therefore be related to label-free principles. A fluorescence intensity refers to the taught added-dye reporter and is the intended exception. This classification does not say that every intensity measurement is labelled, or that autofluorescence is impossible. “No label” means no added reporter of that kind, not no light, no excitation, or no antibody receptor.

**中文：** 因此样题Q3中，B吸收、C折射率、D波长、E频率，都可与无标记原则有关。A荧光强度针对课程所讲的外加染料报告物，是题目预期的例外。这个分类不表示所有强度测量都是有标记，也不否认自发荧光。“无标记”是没有添加这一类报告物，不是不用光、不激发或不用抗体受体。

Source / 来源：Week 3 PDF p15–18; T3-0016–0029; Sample quiz p1 Q3

## B09 • [P2] Quantum dots and flow cytometry / 量子点与流式细胞术

Priority / 复习标记：P2 Important foundations/重要基础 · Repeated explanation/反复讲解

**EN: Key points**

- Quantum dots are semiconductor fluorescent labels with size-dependent emission;
- larger dots commonly emit at longer wavelengths within the same material family.
- Flow cytometry measures cells/particles passing a detection region: forward scatter (FSC) mainly indicates size, side scatter (SSC) internal complexity/granularity, and fluorescence label-associated molecular information.
- A scatter plot separates populations;
- it is not a tissue image.

**中文：必会知识点**

- Quantum dot量子点是半导体荧光标记，发射随尺寸变化；
- 同材料系列中，较大颗粒通常发射更长波长。
- Flow cytometry流式细胞术测量流经检测区的细胞/颗粒：FSC前向散射主要反映大小，SSC侧向散射反映内部复杂度/颗粒度，荧光反映标记所对应的分子信息。
- 散点图用于区分群体，不是组织图像。

**EN:** Start from zero: A quantum dot is a very small semiconductor particle that can fluoresce. Confinement at this size changes the available electronic energies, so particle size can affect emitted color. Within a comparable material family, larger dots commonly emit longer wavelengths. Flow cytometry is a separate method: it passes cells or particles through a detection region and measures each event, often using light scattering and fluorescence.

**中文：** 从零理解：Quantum dot（量子点）是可以发荧光的极小半导体颗粒。这种尺寸下的约束改变可用电子能量，因此颗粒大小可以影响发射颜色。同一个可比材料系列中，较大的量子点通常发射更长波长。Flow cytometry（流式细胞术）是另一种方法：让细胞或颗粒通过检测区域，逐个测量事件，常用光散射和荧光。

**EN:** Follow the mechanism: Imagine cells passing a checkpoint one by one. Forward scatter, FSC, mainly provides information related to size. Side scatter, SSC, relates to internal complexity or granularity. Fluorescent labels can indicate particular molecules. Optical setup and cell properties also affect these signals; FSC is not a perfect ruler and SSC is not a complete picture of the cell interior.

**中文：** 跟着机制走：想象细胞一个接一个经过检查点。Forward scatter（FSC，前向散射）主要提供与大小有关的信息；side scatter（SSC，侧向散射）与内部复杂度或颗粒度有关；荧光标记可以提示某些分子。光学配置和细胞性质也影响这些信号：FSC不是完美的尺，SSC也不是细胞内部的完整照片。

**EN:** Try an example: Teaching example: a scatter plot places each measured cell according to its FSC and SSC values; clusters can suggest different populations. Adding a fluorescent antibody supplies another measurement for the matching target. This plot is not a tissue image showing neighboring cell positions. If asked which signal reports labelled molecular information, choose fluorescence; if asked which mainly relates to size, choose FSC, with the stated approximation.

**中文：** 用例子理解：教学例子：散点图根据每个细胞的FSC与SSC数值放置点，点群可以提示不同细胞群体。加入荧光抗体，又提供一个针对匹配目标的测量量。这个图不是显示邻近细胞位置的组织图像。若问哪个信号反映带标记的分子信息，选荧光；若问哪个主要与大小有关，按上述近似选FSC。

Source / 来源：W2 PDF p89–95; W3 PDF p5–9; W3 PDF p10–12 | T2-0108–0111, T3-0003–0015

## S113 • [P3] Quantum dots compared with molecular dyes / 量子点与分子染料的比较

Priority / 复习标记：P3 Supporting reading/辅助阅读 · Supporting concept/辅助理解

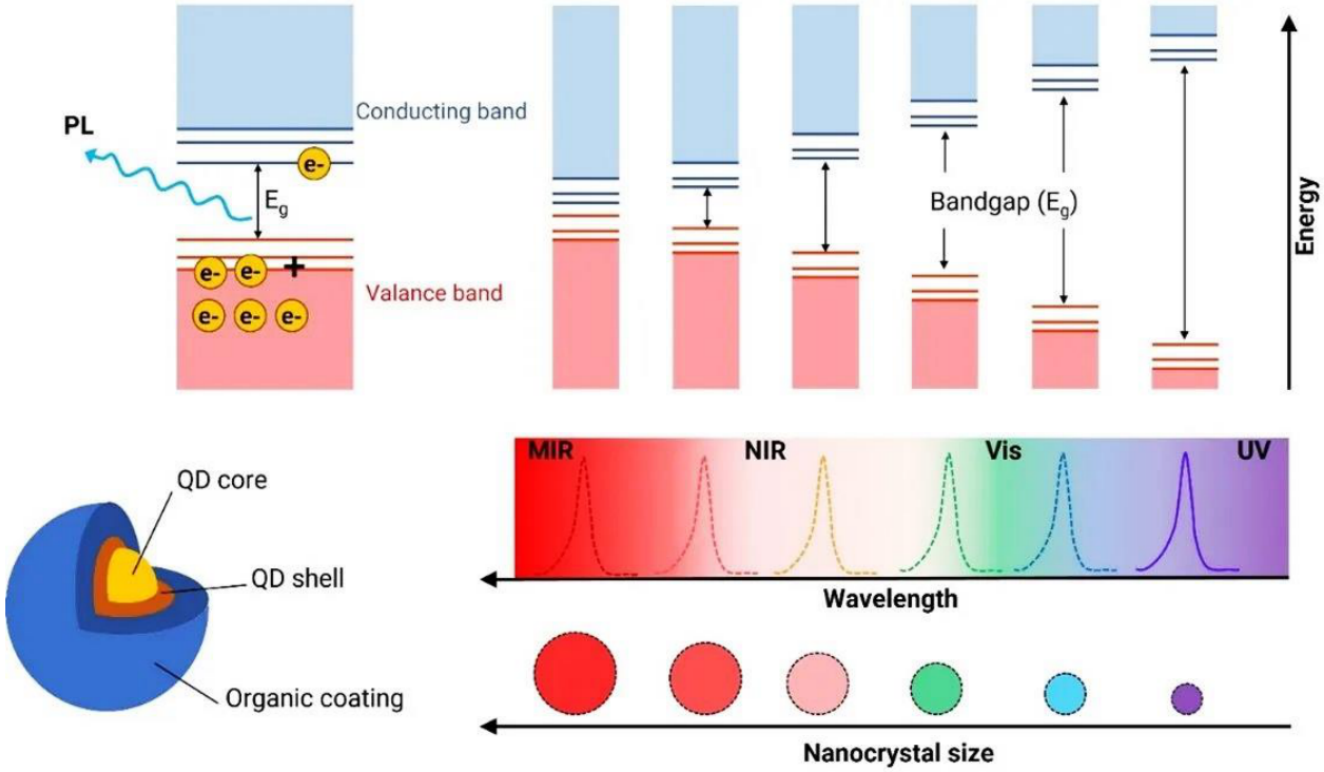
**EN:** Within a comparable quantum-dot material family, smaller dots have a larger band gap and tend to emit shorter-wavelength light; larger dots tend to emit longer wavelengths. Different sizes can supply different emission colors. Many dot colors can be excited by one suitable shorter-wavelength source whose photon energy is sufficient for their absorption bands, simplifying multicolor detection. This does not mean any source excites every dot.

**中文：** 同一个可比量子点材料系列中，较小颗粒有较大band gap（带隙），倾向发射较短波长；较大颗粒倾向发射较长波长。不同尺寸可提供不同发射颜色。能量足够、且波段匹配的一种较短波長光源可激发多种量子点颜色，简化multicolor detection（多色检测），但不是任何光源都能激发所有量子点。

**EN:** The course comparison presents quantum dots as potentially brighter and more photostable than its molecular-dye examples, with more distinguishable color channels. Photostability means maintaining performance under illumination; it is not the same as excited-state lifetime. Such advantages can help reproducibility and multiplexing, but actual yield and stability depend on material and conditions: do not memorize the example percentages as universal values for every dye or quantum dot.

**中文：** 课程比较表把量子点描述为相对其分子染料示例可能更亮、更photostable（耐光照稳定），并能提供更多可区分颜色通道。Photostability指光照下维持性能，不是激发态寿命。这些优势有助于再现性和多通道检测，但实际产率与稳定性取决于材料、条件，不能把表中示例百分比当成所有染料或量子点的普遍值。

Source / 来源：Week 3 PDF p6–9; T3-0003–0007




Quantum-dot size changes the bandgap and therefore the emission color. / 量子点尺寸改变带隙，进而改变发光颜色。

**EN:** Read the figure: Align each dot along the bottom with its bandgap above. For the same semiconductor family, a smaller nanocrystal generally has a larger bandgap and emits a higher-energy, shorter-wavelength photon; a larger one shifts toward longer wavelengths. The arrows point left toward increasing wavelength and size in this original figure, so check the axis direction. The core–shell drawing shows structure, not two separate colors that must both be emitted.

**中文：** 读图：把底部每个小球与上方带隙对应起来看。同一半导体材料体系中，纳米晶越小，Bandgap（带隙）通常越大，发出的光子能量更高、波长更短；越大则趋向较长波长。原图波长和尺寸轴向左增加，读图时别把方向看反。左下Core–shell（核–壳）图表示结构，不表示必定同时发出两种颜色。

Source / 来源：Week 3 PDF p7; original slide figure / 原课件截图

## C10 • [P1] Optical fibers and total internal reflection / 光纤与全内反射

Priority / 复习标记：P1 First pass/先掌握 · Teacher emphasis/课堂强调 · Sample Q9/样卷Q9

**EN: Key points**

- A conventional guiding fiber has a higher-index core and lower-index cladding.
- Total internal reflection requires light travelling from higher to lower refractive index and an incidence angle above the critical angle.
- Point sensing measures locally;
- multiplexed sensing uses several sites;
- distributed sensing reads along the fiber.

**中文：必会知识点**

- 常规导光光纤的纤芯折射率高于包层。
- 全内反射要求光从高折射率介质到低折射率介质，且入射角大于临界角。
- Point sensing在局部测量；
- multiplexed sensing有多个位置；
- distributed sensing沿光纤分布测量。

**EN:** Start from zero: An optical fibre carries light along a thin core surrounded by cladding. Refractive index describes a material's effect on light propagation; a larger index means a lower phase speed than in vacuum. In a conventional guiding fibre, the core has the higher index. The boundary can keep suitable light inside through total internal reflection, allowing it to travel around gentle bends.

**中文：** 从零理解：Optical fibre（光纤）让光沿细小的core（纤芯）传播，外面包着cladding（包层）。Refractive index（折射率）描述材料对光传播的影响，折射率更大表示光的相速度比真空更低。常规导光光纤的纤芯折射率较高。边界可以通过total internal reflection（全内反射）把符合条件的光留在内部，使光能经过适度弯曲的路径。

**EN:** Follow the mechanism: Total internal reflection needs two conditions: light approaches the boundary from the higher-index side, and its incidence angle exceeds the critical angle. The incidence angle is measured from the normal, a line perpendicular to the surface. Think of repeated boundary reflections guiding a beam, but remember the wave field is not perfectly confined: an evanescent field extends a short distance outside and decays rapidly.

**中文：** 跟着机制走：全内反射需要两个条件：光从较高折射率一侧来到边界，而且入射角超过临界角。入射角从normal（法线，即垂直表面的线）开始测量。可以先想象光通过反复的边界反射被引导，但波的场并非完全关在里面：evanescent field（倏逝场）会向外延伸一小段，并迅速衰减。

**EN:** Try an example: Teaching example: molecules near an exposed sensing section can interact with the evanescent field and change the optical signal. A question saying that all external fields vanish under total internal reflection is therefore wrong. Point sensing measures one location; multiplexed sensing reads several separate sites; distributed sensing provides information along the fibre. These describe where measurements occur, not whether the target is biologically specific.

**中文：** 用例子理解：教学例子：靠近暴露传感段的分子，可以与倏逝场相互作用，改变光信号。因此，“全内反射时外部所有场都消失”是错误的。Point sensing（点式传感）测一个位置，multiplexed（复用）测多个分开的位点，distributed（分布式）沿光纤提供信息。这些术语描述测量位置，并不描述目标是否有生物特异性。

**EN:** Exam wording: An evanescent field outside the guiding region can interact with the surrounding sample. Total internal reflection does not mean the electromagnetic field is exactly zero outside the core.

**中文：** 英文读题抓手：导光区域外的evanescent field可与周围样品相互作用。全内反射并不表示纤芯外电磁场严格为零。

Source / 来源：W3 PDF p19–24; W3 PDF p23–28; W3 PDF p31–38 | T3-0030–0045, T3-0052–0057

## S114 • [P2] Snell’s law, critical angle and fibre acceptance / 斯涅尔定律、临界角与光纤接受角

Priority / 复习标记：P2 Important foundations/重要基础 · Teacher emphasis/课堂强调 · Worked example/例题练习 · Final emphasis/仅期末强调

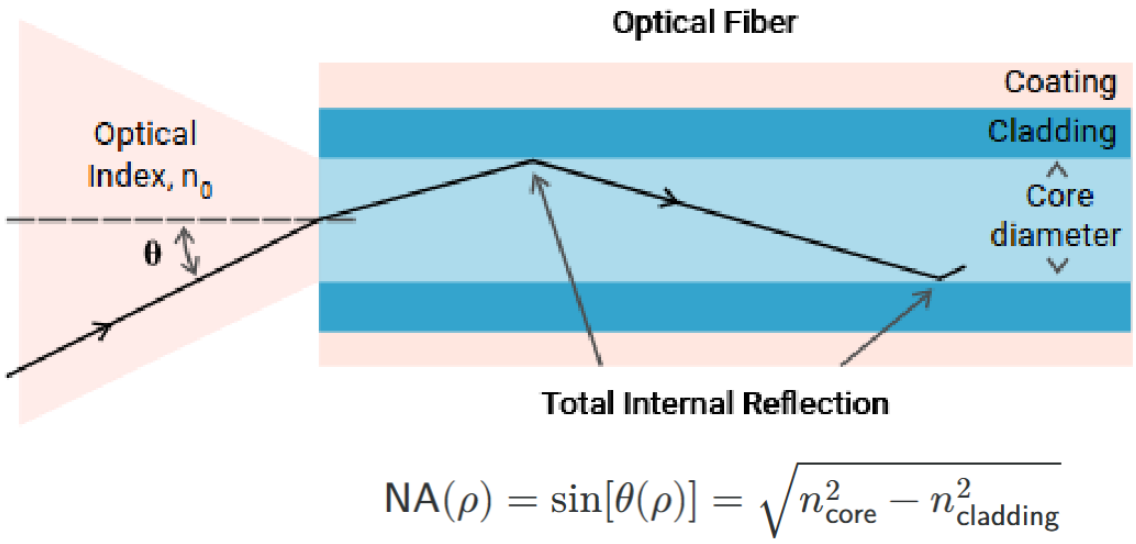
**EN:** Snell’s law is $n_1 \sin\theta_1 = n_2 \sin\theta_2$; angles are measured from the normal. Entering a higher-index medium bends a ray toward the normal; entering a lower-index medium bends it away. For $n_1 > n_2$, the critical angle satisfies $\sin\theta_c = n_2/n_1$. At $\theta_c$ the ideal transmitted ray is tangent to the interface; angles above it give total internal reflection. With $n_1 = 1.6$ and $n_2 = 1.38$, $\theta_c$ is about 59.6°.

**中文：** Snell’s law（斯涅尔定律）为$n_1 \sin\theta_1 = n_2 \sin\theta_2$，角度从法线测量。进入较高折射率介质，光线向法线弯折；进入较低折射率介质，远离法线。$n_1 > n_2$时，临界角满足$\sin\theta_c = n_2/n_1$。$\theta_c$处理想透射光沿界面；超过它发生全内反射。$n_1 = 1.6$、$n_2 = 1.38$时，$\theta_c$约59.6°。

**EN:** The fibre’ s external acceptance angle is different from the internal incidence angle at the core–cladding wall. For an ideal step-index fibre, NA=$n_0$ sinθ_accept=√(n_core$^2$−n_cladding$^2$), where θ_accept is the maximum accepted half-angle and $n_0$ the outside index. A larger NA accepts a wider cone. Use the right indices and angle definition; the refractive index of a lens is not automatically the index to insert in a fibre formula.

**中文：** 光纤外部的acceptance angle（接受角）不同于光线在纤芯—包层壁面的内部入射角。理想阶跃折射率光纤满足NA=$n_0$ sinθ_accept=√(n_core$^2$−n_cladding$^2$)，θ_accept是最大接受半角，$n_0$是外部介质折射率。NA更大，能接受的光锥更宽。要使用正确折射率与角度定义，不能自动把透镜折射率代入光纤公式。

Source / 来源：Week 3 PDF p20–24; Week 1 PDF p86; T3-0030–0041




Fibre acceptance angle and numerical aperture (original lecture slide). / 光纤接受角与数值孔径（原课件）。

**EN:** Read the figure: The marked entry angle is an external acceptance half-angle. The internal reflection angle at the core–cladding boundary is measured from that boundary’ s normal; do not interchange the two. The equation printed on this slide assumes entry from air ($n_0$≈1); in general NA=$n_0$ sinθ_accept.

**中文：** 读图：图中入纤角是外部接受半角；纤芯—包层壁面内部反射角从该壁面的法线测量，两个角不能互换。 原图NA=sinθ默认从空气入射（$n_0$≈1）；一般应写NA=$n_0$ sinθ_accept。

Source / 来源：Week 3 PDF p23; original slide figure / 原课件截图

## S115 • [P1] Evanescent sensing is confined near an interface / 倏逝传感局限在界面附近

Priority / 复习标记：P1 First pass/先掌握 · Sample Q9/样卷Q9 · Repeated explanation/反复讲解

**EN:** An evanescent field extends into the lower-index side of a totally reflecting interface and decays rapidly away from it. In the slide’ s intensity model, $I(d)/I(0)=\exp(-d/L_0)$. At one intensity decay length $L_0$, intensity is about 37% of its boundary value; at two it is about 14%. Amplitude and intensity decay lengths use different conventions, so use the definition printed with the equation.

**中文：** Evanescent field（倏逝场）延伸进入全反射界面较低折射率的一侧，并随离界面距离迅速衰减。课件强度模型为$I(d)/I(0)=\exp(-d/L_0)$。离界面一个强度衰减长度$L_0$时，强度约为界面值37%；两个时约14%。场振幅与强度的衰减长度定义不同，应以方程所注明的定义为准。

**EN:** Penetration depends on wavelength, refractive-index contrast and incidence angle. The sensor is especially sensitive to molecules within this nearby field, which is why immobilized receptors at the surface are useful. It is not uniformly sensitive to an entire thick sample. A strong near-boundary field in sample Q9 A refers to this surface interaction; it does not mean all evanescent fields are automatically amplified or that light propagates deeply through the outside medium.

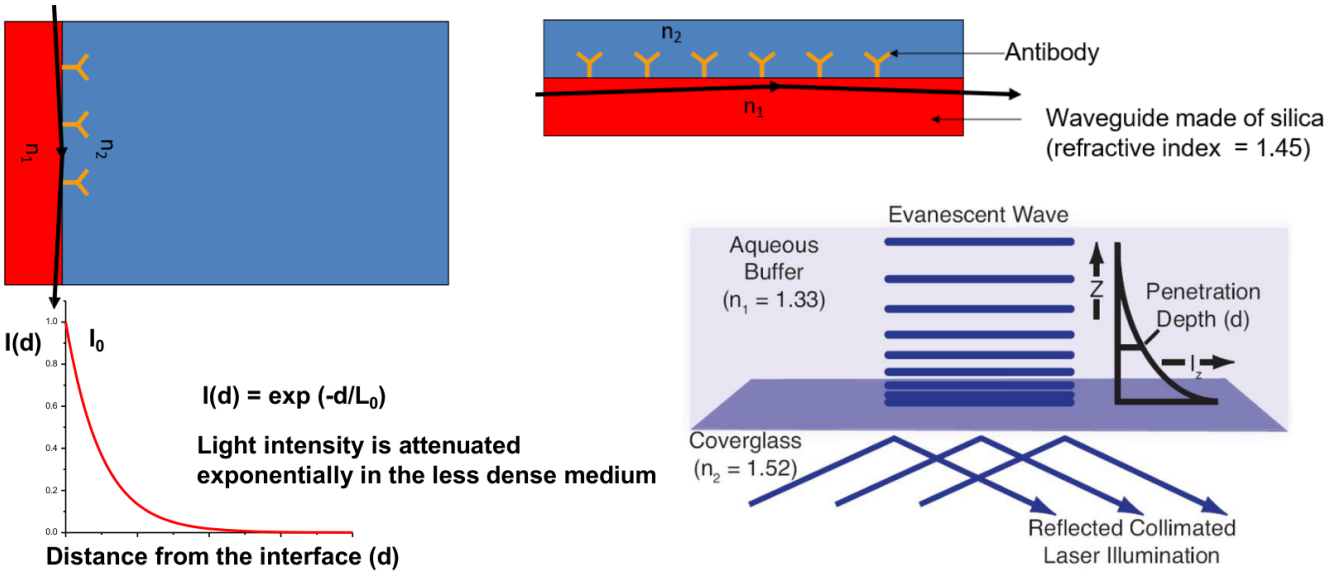
**中文：** Penetration depth（穿透深度）与波长、两介质折射率差及入射角有关。传感器特别感受附近场范围内的分子，因此把受体固定在表面有用；它并非对整个厚样品同样敏感。样题Q9 A的“界面附近强场”指这种近表面相互作用，不意味着所有倏逝场自动被放大，或光会在外部介质深处持续传播。

Source / 来源：Week 3 PDF p25–29; T3-0042–0050

**EN:** Clarify the concept: The wave description of total internal reflection also gives a small lateral displacement of a finite reflected beam relative to the ideal ray reflection point: the Goos–Hänchen shift. The slide uses this with the evanescent-field diagram to show why a purely geometric picture is incomplete. Keep the near-interface exponential decay and its dependence on wavelength, indices and angle; do not confuse a lateral beam shift with penetration depth normal to the interface.

**中文：** 把概念讲清：全内反射的波动描述还给出有限宽反射光束相对理想光线反射点的小横向位移，叫Goos–Hänchen shift（古斯—汉欣位移）。课件配合倏逝场图说明纯几何图像不完整。仍以界面附近指数衰减及其波长、折射率、角度依赖为主；横向光束位移不同于垂直界面的穿透深度。

Source / 来源：Week 3 PDF p25–27 explicitly labels Goos–Haenchen shift; current S115 supplies decay and convention; recognition-level addition only




The evanescent field decays away from the interface, making sensing surface-sensitive. / 倏逝场离开界面后迅速衰减，因此检测对表面附近更敏感。

**EN:** Read the figure: The light remains guided, but its evanescent field extends into the adjoining sample. Surface antibodies place bound targets within this short sensing region. Read the distance axis below: intensity decreases exponentially with distance from the interface. The slide writes exp(−d/$L_0$); interpret this as normalized intensity I(d)/$I_0$. Thus targets close to the surface affect the signal more strongly; total internal reflection does not mean zero field outside the guide.

**中文：** 读图：光仍被波导导引，但Evanescent field（倏逝场）会伸进邻近样品。表面抗体把结合的目标带到这段很薄的感应区。看下方距离轴：离开界面越远，光强越弱，呈指数衰减。课件写exp(−d/$L_0$)，应理解为归一化光强I(d)/$I_0$。因此靠近表面的目标影响更大；Total internal reflection（全内反射）不表示波导外完全没有场。

Source / 来源：Week 3 PDF p27; original slide figure / 原课件截图

## B10 • [P1] Fibre modes and cavity recognition / 光纤模式与共振腔辨识

Priority / 复习标记：P1 First pass/先掌握 · Sample Q2, Q9/样卷Q2、Q9 · Repeated explanation/反复讲解

**EN: Key points**

- A mode is an allowed field pattern;
- single-mode and multimode fibres differ in supported patterns.
- Evanescent fields decay outside the guiding region and enable surface sensing.
- Fabry–Pérot cavities use facing reflectors;
- whispering-gallery modes circulate near a boundary;
- photonic-crystal cavities confine light using a structured refractive-index arrangement.
- Multiplexing uses distinct sensing sites;
- distributed sensing samples along the fibre.

**中文：必会知识点**

- Mode模式是允许存在的场分布；
- single-mode单模与multimode多模光纤支持的模式数不同。
- Evanescent field倏逝场在导光区域外衰减，可用于表面传感。
- Fabry–Pérot腔使用相对反射面；
- whispering-gallery mode回音壁模式沿边界循环；
- 光子晶体腔利用结构化折射率分布约束光。
- Multiplexed在多个离散位点检测；
- distributed沿光纤分布检测。

**EN:** Start from zero: A mode is an allowed pattern of the light field in a structure. A single-mode fibre supports one guided spatial pattern at the operating wavelength; a multimode fibre supports several. This does not mean single-mode light contains only one photon. A cavity confines light so selected patterns can persist and resonate. Its geometry helps determine the allowed patterns, much as an instrument's shape influences its supported notes.

**中文：** 从零理解：Mode（模式）是光场在结构中允许存在的分布。Single-mode fibre（单模光纤）在工作波长处支持一个导光空间模式，multimode fibre（多模光纤）支持多个，并不是说单模光里只有一个光子。Cavity（腔）约束光，让选定模式可以持续并共振。几何形状帮助决定允许模式，类似乐器形状影响可支持的音调。

**EN:** Follow the mechanism: A Fabry–Pérot cavity uses facing reflectors, letting light travel back and forth. A whispering-gallery mode circulates near a boundary, resembling sound traveling around a round gallery. A photonic-crystal cavity uses a structured refractive-index arrangement to confine light. These are recognition aids, not exact mechanical pictures: all involve electromagnetic field patterns. An evanescent field outside a guiding boundary can sample nearby molecules without requiring them to enter the whole cavity.

**中文：** 跟着机制走：Fabry–Pérot（法布里—珀罗）腔使用相对的反射面，使光往返传播；whispering-gallery mode（回音壁模式）沿边界循环，类似声音沿圆形回廊传播；photonic-crystal cavity（光子晶体腔）利用结构化折射率排列约束光。这些只是辨认辅助，不是严格的机械图像，三者都涉及电磁场模式。导光边界外的倏逝场可以感受附近分子，不要求分子进入整个腔。

**EN:** Try an example: Teaching example: identify facing mirrors, a circular rim, or a patterned lattice before naming the cavity. For fibre sensing, distinguish several separate sensing locations, multiplexed sensing, from information continuously resolved along a length, distributed sensing. Neither describes the number of optical modes. Mode describes a field pattern; sensing arrangement describes where information is obtained.

**中文：** 用例子理解：教学例子：先寻找相对镜面、圆形边缘或图案化晶格，再判断腔类型。光纤传感中，多个分开的传感位置属于复用传感，沿长度连续分辨信息属于分布式传感；两者都不是光学模式数。模式描述场分布，传感安排描述从哪里获得信息。

Source / 来源：W3 PDF p19–24; W3 PDF p23–28; W3 PDF p30; W3 PDF p31–38; W3 PDF p31–39; W3 PDF p39–45; W3 PDF p48–50 | T3-0030–0045, T3-0051–0072, T3-0079–0083

## S116 • [P1] Modes, dispersion and spatial sensing arrangements / 模式、色散与空间传感安排

Priority / 复习标记：P1 First pass/先掌握 · Sample Q9/样卷Q9 · Repeated explanation/反复讲解 · Lower priority/降低优先级

**EN:** Different modes can travel with different delays. A multimode fibre can collect more light but a pulse may spread in time because its modes arrive at different times: modal dispersion. Single-mode operation avoids this particular source of spreading; it does not eliminate every loss or every kind of dispersion. Mode count describes field patterns, not target count or number of sensors.

**中文：** 不同模式可具有不同传播延时。Multimode fibre（多模光纤）能收集较多光，但不同模式抵达时间不同，脉冲可能在时间上展宽，这叫modal dispersion（模式色散）。单模操作避免这一类展宽，并不消除所有损耗或所有色散。模式数描述场分布，不是目标数或传感器数量。

**EN:** Point sensing reads locally. Multiplexed sensing separately reads several localized sites/channels along one fibre; the taught example assigns different targets to different sites. Distributed sensing resolves a measurand along the fibre, such as temperature at many positions. Thus sample Q9 D/E fit the classroom examples. In the general definition, multiplexed sites may measure the same target, and distributed systems can have multiple parameters; do not turn the examples into universal restrictions.

**中文：** Point sensing（点式）在局部读出。Multiplexed sensing（复用）分别读取一根光纤上的多个离散位点/通道，课程例子在不同位点安排不同目标。Distributed sensing（分布式）沿光纤分辨某一被测量，例如各位置温度。因此样题Q9 D/E符合课堂示例。通用定义中，复用位点也可测同一目标，分布式系统也可包含多参数，不能把示例扩大成普遍限制。

Source / 来源：Week 3 PDF p24, p30–38; T3-0051–0058; Sample quiz p2 Q9 D/E

## C11 • [P1] Resonators, SPR and LSPR / 共振器、表面等离激元与局域表面等离激元

Priority / 复习标记：P1 First pass/先掌握 · Sample Q2/样卷Q2 · Teacher emphasis/课堂强调 · Repeated explanation/反复讲解 · Lower priority/降低优先级

**EN: Key points**

- Optical resonators enhance repeated light–sample interaction.
- SPR is a light-coupled collective electron oscillation at a metal interface;
- LSPR is localized around metal nanoparticles or nanostructures.
- Surface binding can shift a resonance because the surrounding refractive index changes.

**中文：必会知识点**

- 光学共振器增强光与样品反复相互作用。
- SPR是金属界面上与光耦合的集体电子振荡；
- LSPR局域于金属纳米颗粒或纳米结构。
- 表面结合改变周围折射率，因此可能引起共振位移。

**EN:** Start from zero: Resonance means a system responds especially strongly at certain frequencies. A playground swing is easier to drive when pushes match its timing. An optical resonator similarly favors particular light frequencies, although its mechanism involves electromagnetic waves rather than a swinging mass. Repeated light interaction with a sensing region can make small changes easier to observe through shifts of a resonance peak.

**中文：** 从零理解：Resonance（共振）表示一个系统在某些频率处响应特别强。推秋千时，如果推的节奏配合秋千的运动节奏，就比较容易越荡越高。光学共振器也会特别响应某些光频率，但它涉及电磁波，而不是摆动的物体。光在传感区域反复作用，可以让很小的变化通过共振峰位移变得更容易观察。

**EN:** Follow the mechanism: In SPR, light drives many mobile electrons in a metal to oscillate together at an interface. In LSPR, this collective response is localized around a nanoparticle or nanostructure. Nearby molecules change the optical environment, often represented by refractive index, and can shift the response peak. A metal plasmonic surface is not automatically a closed cavity with two mirrors; the course groups these mechanisms broadly as resonance sensing.

**中文：** 跟着机制走：在SPR（表面等离激元共振）中，光驱动金属中许多可移动电子在界面处一起振荡。在LSPR（局域表面等离激元共振）中，这种集体响应局域在纳米颗粒或纳米结构周围。附近的分子改变光学环境，常用折射率描述，因而可能改变响应峰的位置。金属等离激元表面并不自动等于由两面镜子组成的闭合腔，课程把这些机制广义归入共振传感。

**EN:** Try an example: Teaching example: a sensor peak moves after protein binds to its surface. This suggests resonance-shift readout; recognition chemistry still determines which protein binds. Identify SPR by its metal interface and LSPR by localized metal structures. Do not decide selectivity from a large peak shift alone: unrelated changes in temperature or surrounding solution can also move a resonance.

**中文：** 用例子理解：教学例子：蛋白质结合传感器表面后，峰的位置发生移动，这提示采用共振位移读出；至于哪种蛋白质会结合，仍由识别化学决定。看到金属界面想到SPR，看到局域金属结构想到LSPR。不能只凭位移大就认定选择性好：温度或周围溶液的其他变化也可能移动共振峰。

**EN:** Exam wording: The course uses a broad resonance-sensing classification. Do not equate every SPR structure with a closed Fabry–Perot cavity. For this Quiz, identify mechanism and shift direction before studying long resonance derivations.

**中文：** 英文读题抓手：课程采用广义“共振传感”分类，但不能把所有SPR结构等同于闭合Fabry–Perot腔。这次Quiz优先辨识机制与位移，再考虑长共振公式。

Source / 来源：W3 PDF p39–45; W3 PDF p51–60; W3 PDF p64–75 | T3-0062–0072, T3-0084–0093, T3-0099–0105

## S118 • [P2] Resonance shifts from index and geometry / 折射率与几何造成的共振位移

Priority / 复习标记：P2 Important foundations/重要基础 · Repeated explanation/反复讲解 · Final emphasis/仅期末强调 · Lower priority/降低优先级

**EN:** Cavity resonators such as Fabry–Pérot and WGM recirculate light so a sample can interact with the field repeatedly, rather than only during one pass. For the illustrated WGM model, $\lambda_r=2\pi R n_{eff}/m$ for a fixed mode order m. Small changes give $\Delta\lambda_r/\lambda_r \approx \Delta R/R+\Delta n_{eff}/n_{eff}$. With radius and mode fixed, a larger effective index shifts this resonance to longer wavelength. A size change can also shift it, so a shift is not proof of binding by itself.

**中文：** Fabry–Pérot、WGM等cavity resonator（腔型共振器）使光循环，让样品反复与场作用，而不是只经过一次。课件WGM模型在模式级次m固定时，$\lambda_r=2\pi R n_{eff}/m$。微小变化满足$\Delta\lambda_r/\lambda_r \approx \Delta R/R+\Delta n_{eff}/n_{eff}$。半径、模式固定时，有效折射率更大，会使该共振移向较长波长；尺寸变化也会造成位移，因此位移本身不是结合的充分证据。

**EN:** Bulk refractive-index calibration uses liquids with known indices and measures the resulting shifts to obtain $\Delta\lambda/\Delta n$. It characterizes the optical device’ s index response, not its ability to distinguish one biomolecule from another. Two facing reflectors identify a Fabry–Pérot cavity, a circulating rim mode identifies WGM, and a periodic index structure identifies a photonic-crystal resonator. Raman scattering reports vibrational energy exchange rather than this resonance-recycling mechanism.

**中文：** Bulk refractive-index calibration（体折射率校准）用折射率已知的液体，测对应位移，求$\Delta\lambda/\Delta n$。它表征光学器件的折射率响应，不是区分不同生物分子的能力。两面相对反射面提示Fabry–Pérot腔，沿边界循环的模式提示WGM，周期性折射率结构提示光子晶体共振器。拉曼报告振动能交换，不是这种循环光的共振机制。

Source / 来源：Week 3 PDF p41–50; T3-0062–0083

## S119 • [P2] SPR coupling, localized plasmons and EOT / SPR耦合、局域等离激元与EOT

Priority / 复习标记：P2 Important foundations/重要基础 · Repeated explanation/反复讲解 · Lower priority/降低优先级

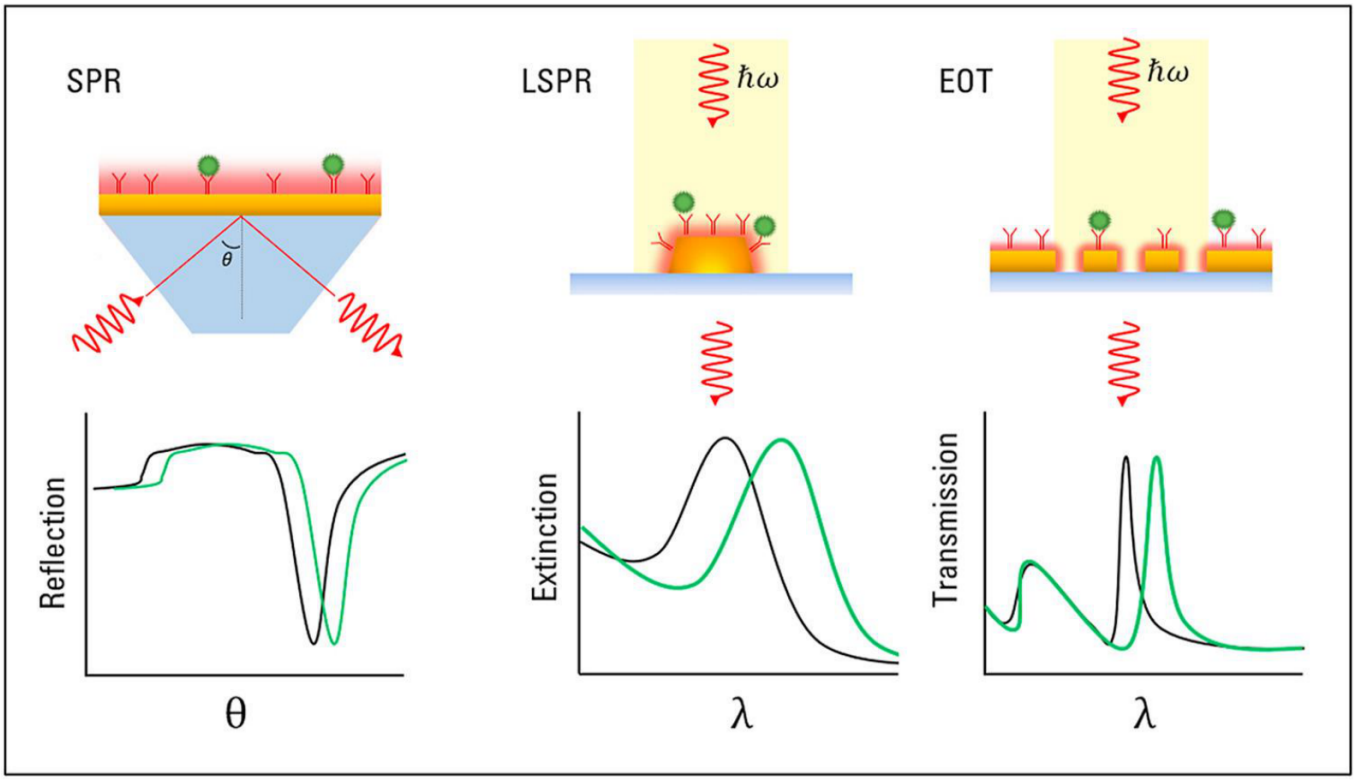
**EN:** A propagating surface plasmon couples an electromagnetic field with collective conduction-electron motion at a metal–dielectric interface. Exciting it requires appropriate coupling; a prism or grating can provide a suitable coupling condition. It is not enough to shine any beam onto any metal and assume an SPR signal. In LSPR the response is localized around a small metal structure; its excitation arrangement is often simpler, but wavelength, structure and polarization can still matter.

**中文：** 传播型surface plasmon（表面等离激元）把电磁场与金属—介质界面的导电电子集体运动耦合。激发它需要合适coupling（耦合），棱镜或光栅可提供相应条件；不能任意光束照任意金属，就假定出现SPR信号。LSPR的响应局域在小金属结构周围，激发布置通常更简单，但波长、结构和偏振仍可能有影响。

**EN:** Extraordinary optical transmission, EOT, refers here to resonantly enhanced transmission through patterned small holes in a metal film. Its spectrum can change when the nearby refractive index changes. Read the plotted observable: traditional SPR often shows a reflectivity dip, LSPR an extinction peak, and EOT a transmission feature. A resonance may appear as a peak or a dip depending on what the detector measures; neither appearance alone identifies the biosensor family.

**中文：** 这里的Extraordinary optical transmission（EOT，异常光透射）指金属膜图案化小孔中的共振增强透射，附近折射率改变可使其光谱变化。要读纵轴：传统SPR常显示reflectivity（反射率）凹谷，LSPR显示extinction（消光）峰，EOT显示透射特征。同一类共振因被测量不同，可以表现为峰或谷；仅靠“峰还是谷”不能确定生物传感器类型。

Source / 来源：Week 3 PDF p54–68, p81; T3-0084–0103

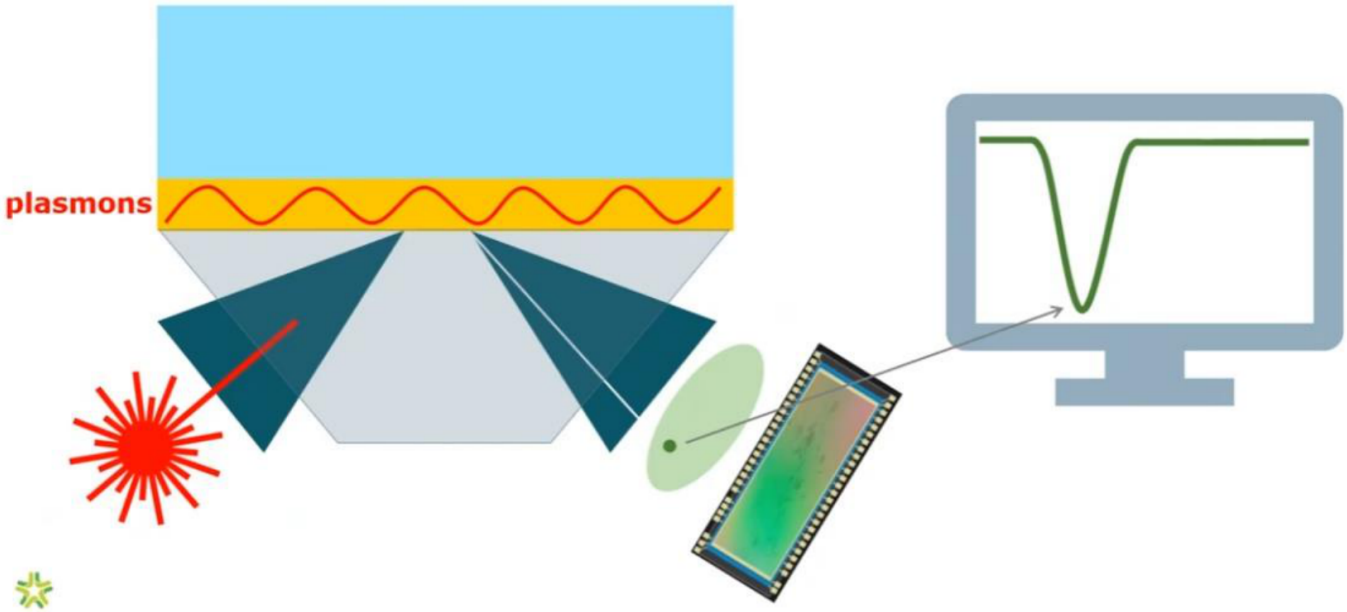



SPR, LSPR and EOT: different measured spectra (original lecture slide). / SPR、LSPR、EOT：不同的被测光谱（原课件）。

**EN:** Read the figure: Read both axes: SPR plots reflection against angle here, LSPR extinction against wavelength, EOT transmission against wavelength. A target-induced resonance change may move a dip or a peak depending on the observable.

**中文：** 读图：先看两条坐标轴：此图SPR为反射对角度，LSPR为消光对波长，EOT为透射对波长。目标引起共振变化时，根据被测量不同，移动的可以是谷，也可以是峰。

Source / 来源：Week 3 PDF p81; original slide figure / 原课件截图




Prism-coupled SPR: energy coupling creates a dip in reflected light. / 棱镜耦合SPR：能量耦合到表面模式，反射光出现谷值。

**EN:** Read the figure: Follow the red incident beam through the prism to the thin metal film. At the resonance condition, energy couples to electron oscillations along the metal–sample interface, so reflected intensity decreases and the detector sees a dip. Binding changes the nearby refractive index and shifts the angle or wavelength where this dip occurs. For conventional propagating SPR, coupling and p/TM polarization matter; simply shining any light onto gold does not guarantee resonance.

**中文：** 读图：沿红色入射光追踪：光经过棱镜到达薄金属膜。在Resonance（共振）条件下，能量耦合到金属–样品界面的电子振荡，反射光减少，探测器读到一个Dip（谷值）。结合改变附近折射率，会移动谷值对应的角度或波长。传统传播型SPR需要合适耦合条件及p/TM偏振；不是任意光照到金上都会产生共振。

Source / 来源：Week 3 PDF p59; original slide figure / 原课件截图

## S120 • [P3] SPR imaging and fixed-wavelength intensity readout / SPR成像与固定波长强度读出

Priority / 复习标记：P3 Supporting reading/辅助阅读 · Supporting concept/辅助理解 · Lower priority/降低优先级

**EN:** SPR imaging illuminates many spots and measures their reflectivity changes in parallel, commonly at a fixed angle and wavelength. If binding shifts a resonance curve, the intensity at that fixed operating point can change. Thus sensing a resonance shift does not always require scanning and measuring the full spectrum in every measurement. The array improves throughput and can support multiplexing when its sites have appropriate recognition elements.

**中文：** SPR imaging（SPR成像）照明多个传感点、并行测它们的反射率变化，常在固定角度和波长工作。结合使共振曲线移动时，固定工作点处的强度也会变化。因此，感测共振位移不要求每次都扫描完整光谱。阵列能提高throughput（通量），各位点有相应识别元件时，还能支持复用检测。

**EN:** A camera records spatial intensity values; a spectrometer resolves optical strength across wavelength. Converting a camera pattern into an estimated wavelength shift or concentration requires calibration or a validated model. A dark or bright spot does not directly supply its wavelength. Machine learning can help map patterns to biological information, but the measurement and calibration still have to establish what the pattern means.

**中文：** Camera（相机）记录空间强度数值，spectrometer（光谱仪）按波长分辨光学强度。把相机图案转换为估计波长位移或浓度，需要校准或验证过的模型。暗点或亮点本身不能直接告诉你波长。机器学习可以帮助把图案映射为生物信息，但仍需测量与校准确定图案的含义。

Source / 来源：Week 3 PDF p81–86; T3-0111–0115

## B11 • [P2] Sensitivity, linewidth, Q and FOM / 灵敏度、线宽、Q值与品质指标

Priority / 复习标记：P2 Important foundations/重要基础 · Final emphasis/仅期末强调 · Repeated explanation/反复讲解 · Worked example/例题练习

**EN: Key points**

- Bulk refractive-index sensitivity $S\lambda=\Delta\lambda/\Delta n$ has units nm/RIU.
- It describes index response, not biochemical selectivity.
- Resonance linewidth is often FWHM (full width at half maximum);
- $Q=f0/\delta f$, equivalently $\lambda0/\delta\lambda$ for a narrow resonance.
- $FOM\lambda=S\lambda/\delta\lambda$ compares sensitivity with linewidth.
- Keep the same spectral coordinate and units;
- narrower lines alone do not determine LOD.

**中文：必会知识点**

- 体折射率灵敏度$S\lambda=\Delta\lambda/\Delta n$，单位nm/RIU，描述折射率响应而非生化选择性。
- 共振线宽常用FWHM半高全宽；
- $Q=f0/\delta f$，窄共振下等价于$\lambda0/\delta\lambda$。
- $FOM\lambda=S\lambda/\delta\lambda$同时考虑灵敏度与线宽。
- 分子分母必须使用相同光谱坐标与匹配单位；
- 线更窄也不能单独确定LOD。

**EN:** Start from zero: A resonance peak has a position and a width. Sensitivity tells us how far the position moves when the measured quantity changes. Linewidth describes how broad the peak is. FWHM, full width at half maximum, is the distance between two points at half the peak height above its baseline. A tall peak can still be broad, so height alone is not sharpness.

**中文：** 从零理解：共振峰同时有位置和宽度。Sensitivity（灵敏度）告诉我们被测量变化时，峰位置移动多少；linewidth（线宽）描述峰有多宽。FWHM，即full width at half maximum（半高全宽），是峰高相对基线的一半处左右两点之间的距离。峰高也可能很宽，因此高度本身不是尖锐程度。

**EN:** Follow the mechanism: For refractive-index sensing, Sλ=Δλ/Δn: peak shift Δλ divided by index change Δn, in nm/RIU. RIU means refractive-index unit. Quality factor Q compares the resonance position with its width; a narrow peak at the same position gives larger Q. FOMλ=Sλ/δλ compares sensitivity with linewidth δλ. Think of a moving pointer with a blurred edge, but actual uncertainty also depends on noise and how the peak is fitted.

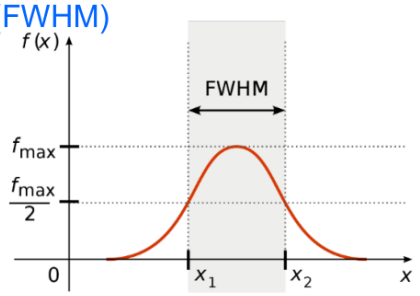
**中文：** 跟着机制走：折射率传感中，Sλ=Δλ/Δn，等于峰位移Δλ除以折射率变化Δn，单位nm/RIU。RIU是refractive-index unit（折射率单位）。Quality factor Q（品质因数）比较共振位置与宽度：位置相同时，峰越窄，Q越大。FOMλ=Sλ/δλ把灵敏度与线宽δλ放在一起比较。可以类比边缘模糊的移动指针，但真实不确定性还取决于噪声和峰的拟合方法。

**EN:** Try an example: Teaching example: a 2 nm shift for Δn=0.01 gives Sλ=200 nm/RIU. If linewidth is 4 nm, FOMλ=50 $RIU^{-1}$. Do not divide this wavelength sensitivity by a width in Hz; convert to one coordinate first. High Q or high FOM does not alone establish LOD or biochemical selectivity. A resonance can respond strongly to any index change, including an unwanted background change.

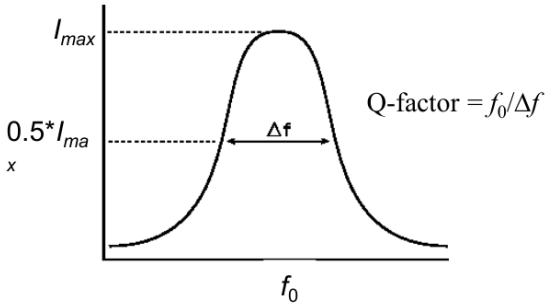
**中文：** 用例子理解：教学例子：Δn=0.01引起2 nm位移，则Sλ=200 nm/RIU；若线宽4 nm，FOMλ=50 $RIU^{-1}$。不能把这个波长灵敏度直接除以Hz单位的宽度，先统一坐标。Q高或FOM高，都不能单独证明检测限低或生化选择性好。共振可以强烈响应任何折射率变化，包括不需要的背景变化。

Source / 来源：W3 PDF p15–18; W3 PDF p25–29; W3 PDF p46–50; W3 PDF p77–80; W3 PDF p81–87 | T3-0021–0029, T3-0046–0050, T3-0073–0078, T3-0107–0114

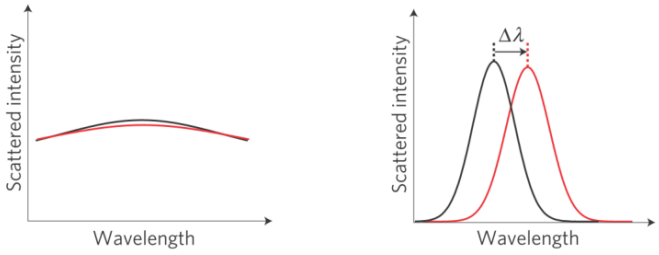
- Full width at half maximum (FWHM)




- Definition of Q-factor




- Importance of full width at half maximum (FWMH)




Na Liu et. al. *Nat. Mater.* **10**, 631-636 (2011)

- Figure of Merit (FOM)

$$FOM = \frac{Sensitivity}{\Delta f}$$

A figure of merit (FOM) is a performance metric that characterizes the performance of a device

FWHM, Q and FOM: distinguish peak width from peak shift. / FWHM、Q与FOM：区分峰宽和峰位移动。

**EN:** Read the figure: For FWHM, first find half the peak height, then measure the horizontal separation of the two intersections; it is neither the full base width nor half the center frequency. Compare the two lower spectra: a narrow peak makes the same wavelength shift easier to distinguish. Q=$f_0$/Δf compares center frequency with linewidth; FOM compares sensitivity with linewidth. Match the units: frequency sensitivity needs Δf, wavelength sensitivity needs Δλ. A tall peak alone does not prove high Q or FOM.

**中文：** 读图：求FWHM（半高全宽）时，先找峰高的一半，再量左右交点的横向距离；它不是整个底宽，也不是中心频率的一半。比较下方光谱：同样的波长移动，窄峰更容易分辨。Q=$f_0$/Δf比较中心频率与线宽；FOM比较灵敏度与线宽。必须匹配单位：频率灵敏度配Δf，波长灵敏度配Δλ。峰很高，不等于Q或FOM就很高。

Source / 来源：Week 3 PDF p80; original slide figure / 原课件截图

## C12 • [P1] Antigen rapid test (ART): gold nanoparticle LSPR / 抗原快速检测：金纳米颗粒LSPR

Priority / 复习标记：P1 First pass/先掌握 · Quiz stated/Quiz明确提及 · Worked example/例题练习

**EN: Key points**

• The lecturer explicitly says the ART principle is a recurring Quiz question.
• In the taught colorimetric rapid-test example, gold nanoparticles produce the visible optical response through LSPR;
• antibody–antigen recognition controls their localization at the test line.
• The control line checks assay operation;
• a missing control line makes the test invalid.

**中文：必会知识点**

• 老师明确说ART原理是反复出现的Quiz问题。
• 课程的比色快检示例中，金纳米颗粒通过LSPR产生可见光学响应，抗体—抗原识别控制其在检测线处的聚集或定位。
• Control line控制线用于检查检测运行；
• 缺少控制线时结果无效。

**EN:** Start from zero: An antigen is a molecular target that an antibody can recognize. In the taught antigen rapid test, antibodies supply the recognition and gold nanoparticles supply a visible reporter. These are separate jobs. A test line appears when the appropriate labelled material is captured there. The gold is not simply a decorative color, and the antibody is not itself an electronic detector.

**中文：** 从零理解：Antigen（抗原）是抗体可以识别的分子目标。在课程讲的antigen rapid test（抗原快速检测，ART）中，抗体负责识别，金纳米颗粒负责提供可见报告信号，这是两个不同工作。当相应的带标记材料被捕获在检测线处，就能形成可见线。金颗粒并非只是装饰颜色，抗体本身也不是电子探测器。

**EN:** Follow the mechanism: The liquid travels along the strip by capillary action, the movement of liquid through small spaces without a pump. Target antigen can associate with labelled antibodies and be captured at the test line. Many gold nanoparticles together create a visible optical response associated with LSPR. Their electrons interact collectively with light. Think of gathering colored markers at one location; actual capture is molecular binding, not deliberate placement.

**中文：** 跟着机制走：液体通过capillary action（毛细作用）沿试纸移动，也就是液体在细小空间内无需泵就能流动。目标抗原可以与带标记抗体结合，并被检测线捕获。大量金纳米颗粒聚在一起，形成与LSPR有关的可见光学响应，其电子会与光发生集体作用。可以类比把彩色标记集中到一个位置，但真实捕获是分子结合，并非人为放置。

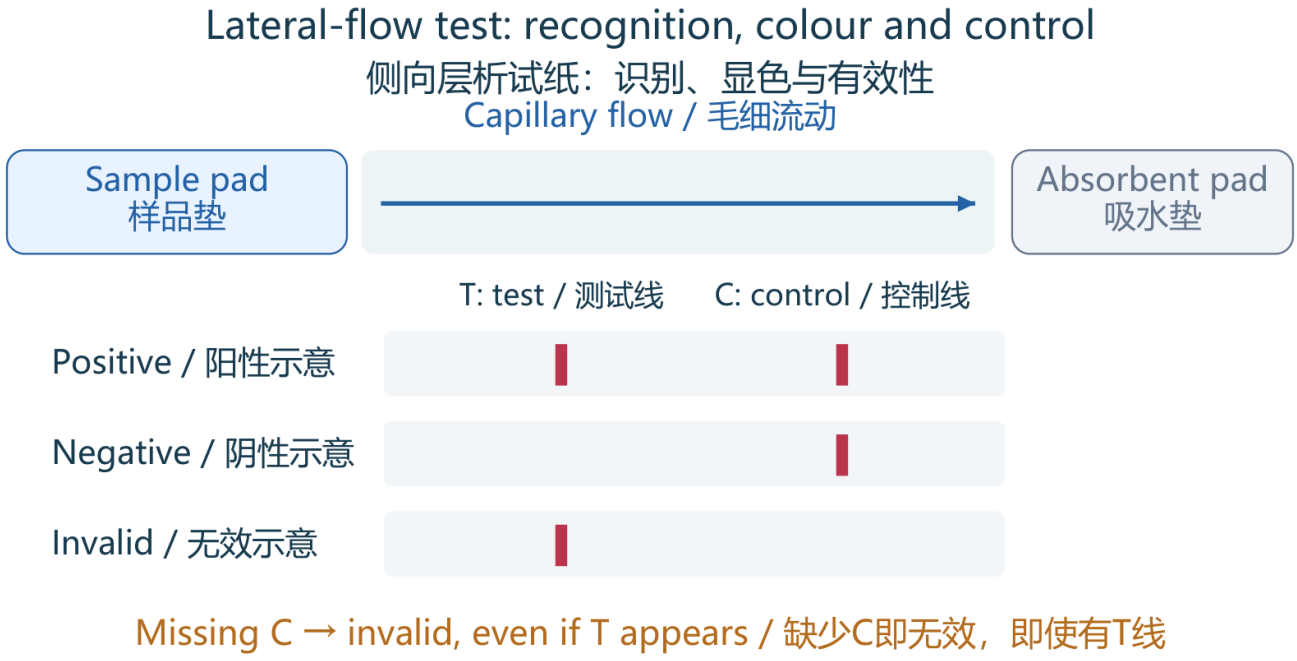
**EN:** Try an example: Teaching example: if a question asks the recognition mechanism, answer antibody–antigen binding; if it asks the gold nanoparticle optical principle, answer LSPR. Do not replace one with the other. This colorimetric ART example is different from PCR, which amplifies nucleic acids, and from ordinary fluorescent reporting. Match the requested step to its mechanism. A control line checks assay operation; a missing control line makes the result invalid, even if a test line appears.

**中文：** 用例子理解：教学例子：如果问识别机制，答抗体—抗原结合；如果问金纳米颗粒的光学原理，答LSPR，两者不能互换。这个比色ART例子不同于扩增核酸的PCR，也不同于普通荧光报告。把题目问的步骤与相应机制对应起来。 Control line（控制线）检查检测是否正常运行；即使出现测试线，缺少控制线时结果仍无效。

**EN:** Exam wording: Separate recognition from readout: antibody binding supplies specificity, while the gold nanoparticles supply the optical signal. Do not answer PCR, electrical membrane potential, or ordinary fluorescence excitation for this taught example.

**中文：** 英文读题抓手：区分识别与读出：抗体结合赋予特异性，金纳米颗粒提供光学信号。该课程示例不能答成PCR、细胞膜电位或普通荧光激发。

Source / 来源：W3 PDF p76; W7 p52–55 (context; classroom question diagram); W3 p76–79; W1 PDF p67 | T1-0062–0064, T3-0106, T7-0047–0050




Lateral-flow test: recognition, colour and control / 侧向层析试纸：识别、显色与有效性

**EN:** Read the figure: Fluid moves along the strip by capillary action. Antibodies provide selective recognition; accumulated gold labels provide a visible colour signal through their optical properties. For the illustrated common assay pattern, C+T means positive, C alone means negative, and no C means invalid. Read within the assay-specific time window and follow its instructions.

**中文：** 读图：液体靠Capillary action（毛细作用）沿试纸流动。Antibody（抗体）负责选择性识别；聚集的Gold labels（金标签）利用自身光学性质显色。图示常见判读模式为C + T阳性、仅C阴性、无C则无效。具体检测须按该试剂说明和规定时间判读。

Source / 来源：Week 1 PDF p67; Week 3 PDF p76; original teaching schematic / 自绘教学示意，非实验数据

## B12 • [P2] LSPR tuning and SERS / 局域等离激元调谐与表面增强拉曼

Priority / 复习标记：P2 Important foundations/重要基础 · Teacher emphasis/课堂强调 · Sample Q2/样卷Q2

**EN: Key points**

- Nanoparticle material, size, shape and surrounding refractive index affect LSPR.
- SERS means surface-enhanced Raman scattering: metal nanostructures create strong local fields, especially at hot spots, and greatly enhance nearby Raman signals.
- Recognize a Raman molecular fingerprint versus an LSPR resonance shift.
- SERS was continued in Week 4;
- its deferral in Week 3 did not remove it from scope.

**中文：必会知识点**

- 纳米颗粒材料、尺寸、形状和周围折射率影响LSPR。
- SERS为surface-enhanced Raman scattering表面增强拉曼散射：金属纳米结构产生强局域场，尤其在hot spot热点处，显著增强附近拉曼信号。
- 区分拉曼分子指纹与LSPR共振位移。
- SERS在Week4续讲，Week3推迟讲解不表示不考。

**EN:** Start from zero: LSPR is the light-driven collective electron response localized around a metal nanostructure. Its preferred wavelengths depend on the material, size, shape, and surrounding optical environment. Thus changing a sphere into a rod can change its spectrum even without adding an analyte. A measured resonance shift needs a controlled comparison, because geometry and environment both influence the signal.

**中文：** 从零理解：LSPR是光驱动的集体电子响应，局域在金属纳米结构周围。它特别响应的波长取决于材料、尺寸、形状和周围光学环境。因此，即使没有添加分析物，把球形颗粒改成棒形也能改变光谱。测得共振位移时需要受控比较，因为几何和环境都影响信号。

**EN:** Follow the mechanism: SERS strengthens normally weak Raman scattering from molecules near suitable metal structures. Structures can concentrate the local electromagnetic field, especially in gaps called hot spots. The stronger field makes nearby Raman signals easier to detect. Imagine a locally intensified light zone; the molecules still contribute their vibration-related spectrum. The analogy does not mean the whole sample becomes equally bright, or that every molecule automatically reaches a hot spot.

**中文：** 跟着机制走：SERS增强位于合适金属结构附近的分子原本很弱的拉曼散射。结构可以集中局部电磁场，尤其是在叫hot spots（热点）的间隙中，使附近的拉曼信号更容易检测。可以想象一个局部光场增强区域，分子仍然贡献与自身振动有关的光谱。这个类比不表示整个样品均匀变亮，也不表示每个分子都会自动来到热点。

**EN:** Try an example: Teaching example: tracking a metal resonance peak moving with the surroundings is LSPR-based resonance readout. Looking for a molecule's shifted vibrational lines, made stronger near metal, is SERS readout. Both may use plasmonic structures but answer different measurement questions. For identification, ask what is plotted: the metal resonance position or a Raman fingerprint. Enhancement does not remove the need for suitable recognition, controls, and interpretation.

**中文：** 用例子理解：教学例子：跟踪金属共振峰随着周围环境移动，是基于LSPR的共振读出；寻找分子的振动位移谱线，并用附近金属增强它们，是SERS读出。两者都可能用等离激元结构，但回答不同的测量问题。辨认时看图里画的是什么：金属共振位置，还是拉曼指纹？增强并不会取消合适识别、对照和解释的需求。

Source / 来源：W3 PDF p64–75; W3 PDF p88–95; W3 p60–94; W3 p88–94 | T3-0099–0105, T3-0115–0116, T4-0001–0014

## S121 • [P2] SERS enhancement and molecular fingerprints / SERS增强与分子指纹

Priority / 复习标记：P2 Important foundations/重要基础 · Teacher emphasis/课堂强调 · Supporting concept/辅助理解

**EN:** SERS means surface-enhanced Raman scattering. The metal structure supplies enhancement while the molecule supplies vibration-related Raman lines. The slide distinguishes electromagnetic enhancement from an intensified near field and chemical enhancement involving molecule–surface interactions. Molecules must be suitably close to the enhancing structure; the entire sample does not receive one uniform enhancement factor.

**中文：** SERS为surface-enhanced Raman scattering（表面增强拉曼散射）。金属结构提供增强，分子提供与振动有关的拉曼谱线。课件区分局域场增强造成的electromagnetic enhancement（电磁增强），以及涉及分子—表面相互作用的chemical enhancement（化学增强）。分子要足够接近增强结构，不能认为整个样品都具有相同增强倍数。

**EN:** For chemical identification, compare the pattern of Raman-shift positions and intensities: this is a molecular fingerprint. The illustrated applications use spectral patterns and analysis to distinguish sample groups. A larger SERS peak alone is not a unique identifier, and the very large enhancement factors shown are examples, not a guaranteed value for every substrate. LSPR-based index sensing follows a metal resonance feature; SERS follows enhanced molecular vibration lines.

**中文：** 化学识别时，要比较拉曼位移位置与强度组成的模式，这叫molecular fingerprint（分子指纹）。图示应用用光谱模式和分析区分样品群体。单一SERS峰更大，不等于唯一识别目标；课件展示的巨大增强倍数属于示例，不保证每个基底都达到。基于LSPR的折射率传感跟踪金属共振特征，SERS跟踪增强后的分子振动谱线。

Source / 来源：Week 3 PDF p88–94; T3-0116 (announces continuation next week)

# Week 4 learning block: Microscopes and optical imaging / 第4学习组：显微镜与光学成像

## Week 4 concept map / 第4周知识框架

**EN:** Chapter logic: Follow light from image formation and contrast to depth selection, detection and spectral information.

**中文：** 本章主线：沿着光的流程，理解成像与对比如何产生，再看怎样选深度、检测光和获取光谱信息。

| English term 中文名词 | Meaning and connection 含义与知识联系 |
|---|---|
| **Imaging modalities 成像模态** | Identify the input energy and detected signal; projection X-ray and reconstructed CT slices differ. 先辨输入能量与检测信号，区分投影X射线与重建CT切片。 → B13 |
| **NA, resolution and sampling NA、分辨率与采样** | d≈0.61λ/NA: larger NA or shorter λ improves lateral separation; pixels must sample the magnified detail. d≈0.61λ/NA：大NA或短λ改善横向间距；像素须充分采样放大后的细节。 → C13 |
| **Optical contrast 光学对比** | Brightfield, darkfield, phase contrast and fluorescence use different sources of image contrast. 明场、暗场、相衬与荧光使用不同对比来源。 → S402 |
| **Confocal versus two-photon 共聚焦与双光子** | Confocal rejects light at detection; two-photon localizes excitation ($F \propto I^2$ before saturation). 共聚焦在检测端拒光；双光子限制激发位置（未饱和时$F \propto I^2$）。 → C14 |
| **OCT mechanism OCT机制** | Sample return + reference light → interference → depth-reflectivity information. 样品返回光+参考光→干涉→深度反射信息。 → C15 |
| **Three depth concepts 三种深度概念** | OCT axial separation follows source bandwidth; lateral detail follows focusing; focused range is another quantity. OCT轴向间距由光源带宽控制，横向细节依赖聚焦，可聚焦范围是另一量。 → B17 |
| **Optical detectors 光探测器** | Photodiode/PMT convert light to current; CCD shifts charge, while CMOS uses pixel-associated circuits. 光电二极管/PMT把光转电流；CCD移电荷，CMOS用像素相关电路。 → C16 |
| **Hyperspectral cube 高光谱立方** | Each position has a spectrum: axes are x,y,λ; scan method determines what is acquired per step. 每个位置有光谱，坐标是x,y,λ；扫描方式决定一步记录什么。 → S406 |

**EN:** Formula links for this chapter: FX401 Thin lens and magnification, FX402 Objective numerical aperture, FX403 Rayleigh separation and Airy radius/diameter, FX404 Detector sampling and magnification, FX405 Two-photon intensity law, FX406 Michelson path, phase and intensity, FX407 OCT coherence-limited axial resolution, FX408 Object-side depth of field with detector term, FX409 Hyperspectral cube data counts.

**中文：** 本章公式定位：FX401 薄透镜与放大率；FX402 物镜数值孔径；FX403 瑞利间距与艾里斑半径 / 直径；FX404 探测器采样与倍率；FX405 双光子强度关系；FX406 迈克尔逊光程、相位与强度；FX407 OCT相干长度限制的轴向分辨率；FX408 含探测器项的物方景深；FX409 高光谱立方的数据数目。

## B13 • [P2] Imaging modality identification / 成像技术辨识

Priority / 复习标记：P2 Important foundations/重要基础 · Repeated explanation/反复讲解

**EN: Key points**

- X-ray imaging measures transmission attenuation;
- CT reconstructs many X-ray projections.
- MRI uses magnetic fields and radiofrequency excitation;
- PET means positron emission tomography and uses a radioactive tracer.
- Ultrasound uses sound;
- optical imaging uses light.
- X-rays/CT and PET involve ionizing radiation.
- Compare contrast mechanism, penetration and resolution rather than assuming one method is best for every task.

**中文：必会知识点**

- X-ray成像测透射衰减；
- CT由多个X射线投影重建。
- MRI磁共振使用磁场及射频激发；
- PET为positron emission tomography正电子发射断层，使用放射性示踪剂。
- Ultrasound超声用声，optical imaging光学成像用光。
- X射线/CT和PET涉及电离辐射。
- 比较对比机制、穿透与分辨率，不能认定某技术对所有任务都最好。

**EN:** Start from zero: Imaging means making a map of where signals come from. Contrast means that two regions give distinguishable signals, so one can be seen against the other. Different imaging methods use different physical signals. A bright region can therefore mean different things in different images. Before comparing pictures, ask what energy is sent in, what is detected, and what tissue property changes that signal.

**中文：** 从零理解：Imaging（成像）就是把信号来自哪里画成图。Contrast（对比）表示两个区域的信号可以区分，才能看出彼此。不同技术使用不同物理信号，所以不同图像中的“亮”可能表示不同含义。比较图像之前，先问：送入什么能量？检测什么信号？组织的哪种性质改变了信号？

**EN:** Follow the mechanism: X-ray imaging measures photons passing through tissue; CT combines many viewing angles to reconstruct slices. MRI uses a strong magnetic field: certain atomic nuclei act like tiny magnets. A radiofrequency (RF) pulse, an oscillating electromagnetic field, disturbs their magnetic state; their subsequent response provides a signal whose relaxation depends on tissue. PET uses a radioactive tracer that releases positrons, particles like electrons but positively charged. A positron meeting an electron produces two gamma photons; detecting pairs helps locate the tracer. Ultrasound uses returning sound; optical imaging uses light. NIH/NIBIB: MRI, NIH/NIBIB: PET.

**中文：** 跟着机制走：X-ray（X射线）成像检测穿过组织的光子；CT合并多个观察角度的信息，重建切片。MRI使用强磁场：某些原子核像微小磁体。Radiofrequency（RF，射频）脉冲是一种振荡电磁场，会扰动它们的磁状态；之后的响应形成信号，其弛豫与组织有关。PET使用放射性示踪剂，释放positron（正电子），即类似电子但带正电的粒子。正电子与电子相遇会产生两个伽马光子；检测成对光子有助于定位示踪剂。超声使用返回的声波；光学成像使用光。NIH/NIBIB：MRI、NIH/NIBIB：PET。

**EN:** Try an example: Ionizing radiation has enough energy to remove electrons from atoms. X-rays, CT, and PET involve ionizing radiation; ordinary diagnostic ultrasound and MRI use different, nonionizing mechanisms. This classification does not mean every nonionizing method is automatically suitable in every circumstance. In English questions, match projection and attenuation to X-rays, magnetic field and RF to MRI, radioactive tracer to PET, and echo to ultrasound. Judge resolution, depth, and contrast separately.

**中文：** 用例子理解：Ionizing radiation（电离辐射）有足够能量把电子从原子中移走。X射线、CT和PET涉及电离辐射；常规诊断超声与MRI采用其他非电离机制。这种分类不表示任何非电离技术都自动适用于任何情况。英文辨认：projection与attenuation对应X射线；magnetic field与RF对应MRI；radioactive tracer对应PET；echo对应超声。分辨率、成像深度及对比机制应分别比较。

Source / 来源：W4 p3–12; W5 p53–58; W5 p64–69 | T4-0015–0021, T5-0073–0080, T5-0088–0095

## B14 • [P1] Optical biopsy, endoscopy and virtual staining / 光学活检、内窥镜与虚拟染色

Priority / 复习标记：P1 First pass/先掌握 · Teacher emphasis/课堂强调 · Supporting concept/辅助理解

**EN: Key points**

• Optical biopsy seeks tissue information through optical imaging, reducing tissue removal in the taught examples.
• Endoscopy brings illumination and detection to internal sites, often with fibre delivery.
• Virtual staining computationally makes an image resemble a stained tissue image;
• it does not physically add a dye.
• Recognize the workflow without assuming every optical method replaces histopathology.

**中文：必会知识点**

• Optical biopsy光学活检通过光学成像获取组织信息，在课程示例中减少取组织操作。
• Endoscopy内窥镜把照明与检测带到体内，常用光纤传输。
• Virtual staining虚拟染色用计算产生类似染色组织的图像，并未实际添加染料。
• 要认得流程，不应推断所有光学方法都能替代组织病理。

**EN:** Start from zero: A conventional tissue biopsy removes a sample for examination. Optical biopsy tries to obtain useful tissue information through optical measurements, sometimes reducing the need to remove tissue in the taught applications. The word biopsy here describes the diagnostic purpose, not a requirement that light cuts tissue out. Different optical methods reveal different information, so an optical image does not automatically provide every answer that laboratory pathology provides.

**中文：** 从零理解：常规tissue biopsy（组织活检）是取出样品进行检查。Optical biopsy（光学活检）尝试通过光学测量取得组织信息，在课程应用中有时可减少取组织操作。这里的biopsy强调检查目的，不表示光必须把组织切出来。不同光学方法显示不同信息，因此光学图像不会自动提供实验室病理检查的全部答案。

**EN:** Follow the mechanism: Endoscopy brings illumination and observation to an internal location using an instrument. Fibres can deliver light or return information through a narrow path, somewhat like extending the instrument's reach. Conventional staining physically adds dyes to make structures easier to distinguish. Virtual staining instead uses computation to produce a staining-like image from measured input. The tissue is not physically dyed by the computer.

**中文：** 跟着机制走：Endoscopy（内窥镜检查）用仪器把照明与观察带到体内。光纤可沿狭窄路径输送光或返回信息，好比延长仪器的观察范围。常规staining（染色）实际加入染料，让结构更容易区分；virtual staining（虚拟染色）则根据测得的输入，用计算生成类似染色效果的图像，电脑并没有实际给组织加染料。

**EN:** Try an example: Imagine a grayscale tissue image converted into a colored image resembling a familiar stain. This can help interpretation, but its colors are calculated from the input and model; they are not directly measured dye signals. Exam clues are internal access for endoscopy, optical tissue assessment for optical biopsy, and computational transformation for virtual staining. Distinguish image acquisition from display and interpretation.

**中文：** 用例子理解：想象一张灰度组织图被转换成类似熟悉染色效果的彩色图：这可能方便阅读，但颜色来自输入和模型的计算，不是新测得的染料信号。英文辨认：internal access提示内窥镜；optical tissue assessment提示光学活检；computational transformation提示虚拟染色。要区分图像如何acquired（采集），与它如何displayed（显示）或interpreted（解释）。

Source / 来源：W4 p13–20; W4 p65–70; W4 p106–112 | T4-0022–0025, T4-0085–0087, T4-0115–0120

## C13 • [P1] Numerical aperture and microscope resolution / 数值孔径与显微分辨率

Priority / 复习标记：P1 First pass/先掌握 · Quiz stated/Quiz明确提及 · Teacher emphasis/课堂强调 · Final emphasis/仅期末强调

**EN: Key points**

- Numerical aperture is NA=n sinθ.
- At a fixed wavelength, the diffraction-limited lateral separation scales as d≈0.61λ/NA.
- Larger NA or shorter wavelength gives a smaller resolvable separation.
- Magnification alone does not guarantee improved resolving ability.

**中文：必会知识点**

- 数值孔径NA=n sinθ。
- 波长固定时，衍射极限横向最小间距约d≈0.61λ/NA。
- NA更大或波长更短，使可分辨间距更小。
- 只有放大倍率增加，并不保证分辨能力提高。

**EN:** Start from zero: Resolution means telling two nearby objects apart. Magnification means making their image larger. Imagine enlarging a blurry phone photograph: the blur becomes bigger, but hidden detail does not appear. A microscope therefore needs both suitable magnification and good resolving ability. In this chapter, lateral resolution concerns separation across the image, rather than separation along its depth.

**中文：** 从零理解：Resolution（分辨率）是把两个靠近的物体区分开的能力；magnification（放大倍率）是把图像变大的程度。想象放大一张模糊的手机照片：模糊区域更大了，但原本看不清的细节不会凭空出现。因此显微镜既需要合适倍率，也需要好的分辨能力。本章的lateral resolution（横向分辨率）指图像平面内的区分能力，而不是沿深度方向。

**EN:** Follow the mechanism: An objective lens collects a cone of light. Numerical aperture, NA=n sinθ, describes this collection ability: n is the surrounding medium's refractive index, and θ is the cone's half-angle. Collecting a wider cone helps capture fine detail. Because light spreads through diffraction, even a tiny point makes a small blurred spot. Larger NA makes that spot smaller at the same wavelength.

**中文：** 跟着机制走：物镜收集的是一束锥形范围内的光。Numerical aperture（数值孔径）NA=n sinθ描述收光能力：n是物镜前介质的折射率，θ是光锥的半角。收集角度更大，有助于保留细小信息。由于diffraction（衍射），光会扩散，因此极小的点也会形成一小团模糊光斑；波长相同时，NA更大会使这个光斑更小。

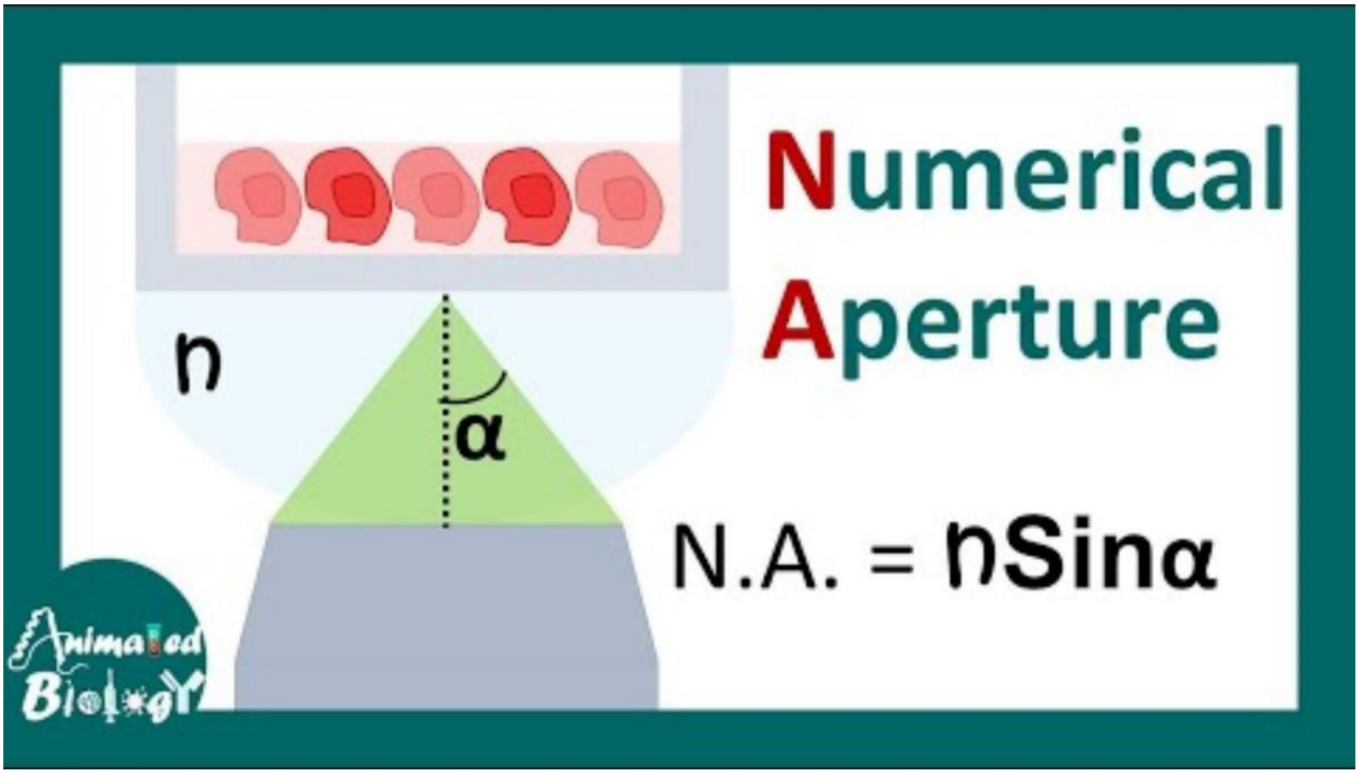
**EN:** Try an example: For the course's Rayleigh criterion, minimum separation d≈0.61λ/NA. With λ=500 nm and NA=0.5, d≈610 nm; raising NA to 1.0 gives about 305 nm. The smaller distance means better resolution. In English questions, higher resolving ability and smaller resolvable separation point to the same improvement. Compare NA while holding wavelength and the criterion fixed; choosing the largest magnification alone is insufficient.

**中文：** 用例子理解：按本课程的Rayleigh criterion（瑞利判据），最小可分辨间距d≈0.61λ/NA。λ=500 nm、NA=0.5时，d约610 nm；NA提高到1.0时，d约305 nm。这个距离越小，分辨能力越好。英文题中的higher resolving ability（更强分辨能力）与smaller resolvable separation（更小可分辨间距）表示同一种改善。比较NA时应固定波长和判据；仅选放大倍率最大的物镜不够。

**EN:** Exam wording: The objective-lens comparison is explicitly named again for the Quiz in Week 5. If NA doubles with the same wavelength and criterion, d halves. “Higher resolution” means a smaller minimum distinguishable distance.

**中文：** 英文读题抓手：Week5明确说Quiz会再问物镜比较。波长与判据相同、NA加倍时，d减半。“Higher resolution”表示最小可分辨距离更小。

Source / 来源：W4 p27–30; W4 p29–34; W4 p29–30 | T4-0030–0038, T5-0059–0061

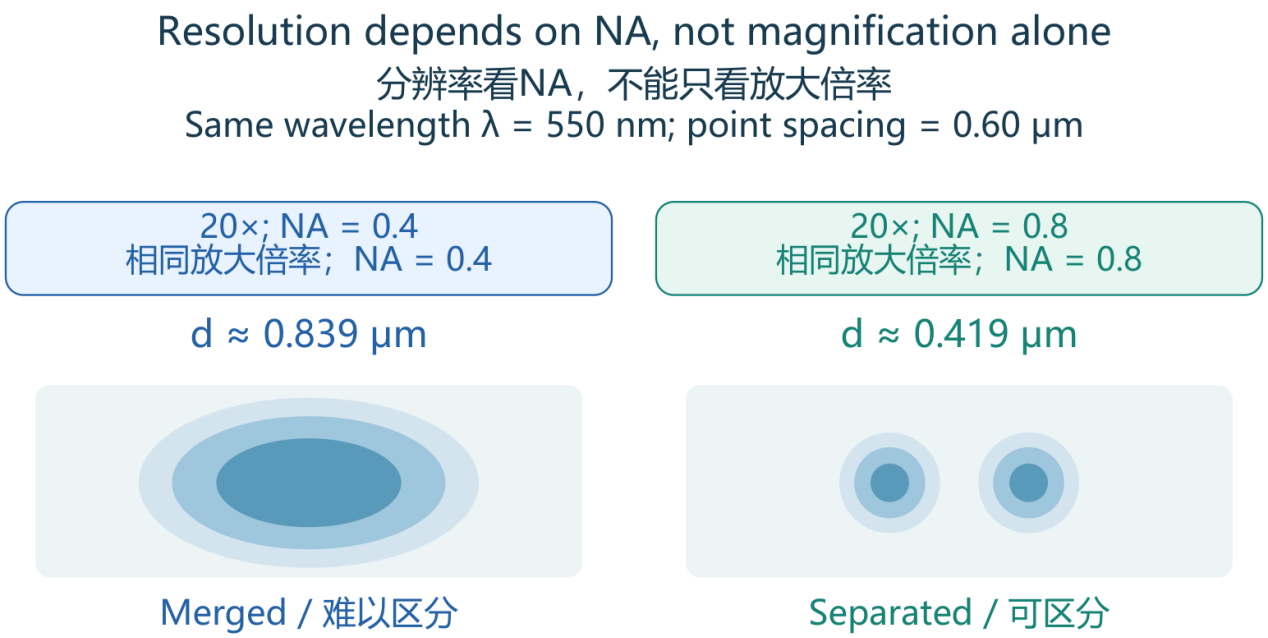



Numerical aperture: refractive index and the collection cone (original lecture figure) / 数值孔径：折射率与收光锥（原课件图）

**EN:** Read the figure: Find the dotted optical axis and angle α: α is the half-angle of the light cone, not its full opening angle. In NA = n sin α, n is the refractive index between specimen and objective. A larger NA collects a wider cone and gives a smaller diffraction-limited separation d at the same wavelength; magnification is a separate property.

**中文：** 读图：找虚线光轴与α：α是收光锥的半角，不是整个张角。NA = n sin α中的n是样品与物镜之间介质的折射率。同一波长下，更大NA收集更宽的光锥，使衍射限制的最小可分辨距离d更小；放大倍率是另一项性质。

Source / 来源：Week 4 PDF p31; original slide figure / 原课件截图




Resolution depends on NA, not magnification alone / 分辨率看NA，不能只看放大倍率

**EN:** Read the figure: Keep wavelength at 550 nm and magnification at 20×. Using d ≈ 0.61λ/NA gives about 0.839 µm for NA 0.4 and 0.419 µm for NA 0.8. A 0.60 µm separation exceeds the second threshold but not the first. The drawn spots illustrate the distinction; they are not calculated point-spread functions.

**中文：** 读图：固定波长550 nm、放大倍率20×。按d ≈ 0.61λ/NA，NA为0.4时约0.839 µm，为0.8时约0.419 µm。间隔0.60 µm超过第二个分辨阈值，却未超过第一个。图中点像只用于示意，没有用真实Point-spread function（点扩散函数）计算。

Source / 来源：Week 4 PDF p27–34; original teaching schematic / 自绘教学示意，非实验数据

## B15 • [P1] NA, Airy disk and detector sampling / NA、艾里斑与探测器采样

Priority / 复习标记：P1 First pass/先掌握 · Quiz stated/Quiz明确提及 · Repeated explanation/反复讲解 · Lower priority/降低优先级

**EN: Key points**

- In NA=n sinθ, θ is the collection cone half-angle;
- immersion increases n.
- Rayleigh separation ≈0.61λ/NA is different from Airy-disk diameter ≈1.22λ/NA.
- Pixels must sample optical features sufficiently after magnification.
- More pixels cannot restore detail absent from the optical image.
- The thin-lens relation 1/f=1/u+1/v uses the stated lens sign convention.

**中文：必会知识点**

- NA=n sinθ中，θ为收集锥半角；
- 浸液可提高n。
- Rayleigh最小分辨间距约0.61λ/NA，与Airy-disk艾里斑直径约1.22λ/NA不同。
- 放大后的光学细节需要充分像素采样；
- 增加像素无法恢复光学成像中本来缺失的细节。
- 薄透镜式1/f=1/u+1/v需配合题目规定的符号约定。

**EN:** Start from zero: Even a point source makes a small diffraction pattern through a circular lens opening. Its central bright region is called the Airy disk. The disk diameter describes the size of one point's pattern; the Rayleigh separation describes how far apart two points must be to distinguish them under that criterion. They answer different questions, so their numerical factors must not be exchanged.

**中文：** 从零理解：点光源经过圆形透镜孔径，也会形成一个小衍射图样，其中中央亮区叫Airy disk（艾里斑）。光斑直径描述“一个点的图样有多大”；Rayleigh separation（瑞利分辨间距）描述按该判据“两个点至少相隔多远才能分开”。两者回答不同问题，因此公式中的数字系数不能互换。

**EN:** Follow the mechanism: NA depends on the collection angle and medium: a suitable immersion liquid can increase the refractive index n. The optics first form an image, then detector pixels sample it at discrete positions. Imagine recording a melody with too few time samples: information can be missed. Spatial undersampling similarly misses image detail. More detector pixels help only if useful optical detail reaches them; they cannot undo optical blur.

**中文：** 跟着机制走：NA取决于收光角度及介质；合适浸液可提高折射率n。先由光学系统形成图像，再由探测器像素在离散位置采样。可以类比录音乐时采样点过少会漏掉信息；空间采样不足也会遗漏图像细节。更多像素只有在有用光学细节到达探测器时才有帮助，无法消除光学系统已经造成的模糊。

**EN:** Try an example: Remember the approximate factors: Rayleigh separation 0.61λ/NA, Airy diameter 1.22λ/NA. Magnification changes detector image size and affects sampling. In 1/f=1/u+1/v, f is focal length, u object distance, and v image distance. Use the stated sign convention; for positive distances of a real object and real image, u=v=20 cm gives f=10 cm. This lens example concerns image formation, not diffraction resolution.

**中文：** 用例子理解：记住近似系数：瑞利间距为0.61λ/NA，艾里斑直径为1.22λ/NA。倍率改变呈现在探测器上的图像尺寸，因此影响采样。薄透镜式1/f=1/u+1/v中，f是焦距、u是物距、v是像距；要遵循题目规定的符号约定。若实物和实像的距离均取正值，u=v=20 cm，则f=10 cm。这个透镜例子讨论成像位置，不是在计算衍射分辨率。

Source / 来源：W4 p21–26; W4 p27–30; W4 p29–34; W4 p31–34 | T4-0026–0038, T4-0045–0047

## S401 • [P2] Image formation, refraction and reflection / 成像、折射与反射

Priority / 复习标记：P2 Important foundations/重要基础 · Supporting concept/辅助理解 · Sample Q8/样卷Q8 · Lower priority/降低优先级

**EN:** An imaging system needs an optical element to collect or focus light, a detector to convert the light pattern into a signal, and processing to form or display an image. A lens bends rays by refraction. Focal length is the lens-to-focus distance for parallel incoming rays; it is not the distance between two objects that can be resolved. For a thin lens with real object and image distances taken as positive, the magnification magnitude is |M|=v/u.

**中文：** 成像系统需要收集或聚焦光的光学部件、把光分布转成信号的探测器，以及形成或显示图像的处理环节。透镜通过refraction（折射）改变光线方向。Focal length（焦距）是平行入射光聚焦时透镜到焦点的距离，不是两个物体的最小可分辨间距。薄透镜实物、实像距离均取正时，放大倍率大小|M|=v/u。

**EN:** In a medium of refractive index n, light speed is c/n and wavelength is $\lambda_0/n$; its frequency stays the same across a stationary interface. Reflection returns light to the original medium; refraction changes its direction in the new medium. Specular reflection gives a mirror-like direction, whereas diffuse reflection spreads light into many directions. Absorption removes energy from the light; light that is not reflected is not necessarily all absorbed, because transmission and scattering are also possible. Polarimetry measures optical polarization, the direction/state of electric-field oscillation explained in S104; spectroscopy instead measures wavelength-dependent signals. Optical polarization differs from electrode polarization in electrochemistry.

**中文：** 折射率为n的介质中，光速为c/n、波长为$\lambda_0/n$；光经过静止界面时频率不变。Reflection（反射）使光回到原介质；refraction使光进入另一介质后改变方向。Specular reflection（镜面反射）方向集中，diffuse reflection（漫反射）向许多方向分散。Absorption（吸收）把能量从光中取走；未被反射的光不一定全部被吸收，也可能透射或散射。 Polarimetry（偏振测量）测光的偏振，即S104解释的电场振动方向/状态；spectroscopy（光谱学）测随波长变化的信号。光的偏振不同于电化学电极polarization（极化）。

**EN:** For a simple, nonabsorbing interface at normal incidence, intensity reflectance is $R=[(n_2-n_1)/(n_2+n_1)]^2$. It is the refractive-index mismatch that matters under this model. Do not turn the slide’ s glass-versus-water example into a universal rule that any material with a larger n must always reflect more at every angle.

**中文：** 简单非吸收界面、垂直入射时，强度反射率$R=[(n_2-n_1)/(n_2+n_1)]^2$。在这个模型中关键是折射率失配。不能把课件玻璃与水的例子推广为“任何折射率更大的材料，在任何角度都一定反射更多”。

Source / 来源：Week 4 PDF p16, p21–26, 92–94; T4-0026–0029

## S403 • [P2] Match detector pixels to optical detail / 让探测器像素匹配光学细节

Priority / 复习标记：P2 Important foundations/重要基础 · Repeated explanation/反复讲解

**EN:** If the detector pixel pitch is Δx and magnification is M, its object-space sampling pitch is Δx/M. The course recommends about 2.3–3 detector pixels across the magnified Rayleigh separation: M d/Δx≈2.3–3. Thus the same camera can under-sample at low magnification even when the objective itself resolves the detail.

**中文：** 探测器像素间距为Δx、倍率为M时，折算到物体上的采样间距是Δx/M。课程建议放大后的瑞利间距内约有2.3–3个像素，即M d/Δx≈2.3–3。因此，同一相机在低倍率下可能采样不足，即使物镜本身能够分辨该细节。

**EN:** For example, d=0.5 μm, M=20 and detector pitch Δx=5 μm give M d/Δx=2: only two pixels span that separation. At M=30, the ratio becomes 3. This improves recording of existing optical detail; it does not make d smaller. Distinguish detector pixel size, object-space pixel size, magnification and optical resolution.

**中文：** 例如d=0.5 μm、M=20、探测器间距Δx=5 μm时，M d/Δx=2，只有两个像素覆盖这个间距。M=30时比例变为3。这改善了已有光学细节的记录，并没有使d变小。要分清探测器像素尺寸、折算到物体的像素尺寸、倍率与光学分辨率。

Source / 来源：Week 4 PDF p32–34; T4-0045–0047

## C14 • [P1] Confocal and two-photon microscopy / 共聚焦与双光子显微镜

Priority / 复习标记：P1 First pass/先掌握 · Sample Q1/样卷Q1 · Repeated explanation/反复讲解

**EN: Key points**

- A confocal pinhole suppresses out-of-focus light and enables optical sectioning.
- Two-photon excitation is nonlinear and largely confined to the focus;
- two longer-wavelength photons together supply the excitation energy.
- These mechanisms improve contrast and sectioning in different ways.

**中文：必会知识点**

- 共聚焦pinhole抑制离焦光，实现光学切片。
- 双光子激发具有非线性，主要限制在焦点；
- 两个长波长光子共同提供激发能量。
- 两种机制以不同方式改善对比度与切片能力。

**EN:** Start from zero: When viewing thick tissue, light from above and below the chosen focus can cover the wanted detail with background. Optical sectioning means selecting a thin depth layer without physically cutting the tissue. Think of trying to read one transparent page while other pages overlap it. Confocal and two-photon microscopes solve this depth problem through different parts of the imaging process.

**中文：** 从零理解：观察较厚组织时，焦点上方和下方的光会成为背景，盖住想看的细节。Optical sectioning（光学切片）就是不实际切开组织，却选择某一薄层深度来成像。可以想象几张透明纸叠在一起，你只想看清其中一张。Confocal（共聚焦）和two-photon（双光子）显微镜在成像流程的不同环节解决这个深度问题。

**EN:** Follow the mechanism: Confocal microscopy illuminates a focused spot and places a small pinhole before the detector. Properly focused return light passes through; much out-of-focus light is blocked. Scanning builds an image. Two-photon microscopy instead needs two photons to supply excitation energy together. This event becomes much more likely where light is most concentrated, so fluorescence is mainly generated near the focus rather than throughout the illuminated path.

**中文：** 跟着机制走：共聚焦用聚焦光点照明，并在探测器前放置很小的pinhole（针孔）。来自正确焦点的返回光可以通过，许多离焦光被挡住；逐点扫描后得到图像。双光子则需要两个光子共同提供激发能量；光越集中，这种事件越容易发生，所以荧光主要在焦点附近产生，而不是沿整个照明路径都大量产生。

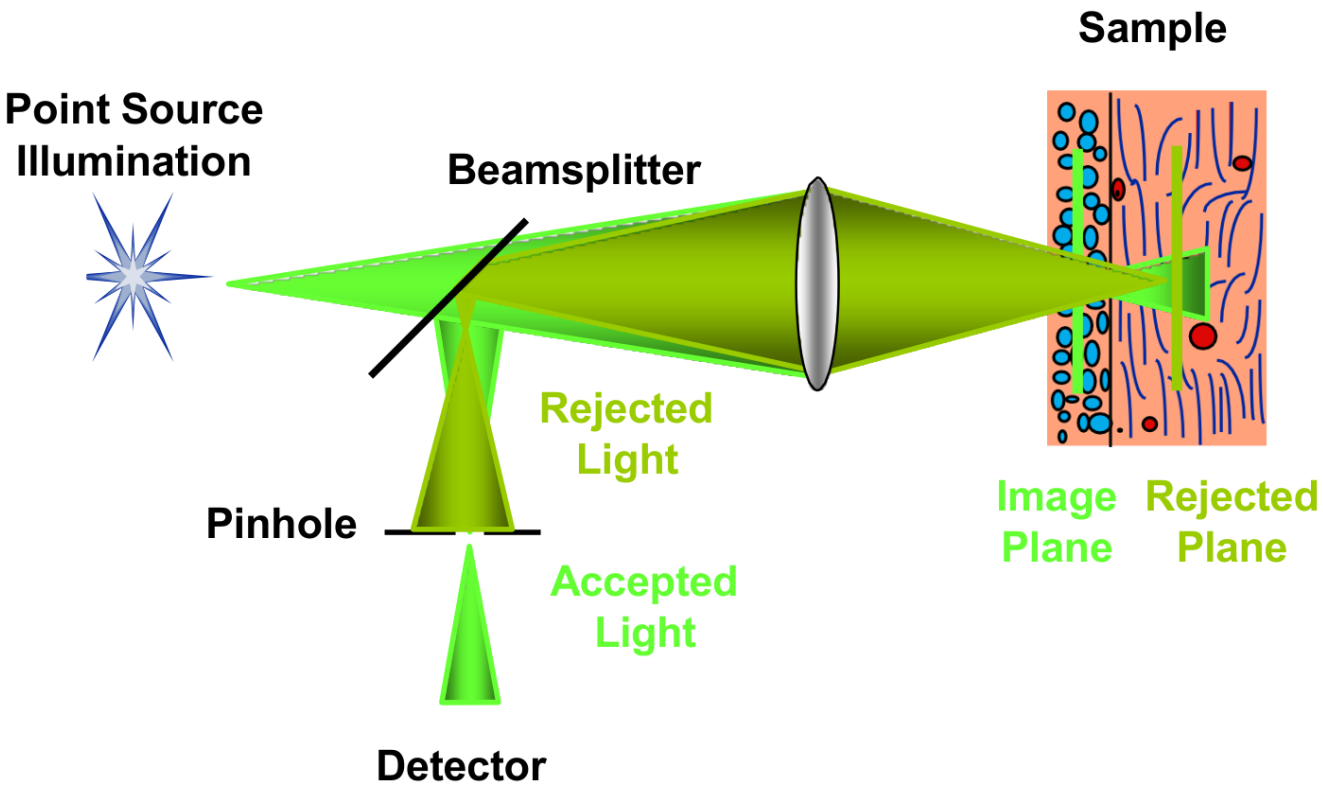
**EN:** Try an example: The exam distinction is detection-side rejection versus excitation-side localization. A pinhole is the signature of confocal imaging; nonlinear, localized excitation is the signature of two-photon imaging. Under the simple unsaturated model, two-photon fluorescence is proportional to $I^2$, where I is light intensity: doubling I gives four times the signal. Longer excitation wavelengths can help tissue penetration, but do not automatically improve diffraction-limited spatial resolution.

**中文：** 用例子理解：考试要区分：共聚焦是在检测端排除不想要的光，双光子是在激发端限制发光位置。看到pinhole应想到共聚焦；看到nonlinear（非线性）和localized excitation（局域激发）应想到双光子。在简单、未饱和模型下，双光子荧光与$I^2$成正比，I是光强：光强加倍，信号变为四倍。更长激发波长可能有利于组织穿透，但不会自动提高衍射极限空间分辨率。

**EN:** Exam wording: Do not call a pinhole the cause of two-photon excitation. The usual two-photon wavelength advantage concerns penetration and localized excitation; it does not erase the dependence of diffraction resolution on wavelength and NA.

**中文：** 英文读题抓手：不要把pinhole说成双光子激发的原因。双光子的常见优势是穿透与局域激发，并不取消衍射分辨率对波长和NA的依赖。

Source / 来源：W4 p47–61; Sample quiz p1 Q1–3; W2 p73–78; W3 p39–50; W4 p54–61 | T4-0062–0069, T7-0051–0053




Confocal optical sectioning: the detection pinhole rejects out-of-focus light (original lecture figure) / 共聚焦光学切片：检测针孔挡掉离焦光（原课件图）

**EN:** Read the figure: Trace illumination from Point Source to Sample, then the returning light toward Pinhole and Detector. Light from the selected focal plane passes the pinhole; light from another plane is rejected. This suppresses background rather than physically cutting the tissue. A point measurement must be scanned to form an image, and changing focus gives different depth slices.

**中文：** 读图：从Point Source光源跟到Sample样品，再看返回光走向Pinhole针孔和Detector探测器。选定焦平面的光通过针孔，其他平面的光被挡掉。它抑制背景，并没有真的切开组织。一次测一点，需要扫描形成图像；改变焦点位置可得到不同深度的切片。

Source / 来源：Week 4 PDF p51; original slide figure / 原课件截图

## S402 • [P2] Brightfield, darkfield, phase contrast and PSF / 明场、暗场、相衬与点扩散函数

Priority / 复习标记：P2 Important foundations/重要基础 · Repeated explanation/反复讲解 · Sample Q1/样卷Q1

**EN:** Brightfield microscopy mainly uses transmitted light: absorbing or scattering structures appear darker against a bright background. Darkfield excludes the direct illumination beam and collects scattered light, so scattering objects can look bright on a dark background. Phase contrast turns phase delays caused by optical-path differences into intensity differences, helping reveal transparent cells without relying on fluorescent labels.

**中文：** Brightfield microscopy（明场显微）主要看透射光：吸收或散射结构在亮背景上较暗。Darkfield（暗场）挡掉直达照明光，收集散射光，因此散射物可在暗背景上发亮。Phase contrast（相衬）把光程差造成的相位延迟转成亮度差，帮助观察透明细胞，并不依赖荧光标记。

**EN:** Fluorescence contrast instead comes from emitted light after excitation. Different labels and detection channels can highlight different structures and show co-localization, meaning that signals appear at the same spatial location. Display colors may be assigned by the software: a red, green or blue region in a pseudo-colored image need not be the sample’ s natural color. A suitable fluorescent probe can report a molecule’ s presence or ion binding. The slide also lists diffusion coefficient, a quantity describing how rapidly molecules spread by random motion; it is not intensity or color itself and needs an appropriate fluorescence measurement/model.

**中文：** 荧光对比来自激发后的发射光。不同标记和检测通道可突出不同结构，也可显示co-localization（共定位），即不同信号出现在同一空间位置。显示颜色可由软件指定：pseudo-colored（伪彩色）图中的红、绿、蓝不一定是样品的天然颜色。 合适荧光探针可报告分子是否存在或是否结合离子。课件还列出diffusion coefficient（扩散系数），描述分子随机运动扩散的快慢；它本身不是光强或颜色，需用相应荧光测量与模型推断。

**EN:** The point spread function, PSF, is the image produced by an ideal tiny point source. It describes how the imaging system spreads that point into a spot, sideways and along depth. A sharper-looking image and stronger contrast do not by themselves prove a smaller PSF or better diffraction resolution. In confocal imaging, the pinhole suppresses out-of-focus contributions; stacking optical sections gives a 3D data set.

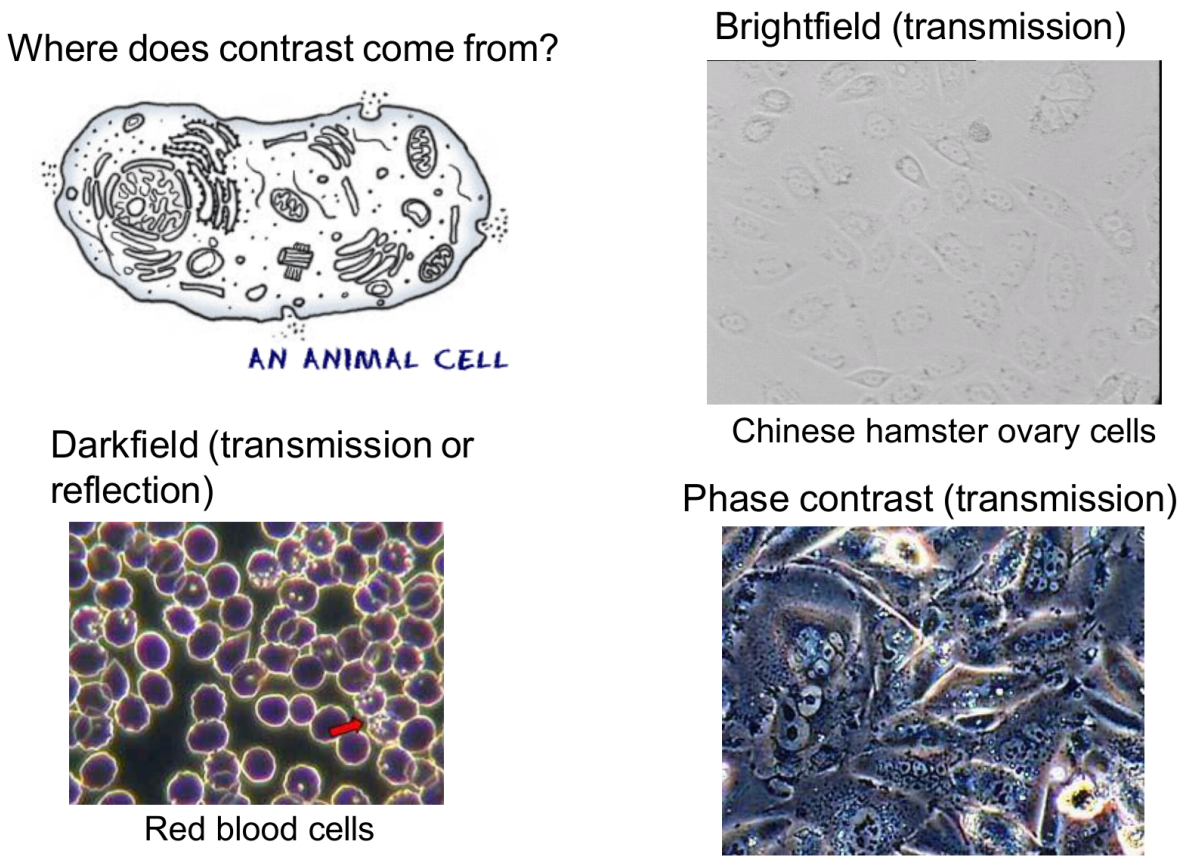
**中文：** Point spread function（PSF，点扩散函数）是理想极小点光源经系统形成的图像，描述一个点在横向和深度方向被扩散成多大的光斑。看起来更锐利、对比更强，不自动证明PSF更小或衍射分辨率更好。共聚焦针孔抑制离焦贡献；把不同深度的光学切片叠起来，可得到三维数据。

Source / 来源：Week 4 PDF p40–46, 51–53; T4-0056–0065

**EN:** Clarify the concept: In the axial-response plot, scan the reflector through focus along z: a narrower I(z) response means a thinner selected optical layer. The smaller pinhole curves are narrower in the shown comparison. Closing the pinhole also rejects useful light, so it is not a free improvement without a signal trade-off. Compare curve width for section thickness; the plot’ s similar peak heights do not establish identical collected photon counts.

**中文：** 把概念讲清：轴向响应图让反射面沿z经过焦点：I(z)响应越窄，选出的光学层越薄。图示比较中，小针孔曲线更窄；缩小针孔也挡掉有用光，因此不是没有信号代价的改善。用曲线宽度判断切片厚度，不要因图中峰值看起来接近就认为收集光子数相同。

Source / 来源：README §4.3 Confocal; Week 4 PDF p53 diagram, re-rendered peer_original_formula_contact.png; T4-0062–0065




Brightfield, darkfield and phase-contrast appearances (original lecture slide) / 明场、暗场与相衬显示差异（原课件图）

**EN:** Read the figure: Compare the image backgrounds and the slide labels. Ask whether contrast comes from transmitted intensity, scattered light or converted phase differences.

**中文：** 读图：对照图像背景与原图标签，问对比来自透射强度、散射光，还是相位差转亮度差。

Source / 来源：Week 4 PDF p40; original slide figure / 原课件截图

## S407 • [P2] Two-photon requirements and practical advantages / 双光子激发条件与实际优势

Priority / 复习标记：P2 Important foundations/重要基础 · Repeated explanation/反复讲解 · Sample Q1/样卷Q1

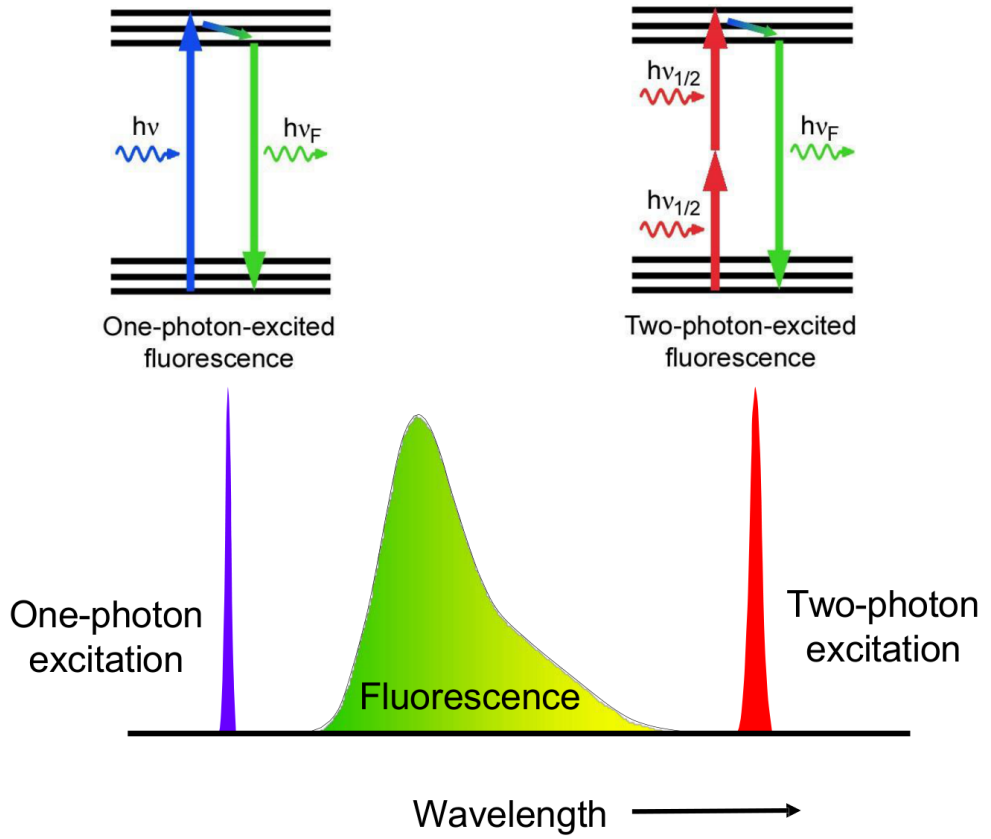
**EN:** Two-photon absorption is rare unless photons are concentrated strongly in space and time. A tightly focused pulsed laser supplies high peak intensity, so simultaneous two-photon excitation is much more likely near the focus. Because the excitation itself selects that small volume, a confocal detection pinhole is not required by this principle. This differs from illuminating a larger volume and then rejecting unwanted light at the detector.

**中文：** 双光子吸收只有在光子于空间和时间上高度集中时才较容易发生。紧密聚焦的pulsed laser（脉冲激光）提供高峰值光强，使两个光子联合激发主要出现在焦点附近。因为激发本身已选择这一小体积，该原理不需要共聚焦那种检测针孔；这不同于先照亮更大体积，再在探测器前挡掉不需要的光。

**EN:** Longer-wavelength excitation and localized fluorescence generation support deeper imaging in scattering tissue and reduce out-of-focus photobleaching and photodamage. Emitted photons scattered on their way out can still be collected because excitation was localized. These are advantages in contrast, collection and penetration, not a promise of zero damage or universally higher diffraction resolution: bright infrared excitation can still heat tissue, and wavelength and NA still matter.

**中文：** 更长波长激发及局域产生荧光，有助于较深散射组织成像，也减少焦外photobleaching（光漂白）和photodamage（光损伤）。因为激发位置局域，发射光在返回途中被散射后仍可被收集。这些是对比、收光和穿透方面的优势，不保证零损伤或普遍更高衍射分辨率：强红外激发仍可造成组织加热，波长与NA仍然重要。

Source / 来源：Week 4 PDF p56–61; T4-0066–0081




One-photon and two-photon excitation: energy diagrams and wavelength order (original lecture figure) / 单光子与双光子激发：能级图与波长顺序（原课件图）

**EN:** Read the figure: Compare the upward arrows: one blue excitation photon or two lower-energy red photons jointly excite the fluorophore; green arrows represent emission. On the lower wavelength axis, right means longer wavelength. The two-photon excitation wavelength can exceed the emission wavelength because two photons supply the energy together. Do not apply the usual single-photon Stokes ordering to each individual two-photon excitation photon.

**中文：** 读图：比较向上的箭头：一个蓝色激发光子，或两个能量较低的红色光子共同激发荧光体；绿色向下箭头表示发射。下方波长轴向右是波长更长。双光子激发波长可以长于发射波长，因为能量由两个光子共同提供；不能把通常的单光子Stokes顺序直接套到每个双光子激发光子上。

Source / 来源：Week 4 PDF p55; original slide figure / 原课件截图

## C15 • [P1] OCT: low-coherence interference and reflectivity / OCT：低相干干涉与反射率

Priority / 复习标记：P1 First pass/先掌握 · Sample Q8/样卷Q8 · Repeated explanation/反复讲解

**EN: Key points**

- Optical coherence tomography combines light returning from a sample with a reference beam.
- Low-coherence interference localizes returning signals in depth.
- Layer brightness represents back-reflection/back-scattering under the course description;
- it is not simply a temperature or fluorescence map.

**中文：必会知识点**

- 光学相干断层成像把样品返回光与参考光合并。
- 低相干干涉把返回信号定位到不同深度。
- 按课程描述，层状亮度反映后向反射或散射，而不是简单的温度图或荧光图。

**EN:** Start from zero: Optical coherence tomography, OCT, builds depth pictures from light returned by tissue. Different layers return different amounts of light, so the picture can show boundaries beneath the surface. Its common grayscale brightness represents the detected return signal, not simply temperature or fluorescence. The practical difficulty is that light travels too quickly for ordinary electronics to time these tiny travel differences directly.

**中文：** 从零理解：Optical coherence tomography（OCT，光学相干断层成像）用组织返回的光构建深度图像。不同层返回的光量不同，因此能显示表面下面的层界；常见灰度图的亮度对应检测到的返回信号，不直接等于温度或荧光。实际困难是光传播太快，普通电子设备不能直接计时这些极小的传播时间差。

**EN:** Follow the mechanism: OCT splits light into a sample arm and a reference arm, then recombines their returning beams. Interference means overlapping light waves reinforce or weaken each other according to their relative phase. With a broad-band, low-coherence source, useful interference is linked to closely matched optical paths. The reference supplies a comparison for identifying return depth; in modern systems this information can also be recovered from a measured interference spectrum.

**中文：** 跟着机制走：OCT先把光分到sample arm（样品臂）和reference arm（参考臂），再把返回的两束光合并。Interference（干涉）是光波叠加时按相对相位互相增强或减弱。使用宽带、低相干光源时，有效干涉与光程接近匹配有关。参考臂提供定位返回深度的比较基准；现代系统也可以从测得的干涉光谱中恢复深度信息。

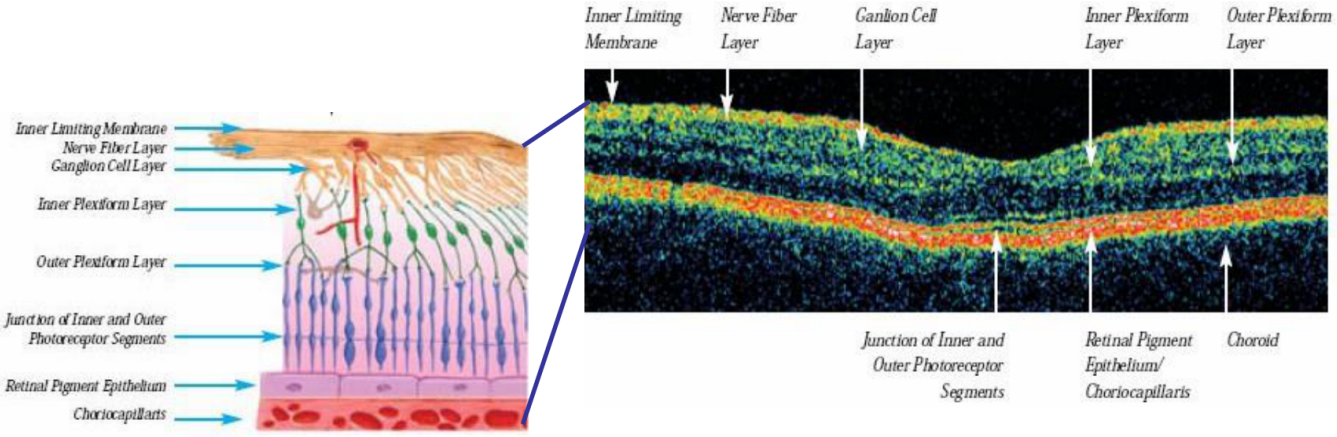
**EN:** Try an example: A useful analogy is comparing an unknown path with a known ruler: the reference is essential, but OCT uses wave interference rather than a physical ruler. In a diagram, look for beam splitting, two arms, and recombination. In a question about OCT contrast, choose back-reflection or back-scattering. Avoid calling OCT ordinary electronic time-of-flight measurement, ultrasound echoes, or a fluorescent-label measurement.

**中文：** 用例子理解：可以类比用已知尺子比较未知距离：参考基准很重要，但OCT实际用的是光波干涉，而不是实体尺子。图中应寻找分光、两个光臂以及返回光合并。题目问OCT的成像对比时，选back-reflection（后向反射）或back-scattering（后向散射）。不要把它理解为普通电子计时的飞行时间测量、超声回波或荧光标记测量。

**EN:** Exam wording: Identify the reference arm, sample arm and interference signal. The long Michelson intensity derivation is lower priority because of the global short-calculation guidance; there is no separate explicit oral exclusion of all OCT equations.

**中文：** 英文读题抓手：要识别reference arm、sample arm和interference signal。依据本次短计算的总要求，Michelson长强度推导降低优先级；没有单独找到“所有OCT公式均不考”的明确声明。

Source / 来源：W4 p71–79; W4 p85–90; Sample quiz p2 Q8–9; W4 p35–38, p73–86; W3 p25–32 | T4-0088–0098, T4-0105–0108, T7-0055–0057




OCT retinal cross-section compared with an anatomical layer diagram (original lecture figure) / OCT视网膜横截面与组织层示意对照（原课件图）

**EN:** Read the figure: Use the labelled arrows to match the anatomical layers on the left with bands in the OCT image on the right. Across the image is lateral position; top-to-bottom separates depths. These display colors encode the strength of returning/scattered light, not the tissue's natural color or a fluorescent stain. Layer separation depends on axial resolution; the figure does not supply a calibrated scale for measuring thickness.

**中文：** 读图：用标注箭头把左侧组织层与右侧OCT条带对应起来。横向是在组织表面不同位置，纵向把不同深度分开。显示颜色编码返回/散射光强，既不是组织天然颜色，也不是荧光染色。能否分开两层取决于轴向分辨率；本图没有可用于测厚度的标定尺度。

Source / 来源：Week 4 PDF p86; original slide figure / 原课件截图

## S405 • [P2] OCT interference, resolution and angiography / OCT干涉、分辨率与血管成像

Priority / 复习标记：P2 Important foundations/重要基础 · Teacher emphasis/课堂强调 · Sample Q8/样卷Q8 · Supporting concept/辅助理解

**EN:** In the slide’ s single-wavelength Michelson example, moving one mirror by L changes the round-trip path by 2L, so the phase difference changes by 4πL/λ. The detected intensity contains a phase-dependent interference term; it is not merely the sum of two unrelated brightness readings. Constructive interference strengthens the signal and destructive interference weakens it. The long derivation is secondary; identify the splitter, sample arm, reference arm and recombination.

**中文：** 课件单波长Michelson示例中，一面镜子移动L，往返光程改变2L，因此相位差改变4πL/λ。检测强度含随相位变化的干涉项，不只是两次互不相关亮度读数相加。Constructive interference（相长干涉）增强信号，destructive interference（相消干涉）减弱信号。长推导次要，先辨认分光器、样品臂、参考臂与重新合束。

**EN:** For the course’ s Gaussian-source approximation in air, $\lambda_0$ is the central wavelength and Δλ is the intensity spectrum’ s FWHM (full width at half maximum); axial separation $\Delta z \approx (2\ln 2/\pi)\lambda_0^2/\Delta\lambda \approx 0.44\lambda_0^2/\Delta\lambda$. Use matching wavelength units: $\lambda_0$=800 nm and bandwidth Δλ=100 nm give about 2816 nm=2.8 μm. At fixed $\lambda_0$, doubling bandwidth halves Δz. This is a depth-resolution relation; changing NA mainly changes lateral focusing and the useful focus range. Physical depth in tissue additionally depends on refractive index.

**中文：** 这里$\lambda_0$是中心波长，Δλ为强度光谱的FWHM（半高全宽），不是任意波段两端之差。课程高斯光源近似下，空气中的轴向间距$\Delta z \approx (2\ln 2/\pi)\lambda_0^2/\Delta\lambda \approx 0.44\lambda_0^2/\Delta\lambda$。波长单位必须统一：$\lambda_0$=800 nm、带宽Δλ=100 nm时，约为2816 nm=2.8 μm。$\lambda_0$固定时，带宽加倍使Δz减半。这是深度方向分辨关系；改变NA主要改变横向聚焦及可用聚焦范围。组织中的实际深度还受折射率影响。

**EN:** An OCT cross-section commonly uses lateral position x and depth z. In the taught display, stronger returning signals receive brighter or red/white display colors, while weak returns receive dark colors. The color is a signal map, not the natural color of each eye layer. Refractive-index inhomogeneity creates reflection/backscatter contrast; absorption along the path can weaken returns but is not the primary answer to sample Q8.

**中文：** OCT剖面常以横向位置x和深度z为坐标。课程显示中，较强返回信号用更亮或红/白色表示，弱返回用暗色。颜色是信号映射，不是眼各层的天然颜色。折射率不均匀产生反射/后向散射对比；沿途吸收可削弱返回光，但它不是样题Q8所问的主要对比答案。

**EN:** OCT angiography, OCT-A, uses changes associated with moving red blood cells to map microvasculature. The example avoids injected vascular dye. It is a flow-related extension of OCT, not fluorescent labeling of every vessel. Structural OCT asks where tissue layers are; OCT-A adds information about where blood-cell motion is detected.

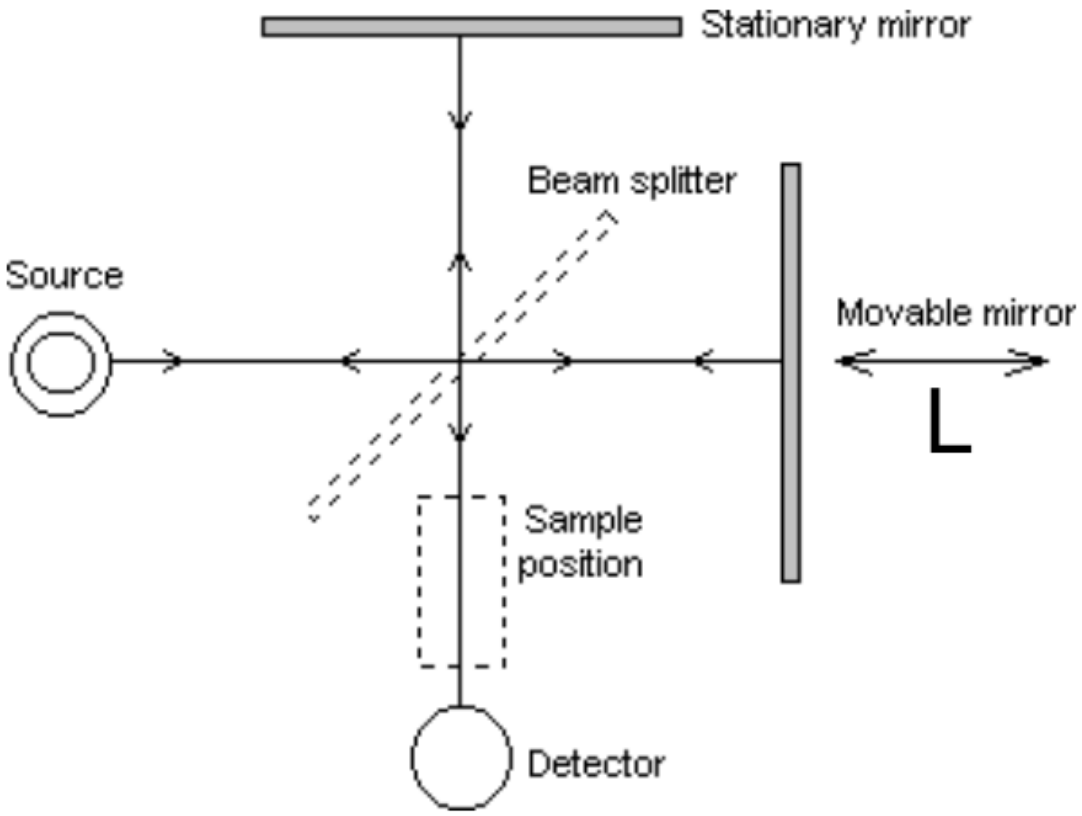
**中文：** OCT angiography（OCT-A，OCT血管成像）利用运动红细胞相关的变化绘制微血管，课件示例不需要注射血管染料。它是OCT的血流相关扩展，不是给每根血管加荧光标签。结构OCT回答组织层在哪里；OCT-A补充哪里检测到血细胞运动。

Source / 来源：Week 4 PDF p75–87; T4-0089–0106

**EN:** Clarify the concept: Spectral/Fourier-domain OCT records the interference spectrum as the sample return combines with the reference. Fourier processing converts the spectral fringe structure into a depth-reflectivity profile, often called an A-scan. Moving laterally and combining depth profiles gives a cross-section. Thus the raw measured horizontal coordinate can be wavelength, while the reconstructed coordinate is depth; neither is a fluorescence emission spectrum used as a label.

**中文：** 把概念讲清：Spectral/Fourier-domain OCT（谱域/傅里叶域OCT）记录样品返回光与参考光形成的干涉光谱。傅里叶处理把光谱条纹结构变为深度反射曲线，常叫A-scan；横向移动并合并深度曲线得到剖面。因此原始测量横轴可为波长，重建后坐标却是深度；这不是把荧光发射光谱当作标记来测。

Source / 来源：README §4.4 Fourier-domain OCT; Week 4 PDF p79–80; T4-0096–0098




Michelson interferometer and the round-trip optical-path difference (original lecture slide) / Michelson干涉仪与往返光程差（原课件图）

**EN:** Read the figure: Identify the beam splitter, fixed mirror, movable mirror, and detector in this Michelson demonstration. Light travels to each mirror and back; mirror displacement L changes the round-trip path by 2L. Interference compares the returning fields, not two independent light counts.

**中文：** 读图：在Michelson演示图中找出beam splitter分光器、固定镜、可动镜与detector探测器。光走到镜面再返回，镜移动L使往返光程改变2L；干涉比较的是两束返回光场，不是两份独立光强。

Source / 来源：Week 4 PDF p76; original slide figure / 原课件截图

## B16 • [P2] Photodiode and photomultiplier tube / 光电二极管与光电倍增管

Priority / 复习标记：P2 Important foundations/重要基础 · Repeated explanation/反复讲解

**EN: Key points**

• A photodiode converts absorbed photons into electron–hole pairs and photocurrent.
• A PMT emits photoelectrons from a photocathode, then multiplies electrons at successive dynodes.
• This internal gain helps detect weak light;
• it does not remove all noise.
• A point detector usually requires scanning for an image, whereas CCD/CMOS arrays provide many spatial detector elements.

**中文：必会知识点**

• Photodiode光电二极管将吸收光子转换为电子—空穴对及光电流。
• PMT光电倍增管由photocathode光阴极发出光电子，再在dynode倍增极逐级倍增电子。
• 内部增益帮助检测弱光，不等于消除所有噪声。
• 点探测器通常需扫描才能成像；
• CCD/CMOS阵列提供许多空间探测单元。

**EN:** Start from zero: A light detector turns arriving light into an electrical signal. A photodiode is a semiconductor device: suitable absorbed photons create electron–hole pairs, which can be separated and collected as photocurrent. An electron carries negative charge; a hole represents a missing electron that behaves as a positive carrier. The current tells us about the detected light under the detector's calibrated response.

**中文：** 从零理解：光探测器把到达的光转换成电信号。Photodiode（光电二极管）是半导体器件：合适光子被吸收后产生electron–hole pairs（电子—空穴对），它们被分离、收集，形成photocurrent（光电流）。电子携带负电荷；空穴代表缺失电子的位置，表现为正电荷载流子。在探测器已校准的响应关系下，电流反映被检测的光。

**EN:** Follow the mechanism: A photomultiplier tube, PMT, first releases photoelectrons from a photocathode. An electric field accelerates them toward successive dynodes. Each impact can release several secondary electrons, producing a cascade before collection at the anode. Think of a branching count that grows at each stage. These stages multiply electrons; they are not tiny mirrors repeatedly reflecting and multiplying light. A PMT requires an appropriate high-voltage supply.

**中文：** 跟着机制走：Photomultiplier tube（PMT，光电倍增管）先由photocathode（光阴极）发出光电子，电场把它们加速到连续的dynodes（倍增极）。每次撞击可释放多个二次电子，逐级级联，最后在阳极收集。可以类比每一级分支后数量增加；这里倍增的是电子，不是小镜子反复反射并倍增光。PMT需要合适的高压供电。

**EN:** Try an example: For illustration, one initial electron multiplied by four at each of three stages gives $4^3$=64 electrons; real gains depend on device and conditions. Internal gain helps weak signals become measurable, but noise remains. A single detector reports light at one measurement location, so scanning is usually needed to build an image. English clues are electron–hole pairs for photodiodes and photocathode, dynodes, and secondary emission for PMTs.

**中文：** 用例子理解：简化举例：一个初始电子经过三级，每级都倍增四倍，则最终为$4^3$=64个电子；实际增益取决于器件及工作条件。内部增益让弱信号更容易测到，但噪声仍存在。单个探测器报告一个测量位置的光，通常需要扫描才能形成图像。英文辨认：electron–hole pairs对应光电二极管；photocathode、dynodes、secondary emission（二次发射）对应PMT。

Source / 来源：W4 p35–38 | T4-0048–0055

## C16 • [P1] CCD, CMOS and hyperspectral imaging / CCD、CMOS与高光谱成像

Priority / 复习标记：P1 First pass/先掌握 · Sample Q9/样卷Q9 · Repeated explanation/反复讲解

**EN: Key points**

- A CCD accumulates photocharge and transfers it through a charge-transfer readout structure.
- CMOS commonly uses pixel-associated circuitry for readout or amplification.
- Hyperspectral imaging records a spectrum at each spatial location, producing a spatial–spectral data cube.

**中文：必会知识点**

- CCD累积光生电荷，并通过电荷转移结构读出；
- CMOS通常通过像素相关电路读出或放大。
- Hyperspectral imaging在每个空间位置记录光谱，形成空间—光谱数据立方。

**EN:** Start from zero: A camera must turn a pattern of light into numbers. A pixel is one small detector element that samples a position in the image. Light produces electrical charge in these elements. Reading all the elements gives a digital image. CCD and CMOS are two detector-array technologies; they differ mainly in how the electrical information is collected and read.

**中文：** 从零理解：相机需要把光的分布转换成数字。Pixel（像素）是对图像某个位置采样的小探测单元；光在这些单元内产生电荷，读出所有单元的信息就得到数字图像。CCD和CMOS是两种探测器阵列技术，主要区别在电信息如何收集和读出。

**EN:** Follow the mechanism: In a CCD, collected charge is transferred through a readout structure, like passing measured packets along a line. CMOS commonly has circuitry associated with pixels for local conversion or readout, like giving positions their own reading stations. These analogies explain architecture only: neither guarantees perfect noise performance. Both still need illumination, optics, and suitable electronics to produce a useful image.

**中文：** 跟着机制走：CCD把收集的电荷通过读出结构转移，可类比为沿队列传递已经装好的测量包。CMOS通常有与像素相关的电路，完成局部转换或读出，可类比为不同位置设有自己的读取站。这些比喻只帮助理解结构，不保证哪种技术噪声永远更小。两者都需要照明、成像光学系统及合适电路才能形成有用图像。

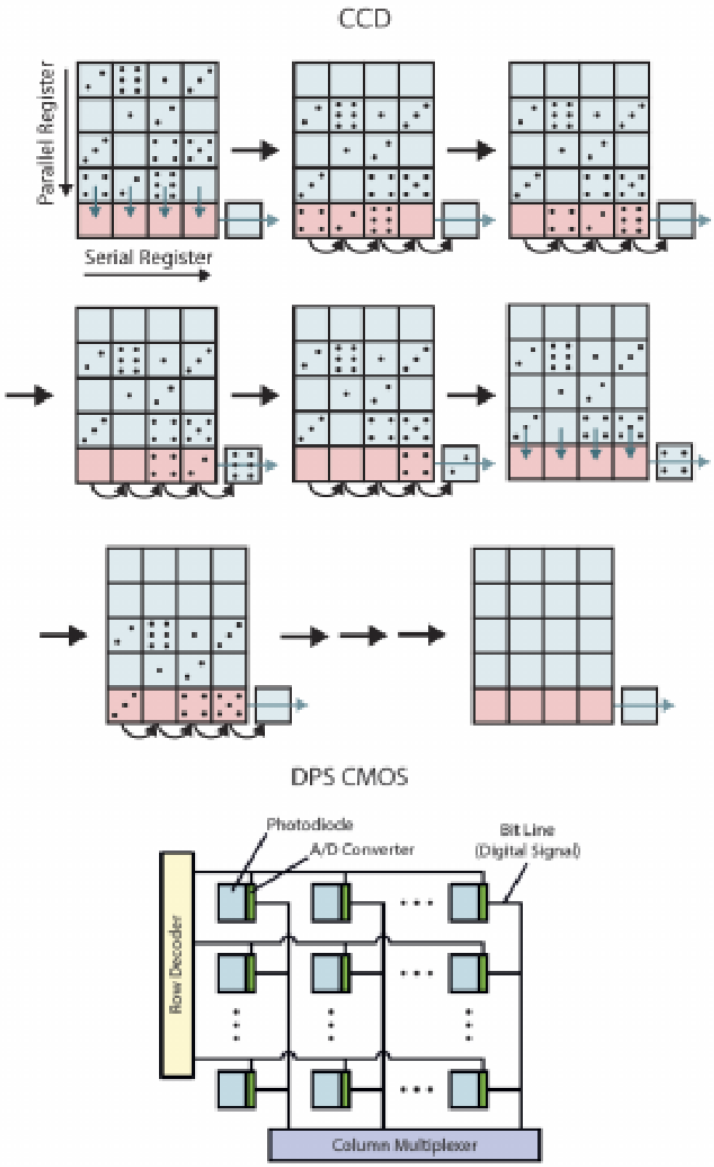
**EN:** Try an example: Hyperspectral imaging adds a wavelength dimension: each position has a spectrum, meaning signal versus wavelength. The result is a cube indexed by x, y, and λ, not a physical cube inside the camera. A line-scan system records one spatial line and its spectra, then moves to collect the next line. English clues are charge transfer for CCD, pixel-associated circuitry for CMOS, and spectrum at each pixel for hyperspectral imaging.

**中文：** 用例子理解：Hyperspectral imaging（高光谱成像）增加了波长维度：每个位置都有一条spectrum（光谱），即信号随波长的变化。结果用x、y、λ三个坐标组成data cube（数据立方），并非相机里有一个实体立方体。Line-scan（线扫描）一次记录一条空间线及其光谱，再移动采集下一条。英文辨认：charge transfer对应CCD，pixel-associated circuitry对应CMOS，spectrum at each pixel对应高光谱。

**EN:** Exam wording: A line-scan instrument records one spatial line plus its spectra at a time; scanning the other spatial axis builds the cube. For sample Q9, B remains a reasoned reference answer; the oral review does not give an unambiguous final letter.

**中文：** 英文读题抓手：Line-scan一次记录一条空间线及其光谱，再沿另一空间轴扫描形成数据立方。样题Q9仍把B作为推导参考答案，但口头回顾没有给出明确的最终字母。

Source / 来源：W4 p35–38; W4 p91–104; Sample quiz p2 Q8–9; W4 p35–38, p73–86; W3 p25–32 | T4-0048–0055, T4-0109–0113, T7-0055–0057

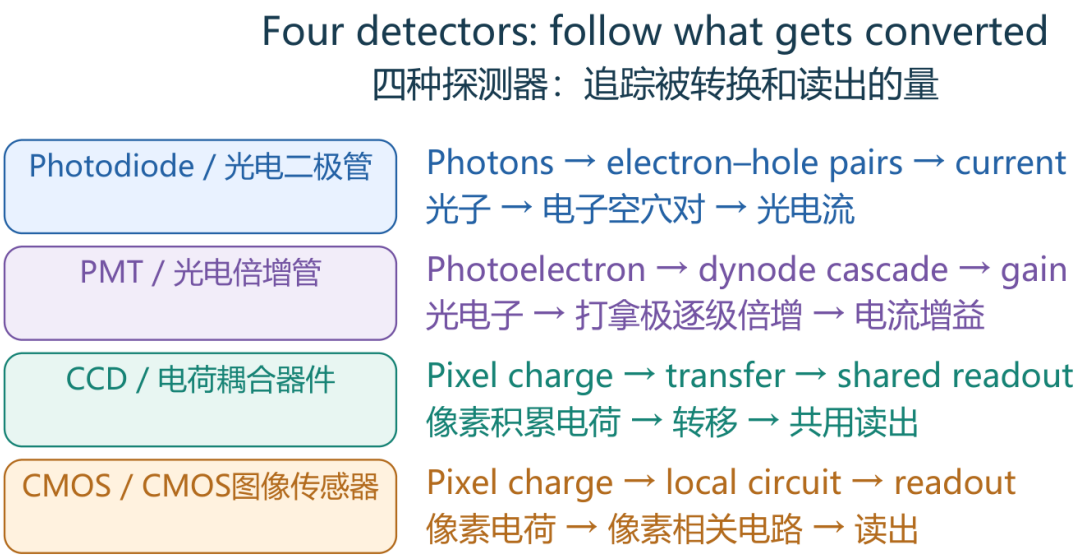



CCD charge transfer and a CMOS pixel-readout example (original lecture figure) / CCD电荷转移与一种CMOS像素读出示例（原课件图）

**EN:** Read the figure: In the upper sequence, follow the arrows: charge packets move down the CCD array, then through the serial register to an output. In the lower CMOS example, each pixel has local conversion circuitry and the row/column circuits select readout. The shown per-pixel ADC is one digital-pixel implementation; it is not a requirement for every CMOS sensor. Both begin with photons generating charge.

**中文：** 读图：上方按箭头看：CCD中的电荷包先沿阵列下移，再通过串行寄存器送到输出。下方CMOS示例在像素附近放转换电路，并用行列电路选择读出。图中的每像素ADC属于一种数字像素实现，并非所有CMOS都必须如此。两者最初都是光子产生电荷。

Source / 来源：Week 4 PDF p36; original slide figure / 原课件截图




Four detectors: follow what gets converted / 四种探测器：追踪被转换和读出的量

**EN:** Read the figure: Photodiodes produce photocurrent. PMTs multiply electrons at dynodes, rather than repeatedly reflecting light. CCDs transfer charge packets for readout. CMOS sensors use pixel-associated circuitry, but analogue-to-digital conversion can be shared at column or other levels; do not generalize one schematic to every CMOS architecture.

**中文：** 读图：Photodiode（光电二极管）产生光电流。PMT在Dynode（打拿极）处倍增电子，不是让光反复反射。CCD转移电荷包再读出。CMOS使用像素相关电路，但Analog-to-digital conversion（模数转换）可以按列或以其他层级共用；不能把单张结构图推广成所有CMOS的必然结构。

Source / 来源：Week 4 PDF p35–38; Sample Quiz Q9 option B; original teaching schematic / 自绘教学示意，非实验数据

## S404 • [P1] Detector architecture and practical trade-offs / 探测器结构与实际取舍

Priority / 复习标记：P1 First pass/先掌握 · Sample Q9/样卷Q9 · Repeated explanation/反复讲解

**EN:** In the course comparison, CCD charge is shifted toward output nodes for conversion and digitization. CMOS uses pixel-associated circuitry and often parallel readout; selected pixels can be addressed. ADC means analog-to-digital converter: it turns an analog voltage into a numerical value. Neither CCD nor CMOS removes the need for optical focusing.

**中文：** 按课程比较，CCD把电荷移向输出节点进行转换和数字化；CMOS使用像素相关电路，常可并行读出，也可选择部分像素读取。ADC是analog-to-digital converter（模数转换器），把模拟电压转成数字数值。CCD和CMOS都仍需要光学聚焦。

**EN:** The slides associate CCD with sensitive, low-noise scientific imaging, and CMOS with faster readout, lower power/cost and on-chip integration. Treat these as architecture trends in the course examples, not absolute rankings for every modern device. A photodiode has no PMT-style dynode multiplication; a PMT has high internal gain but still has noise and a wavelength-dependent response.

**中文：** 课件把CCD与灵敏、低噪声的科学成像联系起来，把CMOS与更快读出、较低功耗/成本及片上集成联系起来。这些是课程示例中的结构趋势，不是所有现代器件的绝对排名。光电二极管没有PMT那种倍增极级联；PMT内部增益高，但仍有噪声，其响应也随波长变化。

Source / 来源：Week 4 PDF p35–38; T4-0048–0055

## B17 • [P1] Sectioning, OCT resolution and hyperspectral data / 切片、OCT分辨率与高光谱数据

Priority / 复习标记：P1 First pass/先掌握 · Teacher emphasis/课堂强调 · Sample Q8/样卷Q8 · Lower priority/降低优先级

**EN: Key points**

- Widefield collects out-of-focus background;
- confocal rejects it;
- two-photon excitation is localized (fluorescence $\propto I^2$ before saturation).
- OCT axial resolution depends on coherence length: broader bandwidth improves it at fixed central wavelength.
- Lateral resolution depends on focusing/NA;
- at fixed wavelength, higher NA reduces object-side depth of field, the usable focused specimen range, not layer separation.
- Slide p83 calls this depth of focus, although standard terminology reserves depth of focus for image space.
- Hyperspectral imaging records many narrow bands at each position, not just RGB.

**中文：必会知识点**

- Widefield宽场会收集离焦背景，共聚焦抑制它，双光子采用局域激发（未饱和时荧光$\propto I^2$）。
- OCT轴向分辨率取决于相干长度：中心波长固定时，带宽越宽、轴向分辨越好。
- 横向分辨率依赖聚焦/NA；
- 固定波长时，NA越大、物侧景深越小。
- Depth of field景深是样品保持清晰的深度范围，不是可分辨层间距；
- 原课件p83称它depth of focus，但标准术语的depth of focus焦深指像侧范围。
- 高光谱在每个位置记录许多窄谱带，不只是RGB。

**EN:** Start from zero: Separate layer selection, resolution and focus range. Optical sectioning selects a layer; axial resolution separates structures along depth, while lateral resolution separates them sideways. Depth of field is the object-side specimen thickness that stays acceptably sharp. Depth of focus is the image-side tolerance for the image plane. Slide p83 labels its object-side range 'depth of focus'; recognize the slide's wording, but keep the standard meanings separate.

**中文：** 从零理解：分清选层、分辨和聚焦范围：Optical sectioning光学切片选择成像层；axial resolution轴向分辨率区分深度方向的结构，lateral resolution横向分辨率区分左右结构。Depth of field景深是样品侧保持清晰的厚度；depth of focus焦深是像侧允许像面移动的范围。原课件p83把物侧范围写成depth of focus；要认得课件用词，同时分清标准含义。

**EN:** Follow the mechanism: Widefield imaging collects substantial out-of-focus background. Confocal imaging rejects much of it with a pinhole; two-photon imaging mainly creates fluorescence at the focus, with signal proportional to $I^2$ before saturation. In OCT, axial separation is governed by source coherence length: broader optical bandwidth improves axial resolution at fixed central wavelength. Lateral resolution instead depends on focusing and NA. Higher NA improves lateral detail but reduces object-side depth of field at fixed wavelength; it does not automatically improve OCT axial resolution.

**中文：** 跟着机制走：Widefield（宽场）会收集较多离焦背景；共聚焦用针孔抑制许多背景；双光子主要在焦点产生荧光，未饱和时信号与$I^2$成正比。OCT的轴向区分由光源相干长度控制：中心波长固定时，光谱带宽越宽，轴向分辨越好。横向分辨则取决于聚焦与NA。NA提高可改善横向细节，在固定波长时却减小物侧景深；并不会自动改善OCT轴向分辨率。

**EN:** Try an example: Hyperspectral imaging answers a different question: what spectrum comes from each location? RGB uses a few broad color channels; hyperspectral data use many narrow wavelength bands. Imagine a small map in which every square contains a wavelength-versus-signal graph. This can help distinguish materials with different spectra, but collecting more bands does not itself sharpen spatial detail. Exam examples: wider OCT source bandwidth improves axial resolution; greater NA affects lateral resolution and focus range; x, y, λ identifies a hyperspectral cube.

**中文：** 用例子理解：高光谱回答另一个问题：每个位置的光谱是什么？RGB只有少数宽颜色通道，高光谱记录许多窄波长带。可以想象一张小地图，每个格子里都放着一条“波长—信号”曲线，这有助于区分光谱不同的材料；但增加谱带不会自动让空间细节更清楚。英文题例：更宽OCT光源带宽改善轴向分辨率；更大NA影响横向分辨率及聚焦范围；x、y、λ指高光谱数据立方。

Source / 来源：Nikon MicroscopyU: depth of field and depth of focus; Week 4 PDF p83

Source / 来源：W4 p39–46; W4 p47–61; W4 p71–79; W4 p80–82; W4 p83; W4 p91–104; W4 p105 | T4-0056–0069, T4-0088–0104, T4-0109–0114

## S406 • [P3] Hyperspectral acquisition and material identification / 高光谱采集与物质辨识

Priority / 复习标记：P3 Supporting reading/辅助阅读 · Lower priority/降低优先级 · Repeated explanation/反复讲解

**EN:** A spectrometer measures signal versus wavelength at a location; hyperspectral imaging adds spatial locations. Its cube has axes x, y and λ, so the third axis is spectral, not anatomical depth z. Reflectance HSI, Raman HSI and fluorescence HSI use different light–matter signals, even though all can provide a spectrum at each position.

**中文：** Spectrometer（光谱仪）测某个位置的信号随波长如何变化；hyperspectral imaging（高光谱成像）再加上空间位置。数据立方坐标为x、y、λ，第三轴是光谱，不是解剖深度z。Reflectance HSI（反射高光谱）、Raman HSI（拉曼高光谱）和fluorescence HSI（荧光高光谱）使用不同光—物质信号，但都可在各位置提供光谱。

**EN:** Read the four acquisition diagrams by asking what is recorded at once. Point scan records one location’ s spectrum; line scan records a spatial line and its spectra, then scans the other spatial axis. Wavelength scan records a 2D image at one band, then changes bands. Snapshot acquisition attempts to collect the spatial–spectral information in one exposure. Line scan does not obtain the entire surface cube in one step.

**中文：** 四种采集图先问“一次记录什么”：point scan（点扫描）记录一个位置的光谱；line scan（线扫描）记录一条空间线及其光谱，再扫另一空间轴。Wavelength scan（波长扫描）一次记录一个谱带的二维图，再改变谱带；snapshot（快照）尝试一次曝光采集空间—光谱信息。线扫描并不是一步就取得整个表面的完整数据立方。

**EN:** Different materials can have different spectral signatures even when an RGB image looks similar. Classification compares these signatures; many spectral bands can therefore help identify tissue or materials. A pixel can contain more than one material, giving a mixed spectrum. More wavelength bands do not automatically improve spatial resolution, and a spectrum-based classification still requires suitable reference data or a trained model.

**中文：** 不同材料即使RGB图看起来相似，也可具有不同spectral signatures（光谱特征）。分类比较这些特征，因此许多谱带可帮助辨识组织或材料。一个像素也可能混有多种材料，形成mixed spectrum（混合光谱）。增加谱带不会自动提高空间分辨率；基于光谱的分类仍需要合适参考数据或训练过的模型。

Source / 来源：Week 4 PDF p95–110; T4-0109–0119

# Week 5 learning block: Ultrasound, photoacoustics and image analysis / 第5学习组：超声、光声与图像分析

## Week 5 concept map / 第5周知识框架

**EN:** Chapter logic: Compare what enters tissue and what returns, then turn measured signals into images and diagnostic decisions.

**中文：** 本章主线：比较送入组织和返回的信号，再理解测量怎样形成图像及诊断判断。

| English term 中文名词 | Meaning and connection 含义与知识联系 |
|---|---|
| **Piezoelectric transducer 压电换能器** | Voltage→deformation transmits sound; pressure→charge receives it. 电压→形变用于发声，压力→电荷用于接收。 → B18 |
| **Acoustic impedance and reflection 声阻抗与反射** | $Z=\rho c$; impedance mismatch creates echoes, and high frequency usually trades depth for detail. $Z=\rho c$；失配形成回波，高频通常以穿透深度换更细细节。 → S501 |
| **Echo versus photoacoustic timing 回波与光声计时** | Echo depth uses ct/2 for the round trip; a stated one-way photoacoustic route uses ct. 回波按往返ct/2，明确单程的光声声路按ct。 → C17 |
| **Photoacoustic contrast 光声对比** | Absorbed pulsed light→heating/expansion→sound; multiple wavelengths can reveal different absorbers. 脉冲光吸收→升温/膨胀→声音，多波长可区分不同吸光物。 → S503 |
| **X-ray and CT X射线与CT** | $I/I_0=\exp(-\Sigma\mu_i L_i)$ for stated uniform layers; CT combines angular projections into slices. 题设均匀多层$I/I_0=\exp(-\Sigma\mu_i L_i)$，CT合并角度投影重建切片。 → S504 |
| **Digital-image representation 数字图像表达** | Sampling discretizes position; b-bit quantization gives $2^b$ intensity levels. 采样离散化位置，b位量化给$2^b$个强度等级。 → B20 |
| **Processing and analysis 处理与分析** | Clean the image, segment regions, measure features, then classify, estimate values or track objects. 整理图像、分割区域、测特征，再分类、估数值或跟踪。 → S505 |
| **Diagnostic metrics 诊断指标** | TP/TN/FP/FN compare predictions with truth; ROC plots TPR against FPR as threshold changes. TP/TN/FP/FN对比预测和真值；ROC在变阈值时画TPR对FPR。 → S505 |

**EN:** Formula links for this chapter: FX410 Acoustic wave, wavelength and frequency, FX411 Echo depth versus one-way photoacoustic path, FX412 Acoustic impedance and pressure/intensity, FX413 Acoustic reflection and normal-incidence optical analogue, FX414 Exponential pressure and intensity attenuation, FX415 dB loss and exponential-coefficient conversion, FX416 Frequency-dependent ultrasound path loss, FX417 Ultrasound pulse length and slide half-depth heuristic, FX418 X-ray single- and multilayer attenuation, FX419 Signed transmission-image contrast, FX420 Photoelectric threshold and absorption trend, FX421 Bit depth, storage and spatial sampling, FX422 Convolution and averaging kernels, FX423 Diagnostic sensitivity, specificity and ROC.

**中文：** 本章公式定位：FX410 声波、波长与频率；FX411 回波深度与光声单程声路；FX412 声阻抗与压力 / 强度；FX413 声反射与正入射光学类比；FX414 压力与强度的指数衰减；FX415 dB损耗与指数系数换算；FX416 频率相关的超声路径损耗；FX417 超声脉冲长度与课件半深度经验式；FX418 X射线单层与分层衰减；FX419 透射图像的带符号对比度；FX420 光电阈值与吸收趋势；FX421 位深、存储与空间采样；FX422 卷积与均值核；FX423 诊断敏感度、特异度与ROC。

## B18 • [P1] Piezoelectric transmission, echoes and attenuation / 压电发射、回波与衰减

Priority / 复习标记：P1 First pass/先掌握 · Teacher emphasis/课堂强调 · Worked example/例题练习 · Lower priority/降低优先级

**EN: Key points**

- Inverse piezoelectric effect: applied voltage produces deformation and sound (transmit).
- Direct effect: returning pressure produces charge (receive).
- Acoustic impedance $Z=\rho c$;
- a larger mismatch produces stronger reflection, so coupling gel reduces the air gap.
- Higher ultrasound frequency usually improves spatial resolution but increases attenuation and reduces penetration.
- Echo attenuation uses the outgoing and returning path.

**中文：必会知识点**

- Inverse piezoelectric effect逆压电效应：施加电压产生形变与声波（发射）；
- direct正压电效应：返回声压产生电荷（接收）。
- 声阻抗$Z=\rho c$;
- 失配越大、反射通常越强，耦合凝胶用于减少空气间隙。
- 超声频率更高通常提高空间分辨率，但增加衰减、降低穿透。
- 回波衰减应计入去程和回程。

**EN:** Start from zero: A piezoelectric material links mechanical deformation and electricity. In ultrasound transmission, an applied voltage makes the material change shape rapidly, creating sound: this is the inverse piezoelectric effect. On reception, returning sound pressure deforms it and produces charge: this is the direct effect. It resembles one device acting as both speaker and microphone, although its physical conversion mechanism is specifically piezoelectric.

**中文：** 从零理解：Piezoelectric（压电）材料把机械形变与电联系起来。超声发射时，施加电压让材料快速形变，产生声波，这叫inverse piezoelectric effect（逆压电效应）。接收时，返回声压让材料变形并产生电荷，这叫direct piezoelectric effect（正压电效应）。可类比同一器件兼做扬声器和麦克风，但这里的具体换能机制是压电效应。

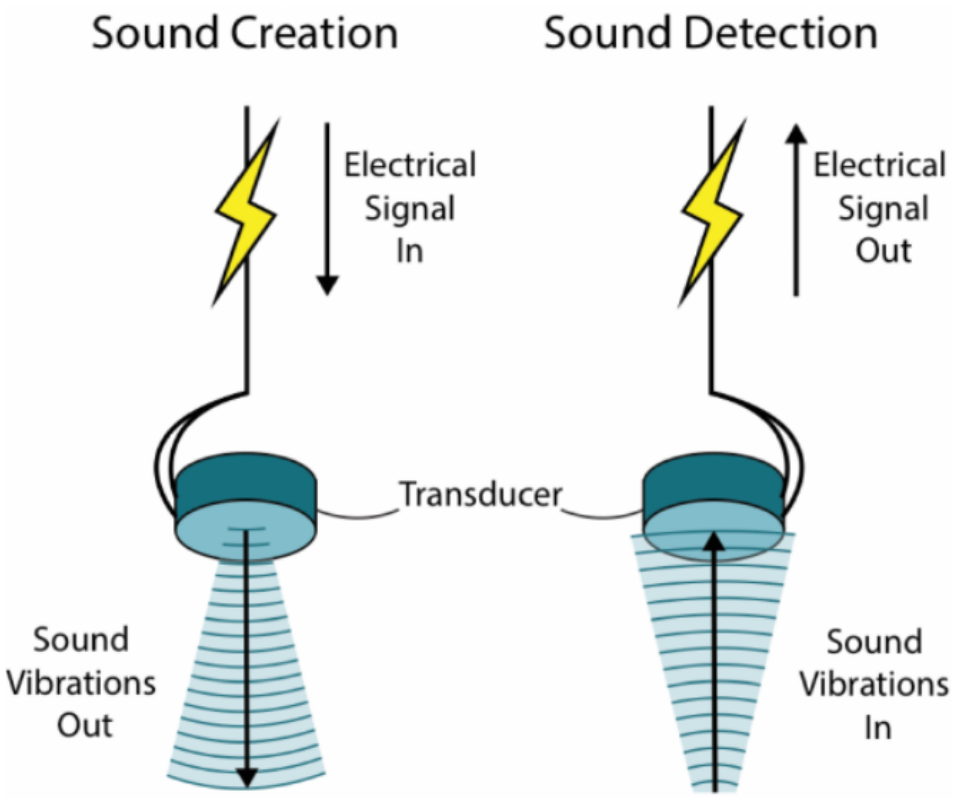
**EN:** Follow the mechanism: Acoustic impedance $Z=\rho c$ combines density $\rho$ and sound speed $c$. It describes a medium's relation between sound pressure and particle motion. When sound meets a boundary with different impedance, some energy reflects. A large mismatch can produce a strong echo and prevent much sound entering the next medium. Coupling gel removes an air gap that would otherwise cause strong reflection at the skin surface.

**中文：** 跟着机制走：Acoustic impedance（声阻抗）$Z=\rho c$由密度$\rho$与声速$c$共同决定，描述介质中声压和质点运动的关系。声遇到声阻抗不同的界面时，部分能量被反射；失配很大时可产生强回波，同时阻止较多声进入下一介质。Coupling gel（耦合凝胶）用于消除空气间隙，否则皮肤表面处会出现很强反射。

**EN:** Try an example: Higher frequency usually means shorter wavelength and finer spatial detail, but also stronger tissue attenuation and reduced penetration. Choose between small shallow features and deeper targets; no frequency is best for everything. An echo also weakens along both outgoing and returning paths. English clues: voltage to deformation is inverse piezoelectric; pressure to charge is direct; mismatch means boundary reflection; attenuation means weakening during travel.

**中文：** 用例子理解：频率更高通常意味着波长更短、能分开更细的空间细节，但组织衰减也通常更强，穿透更浅。可理解为在“看清浅处小结构”和“到达深处目标”之间取舍，没有一种频率适合全部情况。回波在去程和回程都会减弱。英文辨认：voltage to deformation为逆压电；pressure to charge为正压电；mismatch失配对应界面反射；attenuation衰减对应传播中的减弱。

Source / 来源：W5 p5–14; W5 p15–18; W5 p19–23; W5 p19–21; W5 p24–26; W5 p27–28; W5 p29 | T5-0004–0032, T5-0037–0040




The same piezoelectric transducer can transmit and receive ultrasound (original lecture figure) / 同一个压电换能器可以发射也可以接收超声（原课件图）

**EN:** Read the figure: Read the arrow directions. On the left, Electrical Signal In produces Sound Vibrations Out: voltage makes the crystal deform/vibrate, the inverse piezoelectric effect. On the right, Sound Vibrations In produces Electrical Signal Out: incoming pressure generates electrical charge, the direct piezoelectric effect. The outgoing and returning ultrasound are mechanical waves, not electromagnetic light.

**中文：** 读图：看箭头方向。左侧Electrical Signal In电信号输入，变成Sound Vibrations Out声振动输出：电压使晶体变形/振动，是逆压电效应。右侧声音输入产生电信号输出：入射声压产生电荷，是正压电效应。发出与返回的超声都是机械波，不是电磁光波。

Source / 来源：Week 5 PDF p15; original slide figure / 原课件截图

## S501 • [P1] Mechanical waves, pressure and reflection coefficients / 机械波、声压与反射系数

Priority / 复习标记：P1 First pass/先掌握 · Teacher emphasis/课堂强调 · Worked example/例题练习

**EN:** Ultrasound is a mechanical wave above roughly 20 kHz, not an electromagnetic wave. It needs a material medium and cannot travel through vacuum. In the taught tissue model it is longitudinal: local particles oscillate along the propagation direction, forming compression and rarefaction. The particles do not all travel from the probe to the organ. Wavelength follows $\lambda=c/f$; diagnostic frequencies are in the MHz range, distinct from the 20 kHz hearing boundary.

**中文：** Ultrasound（超声）是频率约高于20 kHz的机械波，不是电磁波；它需要物质介质，不能在真空中传播。课程组织模型中它是longitudinal（纵波）：局部质点沿传播方向振动，形成compression（压缩）与rarefaction（稀疏），并不是所有质点从探头一路移动到器官。波长$\lambda=c/f$；诊断频率在MHz量级，要与20 kHz听觉边界区分。

**EN:** Sound pressure is the pressure variation; intensity is power per area. In the travelling-wave model, $I(t)=p(t)^2/Z$, with acoustic impedance $Z=\rho c$. For a sinusoid, mean intensity uses RMS pressure: $\langle I\rangle=p_{rms}^2/Z=p_{peak}^2/(2Z)$. At fixed Z, doubling pressure amplitude gives four times the intensity, not twice. Do not substitute peak pressure into the RMS formula without the factor of two.

**中文：** 声压是压力的变化；intensity（强度）是单位面积上的功率。行波模型中$I(t)=p(t)^2/Z$，声阻抗$Z=\rho c$。正弦波的平均强度应使用RMS（均方根）声压：$\langle I\rangle=p_{rms}^2/Z=p_{peak}^2/(2Z)$。Z固定时，声压幅度加倍，强度变为四倍，不是两倍。峰值声压不能不加系数地直接代入均方根公式。

**EN:** At a lossless boundary with normal incidence, the intensity reflection coefficient is $R=[(Z_2-Z_1)/(Z_2+Z_1)]^2$ and transmission coefficient $T=1-R$. Equal impedances give R=0 in this simple model; a strong mismatch gives a large return and poor onward transmission. These are intensity fractions, not signed pressure amplitudes. Attenuation is loss along the path, including absorption/scattering; an echo path has outgoing and returning distance. Keep pressure and intensity attenuation conventions distinct.

**中文：** 无损界面、垂直入射时，强度反射系数$R=[(Z_2-Z_1)/(Z_2+Z_1)]^2$，透射系数$T=1-R$。简单模型中声阻抗相等时R=0；失配强时返回较多，继续透入较少。这里是强度比例，不是带正负号的声压幅度。Attenuation（衰减）是沿途减弱，含吸收/散射；回波要计去程与回程距离，并区分声压衰减和强度衰减的约定。

Source / 来源：Week 5 PDF p9–25; T5-0008–0032

**EN:** Clarify the concept: At fixed acoustic impedance, intensity is proportional to pressure squared. If pressure amplitude decays as $p=p_0e^{(-\alpha_p x)}$, intensity decays as $I=I_0e^{(-2\alpha_p x)}$. If instead $I=I_0e^{(-\alpha_I x)}$, then $\alpha_I=2\alpha_p$. A dB loss is $10\log_{10}(I_0/I)=20\log_{10}(p_0/p)$; multiply a stated dB-per-length coefficient by path length. The same α cannot mean all three conventions. For a reflected echo, count both outward and return paths.

**中文：** 把概念讲清：声阻抗固定时，强度与声压平方成正比。若声压幅度$p=p_0e^{(-\alpha_p x)}$，强度$I=I_0e^{(-2\alpha_p x)}$；若定义强度式$I=I_0e^{(-\alpha_I x)}$，则$\alpha_I=2\alpha_p$。dB损失为$10\log_{10}(I_0/I)=20\log_{10}(p_0/p)$，给定dB/长度系数应乘传播路径长度。同一个α不能同时代表这三种约定。反射回波应计去程加回程。

Source / 来源：README §5.1 Attenuation; Week 5 PDF p14, 23–25; T5-0027–0032 (especially total path); derived directly from $I=p^2/Z$

## C17 • [P2] Ultrasound versus photoacoustic imaging / 超声与光声成像

Priority / 复习标记：P2 Important foundations/重要基础 · Repeated explanation/反复讲解 · Lower priority/降低优先级

**EN: Key points**

• Conventional ultrasound uses acoustic transmission and returning echoes.
• Photoacoustic imaging starts with optical absorption, followed by heating, thermoelastic expansion and acoustic-wave generation;
• an ultrasound transducer detects the sound.
• Optical absorption supplies contrast, while acoustic detection supports depth.

**中文：必会知识点**

• 常规超声发射声波并接收回波。
• 光声成像从光吸收开始，随后升温、热弹性膨胀并产生声波，由超声换能器检测。
• 光学吸收提供对比，声学检测支持较深成像。

**EN:** Start from zero: Ultrasound uses sound above the human hearing range to look inside tissue. A transducer sends a pulse and listens for returning echoes. Different boundaries return different amounts of sound, allowing an image to be formed. This is similar to listening for an echo in a room, but a medical instrument uses controlled pulses, sensitive detectors, and known sound-speed assumptions to estimate positions.

**中文：** 从零理解：Ultrasound（超声）使用超过人耳听觉范围的声波观察组织内部。Transducer（换能器）发出声脉冲，再接收返回的echo（回波）；不同界面返回声波的强弱不同，从而形成图像。这有点像听房间里的回声，但医学仪器使用受控脉冲、灵敏探测器及已知声速假设来估计位置。

**EN:** Follow the mechanism: Photoacoustic imaging begins differently: a light pulse is absorbed, producing a small temperature rise and rapid thermoelastic expansion, which generates sound. The detector then listens to that sound. Think of converting an optical event into an acoustic message. The analogy does not mean ordinary thermal expansion always makes a useful imaging signal; pulse timing and detection conditions matter.

**中文：** 跟着机制走：Photoacoustic imaging（光声成像）的起点不同：光脉冲被吸收，造成微小升温和快速thermoelastic expansion（热弹性膨胀），从而产生声波，探测器再接收声波。可以把它理解为将光学事件转换成声学信息；但不是任何普通热膨胀都能产生有用成像信号，脉冲时间和探测条件也很重要。

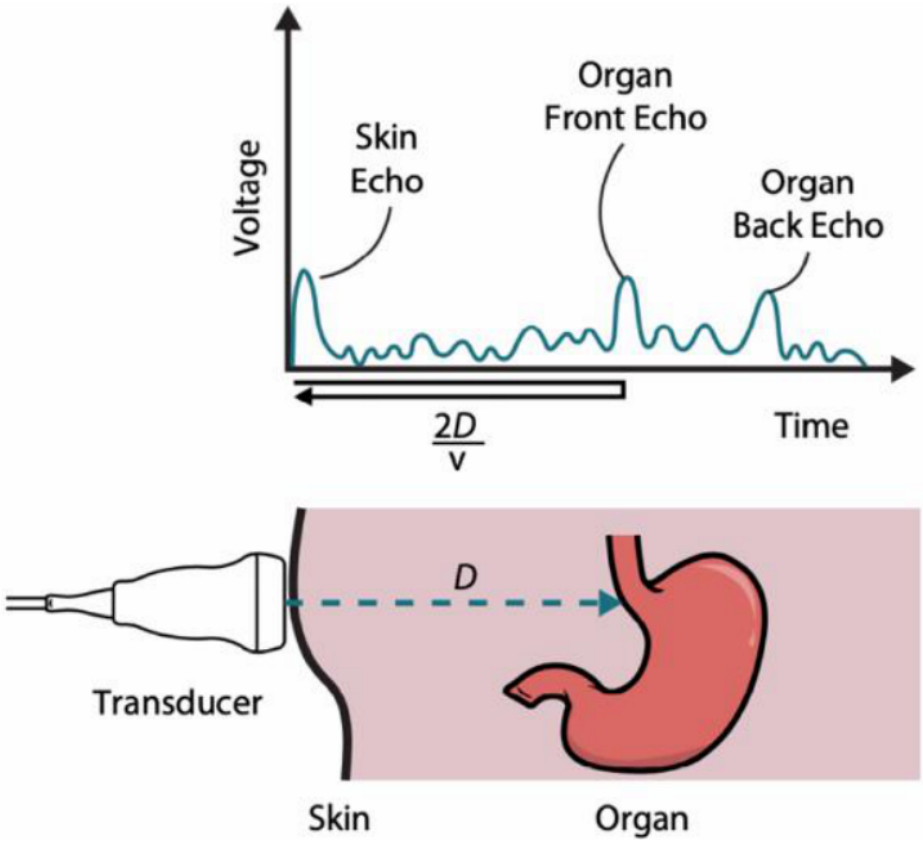
**EN:** Try an example: For an ultrasound echo, sound travels to the boundary and back, so depth z=vt/2. With speed v=1540 m/s and echo time t=40 μs, z=0.0308 m=3.08 cm. For a stated one-way photoacoustic travel time, use z=vt. English clues: transmission and echo indicate conventional ultrasound; optical absorption, heating, expansion, and acoustic detection indicate photoacoustics. The factor of two comes from the path, not the name alone.

**中文：** 用例子理解：超声回波中，声波先到界面再回来，所以深度z=vt/2。若声速v=1540 m/s、回波时间t=40 μs，则z=0.0308 m=3.08 cm。若题目明确给光声的单程传播时间，应使用z=vt。英文中transmission与echo提示常规超声；optical absorption、heating、expansion、acoustic detection提示光声。是否除以2取决于实际传播路径，不能只凭技术名称判断。

**EN:** Exam wording: Echo depth uses a round-trip time: z=vt/2. For a photoacoustic one-way propagation time, z=vt. The problem must state which timing it measures. Detailed ultrasound scan and resolution examples are lower priority under the local lecture exclusions.

**中文：** 英文读题抓手：回波深度按往返时间z=vt/2；若给光声单程传播时间，则z=vt。必须先看题目定义的时间。根据课堂局部排除，复杂超声扫描与分辨率例题降低优先级。

Source / 来源：W5 p5–14; W5 p15–18; W5 p27–28; W5 p29; W5 p34–43 | T5-0004–0017, T5-0037–0040, T5-0044–0051




Ultrasound echoes: arrival time locates depth and peak height gives echo strength (original lecture figure) / 超声回波：到达时间定位深度，峰高表示回波强弱（原课件图）

**EN:** Read the figure: The horizontal axis is Time and the vertical axis is detected Voltage. The early peak is the skin echo; later peaks come from the organ's front and back boundaries. Match the lower distance D to the marked 2D/v delay: sound goes to the boundary and returns, so D = vt/2. Peak height indicates echo strength and is also affected by attenuation; it is not directly the depth.

**中文：** 读图：横轴Time是时间，纵轴Voltage是检测到的电压。早到的峰来自皮肤，稍后的峰来自器官前、后边界。把下方距离D与上方标出的2D/v时延对应：声走到边界还要返回，所以D = vt/2。峰高表示回波强弱，也会受衰减影响；它不直接等于深度。

Source / 来源：Week 5 PDF p21; original slide figure / 原课件截图

## S502 • [P3] Read ultrasound displays: supporting vocabulary / 看懂超声显示：辅助词汇

Priority / 复习标记：P3 Supporting reading/辅助阅读 · Lower priority/降低优先级

**EN:** Supporting, lower-priority vocabulary: A-mode plots echo amplitude against arrival time/depth along one line. B-mode encodes echo strength as brightness and combines lines into a cross-section. M-mode repeatedly observes one line to show how structures move over time. A, B and M here mean amplitude, brightness and motion; they are not three different biological recognition molecules.

**中文：** 辅助、低优先级词汇：A-mode把一条线上的回波幅度画成随到达时间/深度变化的曲线；B-mode把回波强度编码为亮度，再合并多条线得到剖面；M-mode反复观察同一条线，显示结构随时间如何运动。这里A、B、M分别是amplitude（幅度）、brightness（亮度）、motion（运动），不是三种生物识别分子。

**EN:** Deeper echoes are weakened by travel, so time gain compensation increases receiver gain for later returns to make the display more useful. This does not undo the physical attenuation or create missing echoes. The lecturer locally excluded the technical scanning/imaging section; keep its vocabulary as context, while the earlier piezoelectric, impedance, reflection and attenuation principles remain central.

**中文：** 更深回波沿途减弱，因此time gain compensation（时间增益补偿）对较晚返回的信号增加接收增益，帮助显示。这不会逆转实际衰减，也不会创造已经缺失的回波。老师局部排除了技术性的扫描/成像段；这里保留名词供理解，前面的压电、声阻抗、反射及衰减原理仍是重点。

Source / 来源：Week 5 PDF p27–31; T5-0037–0041

## B19 • [P2] Photoacoustic contrast and X-ray attenuation / 光声对比与X射线衰减

Priority / 复习标记：P2 Important foundations/重要基础 · Repeated explanation/反复讲解 · Worked example/例题练习

**EN: Key points**

• Photoacoustic contrast depends on optical absorption, so excitation wavelength can emphasize different absorbers.
• An X-ray tube accelerates electrons toward a target to generate X-ray photons;
• the patient is imaged by photons, not the electron beam.
• X-ray transmission follows exponential attenuation;
• successive layers multiply transmissions.
• Absorption contrast and acoustic reflection contrast are different mechanisms.

**中文：必会知识点**

• 光声对比依赖光吸收，因此激发波长可突出不同吸收物。
• X射线管把电子加速到靶材以产生X射线光子；
• 用于成像的是光子，不是射向患者的电子束。
• X射线透射遵循指数衰减；
• 连续多层的透过率相乘。
• 吸收对比与声学反射对比属于不同机制。

**EN:** Start from zero: Different contrast mechanisms tell us different properties. A conventional ultrasound echo reflects a sound-impedance boundary. Photoacoustic sound reports optical absorption that caused local heating and expansion. Changing the excitation wavelength can therefore emphasize different light absorbers. A strong photoacoustic signal does not necessarily mean the strongest acoustic reflector: the signal started from absorption, rather than from a reflected ultrasound pulse.

**中文：** 从零理解：不同对比机制显示不同性质。常规超声回波反映声阻抗界面；光声信号反映光吸收造成的局部升温和膨胀。因此改变激发波长，可以突出不同吸光物。光声很强的位置，不一定是声反射最强的位置：它的信号起点是吸收光，不是反射超声脉冲。

**EN:** Follow the mechanism: An X-ray tube accelerates electrons toward a solid target. Their interactions with the target generate X-ray photons; these photons pass toward the patient and detector. Do not confuse electrons inside the source with photons used for imaging. Tissues weaken the transmitted beam by different amounts. X-rays are ionizing radiation, meaning individual photons can have enough energy to remove electrons from atoms.

**中文：** 跟着机制走：X射线管把电子加速撞向固体靶材；电子与靶材相互作用产生X射线光子，再由这些光子照向患者及探测器。不要把光源内部的电子与用于成像的光子混淆。不同组织使透射光束衰减的程度不同。X射线属于电离辐射，即单个光子能量可以足以把电子从原子中移走。

**EN:** Try an example: For a uniform layer, transmission $T=I/I_0$ follows $T=\exp(-\mu L)$, where μ is an attenuation coefficient and L the layer thickness in matching units. Successive layers multiply their transmissions. In a simple illustration, two layers each passing 50% leave 0.5×0.5=25% overall, not zero. English questions may contrast absorption-based photoacoustics, reflection-based ultrasound, and transmission attenuation in X-rays. State which mechanism is measured before deciding what image brightness means.

**中文：** 用例子理解：均匀一层的透射率$T=I/I_0$满足$T=\exp(-\mu L)$，μ是衰减系数，L是层厚，两者单位需匹配。连续多层的透射率相乘。简单举例：两层各通过50%，总透过率为0.5×0.5=25%，不是零。英文题可比较基于吸收的光声、基于反射的超声以及X射线透射衰减。判断图像亮度意义前，要先说明测的是哪种机制。

Source / 来源：W5 p34–43; W5 p44–51; W5 p53–58; W5 p59–61; W5 p62; W5 p63 | T5-0044–0051, T5-0063–0087

## S503 • [P2] Photoacoustic spectra and reconstruction / 光声光谱与重建

Priority / 复习标记：P2 Important foundations/重要基础 · Repeated explanation/反复讲解 · Supporting concept/辅助理解

**EN:** Photoacoustic imaging records sound generated inside an absorber after light absorption; the received sound is not simply the incident ultrasound reflecting back. Multiple detectors receive a wave from the same source at different times. With known detector positions and an assumed sound speed, back-projection combines those arrival times to estimate the absorber’ s location. Here the timing describes the source-to-detector acoustic route.

**中文：** 光声成像记录吸光物内部在吸收光后产生的声音，接收声波不是单纯入射超声反射回来。同一声源的波到达不同探测器的时间不同。已知探测器位置并假设声速后，back-projection（反投影）合并到达时间来估计吸光物位置。这里的时间描述声源到探测器的声学路径。

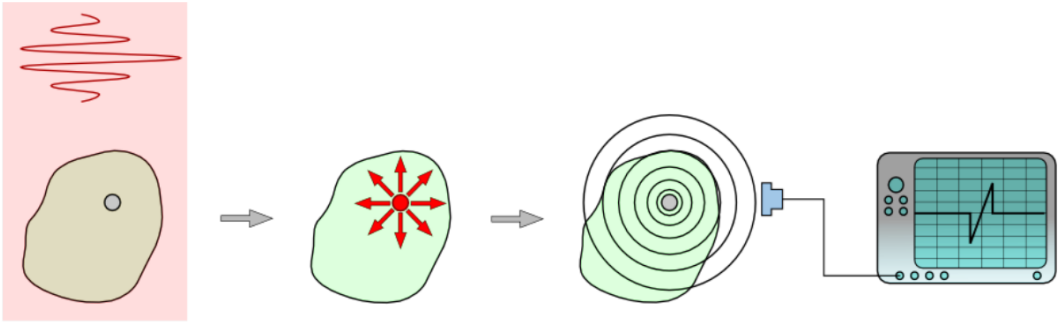
**EN:** Changing laser wavelength changes how strongly a particular molecule absorbs. Oxygenated and deoxygenated hemoglobin have different spectra, so multiwavelength photoacoustic signals can help estimate oxygenation and visualize vessels. This is absorption-based molecular/functional contrast, not fluorescence emission. Signal also depends on local light delivery and detection, so a larger signal is not a universal direct concentration measurement without calibration.

**中文：** 改变激光波长，会改变某种分子的吸收强弱。氧合与脱氧血红蛋白光谱不同，因此多波长光声信号可帮助估计氧合状态、显示血管。这是基于吸收的分子/功能对比，不是荧光发射。信号还受局部光输送和检测条件影响，不能不经校准就普遍把“信号更大”等同于“浓度更高”。

**EN:** In the slide example, a smaller absorber generates a shorter acoustic time feature and a larger absorber a broader feature. The waveform therefore carries size information as well as arrival-time information. Photoacoustics combines optical absorption contrast with acoustic detection, but its resolution and depth still depend on the illumination, acoustic bandwidth and geometry; it is not automatically superior at every depth and scale.

**中文：** 课件示例中，较小吸光物产生较窄的声学时间特征，较大吸光物产生较宽特征，因此波形还携带尺寸信息，不只是到达时间。光声结合光吸收对比与声学检测，但分辨率和深度仍受照明、声学带宽及几何影响，并非在任何深度和尺度都一定更优。

Source / 来源：Week 5 PDF p42–51; T5-0063–0073




Photoacoustic conversion chain: light absorption to detected sound (original lecture slide) / 光声转换链：光吸收到检测声音（原课件图）

**EN:** Read the figure: Follow the arrows from laser pulse through absorption, expansion and acoustic waves to ultrasound detection. The incoming excitation is light; the outgoing detected signal is sound.

**中文：** 读图：按箭头跟随激光脉冲→吸收→膨胀→声波→超声检测。进入的激发是光，最终检测的信号是声音。

Source / 来源：Week 5 PDF p41; original slide figure / 原课件截图

## S504 • [P2] X-ray contrast, scattering and CT reconstruction / X射线对比、散射与CT重建

Priority / 复习标记：P2 Important foundations/重要基础 · Worked example/例题练习 · Repeated explanation/反复讲解 · Lower priority/降低优先级

**EN:** The photoelectric effect absorbs an X-ray photon and ejects a bound electron when enough energy is available. In the course comparison, higher atomic-number materials such as bone give stronger photoelectric attenuation; the approximate trend decreases as photon energy rises, with absorption-edge exceptions. Compton scattering changes the photon direction and energy. Scattered photons reaching the detector can produce fog and reduce useful projection contrast.

**中文：** Photoelectric effect（光电效应）在能量足够时吸收X射线光子并击出束缚电子。课程比较中，骨等原子序数较高材料产生更强光电衰减；近似趋势随光子能量增加而减小，但有absorption edge（吸收边）的例外。Compton scattering（康普顿散射）改变光子的方向和能量；散射光子到达探测器可形成fog（雾状背景），降低有效投影对比。

**EN:** A projection radiograph combines attenuation along each beam path; structures at different depths overlap. In the conventional display taught, more transmission looks darker and less transmission looks whiter, although display inversion is possible. For successive uniform layers, $I/I_0=\exp(-\Sigma\mu_i L_i)$. A single projection does not separately locate every layer in depth.

**中文：** Projection radiograph（投影X射线图）把每条光路沿途的衰减合在一起，不同深度结构会重叠。课程传统显示中，透过更多较暗、透过更少较白；实际也可反转显示。连续均匀多层满足 $I/I_0=\exp(-\Sigma\mu_i L_i)$。单张投影不会单独定位每一层的深度。

**EN:** Contrast agents containing high-atomic-number elements such as iodine or barium can increase attenuation differences between otherwise similar tissues. Beam collimation and grids reduce the contribution of scatter. Lower X-ray energy can improve attenuation contrast but may penetrate poorly; higher energy can reach thicker regions. Attenuation, contrast and penetration are related choices, not interchangeable definitions.

**中文：** 含iodine（碘）或barium（钡）等高原子序数元素的contrast agents（对比剂）可增大原本相似组织的衰减差异。Collimation（限束）及栅格可减少散射贡献。较低X射线能量可改善衰减对比，却可能穿透不足；较高能量可到达更厚区域。衰减、对比、穿透是相关取舍，不是同一个定义。

**EN:** CT collects projections at many angles and reconstructs a map of attenuation inside a slice. Simple back-projection spreads each projection back along its ray paths and adds the contributions; the slide shows that this can blur the result. Rotating tube/detector systems collect angular views; helical scanning combines rotation with patient translation. CT still uses X-rays and therefore still involves ionizing radiation.

**中文：** CT采集许多角度的投影，重建切片内部的衰减分布。Simple back-projection（简单反投影）把各投影沿光线路径分摊回去，再把贡献相加；课件显示这种做法可产生模糊。旋转的射线管/探测器采集角度信息；helical scanning（螺旋扫描）把旋转与患者平移结合。CT仍使用X射线，仍涉及电离辐射。

Source / 来源：Week 5 PDF p54–68; T5-0073–0095

## B20 • [P2] Digital images and analysis vocabulary / 数字图像与分析术语

Priority / 复习标记：P2 Important foundations/重要基础 · Repeated explanation/反复讲解 · Teacher emphasis/课堂强调 · Lower priority/降低优先级

**EN: Key points**

- Sampling discretizes position;
- quantization discretizes intensity ($2^b$ levels for b bits).
- Preprocessing includes denoising/deblurring;
- convolution combines neighbouring values with kernel weights.
- Semantic segmentation labels classes;
- instance segmentation separates objects.
- Classification assigns a class using features such as shape/intensity.
- TP/TN mean true positive/negative;
- FP/FN mean false positive/negative.
- Diagnostic sensitivity=TP/(TP+FN), specificity=TN/(TN+FP);
- these differ from calibration slope.

**中文：必会知识点**

- Sampling采样离散化位置；
- quantization量化离散化强度（b位对应$2^b$级）。
- Preprocessing预处理含去噪/去模糊；
- convolution卷积用核权重组合邻域值。
- Semantic segmentation语义分割标类别；
- instance segmentation实例分割区分对象。
- Classification分类依据形状/强度等特征给出类别。
- TP/TN为真阳性/真阴性，FP/FN为假阳性/假阴性。
- 诊断敏感度=TP/(TP+FN)，特异度=TN/(TN+FP)，不同于校准斜率。

**EN:** Start from zero: A digital image divides space into pixels and brightness into numerical levels. Sampling chooses spatial positions; quantization assigns intensity levels. These are separate choices. With b bits, there are $2^b$ possible levels: eight bits give 256. In a tiled picture, tile size resembles spatial sampling; available colors resemble quantization. Brightness levels do not create finer spatial resolution.

**中文：** 从零理解：数字图像把空间划分为像素，把亮度划分为数字等级。Sampling（采样）选择空间位置；quantization（量化）规定强度等级，这是两个独立选择。b位有$2^b$个等级，8位为256级。可以类比用有限颜色的瓷砖拼图：砖块尺寸对应空间采样，颜色表对应量化；增加亮度等级不会创造更细的空间分辨率。

**EN:** Follow the mechanism: Preprocessing cleans the input, for example by reducing noise or blur. Convolution forms an output using weighted neighboring values; the weights form a kernel. Segmentation marks regions: semantic segmentation says which class each pixel belongs to, while instance segmentation separates individual objects. Classification assigns a category to an image or object, often using shape, intensity, or learned features. Detecting three separate cells requires instance information, not merely a cell-class label.

**中文：** 跟着机制走：Preprocessing（预处理）整理输入，例如减小噪声或模糊。Convolution（卷积）把邻近数值加权组合，权重组成kernel（核）。Segmentation（分割）标出区域：semantic segmentation（语义分割）标每个像素属于哪类，instance segmentation（实例分割）区分一个个对象。Classification（分类）根据形状、强度或学得的特征，给图像或对象分配类别。要认出三个独立细胞，需要实例信息，仅有“细胞类”标签不够。

**EN:** Try an example: For diagnostic decisions, positive means the method predicts the condition is present. TP is a correctly positive case; FN is a missed positive case. Sensitivity=TP/(TP+FN), so detecting eight of ten actual positives gives 80%. TN is correctly negative, and FP a false alarm; specificity=TN/(TN+FP). English sensitivity here means finding actual positive cases, unlike the calibration slope used for a sensor's response versus concentration.

**中文：** 用例子理解：诊断判断中，positive（阳性）表示方法预测目标情况存在。TP（真阳性）是正确报阳性，FN（假阴性）是漏掉真实阳性。Sensitivity=TP/(TP+FN)，所以10个真实阳性中检出8个，诊断敏感度是80%。TN（真阴性）是正确报阴性，FP（假阳性）是误报；specificity=TN/(TN+FP)。这里sensitivity表示找出真实阳性的能力，不是传感器响应随浓度变化的校准斜率。

Source / 来源：W5 p70–74; W5 p75–83; W5 p84–86 | T5-0096–0106

## S505 • [P3] ROC, processing tasks and sampling rate / ROC、处理任务与采样频率

Priority / 复习标记：P3 Supporting reading/辅助阅读 · Lower priority/降低优先级 · Final emphasis/仅期末强调

**EN:** A ROC curve plots true-positive rate, TP/(TP+FN), against false-positive rate, FP/(FP+TN)=1−specificity, while the decision threshold varies. Moving toward the upper-left corner means more true positives and fewer false alarms. AUC is the area under this curve: the slides show 1 for ideal discrimination and about 0.5 for random ranking. This diagnostic sensitivity differs from a sensor’ s calibration slope.

**中文：** ROC曲线在改变判断threshold（阈值）时，画true-positive rate（真阳性率）TP/(TP+FN)，对false-positive rate（假阳性率）FP/(FP+TN)=1−specificity（特异度）。越靠左上角，真阳性多、误报少。AUC是曲线下面积；课件把1画作理想区分、约0.5画作随机排序。这个诊断sensitivity不同于传感器校准斜率。

**EN:** Read image-analysis questions by the output: object detection finds locations such as bounding boxes; segmentation assigns pixel regions; classification gives a category; regression predicts a numerical value; tracking follows an object over time. Feature extraction measures properties such as area, shape or intensity. Enhancement/denoising aims to improve the input image, whereas classification interprets it. ML-assisted processing still depends on suitable sensor and image data.

**中文：** 图像分析题先看输出：object detection（目标检测）找位置，例如边界框；segmentation（分割）分配像素区域；classification（分类）给类别；regression（回归）预测数值；tracking（跟踪）追踪对象随时间的变化。Feature extraction（特征提取）量出面积、形状或强度等性质。Enhancement/denoising（增强/去噪）改善输入图像，分类则解释图像。机器学习辅助处理仍依赖合适的传感器与图像数据。

**EN:** Sampling frequency is samples per unit time, $f_s=1/\Delta t$, not the number of times an image is averaged. Repeating and averaging suitably independent noisy measurements can reduce random noise, but this is a separate operation. Bit depth controls the number of intensity levels; spatial pixel spacing controls spatial sampling. Increasing any one of these does not automatically improve all definitions of resolution or eliminate systematic error.

**中文：** Sampling frequency（采样频率）是单位时间的采样数，$f_s=1/\Delta t$，不是把图像平均了多少次。重复并平均合适的独立噪声测量，可减小随机噪声，但这是另一个操作。Bit depth（位深）控制强度等级数，空间像素间距控制空间采样。增加其中一项不会自动改善所有“分辨率”，也不会消除系统误差。

Source / 来源：Week 5 PDF p76–91; T5-0100–0106

**EN:** Clarify the concept: Use the denominator to distinguish diagnostic metrics: recall/sensitivity=TP/(TP+FN), among actual positives; precision=TP/(TP+FP), among predicted positives; specificity=TN/(TN+FP), among actual negatives; accuracy=(TP+TN)/all cases. A high value of one metric does not guarantee all others are high. The course directly emphasizes the confusion matrix and sensitivity/specificity; precision and accuracy here are supporting terminology.

**中文：** 把概念讲清：看分母辨认诊断指标：recall/sensitivity（召回率/敏感度）=TP/(TP+FN)，看真实阳性；precision（精确率）=TP/(TP+FP)，看预测阳性；specificity（特异度）=TN/(TN+FP)，看真实阴性；accuracy（准确率）=(TP+TN)/全部案例。一项高不保证其他项都高。课程直接介绍混淆矩阵与敏感度/特异度，precision与accuracy这里作为辅助词汇。

Source / 来源：README §5.4; Week 5 PDF p84–86; T5-0103–0105; precision is README supporting extension, formula checked by TP/FP definitions

# Week 6 learning block: Electrochemical biosensors / 第6学习组：电化学生物传感器

## Week 6 concept map / 第6周知识框架

**EN:** Chapter logic: Link chemical recognition or molecular charge to the electrical variable measured by the sensor.

**中文：** 本章主线：把化学识别或分子电荷，与传感器实际测得的电量连起来。

| English term 中文名词 | Meaning and connection 含义与知识联系 |
|---|---|
| **Redox and reference 氧化还原与参比** | Oxidation loses electrons, reduction gains them; electrode potential is relative to a stated reference. 氧化失电子、还原得电子；电极电位相对给定参比。 → B21 |
| **Potentiometric/ion-selective sensing 电位型/离子选择感测** | Read potential at negligible current; an ideal Nernst response is logarithmic in ion activity. 在极小测量电流下读电位，理想Nernst响应与离子活度呈对数关系。 → S603 |
| **Amperometry and glucose 安培型与葡萄糖** | Controlled potential measures reaction current; distinguish enzyme recognition from electrode readout and GOx generations. 控制电位测反应电流，区分酶识别、电极读出与GOx各代路线。 → C19 |
| **Charge and stoichiometry 电荷与反应计量** | For constant current Q=It; with complete charge collection $Q=z\,n_a F$. 恒定电流Q=It；完全收集电荷时$Q=z\,n_a F$。 → B24 |
| **Conductance and impedance 电导与阻抗** | G=1/R; AC impedance adds magnitude, phase and frequency dependence. G=1/R；交流阻抗还包含幅值、相位及频率依赖。 → B26 |
| **FET, nanopore and capacitance FET、纳米孔与电容** | Gate charge changes a channel; pore passage changes ionic current; dielectric/geometry changes capacitance. 栅电荷改变沟道，穿孔改变离子电流，介电/几何变化改变电容。 → B25 |
| **Molecular charge 分子电荷** | DNA is typically negative; protein net charge varies with pH/pI, so bound charge is not universally positive or negative. DNA通常负电，蛋白净电荷随pH/pI变，因此结合电荷并非普遍同正或同负。 → S605 |
| **Device fabrication 器件制造** | Photolithography patterns resist; etching removes, deposition adds, and PDMS molding copies microstructures. 光刻形成胶图案，刻蚀去材料、沉积加材料，PDMS模制复制微结构。 → B22 |

**EN:** Formula links for this chapter: FX424 Cell potential and electrochemical Nernst equation, FX425 Ideal ion-selective response andpH, FX426 Reference-electrode conversion, FX427 Faraday charge, electron moles and counts, FX428 Analyte stoichiometry, mass and current efficiency, FX429 Glucose-strip background subtraction, FX430 Conductance, conductivity and geometry, FX431 Parallel-plate capacitance model, FX432 R,C,L complex impedances, FX433 Combining AC impedances, FX434 SeriesRC capacitor voltage divider, FX435 Pure parallelRC and Bode conventions, FX436 Series resistance with a parallelRC/Nyquist semicircle.

**中文：** 本章公式定位：FX424 电池电势与电化学能斯特式；FX425 理想离子选择响应与pH；FX426 参比电极换算；FX427 法拉第电荷、电子摩尔与个数；FX428 分析物计量、质量与电流效率；FX429 葡萄糖试纸背景扣除；FX430 电导、电导率与几何；FX431 平行板电容模型；FX432 电阻、电容、电感复阻抗；FX433 交流阻抗串并联；FX434 串联RC的电容分压；FX435 纯并联RC与Bode约定；FX436 串联电阻加并联RC / Nyquist半圆。

## B21 • [P2] Redox and reference-relative potential / 氧化还原与相对电位

Priority / 复习标记：P2 Important foundations/重要基础 · Repeated explanation/反复讲解 · Lower priority/降低优先级

**EN: Key points**

- Oxidation loses electrons;
- reduction gains electrons.
- Oxidation occurs at the anode and reduction at the cathode;
- electrode polarity depends on cell type.
- Electrode potential is measured relative to a reference.
- Nernst-type equilibrium potentials depend logarithmically on ion activities, not linearly on concentration.
- When using reduction-potential tables, Ecell=Ecathode−Eanode;
- do not mix oxidation and reduction signs.

**中文：必会知识点**

- Oxidation氧化失电子；
- reduction还原得电子。
- Anode阳极发生氧化、cathode阴极发生还原，但正负极性取决于电池类型。
- 电极电位相对参比测量。
- Nernst型平衡电位与离子活度的对数有关，并非与浓度直接成正比。
- 使用还原电位表时Ecell=Ecathode−Eanode，不要混用氧化与还原电位符号。

**EN:** Start from zero: An oxidation–reduction reaction transfers electrons. Oxidation loses electrons; reduction gains them. The two occur together because electrons need an acceptor. Use the memory aid OIL RIG: Oxidation Is Loss, Reduction Is Gain. The anode is where oxidation occurs, and the cathode where reduction occurs. Their positive or negative polarity depends on the cell type, so polarity alone is a poor universal definition.

**中文：** 从零理解：Oxidation–reduction（氧化还原）反应转移电子：oxidation氧化失去电子，reduction还原得到电子。两者同时发生，因为释放的电子必须被某处接受。可记OIL RIG：Oxidation Is Loss，Reduction Is Gain。Anode（阳极）发生氧化，cathode（阴极）发生还原；它们的正负极性取决于电池类型，因此不能只靠正负极性作普遍定义。

**EN:** Follow the mechanism: An electrode potential tells us its electrical potential relative to a reference, just as a height needs a chosen ground level. Specify the reference before comparing readings. Ion-selective electrodes respond to particular ions, and their equilibrium potential follows a logarithmic activity relationship. Activity is an effective chemical amount; in suitable dilute approximations it relates to concentration. It is not a universally straight-line voltage-versus-concentration response.

**中文：** 跟着机制走：电极电位是相对于reference（参比）的电位，好比高度需要选一个地面基准。比较读数前必须知道参比。Ion-selective electrode（离子选择电极）响应特定离子，其平衡电位遵循与activity（活度）的对数关系。活度可理解为有效化学量，在合适稀溶液近似下与浓度关联，但电压对浓度并非普遍直线关系。

**EN:** Try an example: If a question supplies both electrode values as reduction potentials, calculate Ecell=Ecathode−Eanode. For example, 0.8 V minus 0.2 V gives 0.6 V. Do not reverse another sign because oxidation occurs there; both values already use a reduction convention. Exam clues are loss of electrons for oxidation, gain for reduction, stable reference for potential measurement, and logarithmic rather than linear dependence for Nernst-type equilibrium sensing.

**中文：** 用例子理解：若题目提供的两电极数值都是reduction potentials（还原电位），计算Ecell=Ecathode−Eanode。例如0.8 V减0.2 V得到0.6 V。不能因为某电极实际发生氧化，就在相减之外再随意翻转一次符号；表格已使用统一的还原约定。英文辨认：loss of electrons是氧化，gain是还原；stable reference是电位测量基准；Nernst型平衡传感是对数而非线性依赖。

Source / 来源：W6 p13–18; W6 p31–35, p38; W6 p36; W6 p37–40; W6 p41–46 | T6-0011–0018, T6-0034–0043

## S601 • [P2] Electron flow, ionic flow and wearable chemical sensing / 电子流、离子流与可穿戴化学检测

Priority / 复习标记：P2 Important foundations/重要基础 · Repeated explanation/反复讲解 · Supporting concept/辅助理解

**EN:** In an electrochemical cell, electrons travel through the metal/electrical circuit and ions carry charge in the solution. A salt bridge or porous junction permits ionic charge balance without requiring the two bulk solutions to mix freely. The bridge does not replace the external wire that carries electrons. Redox links the chemical reactions at the electrodes to the electrical measurement.

**中文：** 电化学电池中，电子通过金属/电路移动，溶液里由离子传递电荷。Salt bridge（盐桥）或 porous junction（多孔接界）允许离子配平电荷，而不必让两份溶液自由混合。盐桥不能代替传递电子的外部导线。氧化还原把电极处化学反应与电测量联系起来。

**EN:** Wearable chemical sensors can analyze ions or metabolites in sweat, tears or saliva; they are different from electrodes recording ECG/EEG voltages. Continuous monitoring requires repeatable responses when the target rises and falls, not only a single irreversible detection. Printed flexible electrodes and wireless electronics make integration convenient, but a chemical signal measured in one body fluid is not automatically the same measurement as blood concentration.

**中文：** 可穿戴化学传感器可分析汗、泪液或唾液中的离子或代谢物；它们不同于记录ECG/EEG电位的电极。Continuous monitoring（连续监测）需要目标上升与下降时都可重复响应，不只是一次不可逆检测。印刷柔性电极与无线电路便于集成，但某一种体液的化学信号并不自动等同于血液浓度的测量。

Source / 来源：Week 6 PDF p3, 6–17, 44–46, 87–89; T6-0002–0016, T6-0047–0049, T6-0102–0105

## S602 • [P3] Standard potential, Nernst equation and reference conversion / 标准电位、Nernst式与参比换算

Priority / 复习标记：P3 Supporting reading/辅助阅读 · Lower priority/降低优先级

**EN:** Standard reduction potential E° compares a half-reaction’ s tendency to gain electrons under specified standard conditions. The standard hydrogen electrode, SHE, is assigned 0 V. In the course table, a more positive reduction potential indicates the oxidized form is a stronger electron acceptor/oxidizing agent. Do not confuse a stronger oxidizing agent with a species that itself loses electrons.

**中文：** Standard reduction potential（标准还原电位）E°在规定标准条件下比较半反应接受电子的倾向。Standard hydrogen electrode（SHE，标准氢电极）被规定为0 V。课程表中，还原电位更正表示对应氧化态是更强的电子接受者/oxidizing agent（氧化剂）。强氧化剂是接受电子的物种，不是“它自己失电子”。

**EN:** For a specified redox reaction, $E=E°-(RT/nF)\ln Q$; at 25 °C, $E=E°-(0.05916\ \text{V}/n)\log_{10}Q$. Here n is electrons transferred in the written reaction, and Q is the reaction quotient built from activities with their stoichiometric powers. It is not the Q used for resonator quality factor or electrical charge. With E° and n fixed, increasing Q by ten lowers E by 0.05916/n V under this convention.

**中文：** 给定氧化还原反应时，$E=E°-(RT/nF)\ln Q$；25 °C时，$E=E°-(0.05916\ \text{V}/n)\log_{10}Q$。n是所写反应转移的电子数，Q是按计量指数用活度组成的reaction quotient（反应商），不是共振器品质因数Q，也不是电荷量Q。在此约定下，E°、n固定，Q增大十倍使E降低0.05916/n V。

**EN:** Changing the reference changes the numerical reading, not the physical indicator electrode: E(indicator versus new reference)=E(indicator versus SHE)−E(new reference versus SHE). In the slide, 0.750 V versus SHE becomes 0.750−0.197=0.553 V versus Ag/AgCl, or 0.750−0.2444≈0.506 V versus saturated calomel. Use the reference values stated in the question, since they depend on reference conditions.

**中文：** 改变参比会改变数值读数，并没有改变指示电极本身：E（指示电极相对新参比）=E（指示电极相对SHE）−E（新参比相对SHE）。课件中，相对SHE的0.750 V换成Ag/AgCl为0.750−0.197=0.553 V；换成饱和甘汞为0.750−0.2444≈0.506 V。应使用题目给定参比值，因为它们依赖参比条件。

Source / 来源：Week 6 PDF p35–38; T6-0036–0039

## C18 • [P1] Electrochemical sensing modes / 电化学传感模式

Priority / 复习标记：P1 First pass/先掌握 · Repeated explanation/反复讲解 · Teacher emphasis/课堂强调

**EN: Key points**

- Potentiometric sensing reads potential, ideally with negligible measurement current.
- Amperometric sensing measures current at a controlled potential.
- Conductometric sensing reads conductance;
- impedimetric sensing reads the response to an AC perturbation.
- Identify the measured variable before doing arithmetic.

**中文：必会知识点**

- Potentiometric测电位，理想情况下测量电流极小；
- amperometric在控制电位下测电流；
- conductometric测电导；
- impedimetric测交流扰动的响应。
- 先识别被测量，再计算。

**EN:** Start from zero: Electrical sensing can measure different things. Voltage, or potential difference, describes electrical energy per unit charge; current describes how much charge passes each second. A water analogy helps: pressure difference resembles voltage, while flow rate resembles current. Electricity is not literally flowing water, but the analogy explains why measuring a driving difference and measuring a flow are different tasks.

**中文：** 从零理解：电传感可以测不同物理量。Voltage（电压，即电位差）描述单位电荷对应的电能差；current（电流）描述每秒通过多少电荷。可用水流类比：压力差像电压，流量像电流。电并不真的等于流动的水，但这个比喻能帮助区分“测推动差异”和“测通过的量”是两种任务。

**EN:** Follow the mechanism: Potentiometric sensing measures potential with very little measurement current, like checking pressure without drawing much water. Amperometric sensing controls electrode potential and measures reaction current. Conductometric sensing measures ease of electrical flow. Impedimetric sensing applies a small alternating signal and measures both response size and timing. Match the method to its readout before thinking about the biological target.

**中文：** 跟着机制走：Potentiometric（电位型）在测量电流极小时测电位，好比测压力而尽量不抽走水。Amperometric（安培型）控制电极电位，测反应电流；conductometric（电导型）测电流通过的容易程度；impedimetric（阻抗型）施加小交流信号，同时看响应大小及时间相位。先对应测量量，再考虑它测哪一种生物目标。

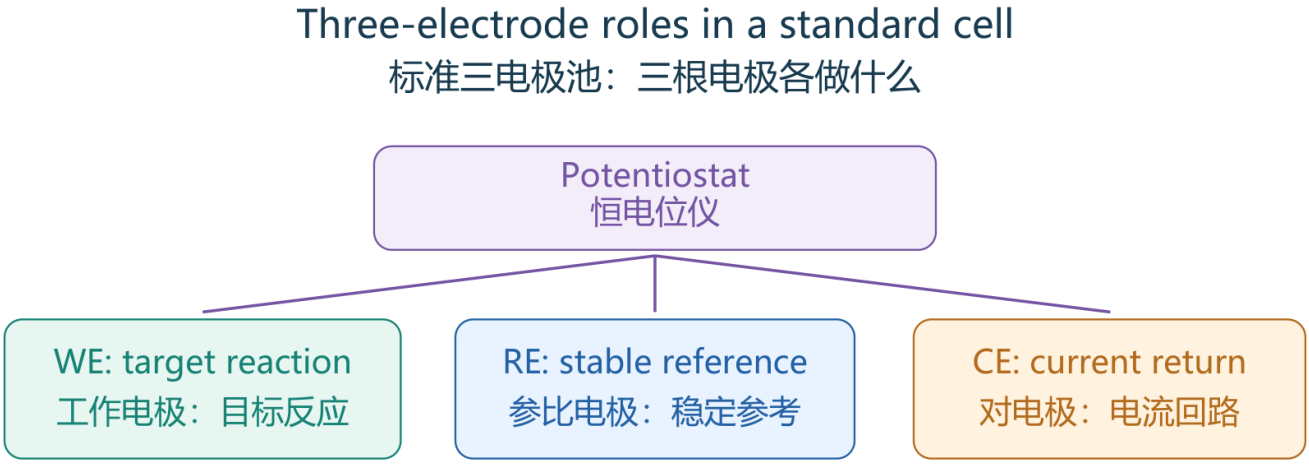
**EN:** Try an example: In a three-electrode system, the working electrode is the sensing reaction site. The reference electrode provides a stable potential baseline and ideally carries negligible current. The counter electrode carries the balancing current. A reference is like a reliable zero mark on a ruler, not the main reaction electrode. Exam clues are potential for potentiometry, current at controlled potential for amperometry, and AC response for impedance.

**中文：** 用例子理解：三电极系统中，working electrode（工作电极）是感测反应位置；reference electrode（参比电极）提供稳定电位基准，理想情况下几乎不承载电流；counter electrode（对电极）承担配平电流。参比像尺子上的可靠零点，不是主要反应电极。英文辨认：potential对应电位型；current at controlled potential对应安培型；AC response（交流响应）对应阻抗型。

**EN:** Exam wording: Working, reference and counter electrodes have different roles. A reference electrode provides a stable potential reference. In a three-electrode arrangement, the counter electrode carries the balancing current while the reference ideally carries negligible current.

**中文：** 英文读题抓手：Working、reference和counter electrode作用不同。参比电极提供稳定电位基准。在三电极结构中，对电极承担配平电流，参比电极理想情况下电流极小。

Source / 来源：W6 p13–18; W6 p19–21; W6 p37–40 | T6-0011–0022, T6-0038–0040




Three-electrode roles in a standard cell / 标准三电极池：三根电极各做什么

**EN:** Read the figure: The potentiostat controls WE potential relative to RE while measuring the reaction current through the WE–CE path. RE provides a stable potential reference and its sensing input draws approximately no current. This standard WE/RE/CE arrangement differs from the pictured glucose strip with two working contacts and one reference contact.

**中文：** 读图：Potentiostat（恒电位仪）控制WE相对RE的电位，并测量WE–CE回路中的反应电流。RE提供稳定电位参考，其测量输入近似不取电流。这个标准WE/RE/CE结构，与后面血糖试纸图中的“两根工作接点＋一根参比接点”要区分。

Source / 来源：Week 6 PDF p37–40; original teaching schematic / 自绘教学示意，非实验数据

## S603 • [P1] Ion-selective membranes, pH and ISFET / 离子选择膜、pH与ISFET

Priority / 复习标记：P1 First pass/先掌握 · Teacher emphasis/课堂强调 · Repeated explanation/反复讲解

**EN:** An ion-selective electrode, ISE, combines a selective membrane response with an approximately open-circuit potential measurement relative to a stable reference. The sensing membrane favors a target ion; it is not simply a perfectly selective size sieve for every molecule. Activity means effective concentration after considering interactions, expressed relative to a standard state as a dimensionless quantity for the logarithm; in a suitable dilute approximation it corresponds to concentration. A bare metal electrode alone does not automatically identify which of many dissolved species caused the signal.

**中文：** Ion-selective electrode（ISE，离子选择电极）把选择膜响应与近似开路电位测量结合，并相对稳定参比读数。感测膜优先响应目标离子，并不只是对所有分子都完美按大小筛选。Activity（活度）可理解为考虑相互作用后的有效浓度，再以标准态归一化为对数中使用的无量纲量；合适的稀溶液近似下与浓度对应。裸金属电极本身也不会自动认出许多溶解物中究竟哪种引起信号。

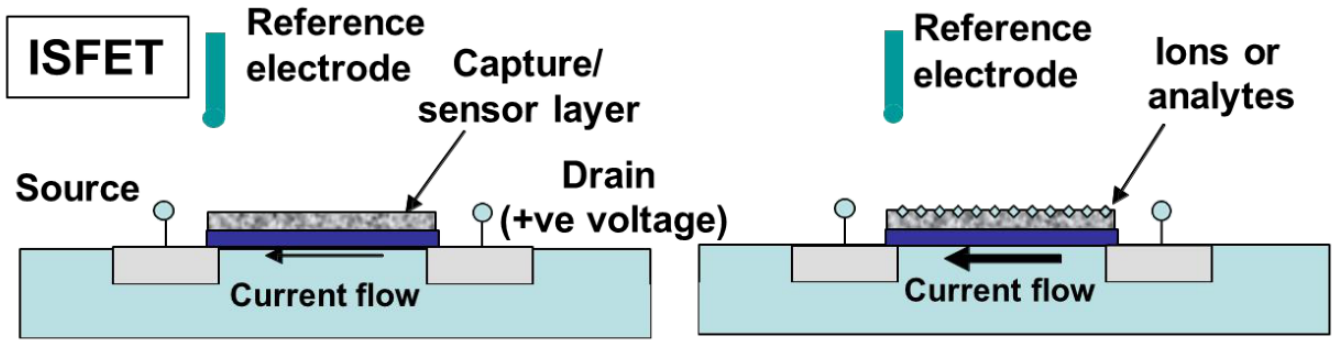
**EN:** A potentiometric pH sensor responds to hydrogen-ion activity, with $pH = -\log_{10} a_{H^+}$. Ten times greater $H^+$ activity lowers pH by one. A glass membrane and a fixed internal reference create a potential related to the outside solution. For an ideal monovalent ion at 25 °C, the potential change has magnitude about 59 mV per tenfold activity change; the sign depends on the defined electrode/potential convention.

**中文：** 电位型pH传感器响应氢离子活度，$pH = -\log_{10} a_{H^+}$。$H^+$活度增大十倍，pH降低1。玻璃膜与固定内部参比形成与外部溶液有关的电位。理想单价离子在25 °C时，每十倍活度变化对应约59 mV的电位变化大小；正负方向取决于电极/电位定义约定。

**EN:** In an ion-sensitive FET, ISFET, the sensing layer’ s ion-dependent potential acts as a gate influence, and the transistor converts it to channel-current change. Reading channel current does not mean that the analyte must undergo amperometric redox at the electrode. Distinguish the chemical sensing mechanism at the gate from the electrical readout of the transistor.

**中文：** Ion-sensitive FET（ISFET，离子敏感场效应管）中，感测层随离子变化的电位形成栅极作用，晶体管把它转成沟道电流变化。读出沟道电流不表示分析物必须在电极上发生安培型氧化还原。要区分栅极处的化学感测机制与晶体管的电读出方式。

Source / 来源：Week 6 PDF p39–46; T6-0040–0049




ISFET: an ion-sensitive surface gates a source–drain current (original lecture figure) / ISFET：离子敏感表面控制源漏电流（原课件图）

**EN:** Read the figure: Find Reference electrode in the solution, the Capture/sensor layer on the transistor, and Source/Drain at the channel ends. Ions or bound analytes change the surface/gate potential; the transistor converts that change into a change in channel current. The reference electrode establishes the solution potential. The arrow is current along the transistor channel, not a stream of target ions entering the electronic circuit.

**中文：** 读图：找溶液中的Reference electrode参比电极、晶体管表面的Capture/sensor layer捕获/传感层，以及沟道两端Source/Drain源漏。离子或结合的目标改变表面/栅电位，晶体管把它变成沟道电流变化。参比电极提供溶液电位基准。箭头是沿晶体管沟道的电流，不是目标离子流进电子电路。

Source / 来源：Week 6 PDF p45; original slide figure / 原课件截图

## B22 • [P3] Microfabrication and PDMS / 微加工与PDMS

Priority / 复习标记：P3 Supporting reading/辅助阅读 · Supporting concept/辅助理解

**EN: Key points**

• Photolithography patterns photoresist using exposure and development;
• etching removes selected material, whereas deposition adds material.
• PDMS (polydimethylsiloxane) soft lithography casts and cures a polymer on a master, then peels and bonds it to form structures such as microchannels.
• Know each step’ s function and the integration advantage;
• detailed recipes and equipment settings are secondary.

**中文：必会知识点**

• Photolithography光刻通过曝光、显影形成光刻胶图案；
• etching刻蚀去除选定材料，deposition沉积添加材料。
• PDMS聚二甲基硅氧烷软光刻在母模上浇注、固化聚合物，再剥离、键合，可形成微通道。
• 认识各步骤功能与集成优势即可，详细配方和设备参数次要。

**EN:** Start from zero: Microfabrication makes small, patterned structures such as electrodes and fluid channels. The challenge is processing selected places rather than covering everything uniformly. Photolithography solves a patterning task using light-sensitive photoresist. A mask determines where exposure reaches it, and development creates the pattern. Which exposed or unexposed parts remain depends on the resist type; there is no single rule for every process.

**中文：** 从零理解：Microfabrication（微加工）制作微小且有图案的结构，例如电极和流体通道。困难在于只在指定位置处理材料，而不是全部均匀覆盖。Photolithography（光刻）用对光敏感的photoresist（光刻胶）形成图案；mask（掩模）决定哪些位置被曝光，development（显影）形成所需图案。曝光或未曝光部分哪种留下取决于光刻胶类型，不存在适用于所有工艺的单一规则。

**EN:** Follow the mechanism: Etching removes selected material; deposition adds material. PDMS is a flexible polymer used in soft lithography. Liquid polymer is placed over a patterned master, cured to become solid, peeled away, and bonded to another surface to make enclosed channels. Think of making a molded stamp and sealing it onto a floor. The analogy explains shape copying; actual bonding and material compatibility still matter.

**中文：** 跟着机制走：Etching（刻蚀）移除指定材料，deposition（沉积）添加材料。PDMS是一种柔性聚合物，常用于soft lithography（软光刻）。把液态聚合物放在有图案的master（母模）上，cure（固化）成为固体，再剥离并与另一表面bond（键合），形成封闭通道。可类比制作模压印章再把它封到地板上；比喻解释复制形状，但实际键合及材料兼容仍很重要。

**EN:** Try an example: A microfluidic channel guides a small fluid volume past a sensing surface, supporting controlled handling and integration. It does not itself guarantee perfect biological recognition or a lower detection limit. For English matching questions, exposure and development point to photolithography; material removal to etching; material addition to deposition; master, casting, curing, and peeling to PDMS molding. Understand each action before memorizing fabrication recipes or equipment settings.

**中文：** 用例子理解：Microfluidic channel（微流控通道）让少量液体沿指定路径流过感测表面，有利于受控处理和集成；但它本身不保证完美生物识别或更低检测限。英文配对题中，exposure与development对应光刻；material removal对应刻蚀；material addition对应沉积；master、casting、curing、peeling对应PDMS模制。先理解各步骤做成什么，再考虑长的工艺配方和设备参数。

Source / 来源：W6 p22–30 | T6-0023–0033

## S604 • [P3] Positive/negative resist, etching and lift-off / 正负光刻胶、刻蚀与剥离

Priority / 复习标记：P3 Supporting reading/辅助阅读 · Supporting concept/辅助理解

**EN:** Photoresist is a light-sensitive material that makes a temporary pattern. After exposure and development, positive resist removes the exposed regions, whereas negative resist retains/crosslinks the exposed regions. A mask therefore produces opposite resist patterns with the two types. Exposure patterns the resist; it does not by itself mean that the underlying silicon has already been etched.

**中文：** Photoresist（光刻胶）是用来制作临时图案的光敏材料。曝光、显影后，positive resist（正胶）去掉曝光区域；negative resist（负胶）使曝光区域保留/交联。同一个掩模因此可产生相反胶图案。曝光先形成胶图案，不等于下面的硅已经被刻掉。

**EN:** Etching removes exposed substrate/material through the patterned mask. Wet etching uses liquid chemical reagents; dry etching uses processes such as plasma. Deposition adds material. In lift-off, material is deposited over a resist pattern, then removing the resist also removes the material on top of it, leaving the material deposited in the intended openings. Lift-off is not simply another word for etching the substrate.

**中文：** Etching（刻蚀）通过图案掩模去除露出的基底/材料。Wet etching（湿法刻蚀）使用液体化学试剂，dry etching（干法刻蚀）可用等离子体等过程。Deposition（沉积）添加材料。Lift-off（剥离）先在胶图案上沉积，再去掉胶及胶上材料，留下预定开口中的沉积物；它不是“刻蚀基底”的同义词。

**EN:** In PDMS soft lithography, a patterned master is the mold; liquid polymer precursor is poured, cured and peeled off to reproduce microstructures. The cured structure can then be assembled/bonded into a microfluidic device. PDMS is a polymer, not silicon; soft lithography copies a master' s pattern rather than exposing every final polymer device as if it were silicon photolithography. Biocompatibility means compatibility with the intended biological sample/tissue use. Visible transparency supports optical readout; chemical modification adds recognition layers. Material choice also balances flexibility and manufacturability.

**中文：** PDMS soft lithography（软光刻）用图案母模作模具，倒入液态聚合物前体、固化、剥离，复制微结构；再装配/键合成微流控器件。PDMS是聚合物，不是硅；软光刻复制母模图案，并非把每个最终聚合物器件都当作硅光刻来曝光。 Biocompatibility（生物相容性）指与预定生物样品或组织用途兼容；可见光透明便于光学读出，chemical modification（化学修饰）便于添加识别层。材料选择还要兼顾柔性与可制造性。

Source / 来源：Week 6 PDF p24–30; T6-0026–0033, T6-0083

## C19 • [P1] Glucose biosensor: recognition and electrical readout / 葡萄糖传感器：识别与电读出

Priority / 复习标记：P1 First pass/先掌握 · Teacher emphasis/课堂强调 · Repeated explanation/反复讲解

**EN: Key points**

• In the taught glucose-oxidase example, an enzymatic reaction links glucose to an electroactive product.
• An amperometric electrode converts the product-related reaction into current.
• A paired background channel can help subtract non-target electroactive interference under matched conditions.

**中文：必会知识点**

• 课程的glucose oxidase示例中，酶反应把葡萄糖与电活性产物联系起来。
• 安培型电极把产物相关反应转为电流。
• 条件匹配时，配对背景通道可帮助扣除非目标电活性干扰。

**EN:** Start from zero: An enzyme is a biological catalyst: it speeds a suitable chemical reaction rather than being the target we want to count. Glucose oxidase, GOx, helps react glucose. This recognition step makes the response linked to glucose, but a reaction alone is not yet a number on a meter. An electrode and electronics are needed to convert chemical activity into a measurable electrical signal.

**中文：** 从零理解：Enzyme（酶）是生物催化剂：它加快适合的化学反应，并不是我们要计数的被测目标。Glucose oxidase（GOx，葡萄糖氧化酶）帮助葡萄糖发生反应；这个识别环节让响应与葡萄糖关联，但单有反应还不是仪表上的数字，还需要电极和电路把化学活动转换为可测电信号。

**EN:** Follow the mechanism: In the taught first-generation example, the enzyme reaction generates hydrogen peroxide, $H_2O_2$. Its electrode reaction transfers electrons, and moving charge produces current. Within a calibrated working range, more glucose can produce a larger current. Think of the enzyme as a selective chemical worker and the electrode as a counter for transferred charge. Neither role alone explains the complete biosensor.

**中文：** 跟着机制走：课程第一代例子中，酶反应生成hydrogen peroxide（$H_2O_2$，过氧化氢）；它在电极上的反应转移电子，电荷移动形成电流。在已校准的有效范围内，更多葡萄糖可对应更大电流。可以把酶比作有选择性的化学工作人员，把电极比作记录转移电荷的计数器；任何一个角色都不能单独解释完整传感器。

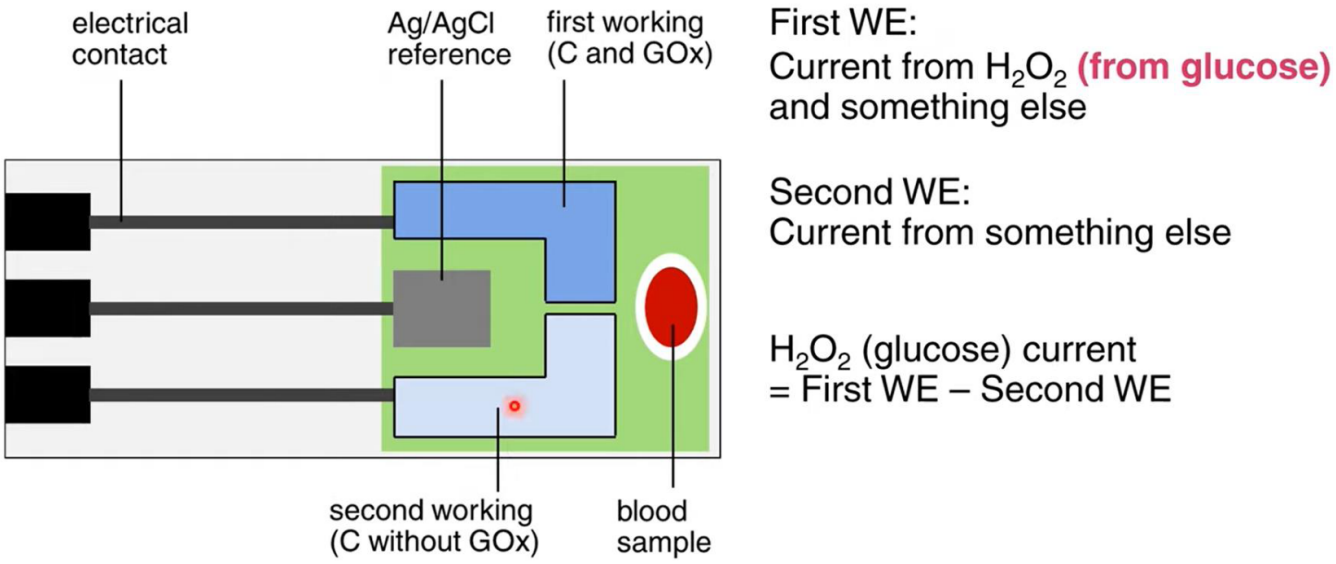
**EN:** Try an example: Other electroactive substances also contribute current. A matched channel without GOx can estimate background for subtraction. For illustration, a sensing channel reading 12 μA and a background channel reading 3 μA gives a corrected signal of 9 μA, assuming matched backgrounds. This does not automatically give glucose concentration without calibration. English clues: enzymatic recognition identifies the target; amperometric readout measures current. Other generations can use different electron-transfer pathways.

**中文：** 用例子理解：其他electroactive（电活性）物质也可能贡献电流。一个条件匹配、但没有GOx的通道可以估计背景，供相减使用。举例：感测通道12 μA、背景通道3 μA，在背景匹配假设下，校正信号是9 μA；没有校准关系，仍不能直接换算葡萄糖浓度。英文辨认：enzymatic recognition指酶识别目标，amperometric readout指测电流；其他代传感器可使用不同电子传递途径。

**EN:** Exam wording: The enzyme recognition step and electrode transduction step are different. The lactate example is called exam-style without a clear Quiz/final distinction. This edition lowers the multi-step mass-to-moles-to-electrons calculation because of the Quiz’ s short-calculation guidance; keep the simple reaction–charge–current chain as supporting knowledge.

**中文：** 英文读题抓手：酶识别与电极换能是不同步骤。乳酸例题被称为考试类型，但没有明确区分Quiz与期末。本版依据Quiz短计算指导降低质量→摩尔→电子多步计量的优先级，保留反应、电荷与电流的简单关系作为辅助知识。

Source / 来源：W6 p55, p57–59; W6 p58–59; W6 p60 | T6-0060–0073




Glucose strip example: enzyme-coated and uncoated working electrodes subtract background (original lecture figure) / 葡萄糖试纸示例：有酶与无酶工作电极相减去背景（原课件图）

**EN:** Read the figure: Follow the labels, not just the three contact pads. This example contains an Ag/AgCl reference plus two working electrodes: the first has GOx, the second has no GOx. The first measures glucose-related $H_2O_2$ current plus interference; the second estimates interference. Subtracting the second current helps isolate the glucose contribution when backgrounds match. These are not the usual working/reference/counter three-electrode roles, and subtraction cannot remove every interference automatically.

**中文：** 读图：按标签辨认，别只数三个接触点。本例是Ag/AgCl参比电极加两个工作电极：第一个有GOx，第二个没有GOx。第一个测到葡萄糖相关$H_2O_2$电流加干扰，第二个估计干扰；背景匹配时两电流相减有助于分离葡萄糖贡献。它不是通常的工作/参比/对电极三种角色组合，也不能自动消除所有干扰。

Source / 来源：Week 6 PDF p58; original slide figure / 原课件截图

## B23 • [P2] Three glucose-sensor generations / 葡萄糖传感器三代原理

Priority / 复习标记：P2 Important foundations/重要基础 · Teacher emphasis/课堂强调 · Supporting concept/辅助理解

**EN: Key points**

- First generation uses the natural oxygen/peroxide pathway;
- second generation uses an electron-transfer mediator;
- third generation aims at direct enzyme–electrode electron transfer.
- In the taught peroxide-oxidation example: $H_2O_2 \rightarrow O_2 + 2H^+ + 2e^-$.
- A matched electrode without GOx measures background for subtraction.
- Do not generalize this example into "all amperometric sensors must measure peroxide".

**中文：必会知识点**

- 第一代使用天然氧气/过氧化氢途径；
- 第二代使用electron-transfer mediator电子传递介体；
- 第三代追求酶—电极直接电子传递。
- 课程的过氧化氢氧化例子为$H_2O_2 \rightarrow O_2 + 2H^+ + 2e^-$。
- 匹配的无GOx电极测背景以便相减。
- 不能推广为"所有安培型传感器都必须测过氧化氢"。

**EN:** Start from zero: A glucose enzyme reacts with glucose, but the sensor also needs a route for the reaction's electrons to communicate with an electrode. The three generations describe different routes, not three release dates. First-generation GOx sensing uses the natural oxygen pathway and can monitor oxygen consumption or hydrogen peroxide. In the taught peroxide example, peroxide oxidation produces electrons that are measured as current.

**中文：** 从零理解：葡萄糖酶与葡萄糖反应，但传感器还需要把反应中的电子信息传到电极。三代描述不同传递路线，不只是三次产品发布日期。第一代GOx传感使用天然氧气途径，可测耗氧或过氧化氢；课程的过氧化氢例子中，过氧化氢氧化释放电子，形成可测电流。

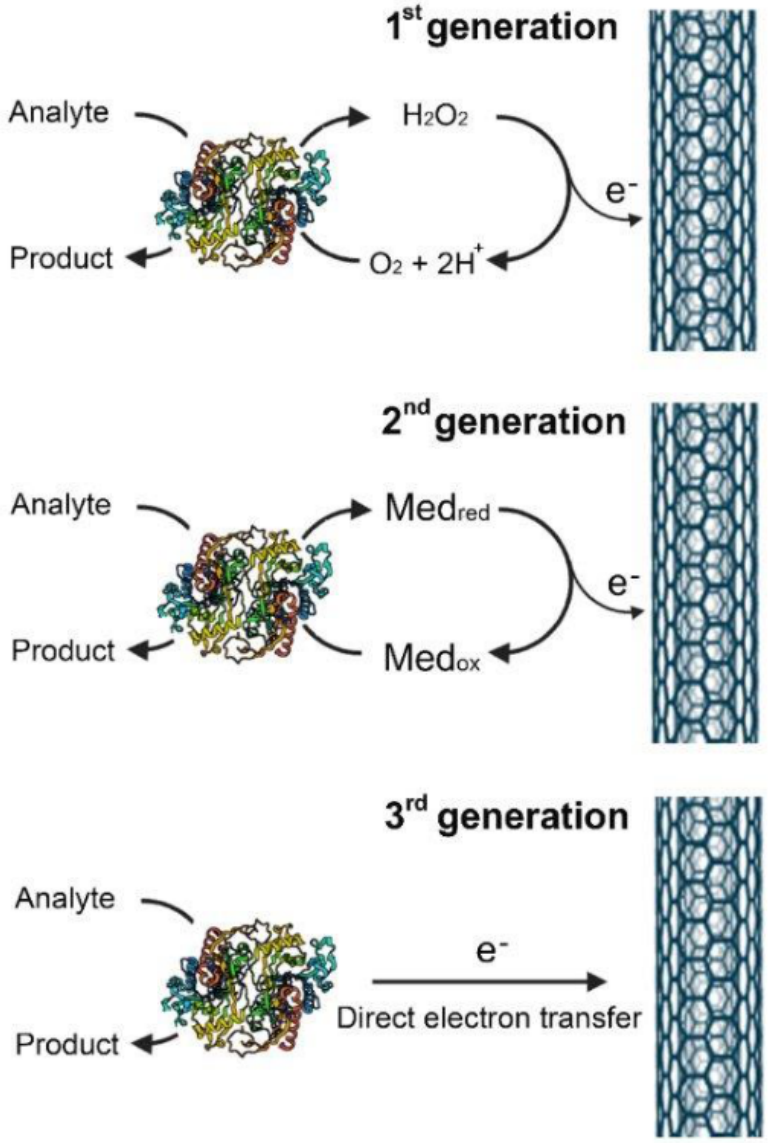
**EN:** Follow the mechanism: Second-generation sensing uses a mediator, a chemical electron carrier, to shuttle electrons between the enzyme and electrode. Think of a delivery worker connecting two sites. Third-generation sensing aims at direct enzyme–electrode electron transfer, without that shuttle. The analogy concerns electron transport, not physical movement of the enzyme. Direct transfer is a design goal whose practical success depends on the enzyme and electrode arrangement.

**中文：** 跟着机制走：第二代使用mediator（介体），即化学电子载体，在酶和电极之间传递电子，可类比连接两处的送货员。第三代追求direct enzyme–electrode electron transfer（酶—电极直接电子传递），不再用这个中间载体。比喻说的是电子传递，不是酶实际被搬运；直接传递是设计目标，能否实际实现取决于酶及电极的布置。

**EN:** Try an example: For the taught reaction $H_2O_2 \rightarrow O_2 + 2H^+ + 2e^-$, each peroxide molecule releases two electrons. A background electrode without GOx helps estimate non-target current when conditions are matched. English clues: oxygen or peroxide for first generation, mediator for second, direct electron transfer for third. Amperometric only tells you current is measured at controlled potential; it does not imply that every amperometric biosensor measures peroxide or uses a glucose enzyme.

**中文：** 用例子理解：课程反应$H_2O_2 \rightarrow O_2 + 2H^+ + 2e^-$中，每个过氧化氢分子释放两个电子。无GOx背景电极在条件匹配时帮助估计非目标电流。英文辨认：oxygen或peroxide提示第一代，mediator提示第二代，direct electron transfer提示第三代。Amperometric只说明在受控电位下测电流，并不代表所有安培型生物传感器都测过氧化氢或使用葡萄糖酶。

Source / 来源：W6 p55, p57–59; W6 p58–59 | T6-0064–0073




Three glucose-sensor generations: different routes from enzyme reaction to electrode (original lecture figure) / 葡萄糖传感器三代：酶反应到电极的不同传递路线（原课件图）

**EN:** Read the figure: Read top to bottom and follow the $e^-$ arrows to the electrode on the right. First generation uses oxygen/$H_2O_2$ chemistry; second generation uses a mediator cycling between reduced and oxidized forms (Medred/Medox); third generation seeks direct electron transfer between enzyme and electrode. The changing feature is the electron-transfer route, not three different targets. Direct transfer is a design objective that requires a suitable enzyme–electrode interface.

**中文：** 读图：从上往下读，跟着$e^-$箭头看到右侧电极。第一代利用氧气/$H_2O_2$化学；第二代用在还原态与氧化态间循环的介体Medred/Medox传递；第三代追求酶与电极直接电子转移。变的是电子传递路线，不是三种不同目标。直接转移是设计目标，需要合适的酶—电极界面。

Source / 来源：Week 6 PDF p59; original slide figure / 原课件截图

## B24 • [P2] Charge, current and reaction stoichiometry / 电荷、电流与反应计量

Priority / 复习标记：P2 Important foundations/重要基础 · Worked example/例题练习 · Teacher emphasis/课堂强调 · Quiz stated/Quiz明确提及

**EN: Key points**

- For constant current, Q=It.
- Faraday’ s relation is $Q=n_eF=z\ n_aF$, where $n_e$ is moles of electrons, $n_a$ moles of analyte, z electrons per analyte molecule and F≈96485 C/mol.
- Obtain z from the stated reaction.
- Mass-to-moles requires molar mass;
- counting molecules requires Avogadro’ s constant.
- Keep the short relations;
- lengthy multi-step examples are lower priority.

**中文：必会知识点**

- 恒定电流时Q=It。
- 法拉第关系$Q=n_eF=z\ n_aF$，其中$n_e$为电子摩尔数，$n_a$为分析物摩尔数，z为每分子转移电子数，F≈96485 C/mol。
- z应由给定反应式获得。
- 质量换摩尔需摩尔质量，分子数换算需阿伏伽德罗常数。
- 保留短关系，长的多步例题降低优先级。

**EN:** Start from zero: Charge Q is the amount of electrical charge transferred; current I is its transfer rate. One ampere means one coulomb each second. If current stays constant, Q=It, where t is time. For example, 1 mA for 10 s transfers 0.001 A×10 s=0.010 C. A large current for a short time can transfer the same charge as a small current for a long time.

**中文：** 从零理解：Charge Q（电荷量）表示转移多少电荷；current I（电流）表示转移速度。1 ampere（安培）是每秒1 coulomb（库仑）。恒定电流时Q=It，t为时间。例如1 mA持续10 s，转移0.001 A×10 s=0.010 C。大电流持续短时间，与小电流持续长时间，可能转移同样电荷量。

**EN:** Follow the mechanism: Chemical reactions count particles, while electrical instruments measure charge. A mole is a counting unit for many particles. Faraday's constant F≈96485 C/mol connects these descriptions: one mole of electrons carries that charge magnitude. If each analyte molecule transfers z electrons, $n_a$ moles of analyte transfer z $n_a$ moles of electrons. Under complete conversion and collection, Q=z $n_a$F.

**中文：** 跟着机制走：化学反应按粒子数量计量，电仪器则测电荷。Mole（摩尔）是表示极大量粒子的计数单位。Faraday's constant（法拉第常数）F≈96485 C/mol把两种描述连起来：一摩尔电子对应这么多电荷量的大小。若每个分析物分子转移z个电子，$n_a$摩尔分析物就转移z $n_a$摩尔电子；在完全反应且全部电荷被收集的假设下，Q=z $n_a$F。

**EN:** Try an example: Read z from the stated reaction, not from the analyte's name. If 1 μmol of analyte transfers two electrons per molecule completely, Q=2×$10^{-6}$×96485≈0.193 C. To start from mass, first divide by molar mass; do not insert grams directly into the mole formula. English clues: stoichiometry determines electron count, current is charge per time, and integrated charge measures total transfer. These are illustration calculations, not claimed actual Quiz questions.

**中文：** 用例子理解：z应从题目给定反应读出，不能只靠分析物名字猜。如果1 μmol分析物完全反应，每个分子转移两个电子，则Q=2×$10^{-6}$×96485≈0.193 C。若起点是质量，先除以molar mass（摩尔质量），不能把克直接代入摩尔公式。英文辨认：stoichiometry（反应计量）决定电子数；current是电荷/时间；integrated charge（积分电荷）测总转移量。这些是讲解例子，不声称实际Quiz会出现。

Source / 来源：W6 p50–52; W6 p50–52, p60; W6 p60; W7 p9–13 | T6-0054–0063, T7-0006–0008

## B25 • [P2] FET, nanopore and capacitive readout / 场效应管、纳米孔与电容读出

Priority / 复习标记：P2 Important foundations/重要基础 · Repeated explanation/反复讲解 · Teacher emphasis/课堂强调

**EN: Key points**

- In a field-effect transistor (FET) sensor, binding near the gate changes the channel’ s electrical response.
- A nanopore reads ionic-current changes/blockades as molecules interact with the pore.
- A capacitive sensor reads changes in capacitance, which depends on geometry and dielectric properties (ideal plates: C=εA/d).
- Conductance G=1/R;
- conductance and material conductivity are related but different quantities.

**中文：必会知识点**

- Field-effect transistor（FET，场效应晶体管）传感器中，栅极附近结合改变沟道电响应。
- Nanopore纳米孔测分子与孔相互作用引起的离子电流变化/阻塞。
- Capacitive电容型传感测电容变化，电容依赖几何与介电性质（理想平板C=εA/d）。
- 电导G=1/R;
- conductance电导与conductivity材料电导率相关，但并非同一物理量。

**EN:** Start from zero: Different electrical sensors turn binding into different circuit changes. In a field-effect transistor, FET, the gate's electrical influence controls current through a channel, like a handle controlling flow. Binding near the sensing gate can alter that influence. The analogy does not mean molecules physically block the channel in every FET sensor; the important idea is electrical control of channel response.

**中文：** 从零理解：不同电传感器把结合转成不同电路变化。Field-effect transistor（FET，场效应晶体管）中，gate（栅极）的电作用控制channel（沟道）电流，可类比控制把手影响流动路径。感测栅极附近的结合可改变这种电作用；但不是所有FET中分子都实际堵住沟道，重点是栅极对沟道响应的电控制。

**EN:** Follow the mechanism: A nanopore is a very small opening carrying ionic current through liquid. An interacting molecule changes current, often causing a blockade. Capacitance instead describes charge stored per voltage. For ideal parallel plates, C=εA/d: ε describes dielectric properties, A plate area, and d spacing. Binding can alter the effective dielectric environment or geometry, changing capacitance without being a nanopore blockade.

**中文：** 跟着机制走：Nanopore（纳米孔）是允许液体中离子电流通过的极小孔。分子进入或与孔相互作用时会改变电流，常表现为blockade（阻塞）。Capacitance（电容）则描述每单位电压能储存多少电荷。理想平板C=εA/d，ε描述介电性质、A是板面积、d是间距；结合可改变有效介电环境或几何，导致电容变化，不等于纳米孔堵塞。

**EN:** Try an example: Conductance G=1/R measures how easily a particular device conducts, in siemens; resistance R is measured in ohms. Conductivity is a material property; a device’ s conductance also depends on its geometry. English matching: gate and channel point to FET; ionic-current blockade to nanopore; stored charge and dielectric changes to capacitive sensing. For ideal plates with ε and A fixed, doubling d halves C. Apply that example only when the stated geometry supports it.

**中文：** 用例子理解：Conductance（电导）G=1/R表示一个具体器件导电有多容易，单位siemens（西门子）；电阻R用ohm（欧姆）。Conductivity（电导率）是材料性质，具体器件测量还受几何影响。英文配对：gate和channel对应FET；ionic-current blockade对应纳米孔；stored charge和dielectric changes对应电容传感。理想平板中，ε、A固定，d加倍时C减半；只有题目几何支持该模型时才这样计算。

Source / 来源：W6 p64–74; W6 p75–77; W6 p78–81, p87–90 | T6-0074–0091, T6-0102–0105

## S605 • [P2] DNA charge, protein pH response and isoelectric point / DNA电荷、蛋白质pH响应与等电点

Priority / 复习标记：P2 Important foundations/重要基础 · Teacher emphasis/课堂强调 · Repeated explanation/反复讲解

**EN:** DNA is normally negatively charged in the taught aqueous setting because of its phosphate backbone. A protein is not always positive or always negative: its net charge depends on ionizable amino-acid groups and the solution pH. Binding can therefore change an electrical sensor in different directions; “all biomolecules have the same charge” is a poor general rule.

**中文：** 课程水溶液背景下，DNA通常因phosphate backbone（磷酸骨架）带负电。蛋白质不是总正电或总负电：净电荷取决于可电离氨基酸基团与溶液pH。因此结合可让电传感信号向不同方向变化，“所有生物分子电荷相同”不是可靠通则。

**EN:** For a simple ionizable group, pKa is the pH where protonated and deprotonated forms are equally populated. Below pKa, the protonated form is favored; above it, the deprotonated form is favored. Deprotonation of an acidic group changes neutral to negative, while deprotonation of a basic group can change positive to neutral. Losing $H^+$ does not always produce the same final charge for every group.

**中文：** 简单可电离基团的pKa，是protonated（带质子）与deprotonated（失去质子）形式各占一半的pH。pH低于pKa时带质子形式较多，高于时失去质子形式较多。酸性基团失去质子可从中性变负电；碱性基团可从正电变中性。失去$H^+$并不让所有基团都得到同一种最终电荷。

**EN:** The isoelectric point, pI, is the pH at which a protein has zero net charge. It can still contain both positive and negative groups. In the usual acid–base trend, pH below pI gives a more positive net charge and pH above pI gives a more negative net charge. pI is a whole-protein net-charge condition; it is not the pKa of every side chain.

**中文：** Isoelectric point（pI，等电点）是蛋白质净电荷为零时的pH；它内部仍可同时存在正、负基团。通常酸碱趋势中，pH低于pI时净电荷更正，高于pI时更负。pI描述整个蛋白质的净电荷条件，不是每个侧链的pKa。

Source / 来源：Week 6 PDF p64–65; T6-0074–0075

## S607 • [P2] FET polarity, nanopore sequences and motion capacitance / FET方向、纳米孔序列与运动电容

Priority / 复习标记：P2 Important foundations/重要基础 · Repeated explanation/反复讲解 · Supporting concept/辅助理解

**EN:** In a FET, source and drain are connected by a conducting channel and the gate controls its carrier population. The same bound charge can affect an n-type electron channel differently from a p-type hole channel. The slides illustrate opposite conductance changes. Therefore, identify carrier type, charge and bias before assigning an increase or decrease; detecting a target does not universally make FET current rise. Under the illustrated ideal electrostatic-gating model at fixed bias, positive bound charge attracts electrons and increases n-type channel conductance, but depletes holes and decreases p-type conductance; negative charge gives the opposite tendency. Screening, bias and device design can modify the actual response.

**中文：** FET中，source（源极）与drain（漏极）之间由导电channel（沟道）连接，gate（栅极）控制载流子数量。同一结合电荷对n-type电子沟道与p-type空穴沟道的影响可不同，课件画出相反电导变化。因此判断增减前要看载流子类型、电荷与偏置；检测到目标不会普遍使FET电流上升。 在图示固定偏置、理想静电门控模型下，结合正电荷吸引电子，使n型沟道电导增大；同时减少空穴，使p型电导减小。负电荷的趋势相反。实际响应还受屏蔽、偏置及器件设计影响。

**EN:** In nanopore sequencing, a voltage drives ionic current through a small opening. A DNA/RNA strand changes that current as it translocates; the time sequence of current levels is interpreted to infer sequence. The measured current is carried by ions through the pore, not by electrons travelling along the DNA as a metal wire. Read blockade amplitude and duration as electrical features, not as fluorescence colors.

**中文：** 纳米孔测序中，电压驱动离子电流通过小孔，DNA/RNA链移过孔时改变该电流；分析随时间的电流水平来推断序列。被测电流由穿孔离子携带，不是电子把DNA当金属线沿链移动。Blockade amplitude（阻塞幅度）与duration（持续时间）是电特征，不是荧光颜色。

**EN:** Capacitive sensing can measure either molecular changes near an interface or mechanical deformation. In the wearable examples, bending or pressure changes effective separation/geometry and therefore capacitance. Such a motion sensor need not use an enzyme or redox reaction. Ask whether the cause is charge/dielectric binding or physical geometry before assigning the sensing principle.

**中文：** 电容传感既可测界面附近分子变化，也可测机械形变。可穿戴示例中，弯曲或压力改变有效间距/几何，从而改变电容；这种运动传感器不一定使用酶或氧化还原。判别原理前，先问变化来自电荷/介电结合，还是来自物理几何。

Source / 来源：Week 6 PDF p45, 71–81; T6-0081–0091

## B26 • [P2] Impedance, reactance and Bode plots / 阻抗、电抗与Bode图

Priority / 复习标记：P2 Important foundations/重要基础 · Worked example/例题练习 · Sample Q10/样卷Q10 · Final emphasis/仅期末强调

**EN: Key points**

- Impedance includes magnitude and phase in AC response.
- Ideal R is frequency-independent;
- |XC|=1/(2πfC) decreases with frequency, while XL=2πfL increases.
- A Bode plot shows magnitude and phase versus logarithmic frequency.
- A capacitor’ s imaginary impedance is not its positive magnitude.
- Long complex-number combinations and full Bode plotting remain secondary for this Quiz.

**中文：必会知识点**

- Impedance阻抗包含交流响应的幅值与相位。
- 理想R不随频率变；
- |XC|=1/(2πfC)随频率升高减小，XL=2πfL随频率升高增大。
- Bode图画幅值和相位随对数频率的变化。
- 电容的虚数阻抗不能等同于其正值幅度。
- 复杂阻抗组合与完整Bode作图仍为本次Quiz次要内容。

**EN:** Start from zero: An alternating signal repeatedly changes direction. Its response can differ not only in size but also in timing. Impedance describes both: magnitude is the voltage-amplitude to current-amplitude ratio, and phase describes their relative position within a cycle. Imagine two people swinging with the same rhythm but reaching the front at different times. That timing difference resembles phase; it is not merely another amount of resistance.

**中文：** 从零理解：交流信号反复改变方向，响应不仅大小可能变化，时间关系也可能变化。Impedance（阻抗）同时描述两者：magnitude（幅值）是电压幅度与电流幅度之比，phase（相位）描述两者在一个周期中的相对位置。可以想象两人荡秋千节奏相同，但到最前方的时刻不同；这个时间差像相位，不只是又多了一个电阻数值。

**EN:** Follow the mechanism: An ideal resistor has frequency-independent resistance. A capacitor stores charge; its reactance magnitude |XC|=1/(2πfC) decreases when frequency f increases. An inductor opposes changing current; XL=2πfL increases with frequency. C and L are capacitance and inductance. In complex-number notation, capacitive impedance has a negative imaginary part, not simply a positive resistance. Doubling f halves capacitive reactance magnitude when C stays fixed.

**中文：** 跟着机制走：理想电阻的阻值不随频率变化。电容储存电荷，其电抗幅值|XC|=1/(2πfC)随频率f提高而减小；电感阻碍电流变化，XL=2πfL随频率提高而增加。C和L分别是电容与电感量。在复数表示中，电容阻抗有负虚部，不是一个单纯的正电阻。C固定时，f加倍，电容电抗幅值减半。

**EN:** Try an example: A Bode plot shows magnitude and phase against logarithmic frequency, revealing behavior across a wide frequency range. Do not add resistance and reactance magnitudes as ordinary numbers. For an ideal series 3 Ω resistor and capacitor impedance −j4 Ω, total magnitude is $\sqrt{(3^2+4^2)}=5$ Ω, not 7 Ω. Here j marks the imaginary direction. For this Quiz, understand trends and phase before lengthy complex-number algebra.

**中文：** 用例子理解：Bode plot（波特图）画响应幅值及相位随对数频率的变化，方便观察很宽频率范围内的行为。不要把电阻和电抗的幅值当普通数字相加。理想串联中，电阻3 Ω、电容阻抗−j4 Ω，总幅值为$\sqrt{(3^2+4^2)}=5$ Ω，不是7 Ω；j标记虚数方向。本次Quiz先理解趋势和相位，再考虑长的复数代数。

Source / 来源：W6 p82–85; W6 p84–86; W6 p86; W7 p70–78 | T6-0092–0101, T7-0073–0081

## S606 • [P2] Interface impedance, Nyquist plots and cell coverage / 界面阻抗、Nyquist图与细胞覆盖

Priority / 复习标记：P2 Important foundations/重要基础 · Repeated explanation/反复讲解 · Supporting concept/辅助理解

**EN:** An impedance biosensor can change because of the liquid’ s bulk conduction or the electrode–liquid interface. In the slide’ s equivalent circuit, R_s represents solution resistance. C_dl is double-layer capacitance: opposite charges accumulate on the two sides of the interface, storing charge like a capacitor. R_ct describes difficulty of charge transfer across the interface; a larger value means harder transfer. W denotes a diffusion contribution, associated with chemical species moving toward or away from the surface. Binding can impede a redox probe’ s access and increase R_ct in this example; not every sensor must change by the same component or in the same direction.

**中文：** 阻抗生物传感器可因液体内部导电改变，也可因电极—液体界面改变。课件等效电路中，R_s是溶液电阻。C_dl是double-layer capacitance（双电层电容）：界面两侧聚集相反电荷，像电容一样储存电荷。R_ct描述跨界面电荷转移的困难程度，越大表示越难转移。W表示diffusion（扩散）贡献，与化学物种移向或离开表面有关。示例中结合可妨碍氧化还原探针接近并提高R_ct；但不是每个传感器都必然改变同一元件或同一方向。

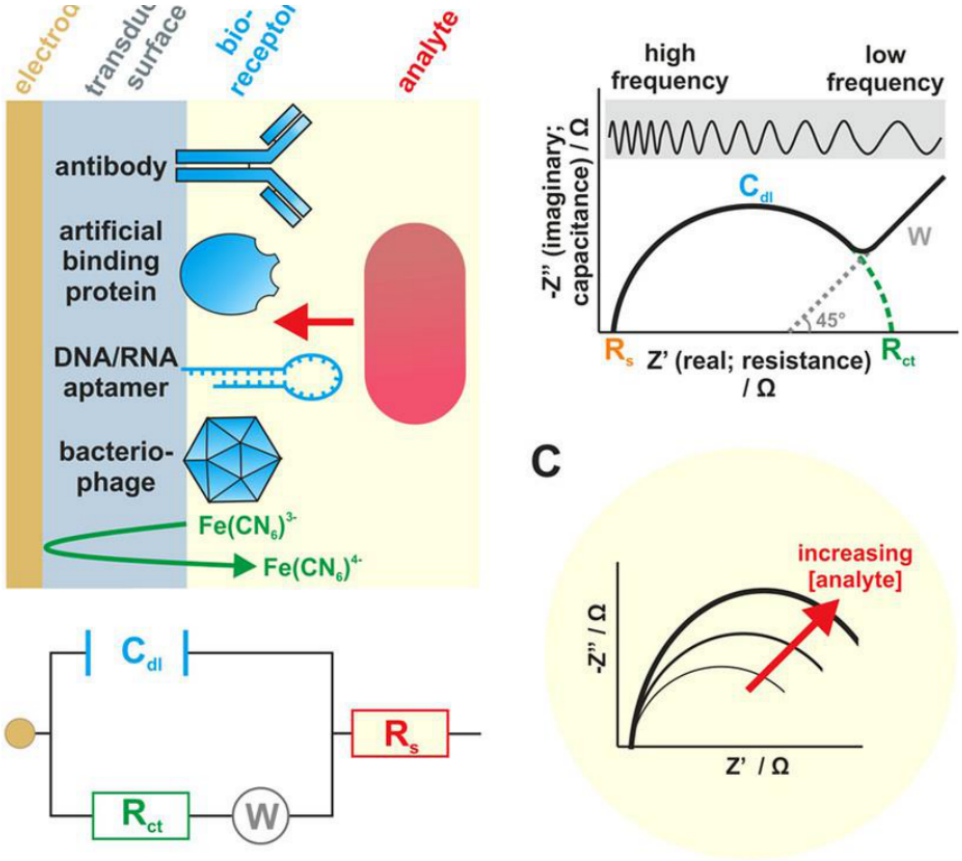
**EN:** A Nyquist impedance plot places real impedance Z′ on the horizontal axis and commonly −Z″ on the vertical axis. Each point corresponds to a different measurement frequency; frequency is not the horizontal coordinate. In the simple semicircle example, its width relates to R_ct, and the real-axis offset relates to R_s. A Bode plot instead shows magnitude and phase versus frequency. The name Nyquist here is distinct from the optical sampling theorem.

**中文：** Nyquist impedance plot（Nyquist阻抗图）横轴是阻抗实部Z′，纵轴常为−Z″，即负虚部。各点对应不同测量频率，但横坐标本身不是频率。简单半圆示例中，半圆宽度与R_ct有关，实轴起始偏移与R_s有关。Bode图则把幅值和相位画成随频率变化。这里的Nyquist图要与光学采样定理区分。

**EN:** The cell-coverage examples use a resistive barrier: more surface coverage obstructs ionic current and lowers conductance under the shown conditions. The measured change can follow cell adhesion, spreading or growth. This is an electrical observation of the interface, not proof that every covered surface has the same biological identity. For a particular sensor, interpret the calibrated direction and background rather than assuming "more analyte always means more current."

**中文：** 细胞覆盖示例利用电阻性屏障：图示条件下，覆盖更多会阻碍离子电流、降低电导。测量可追踪细胞黏附、铺展或生长。这是在电学上观察界面，不自动证明所有覆盖表面都具有同一生物身份。具体传感器要按已校准的方向及背景解释，不能假设"目标越多，电流一定越大"。

Source / 来源：Week 6 PDF p66–70, 72, 84–89; T6-0075–0080, T6-0092–0101




Electrode interface circuit and Nyquist impedance response (original lecture slide) / 电极界面电路与Nyquist阻抗响应（原课件图）

**EN:** Read the figure: Find R_s, R_ct and C_dl and the real/negative-imaginary axes. Increasing analyte widens the semicircle in this illustrated mechanism. Frequency labels indicate which point was measured; they are not the horizontal axis.

**中文：** 读图：找出R_s、R_ct、C_dl及实部/负虚部坐标。图示机制中目标增多使半圆变宽；频率标签表示各点的测量频率，不是横坐标。

Source / 来源：Week 6 PDF p66; original slide figure / 原课件截图

## S608 • [P2] Peak, RMS and voltage–current phase / 峰值、均方根与电压电流相位

Priority / 复习标记：P2 Important foundations/重要基础 · Worked example/例题练习 · Repeated explanation/反复讲解 · Final emphasis/仅期末强调

**EN:** For a sinusoidal voltage, V_rms=V_peak/√2, and similarly I_rms=I_peak/√2. RMS describes the effective magnitude for power in a resistor; it is neither peak-to-peak magnitude nor the signed cycle average, which is zero for a symmetric sinusoid. Apply this factor only to the stated sinusoidal waveform, not to every possible signal.

**中文：** 正弦电压V_rms=V_peak/√2，电流同样I_rms=I_peak/√2。RMS（均方根）描述电阻上功率对应的有效大小，不是peak-to-peak（峰峰值），也不是带正负的周期平均值；对称正弦的后者为零。这个系数只用于给定正弦波，不适用于任何波形。

**EN:** For ideal components using the voltage phase relative to current, a resistor has zero phase difference, a capacitor has negative impedance phase (current leads voltage), and an inductor has positive impedance phase (current lags voltage). At DC steady state an ideal capacitor blocks continuous conduction; as frequency rises its reactance magnitude falls. An ideal inductor has the opposite frequency trend. Phase sign depends on the stated convention, so check which signal is the reference.

**中文：** 理想元件以“电压相对电流的相位”为约定时，电阻相位差为零；电容阻抗相位为负（电流领先电压）；电感为正（电流落后电压）。直流稳态下理想电容不持续导通；频率提高时它的电抗幅值减小。理想电感的频率趋势相反。相位正负依赖约定，要先看用哪个信号作参考。

Source / 来源：Week 6 PDF p82–86; T6-0092–0101

# Week 7 learning block: Bioelectricity, ECG and EEG / 第7学习组：生物电、心电与脑电

## Week 7 concept map / 第7周知识框架

**EN:** Chapter logic: Ionic gradients and selective channels create cell voltages; tissue activity reaches electrodes as potential differences, which are interpreted as EMG, ECG or EEG patterns.

**中文：** 本章主线：离子梯度与选择性通道形成细胞电压；组织活动以电位差到达电极，再根据来源与时间模式解释为EMG、ECG或EEG。

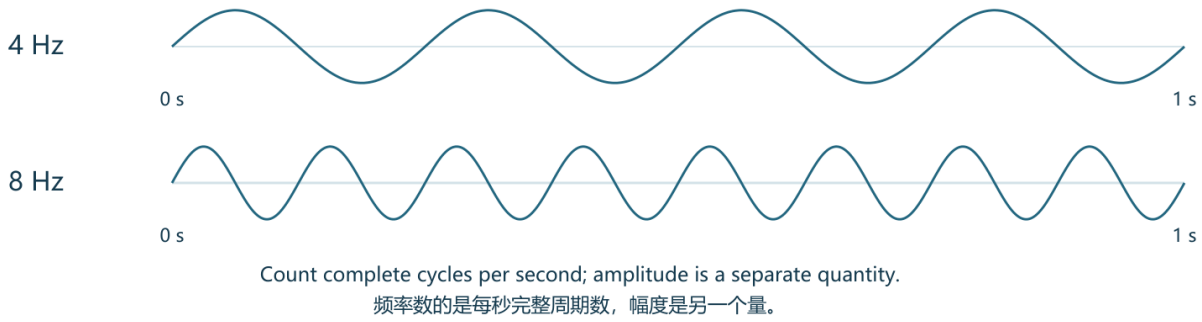
| English term 中文名词 | Meaning and connection 含义与知识联系 |
|---|---|
| **Bioelectricity / biopotential 生物电 / 生物电位** | Recording measures potential differences; stimulation deliberately applies an electrical input. ECG, EMG and EEG identify heart, muscle and brain-related recordings. 记录测电位差；刺激主动施加电输入。ECG、EMG、EEG分别对应心脏、肌肉及脑相关记录。 → C20 |
| **Membrane potential and AP phases 膜电位与动作电位阶段** | Vm=inside−outside. $Na^+$ entry raises the taught neuronal voltage; later $K^+$ exit lowers it, with a possible dip below rest. Vm=胞内−胞外。课程神经元中$Na^+$内流使电压升高，随后$K^+$外流使其降低，可短暂低于静息。 → C21 |
| **Threshold / all-or-none / firing rate 阈值 / 全或无 / 发放频率** | Threshold triggers a spike; stronger input commonly produces more similar spikes per second, not proportionally taller spikes. 阈值决定是否触发；更强输入通常增加每秒相似尖峰数量，不是让单个尖峰成比例变高。 → B27 |
| **Nernst versus GHK Nernst与GHK** | Nernst balances one ion; GHK combines ions with different permeabilities. Zero summed current does not require every ionic current to vanish. Nernst平衡一种离子，GHK综合不同通透性的多种离子。电流总和为零不要求每项都为零。 → B28 |
| **Electrode–electrolyte interface 电极—电解质界面** | Ionic tissue current meets electronic metal conduction; the interface adds voltage offsets and frequency-dependent impedance. 组织离子载流与金属电子载流在界面交接；界面加入电压偏移及随频率变化的阻抗。 → B29 |
| **P / QRS / T P波 / QRS波群 / T波** | P: atrial depolarization; QRS: ventricular depolarization; T: ventricular repolarization. These are waveform events, not separate leads. P：心房去极化；QRS：心室去极化；T：心室复极化。这是曲线事件，不是三个导联。 → C22 |

| English term 中文名词 | Meaning and connection 含义与知识联系 |
|---|---|
| **ECG leads and RR interval 心电导联与RR间期** | A lead defines a voltage-difference view; II=I+III for the given limb definitions. With RR in seconds, heart rate=60/RR. 导联定义一种电位差视角；所给肢体定义满足II=I+III。RR用秒时，心率=60/RR。 → B30 |
| **Scalp EEG and wave frequency 头皮脑电与脑波频率** | EEG reflects combined neuronal activity at standardized sites. Frequency comes from complete cycles/elapsed seconds; amplitude is a different quantity. EEG在标准位置记录神经元共同活动。频率=完整周期数/实际秒数；幅度是另一个量。 → B31 |

**EN:** Formula links for this chapter: FX701 Cell Nernst potential: valence, sign and temperature, FX702 GHK: permeability weights and reversed chloride placement, FX703 Ionic branch: conductance and driving force, FX704 Einthoven limb leads: write the polarity first, FX705 Frequency, peak intervals and heart rate, FX708 Loading and contact mismatch: voltage-divider model, FX709 Electrode-interface RC impedance and limits, FX713 Sampling Nyquist frequency is not an impedance-plot axis.

**中文：** 本章公式定位：FX701 细胞 Nernst 电位：价态、符号与温度；FX702 GHK：通透性权重与 $Cl^-$ 的反向位置；FX703 离子支路：电导与驱动力；FX704 Einthoven 肢体导联：先写清极性；FX705 频率、峰间隔与心率；FX708 加载与接触不匹配：分压模型；FX709 电极界面 RC 阻抗与频率极限；FX713 采样 Nyquist 频率不是阻抗图的横轴。

Same amplitude, different frequency / 幅度相同，频率不同




Frequency and amplitude / 频率与幅度教学示意

## C20 • [P1] ECG, EMG and EEG: match the organ and signal / ECG、EMG与EEG：组织来源对应

Priority / 复习标记：P1 First pass/先掌握 · Sample Q4/样卷Q4 · Quiz stated/Quiz明确提及 · Repeated explanation/反复讲解

**EN: Key points**

- ECG records cardiac electrical activity;
- EMG records muscle electrical activity;
- EEG records scalp potential differences associated with brain activity.
- These are biopotentials measured relative to a reference.
- A smartwatch can estimate sleep from motion and heart-related signals without directly recording scalp EEG.

**中文：必会知识点**

- ECG记录心脏电活动，EMG记录肌肉电活动，EEG记录与脑活动有关的头皮电位差。
- 它们都是相对于参照测得的生物电位。
- 智能手表可以用运动与心脏相关信号估计睡眠，而不直接记录头皮EEG。

**EN:** Start from zero: Start with biopotential: a voltage difference produced by biological activity. Voltage compares two locations; it is not the amount of electricity stored in an organ. Electrodes are conductive contacts that let an instrument measure that difference. ECG means electrocardiogram, recording heart activity; EMG means electromyogram, recording muscle activity; EEG means electroencephalogram, recording brain-related activity at the scalp.

**中文：** 从零理解：先理解biopotential生物电位：生物活动产生的电压差。电压比较两个位置，并不是一个器官里储存了多少“电”。Electrode电极是导电的接触部件，让仪器测出这个差值。ECG是electrocardiogram心电图，记录心脏活动；EMG是electromyogram肌电图，记录肌肉活动；EEG是electroencephalogram脑电图，在头皮记录与脑活动有关的信号。

**EN:** Follow the mechanism: Cells move charged particles called ions across their membranes. Many cells acting together create small voltage changes that can reach recording electrodes. The instrument compares a recording location with another location or reference, then amplifies the difference. Think of listening to a crowd rather than one person: a surface recording combines nearby activity. This analogy explains summation, not literal sound transmission.

**中文：** 跟着机制走：细胞让带电的粒子——ion离子——跨过细胞膜移动。许多细胞一起活动，就会产生能被记录电极检测到的微小电压变化。仪器把记录位置与另一个位置或reference参照比较，再放大差值。可以想成“听一群人，而不是只听一个人”：体表记录包含附近活动的组合。这个类比用于理解信号叠加，并不是说电信号真的通过声音传播。

**EN:** Try an example: In an English question, match cardiac with ECG, muscular with EMG, and scalp or brain with EEG. A watch using an accelerometer measures motion; its heart sensor provides heart-rate information. Combining these signals can estimate sleep patterns. That does not mean the watch directly records EEG or measures brain-wave frequency. First identify what is physically measured, then identify what the software estimates.

**中文：** 用例子理解：遇到英文题，先把cardiac心脏对应ECG、muscular肌肉对应EMG、scalp头皮或brain脑对应EEG。手表的accelerometer加速度计测运动，心脏传感器提供心率信息。把这些信号结合，可以估计睡眠模式，但不代表手表直接记录EEG或测出了脑波频率。先认清“传感器实际测什么”，再认清“软件根据这些数据推测什么”。

**EN:** Exam wording: Do not infer that every wearable measures EEG. The sample smartwatch combination is accelerometer plus heart-rate/heart-rate-variability information. EMG numerical ranges labelled “for reference” do not remove basic EMG recognition from the review.

**中文：** 英文读题抓手：不能认为所有可穿戴设备都测EEG。样题手表对应accelerometer加heart rate/heart rate variability。EMG某些数值范围被标为“供参考”，并不等于基础EMG识别全部不用复习。

Source / 来源：W7 p5–8; Sample quiz p1 Q4; W1 p16, p48, p69–75 | T7-0002–0006, T7-0053

## S706 • [P1] Watch sensors: PPG, HRV and oxygen saturation / 手表传感器：PPG、HRV与血氧

Priority / 复习标记：P1 First pass/先掌握 · Sample Q4/样卷Q4 · Quiz stated/Quiz明确提及

**EN:** In an optical pulse sensor, an LED illuminates tissue and a photodetector records changes in received light as blood volume changes with each pulse: photoplethysmography, PPG. Pulse timing can provide heart rate and pulse-interval variability, used as a proxy for HRV in suitable conditions. An accelerometer separately measures movement. Software combines such clues to estimate sleep.

**中文：** 光学脉搏传感器用LED照组织，photodetector光探测器记录每次脉搏引起血容量变化时的收光变化，这叫photoplethysmography（PPG，光电容积描记）。脉搏时间可提供心率及脉搏间隔变化，适当条件下用作HRV心率变异性的代理指标。Accelerometer加速度计另行测运动，软件把这些线索结合来估计睡眠。

**EN:** PPG pulse sensing, $SpO_2$ oxygen-saturation estimation and EEG are different measurements. Oxygen saturation estimates blood oxygenation using wavelength-dependent absorption; it does not directly count brain-wave cycles. EEG electrodes record scalp electrical potentials; EMG electrodes record muscle electrical activity. For sample Q4, motion plus optical pulse/variability supports A. The fact that LEDs can also be used for $SpO_2$ does not establish D’s proposed brain-activity mechanism.

**中文：** PPG脉搏检测、$SpO_2$血氧饱和度估计与EEG是不同测量。血氧根据与波长有关的吸收来估计血液含氧，不直接数脑波周期。EEG电极记录头皮电位，EMG电极记录肌肉电活动。样题Q4中，“运动＋光学脉搏/间隔变化”支持A；LED也能用于$SpO_2$，不等于D提出的脑活动预测机制就成立。

Source / 来源：Sample quiz PDF p1 Q4; Week 7 PDF p7, p43–44, p79–85; Week 1 PDF p16, p48, p69–75; Week 2 pulse-oximetry section; T7-0053

## S707 • [P1] Elementary charge and electrical units / 元电荷与基础电学单位

Priority / 复习标记：P1 First pass/先掌握 · Quiz stated/Quiz明确提及

**EN:** Electrical charge is quantized: the charge of an electron is −e and that of a proton is +e, where e≈1.602×$10^{-19}$ C. One coulomb corresponds to about 6.24×$10^{18}$ elementary charge magnitudes. e is a charge magnitude; eV is an energy unit. The same numerical factor appears in converting 1 eV to joules, but coulombs and joules are different units.

**中文：** 电荷具有quantized量子化特性：电子带−e，质子带+e，其中e≈1.602×$10^{-19}$ C。1库仑约对应6.24×$10^{18}$个元电荷的大小。e是电荷量的大小，eV是能量单位；把1 eV换为焦耳时也出现同样的数值因子，但coulomb库仑与joule焦耳是不同单位。

**EN:** Current is the rate of charge transfer: I=ΔQ/Δt for an average, or dQ/dt instantaneously; 1 A=1 C/s. Voltage is energy per unit charge: 1 V=1 J/C. Resistance has units Ω=V/A. For example, transferring 2 μC in 1 ms gives an average current of 2 mA, not 2 μA.

**中文：** Current电流是电荷转移速率：平均电流I=ΔQ/Δt，瞬时电流为dQ/dt；1 A=1 C/s。Voltage电压是单位电荷对应的能量，1 V=1 J/C。电阻单位Ω=V/A。例如1 ms内转移2 μC，平均电流为2 mA，不是2 μA。

**EN:** Conventional current follows positive-charge motion. Thus an electron moving in one direction contributes current in the opposite direction. Positive charges tend toward lower electrical potential under an electric field; negative charges respond oppositely. Biological fluids carry charge mainly through ions, while metal wires carry it through electrons. The lecturer explicitly identified these basic charge/unit facts as items that may need recall rather than being supplied.

**中文：** Conventional current约定电流方向沿正电荷运动方向，因此电子向一个方向移动，对应的电流方向相反。在电场下，正电荷倾向移向较低电位，负电荷的响应相反。生物体液主要靠离子载流，金属导线主要靠电子载流。老师明确提到这些基础电荷与单位事实可能需要自行记住，不一定在题中提供。

Source / 来源：Week 7 PDF p9–13; T7-0007–0008

## B27 • [P1] Neurons, threshold and all-or-none firing / 神经元、阈值与全或无发放

Priority / 复习标记：P1 First pass/先掌握 · Repeated explanation/反复讲解 · Teacher emphasis/课堂强调

**EN: Key points**
• Dendrites receive input;
• the axon conducts action potentials.
• Beyond threshold, a neuronal action potential is approximately all-or-none;
• stronger stimulation commonly increases firing rate, not single-spike amplitude proportionally.
• Below threshold does not mean zero voltage: a typical neuronal resting value is about −70 mV, varying by cell.
• Passive recording measures activity;
• active stimulation applies an external electrical input.

**中文：必会知识点**
• Dendrite树突接收输入，axon轴突传导动作电位。
• 达到threshold阈值后，神经元动作电位近似all-or-none全或无；
• 更强刺激通常提高firing rate发放频率，而非成比例增大单个尖峰幅度。
• 未达阈值不等于零电压：典型神经元静息约−70 mV，不同细胞可不同。
• Passive recording被动记录测活动；
• active stimulation主动刺激施加外部电输入。

**EN:** Start from zero: A neuron is a signalling cell. Dendrites receive many inputs; the cell body integrates them; the axon carries action potentials toward other cells. Threshold is the level of excitation needed to trigger a spike. Below threshold, a neuron can still have a resting voltage and small voltage changes. For example, moving from −70 to −65 mV is electrical change even if it does not trigger an action potential.

**中文：** 从零理解：Neuron神经元是传递信号的细胞。Dendrites树突接收许多输入，cell body细胞体整合输入，axon轴突把动作电位传向其他细胞。Threshold阈值是触发尖峰所需的兴奋水平。未达到阈值时，神经元仍有静息电压，也可以有小幅电压变化。例如−70变成−65 mV，虽然未触发动作电位，却仍是电变化。

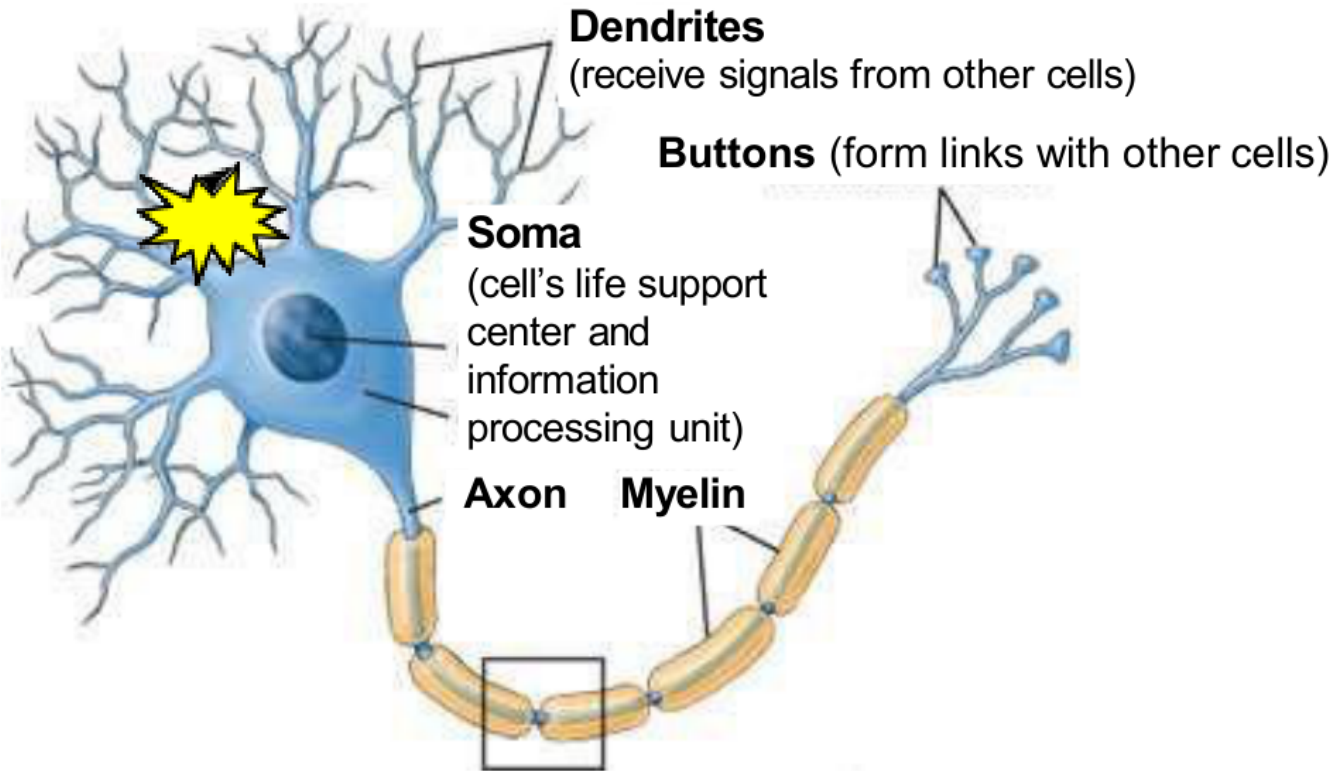
**EN:** Follow the mechanism: Beyond threshold, the taught neuronal spike is approximately all-or-none: it fires rather than being proportionally smaller. Stronger stimulation commonly produces more spikes per second, a higher firing rate. Think of sending more similar pulses to communicate stronger input. This describes the course model, not a promise of identical numerical spike heights in every cell and recording.

**中文：** 跟着机制走：达到阈值后，课程中的神经元尖峰近似all-or-none全或无：触发一个尖峰，而不是产生一个成比例缩小的版本。更强的刺激通常让每秒尖峰更多，这叫更高的firing rate发放频率。可以理解为“发送更多相似的脉冲”来表达更强输入。这针对课程模型，不保证每个细胞、每次记录的尖峰数值高度都完全相同。

**EN:** Try an example: Compare two teaching traces: both have similar spike heights, but one contains 5 spikes and the other 10 in one second. Their firing rates are 5 and 10 Hz; the second is more frequent, not necessarily larger in amplitude. Passive recording observes activity already occurring. Active stimulation deliberately applies an input to influence activity. In English questions, do not read “no spike” as “zero membrane voltage”.

**中文：** 用例子理解：比较两条教学记录：尖峰高度相似，一条在1秒有5个尖峰，另一条有10个。发放频率分别为5和10 Hz，后者更频繁，并不一定幅度更大。Passive recording被动记录观察已发生的活动；active stimulation主动刺激是人为施加输入来影响活动。英文题中的no spike“没有尖峰”，不能理解成“膜电压为零”。

Source / 来源：W7 p5–8; W7 p19–21; W7 p32–39 | T7-0002–0006, T7-0011–0012, T7-0023–0025, T7-0028–0030




A neuron's parts: dendrites, soma, axon, myelin and terminal buttons. / 神经元各部分：树突、胞体、轴突、髓鞘与末梢。

**EN:** Read the figure: Start at the branched dendrites, which receive inputs, then locate the soma and the long axon leading to the terminal buttons. The yellow sleeves are myelin around the axon; they are not extra electrodes. This is a single-cell structure diagram: an action potential propagates along its axon, whereas scalp EEG records activity contributed by many cells through tissue.

**中文：** 读图：先找分叉的dendrites树突，它们接收输入；再找到soma胞体，以及通往terminal buttons末梢的长axon轴突。轴突外黄色的一节节套层是myelin髓鞘，不是另加的电极。这是单细胞结构图：动作电位沿轴突传播，而头皮EEG是许多细胞的活动经组织共同形成的记录。

Source / 来源：Week 7 PDF p19; original slide figure / 原课件截图

## S701 • [P2] Ion channels, pumps and propagation / 离子通道、离子泵与动作电位传导

Priority / 复习标记：P2 Important foundations/重要基础 · Repeated explanation/反复讲解 · Supporting concept/辅助理解

**EN:** The sodium–potassium pump maintains the concentration gradients: it moves $Na^+$ outward and $K^+$ inward using metabolic energy. Ion channels instead allow selected ions to flow when their gates open. The rapid rising and falling phases of the taught neuronal action potential are mainly channel events. Repolarization is not simply the pump suddenly switching on.

**中文：** Sodium–potassium pump钠钾泵用代谢能维持浓度梯度：把$Na^+$向外运、$K^+$向内运。Ion channel离子通道则在门打开时让特定离子流动。课程神经元动作电位快速上升、下降，主要由通道的变化造成。Repolarization复极化不是“钠钾泵突然启动”的简单结果。

**EN:** Ligand-gated means that a chemical messenger opens a channel; voltage-gated means that membrane voltage controls the gate. Dendritic inputs can produce small graded changes. At the axon hillock or trigger zone, a sufficiently strong combined input reaches threshold and activates a spike. Threshold is the trigger level, not the height of the completed spike.

**中文：** Ligand-gated配体门控表示化学信使使通道打开；voltage-gated电压门控表示膜电压控制通道。树突输入可产生小幅graded change渐变电位。输入在axon hillock轴丘或trigger zone触发区综合，足够强、达到threshold阈值时触发尖峰。阈值是“启动所需的水平”，不是最终尖峰的高度。

**EN:** Propagation means that excitation triggers the next section of axon membrane, so the action potential is regenerated along the axon rather than merely fading like a passive voltage disturbance. In the course model, stronger stimulation produces more similar spikes, not a proportionally larger spike at every position. A spike travelling along an axon and a surface EEG trace are different recording scales.

**中文：** Propagation传导是指兴奋触发下一段轴突膜，因此动作电位沿轴突不断再生，而不只是像被动电压扰动那样逐渐变弱。课程模型中，更强刺激通常产生更多相似尖峰，不是让每个位点的单个尖峰成比例变大。沿轴突传播的尖峰与体表EEG记录，属于不同测量尺度。

**EN:** Supporting explanation: after a spike, channel inactivation and recovery produce a refractory period during which the membrane briefly cannot fire again normally, helping explain why spikes remain separate events. This term was not separately emphasized as a Quiz memorization requirement in the supplied slides.

**中文：** 辅助理解：尖峰后，通道失活与恢复带来refractory period不应期，使膜短暂不能正常再次发放，帮助解释尖峰为什么是分开的事件。本组课件未把此术语单独强调为Quiz必背要求。

Source / 来源：Week 7 PDF p19–21, p34–38; T7-0023–0025; 不应期辅助核实：OpenStax §12.4

## C21 • [P1] Membrane potential and action-potential phases / 膜电位与动作电位阶段

Priority / 复习标记：P1 First pass/先掌握 · Teacher emphasis/课堂强调 · Repeated explanation/反复讲解

**EN: Key points**

- Use $V_m$=Vinside−Voutside.
- A resting membrane is usually negative inside.
- Depolarization means becoming less negative and may include a positive overshoot;
- repolarization returns toward rest;
- hyperpolarization becomes more negative than rest.
- The taught sequence emphasizes $Na^+$ influx and subsequent $K^+$ efflux.

**中文：必会知识点**

- 使用$V_m$=Vinside−Voutside。
- 静息时细胞内通常为负。
- Depolarization表示变得不那么负，并可能出现正向超射；
- repolarization返回静息；
- hyperpolarization比静息更负。
- 课程重点顺序为$Na^+$内流，随后$K^+$外流。

**EN:** Start from zero: A cell membrane separates intracellular fluid, inside the cell, from extracellular fluid, outside. Ions are charged atoms or molecules: $Na^+$ is a positive sodium ion and $K^+$ a positive potassium ion. Membrane potential is $V_m$=Vinside−Voutside. If the outside is our zero reference and the inside is −70 mV, $V_m$ is −70 mV. Negative means lower electrical potential, not absence of electrical activity.

**中文：** 从零理解：Cell membrane细胞膜把intracellular细胞内液与extracellular细胞外液分开。Ion离子是带电的原子或分子：$Na^+$是带正电的钠离子，$K^+$是带正电的钾离子。Membrane potential膜电位定义为$V_m$=Vinside−Voutside。把细胞外作为零参考，细胞内为−70 mV时，$V_m$就是−70 mV。负值表示电位更低，不表示没有电活动。

**EN:** Follow the mechanism: In the taught neuronal action potential, sodium channels open and positive $Na^+$ enters. The inside becomes less negative: depolarization. Voltage can briefly become positive. Later sodium entry declines and $K^+$ leaves, bringing voltage downward: repolarization. Continued potassium movement can briefly make voltage more negative than rest: hyperpolarization. Channels are selective gates, but the analogy does not mean every ion can use every gate.

**中文：** 跟着机制走：课程的神经元action potential动作电位中，钠通道打开，正电$Na^+$进入，细胞内变得不那么负，这叫depolarization去极化；电压甚至会短暂变成正值。随后钠内流减弱，$K^+$向外移动，使电压下降，这叫repolarization复极化。钾继续外流可让电位暂时比静息更负，这叫hyperpolarization超极化。通道像有选择性的门，但不是每种离子都能通过每扇门。

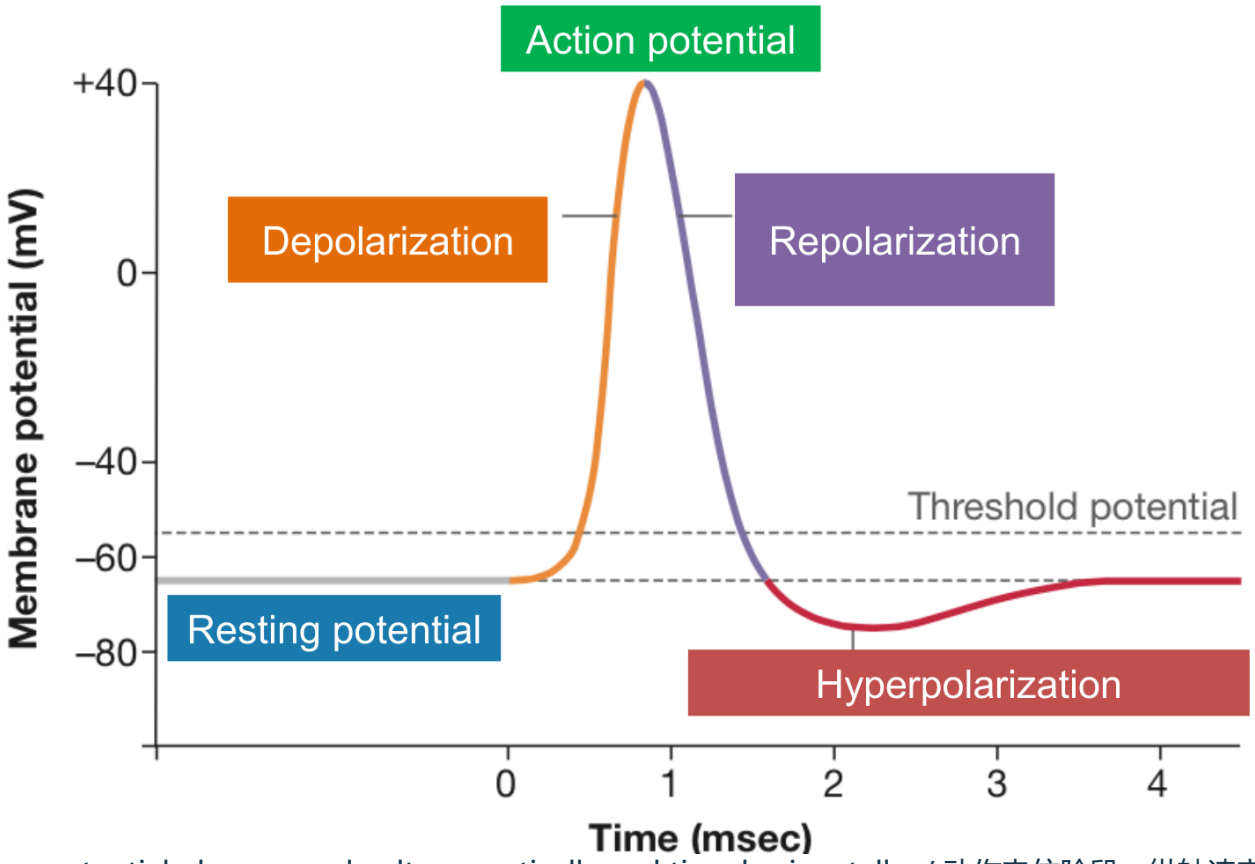
**EN:** Try an example: Follow the numbers: −70→−40 mV is depolarization, even though both values remain negative. +30→−70 mV is repolarization; −70→−80 mV is hyperpolarization relative to a −70 mV resting level. Thus becoming less negative is the key phrase. The statement “inside is always negative” fails during a positive overshoot. Remember 1 mV=0.001 V when calculating or reading axes.

**中文：** 用例子理解：沿数值判断：−70→−40 mV是去极化，虽然两个数都仍为负；+30→−70 mV是复极化；以−70 mV为静息基准，−70→−80 mV是超极化。因此becoming less negative“变得不那么负”是关键表达。“细胞内始终为负”遇到动作电位的正向超射就不成立。计算或读坐标时，记住1 mV=0.001 V。

**EN:** Exam wording: “The inside is always negative” is false during an action potential. Keep mV separate from V. Long multi-ion membrane equations and derivations are secondary to the phase names, ion directions and sign convention.

**中文：** 英文读题抓手：“细胞内始终为负”在动作电位中是错误的。mV与V不能混。长的多离子膜方程及推导，优先级低于阶段名称、离子流向和符号约定。

Source / 来源：W7 p22–27; W7 p19–21; W7 p32–39 | T7-0011–0012, T7-0014–0018, T7-0023–0025, T7-0028–0030




Action-potential phases: read voltage vertically and time horizontally. / 动作电位阶段：纵轴读电压，横轴读时间。

**EN:** Read the figure: Start near the resting negative voltage; rising is depolarization, falling is repolarization, and the dip below rest is hyperpolarization. The dashed threshold is a trigger level, not peak height. The drawn numerical values illustrate this model rather than every cell.

**中文：** 读图：从负的静息值开始，上升是去极化、下降是复极化、降到静息以下是超极化。虚线threshold是启动阈值，不是峰高。图中数值示范该模型，不代表所有细胞都相同。

Source / 来源：Week 7 PDF p35; original slide figure / 原课件截图

## B28 • [P2] Ionic gradients: Nernst versus GHK / 离子梯度：Nernst与GHK

Priority / 复习标记：P2 Important foundations/重要基础 · Teacher emphasis/课堂强调 · Lower priority/降低优先级

**EN: Key points**

- Membrane voltage arises from ion gradients and selective permeability: typically more $K^+$ inside and $Na^+$ outside.
- Nernst describes the equilibrium potential for one ion;
- Goldman–Hodgkin–Katz (GHK) includes several ions weighted by permeability.
- Zero net membrane current does not require each ionic current to be zero.
- Learn the meaning of the equations without prioritizing their long derivations.

**中文：必会知识点**

- 膜电压来自离子梯度与选择性通透：通常$K^+$胞内更多、$Na^+$胞外更多。
- Nernst能斯特描述单种离子的平衡电位；
- Goldman–Hodgkin–Katz（GHK）按通透性考虑多种离子。
- 跨膜净电流为零不要求每种离子电流都为零。
- 记住方程含义，长推导不作为优先任务。

**EN:** Start from zero: An ion concentration gradient means unequal amounts per unit volume on two sides. Typically $K^+$ is more concentrated inside a neuron and $Na^+$ outside. An open channel permits selected ions to cross: this is permeability. For $K^+$, concentration favours outward movement, while a negative interior attracts positive $K^+$ inward. The combination of concentration and electrical effects determines movement, not concentration alone.

**中文：** 从零理解：Ion concentration gradient离子浓度梯度，表示两侧单位体积内的离子数量不同。神经元通常$K^+$在细胞内更多，$Na^+$在细胞外更多。打开的通道允许特定离子通过，这涉及permeability通透性。对$K^+$而言，浓度差推动其向外，而负的细胞内部会吸引正$K^+$向内。运动方向由浓度与电效应共同决定，不能只看浓度。

**EN:** Follow the mechanism: Nernst asks a single-ion question: what voltage balances the concentration drive for this ion, giving zero net flow of that ion under its equilibrium assumptions? GHK considers several ions and their different permeabilities. Imagine several selective doors: how open each door is affects the combined outcome. This is a guide to weighting; GHK voltage is not simply an ordinary arithmetic average of the individual Nernst voltages.

**中文：** 跟着机制走：Nernst能斯特回答单离子问题：什么电压能平衡这一种离子的浓度驱动力，让它在相应平衡假设下的净流为零？GHK同时考虑多种离子及其不同通透性。可以想成多扇有选择性的门：每扇门开得多大，会影响综合结果。这用于理解“权重”，但GHK膜电压不是把各离子的Nernst电压直接做普通算术平均。

**EN:** Try an example: For conceptual questions, “one ion’s equilibrium potential” points to Nernst; “multiple ions weighted by permeability” points to GHK. Zero total current does not require every ionic current to be zero. In a simple teaching example, +2 units from one contribution and −2 from another sum to zero while both remain nonzero. You should understand this balance before trying a long equation or memorizing a particular resting voltage.

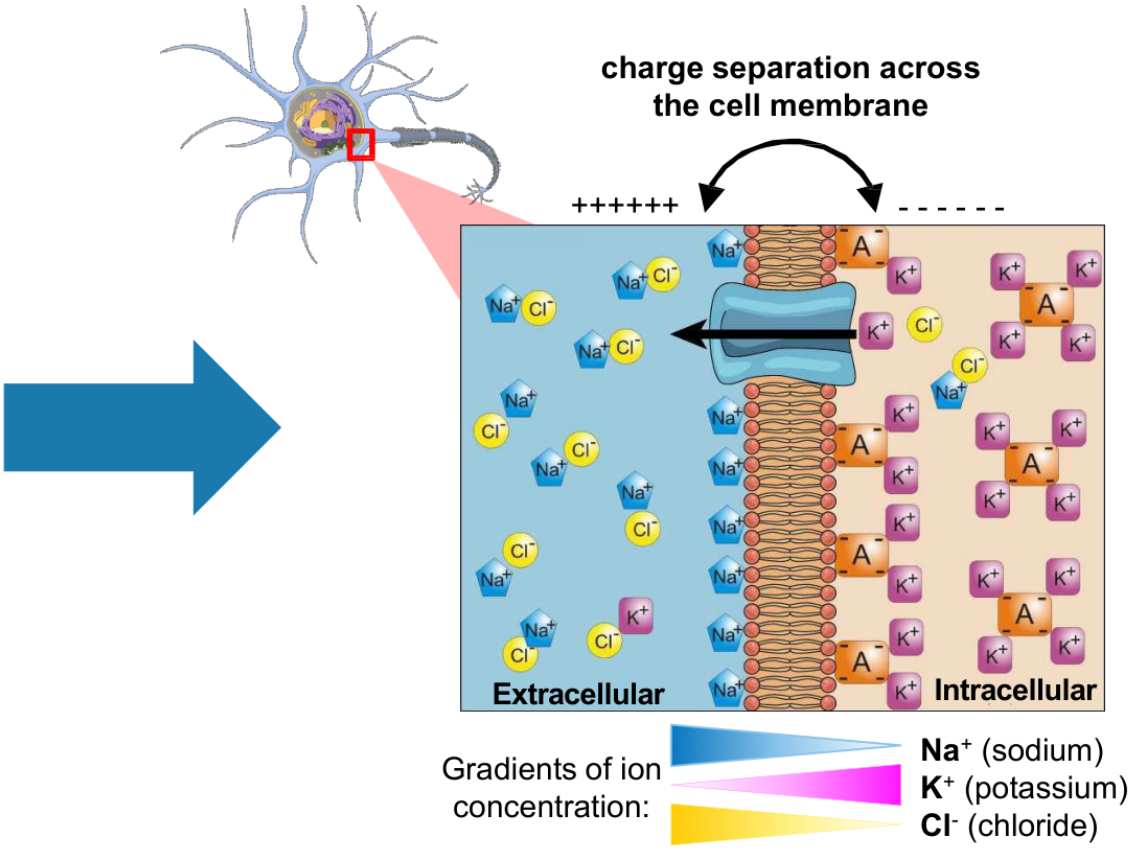
**中文：** 用例子理解：概念题看到one ion’s equilibrium potential“单种离子的平衡电位”，对应Nernst；看到multiple ions weighted by permeability“按通透性考虑多种离子”，对应GHK。总电流为零，不要求每种离子的电流都为零。一个简单例子是：某项贡献+2单位，另一项−2单位，合计为零，但两项各自都不为零。先理解这种平衡，再去看长公式或背某个静息电压数值。

**EN:** Clarify the concept: In the slide’s equivalent-circuit branch, Iion=gion(Vm−Eion), with conductance gion=1/Rion. Vm−Eion expresses the electrochemical driving force as a voltage relative to that ion’s equilibrium potential. If Vm=Eion, this branch’s net current is zero in the model, even with an open channel. That is different from GHK’s zero sum of several nonzero ionic currents. Treat this as supporting interpretation, not a required long derivation.

**中文：** 把概念讲清：课件等效电路单支路满足Iion=gion(Vm−Eion)，gion=1/Rion为电导。Vm−Eion把相对该离子平衡电位的电化学驱动力表示成电压。模型中Vm=Eion时，该支路净电流为零，即使通道打开；这不同于GHK中多个非零离子电流相加为零。作为辅助读式即可，不增加长推导要求。

Source / 来源：Week 7 PDF p25, p30; T7-0017–0021明确reference；当前B28未出现该短式；README行578

Source / 来源：W7 p15–16; W7 p22–27; W7 p25–30 | T7-0008–0010, T7-0014–0021, T7-0030




Inside and outside the membrane: ion concentration gradients and a selective channel. / 细胞膜内外：离子浓度梯度与选择性通道。

**EN:** Read the figure: The enlarged membrane separates extracellular fluid on the left from the intracellular region on the right. Read the colored wedges below: Na+ and Cl− are more concentrated outside in this model, while K+ is more concentrated inside. The black arrow illustrates K+ movement through a selective channel; actual net ionic flow depends on both concentration and electrical driving forces. The +/− signs show charge separation near the membrane, not that the whole intracellular fluid contains only negative ions.

**中文：** 读图：放大的细胞膜把左侧extracellular细胞外与右侧intracellular细胞内分开。看下面彩色楔形：此模型中Na+和Cl−在外面较多，K+在里面较多。黑箭头示意K+通过选择性通道；实际净流动由浓度驱动力和电驱动力共同决定。膜两侧“+” “−”表示膜附近的电荷分离，不表示整个细胞内液只有负离子。

Source / 来源：Week 7 PDF p22; original slide figure / 原课件截图

## B29 • [P2] Electrode–electrolyte interface / 电极—电解质界面

Priority / 复习标记：P2 Important foundations/重要基础 · Repeated explanation/反复讲解 · Final emphasis/仅期末强调

**EN: Key points**

• Tissue carries ionic current;
• metal leads carry electronic current.
• The interface introduces a half-cell potential and frequency-dependent polarization/contact impedance.
• The course equivalent circuit contains a half-cell source, an interfacial resistance/capacitance branch and solution resistance.
• Poor or unequal contact can distort recorded signals;
• a pure series-resistor exercise is only a simplified model.

**中文：必会知识点**

• 组织中为离子电流，金属导线中为电子电流。
• 界面引入half-cell potential半电池电位及随频率变化的极化/接触阻抗。
• 课程等效电路含半电池电源、界面电阻/电容支路及溶液电阻。
• 接触差或两侧不匹配可能使记录失真；
• 纯串联电阻练习只是简化模型。

**EN:** Start from zero: An electrolyte is a fluid containing mobile ions, such as dissolved salts. Ions carry electrical charge in tissue fluids; electrons carry charge in metal wires. At the electrode–electrolyte interface, these two conducting systems meet. The electrode is therefore more than a perfect wire touching skin: its surface interacts with the fluid and adds its own electrical behaviour to the measurement.

**中文：** 从零理解：Electrolyte电解质是含有可移动离子的流体，例如溶有盐的液体。组织液里由离子携带电荷，金属导线里由电子携带电荷。Electrode–electrolyte interface电极—电解质界面，就是这两种导电系统的交界。因此，电极不只是“贴着皮肤的一根完美导线”：它的表面会与流体相互作用，把自身的电学特性也加入测量。

**EN:** Follow the mechanism: The course represents this interface with an equivalent circuit, a simplified electrical model. Ehc is the half-cell potential, a surface-related voltage offset. Rs represents resistance in the fluid. Rd and Cd represent interfacial resistance and capacitance associated with charge transfer and polarization. Capacitance allows charge separation at the surface, so response depends on changing frequency. These components explain behaviour; they are not literal separate parts hidden under the electrode.

**中文：** 跟着机制走：课程用equivalent circuit等效电路表示界面，即用简化的电学模型描述它。Ehc是half-cell potential半电池电位，代表与表面相关的电压偏移；Rs代表液体电阻。Rd、Cd代表与电荷转移及极化有关的界面电阻、电容。表面可以形成电荷分离，因此响应会随变化频率而变。这些元件用于解释行为，不是说电极下面真的藏着几个独立电路零件。

**EN:** Try an example: Suppose the biological voltage stays unchanged but electrode contact shifts. The recorded voltage can change because contact impedance or surface potential changed: this is an artifact. Two unequal contacts can also introduce unequal errors that differential subtraction cannot fully cancel. Thus a pure series-resistor problem is useful for practising V=IR, but it omits the full interface. In English, look for contact impedance, polarization and half-cell potential as interface terms.

**中文：** 用例子理解：假设生物电压没变，但电极接触发生改变。记录仍可能变化，因为contact impedance接触阻抗或表面电位变了，这叫artifact伪迹。两侧接触不同，还会引入不同误差，差分相减不能完全抵消。因此，纯串联电阻题适合练V=IR，却没有包括完整界面。英文中，contact impedance、polarization极化、half-cell potential都是与界面有关的词。

Source / 来源：W7 p41–42; W7 p52–54; W7 p70–78 | T7-0035–0038, T7-0043–0046, T7-0073–0081

## S702 • [P2] Skin contact and interface impedance / 皮肤接触与界面阻抗

Priority / 复习标记：P2 Important foundations/重要基础 · Repeated explanation/反复讲解 · Final emphasis/仅期末强调

**EN:** Surface electrodes do not have to touch the muscle itself. Ionic currents in conductive tissue fluids produce extracellular voltage differences that can be detected at the skin: this is volume conduction. Conductive gel improves electrode–skin contact. Changing contact pressure, drying gel or motion can change contact impedance and add an artifact even if muscle activity stays the same.

**中文：** 表面电极不必直接碰到肌肉。导电组织液中的离子电流产生细胞外电压差，可在皮肤检测，这涉及volume conduction容积传导。Conductive gel导电凝胶改善电极—皮肤接触。接触压力变化、凝胶变干或运动，都可能改变接触阻抗，加入artifact伪迹，即使肌肉活动没有改变。

**EN:** For the slide model Rs+(Rd in parallel with Cd), low frequency makes the capacitor branch difficult to pass, so the impedance approaches Rs+Rd. At high frequency the capacitor branch becomes easier to pass and bypasses Rd, so the impedance approaches Rs. Thus a single fixed resistor cannot describe this interface at every frequency. Ehc is a voltage offset and is not the same quantity as impedance.

**中文：** 课件模型为Rs与“Rd、Cd并联支路”串联。低频时，电容支路不易通过，阻抗趋近Rs+Rd；高频时，电容支路更易通过、旁路Rd，阻抗趋近Rs。因此，不能用一个固定电阻描述所有频率下的界面。Ehc是电压偏移，与impedance阻抗不是同一个量。

**EN:** Differential measurement cancels interference only to the extent that it reaches both inputs equally. Unequal electrode contact impedances can turn shared mains interference into unequal input voltages, leaving a residual after subtraction. This explains the ECG interference example conceptually; its long resistor-network calculation is secondary to good contact, matched inputs and common-mode rejection.

**中文：** 差分测量能消除干扰，前提是干扰到达两输入端时足够相同。两侧电极接触阻抗不匹配，可把共有工频干扰变成不同的输入电压，相减后仍有残留。这解释了ECG干扰例题的基本原理；长的电阻网络计算，优先级低于良好接触、输入匹配与common-mode rejection共模抑制。

Source / 来源：Week 7 PDF p41–42, p49–54, p74–78; T7-0035–0046, T7-0073–0081

## C22 • [P1] ECG morphology and reading frequency from a trace / ECG波形与从时轴读频率

Priority / 复习标记：P1 First pass/先掌握 · Sample Q5/样卷Q5 · Teacher emphasis/课堂强调 · Repeated explanation/反复讲解

**EN: Key points**

- The P wave relates to atrial depolarization, the QRS complex to ventricular depolarization, and the T wave to ventricular repolarization.
- For a reasonably periodic trace, frequency f=N/Δt or f=1/T.
- Always read the horizontal-axis unit first.

**中文：必会知识点**

- P wave对应心房去极化，QRS complex对应心室去极化，T wave对应心室复极化。
- 近似周期波形的频率f=N/Δt或f=1/T。
- 必须先读横轴单位。

**EN:** Start from zero: An ECG is a voltage-versus-time trace. The atria are the heart’ s upper chambers; the ventricles are its lower pumping chambers. P marks atrial depolarization, QRS ventricular depolarization, and T ventricular repolarization. These labels describe electrical stages, which are related to contraction but are not identical to mechanical movement. P, QRS and T are parts of a waveform, not three separate ECG leads.

**中文：** 从零理解：ECG心电图是“电压随时间变化”的曲线。Atria心房是心脏上方的腔室，ventricles心室是下方负责泵血的腔室。P表示心房去极化，QRS表示心室去极化，T表示心室复极化。这些标记描述电活动阶段，虽然与收缩有关，却不直接等同于机械运动。P、QRS、T是波形中的部分，不是三个独立的心电导联。

**EN:** Follow the mechanism: Frequency asks how many complete repetitions occur each second; its unit is hertz, Hz. Period T asks how long one repetition takes. Therefore f=1/T, or f=N/Δt for N complete cycles in a time window Δt. Picture repeatedly walking the same loop: count completed loops, not every step. For a waveform, do not count both its upward and downward portions as separate cycles.

**中文：** 跟着机制走：Frequency频率问的是“每秒完整重复几次”，单位hertz赫兹，记作Hz。Period周期T问的是“一次完整重复要多久”。所以f=1/T；若时间窗口Δt里有N个完整周期，则f=N/Δt。可以想成反复走同一圈：数走完的圈数，不数每一步。对应波形时，不能把向上和向下两段各算成一个周期。

**EN:** Try an example: Suppose a teaching trace shows 10 complete cycles in 200 ms. Convert 200 ms to 0.200 s, then calculate f=10/0.200=50 Hz. Eleven successive matching peaks contain ten peak-to-peak intervals. An irregular EEG trace may support only a rough estimate: choose a representative window and state its uncertainty. A taller wave means larger voltage amplitude; it does not automatically mean higher frequency.

**中文：** 用例子理解：假设教学波形在200 ms中有10个完整周期。先把200 ms换成0.200 s，再算f=10/0.200=50 Hz。连续11个同类波峰之间，只有10个峰到峰的间隔。实际EEG不规则时，可能只能估计频率：选代表性的时间窗口，并承认估读误差。波形更高表示电压幅度更大，并不自动表示频率更高。

**EN:** Exam wording: Sample Q5 is a rough estimate from a low-resolution, irregular EEG trace, around 45 Hz with C (40–48 Hz) only a provisional reference, not a confirmed answer. Treat it as graph-reading practice rather than an exact numerical standard.

**中文：** 英文读题抓手：样题Q5来自低分辨率且不规则的EEG波形，仅能估读约45 Hz，C（40–48 Hz）仅为暂定参考，不是已确认答案。应把它作为读图训练，而非精确数值标准。

Source / 来源：W7 p62–69; W7 p79–87; Sample quiz p1 Q5 | T7-0068–0073, T7-0084–0087, T7-0091–0095

## S703 • [P2] Heart pumping and the electrical sequence / 心脏泵血与电活动顺序

Priority / 复习标记：P2 Important foundations/重要基础 · Repeated explanation/反复讲解 · Lower priority/降低优先级

**EN:** Atria are receiving chambers and ventricles are pumping chambers. The right side sends oxygen-poor blood from the body to the lungs; the left side sends oxygen-rich blood from the lungs to the body. Heart valves promote one-way flow. Left and right describe the person’ s anatomy, not necessarily the left and right of a drawing.

**中文：** Atria心房主要接收血液，ventricles心室负责泵血。右侧把身体返回的低氧血送往肺，左侧把肺返回的富氧血送往全身。Heart valve心脏瓣膜帮助维持单向血流。Left/right指人体解剖学左右，未必等于图纸的左右。

**EN:** Systole is the contraction or pumping phase; diastole is the relaxation and filling phase. An ECG records the electrical activation associated with this cycle, not blood flow itself. Electrical depolarization helps initiate contraction, but the voltage trace and the mechanical contraction are different quantities.

**中文：** Systole收缩期是收缩、泵血阶段；diastole舒张期是放松、充盈阶段。ECG记录与这个周期有关的电激活，并不直接测血流。电去极化帮助启动收缩，但电压曲线与机械收缩是不同的量。

**EN:** The SA node, sinoatrial node, is the usual pacemaker that starts the normal heartbeat. Activation spreads through the atria and then through the AV node, atrioventricular node, before ventricular activation. The ordered timing helps the atria and ventricles work in sequence. Pacemaker cells set the timing; the larger muscle tissue performs most of the pumping.

**中文：** SA node是sinoatrial node窦房结，通常作为pacemaker起搏点启动正常心搏。电激活经过心房，再经过AV node（atrioventricular node房室结），随后激活心室。这个有顺序的时间安排让心房、心室依次工作。起搏细胞提供节律，较大的肌肉组织完成主要泵血。

Source / 来源：Week 7 PDF p57–65; T7-0058–0069

## S704 • [P2] ECG waves, segments and intervals / 心电的波、段与间期

Priority / 复习标记：P2 Important foundations/重要基础 · Supporting concept/辅助理解 · Lower priority/降低优先级

**EN:** A wave is a visible deflection; a segment is the portion between waves; an interval includes a specified time span containing one or more waves. Do not confuse a flat-looking segment with a zero membrane potential in every heart cell. The ECG is a difference recorded at the body surface, and different cell activities can cancel in that difference.

**中文：** Wave波是曲线可见的偏转；segment段是波与波之间的部分；interval间期是包含一个或多个波的指定时间跨度。看起来平的心电段，不表示每个心肌细胞的膜电位都为零。ECG是体表记录的电位差，不同细胞活动在这个差值中可以相互抵消。

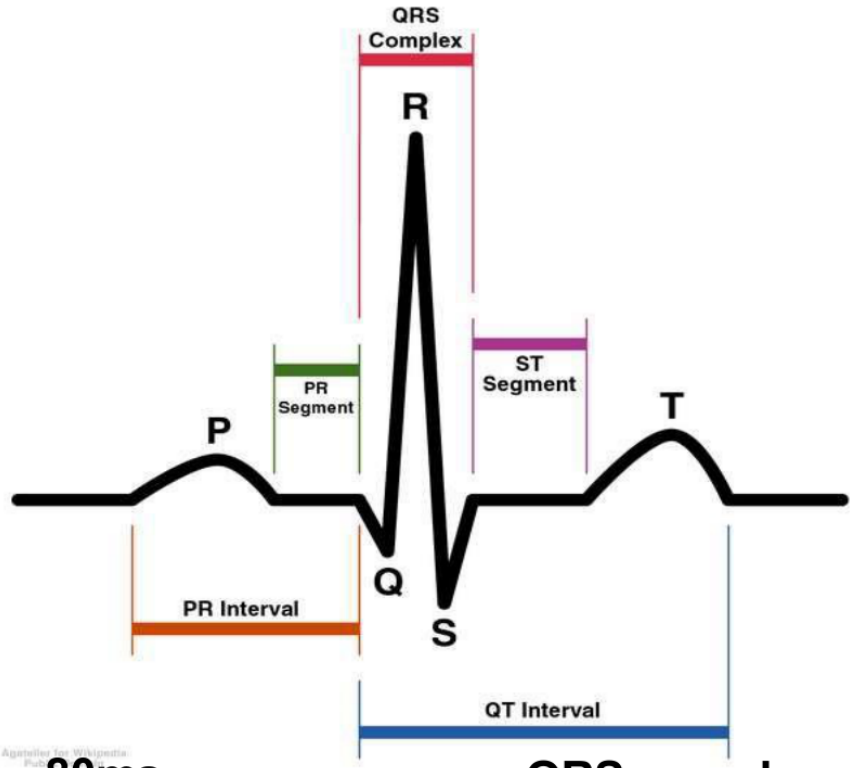
**EN:** PR interval runs from the start of P to the start of QRS: from the start of atrial depolarization to the start of ventricular depolarization. ST segment lies after QRS and before T. QT interval runs from the start of QRS to the end of T, covering ventricular depolarization through repolarization. RR is instead the spacing between successive R peaks, used for heart-rate calculation.

**中文：** PR interval从P波开始到QRS开始，对应心房开始去极化到心室开始去极化。ST segment在QRS之后、T波之前。QT interval从QRS开始到T波结束，包含心室去极化至复极化。RR则是连续R峰的间隔，用来计算心率。

**EN:** Atrial repolarization is usually hidden by the larger QRS complex; the T wave is ventricular repolarization. The slide shows approximate P, PR, QRS and T durations for orientation. Understanding which event and time span each label represents is more useful here than treating those approximate durations as universal constants or as a diagnosis.

**中文：** 心房复极化通常被较大的QRS波群遮住；T波表示心室复极化。课件给出的P、PR、QRS、T持续时间是帮助认识时间尺度的近似值。这里更应理解每个标签表示什么事件、覆盖哪段时间，而非把近似时长当成普适常数或诊断结论。

Source / 来源：Week 7 PDF p65–67; T7-0068–0070; 图示定义核实：OpenStax §19.2




ECG wave labels and intervals from the lecture slide. / 原课件的心电波与间期标记。

**EN:** Read the figure: Locate P, QRS and T first, then follow the horizontal spans of PR, ST and QT. These are features within a trace, not limb lead I/II/III. Durations on this slide are approximate reference values.

**中文：** 读图：先找到P、QRS、T，再沿横向看PR、ST、QT覆盖的时间。它们是同一曲线内的特征，不是肢体导联I/II/III。页上时长属于近似参考值。

Source / 来源：Week 7 PDF p66; original slide figure / 原课件截图

## B30 • [P2] ECG leads, heart rate and EMG / 心电导联、心率与肌电

Priority / 复习标记：P2 Important foundations/重要基础 · Repeated explanation/反复讲解 · Supporting concept/辅助理解 · Lower priority/降低优先级

**EN: Key points**

- An ECG lead is a potential-difference view;
- P, QRS and T are features within a waveform, not three different leads.
- For standard limb leads, II=I+III.
- Heart rate=60/RR when RR is in seconds;
- RR=0.8 s gives 75 beats/min.
- EMG measures muscle electrical activity, with electrode placement and reference affecting the recorded signal.

**中文：必会知识点**

- ECG lead心电导联是一种电位差观察视角；
- P、QRS、T是同一波形内的特征，不是三个不同导联。
- 标准肢体导联满足II=I+III。
- RR用秒时，heart rate心率=60/RR；
- RR=0.8 s对应75次/分。
- EMG肌电测肌肉电活动，电极位置与参考位置影响记录。

**EN:** Start from zero: An electrode is a physical contact; an ECG lead is a defined voltage difference between contacts or references. Think of viewing the same event from different camera angles: different leads view the same heart activity differently. P, QRS and T are stages visible within a trace. They are not lead I, lead II and lead III, and changing the lead can change waveform appearance.

**中文：** 从零理解：Electrode电极是实际接触部件；ECG lead心电导联是按定义取得的电极或参考之间的电压差。可以类比不同相机角度拍同一事件：不同导联从不同电位差视角观察同一心脏活动。P、QRS、T是曲线里的阶段，不是lead I、lead II、lead III三个导联；换导联后，波形外观可能改变。

**EN:** Follow the mechanism: For the three limb contacts in the slide, $I=V_2-V_1$, $II=V_3-V_1$ and $III=V_3-V_2$. Adding I and III cancels $V_2$, leaving II. Example: $V_1$=1.0, $V_2$=1.2 and $V_3$=1.5 mV give I=0.2, III=0.3 and II=0.5 mV. This is a voltage identity for the stated lead definitions, not a claim that three different heart stages are being added together.

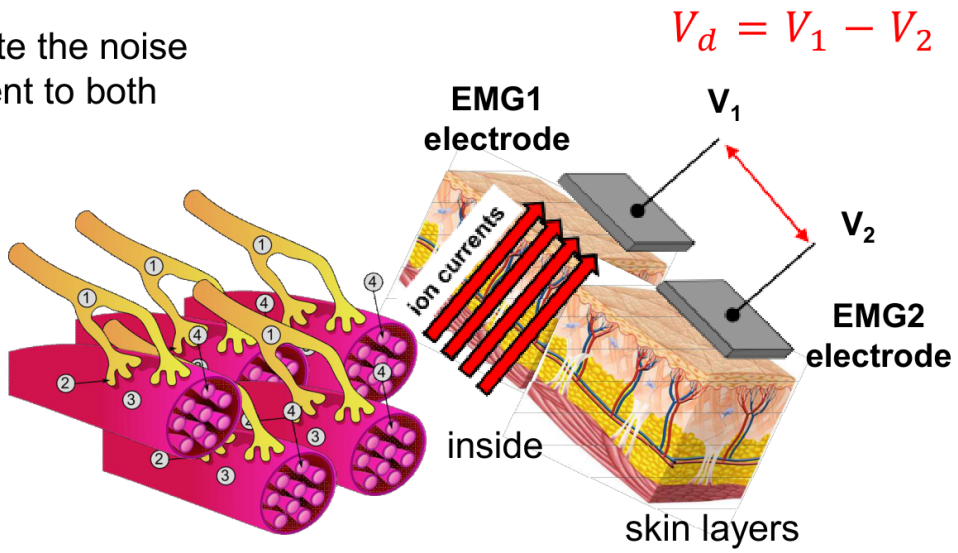
**中文：** 跟着机制走：课件三个肢体接触点的定义为$I=V_2-V_1$、$II=V_3-V_1$、$III=V_3-V_2$。把I与III相加，$V_2$抵消，得到II。例如$V_1$=1.0、$V_2$=1.2、$V_3$=1.5 mV时，I=0.2、III=0.3、II=0.5 mV。这是按所给导联定义成立的电压关系，并不是把三个不同心脏阶段相加。

**EN:** Try an example: The RR interval is the time from one R peak to the next. If RR=0.8 s, one beat takes 0.8 s: 1/0.8=1.25 beats per second, then ×60 gives 75 beats per minute. This is a teaching calculation, not a diagnosis. EMG instead records muscle electrical activity; which muscle and where the recording and reference electrodes sit affect the signal. Heart-rate arithmetic does not automatically apply to arbitrary EMG spikes.

**中文：** 用例子理解：RR interval是一个R峰到下一个R峰的时间。如果RR=0.8 s，一次心搏用0.8秒，那么1/0.8=1.25次/秒，再乘60，得到75次/分。这是教学计算，不是诊断。EMG记录的是肌肉电活动，测哪块肌肉、记录电极与参照电极放哪里，都会影响信号。不能把心率算法自动套到任意肌电尖峰上。

Source / 来源：W7 p47–48; W7 p52–54; W7 p62–69 | T7-0038–0039, T7-0043–0046, T7-0068–0073

In this way, we attenuate the noise that is commonly present to both electrodes.




Surface EMG: two electrode potentials form the differential signal Vd=V1−V2. / 表面肌电：两个电极的电位形成差分信号Vd=V1−V2。

**EN:** Read the figure: Find the two black electrode points above the skin; they measure local potentials V1 and V2 arising from muscle electrical activity conducted through tissue. The red voltage arrow and formula indicate subtraction, not two separate kinds of muscle. Noise that reaches both inputs with the same value cancels ideally; unequal interference remains. Surface EMG generally contains contributions from multiple muscle fibers, rather than the intracellular voltage of one fiber.

**中文：** 读图：找皮肤上方两个电极的黑点：它们测V1与V2两个局部电位，来自肌肉电活动经导电组织的传播。红色电压箭头与公式表示相减，不是两种不同肌肉。相同数值同时进入两输入的噪声在理想相减中抵消；不相同的干扰仍可留下。表面EMG通常包含多条肌纤维的贡献，不是一条肌纤维的细胞内电压。

Source / 来源：Week 7 PDF p53; original slide figure / 原课件截图

## B31 • [P2] EEG placement and brain-wave vocabulary / 脑电位置与脑波词汇

Priority / 复习标记：P2 Important foundations/重要基础 · Teacher emphasis/课堂强调 · Repeated explanation/反复讲解

**EN: Key points**

- Scalp EEG reflects population activity;
- the 10–20 system standardizes electrode positions.
- Slow-to-fast order: delta, theta, alpha, beta, gamma.
- W7 p84 lists delta 1–3, theta 4–7, alpha 8–12, beta 13–25 Hz;
- exact boundaries vary by convention, and its gamma <25 label is erroneous.
- Alpha often accompanies relaxed wakefulness, delta slow-wave sleep;
- these are associations, not diagnoses.

**中文：必会知识点**

- 头皮EEG反映群体活动，10–20 system规范电极位置。
- 由慢到快为delta δ、theta θ、alpha α、beta β、gamma γ。
- W7 p84列出δ 1–3、θ 4–7、α 8–12、β 13–25 Hz；
- 具体边界随约定变化，其中γ<25为错误标注。
- Alpha常伴随放松清醒，delta常见于慢波睡眠；
- 这是相关表现，不是诊断规则。

**EN:** Start from zero: Scalp EEG records small voltage differences associated with the combined activity of many neurons, especially coordinated synaptic activity. It is not simply a direct recording of one neuron’ s action potential. The international 10–20 system gives standard scalp positions, so recordings can use comparable locations. Imagine microphones placed consistently around a crowd; placement matters. This analogy describes collective measurement, not actual sound or precise identification of every individual neuron.

**中文：** 从零理解：头皮EEG记录与许多神经元共同活动有关的微小电压差，尤其与协调的突触活动有关，并不只是直接记录单个神经元的动作电位。国际10–20 system给出标准头皮位置，方便在可比较的位置记录。可以类比把麦克风按固定位置放在人群周围：位置很重要。但这个类比只用于理解群体测量，不是实际声音，也不表示能准确辨认每一个神经元。

**EN:** Follow the mechanism: Brain-wave bands group oscillations by frequency: delta, theta, alpha, beta, gamma run from slower to faster. Read time on the horizontal axis and count complete cycles to estimate frequency; vertical height describes voltage amplitude. The slide lists delta 1–3, theta 4–7, alpha 8–12 and beta 13–25 Hz. These are course conventions, not universal boundaries. Its gamma “<25 Hz” label is inconsistent and should not be memorized.

**中文：** 跟着机制走：脑波频带按频率分组，从慢到快是delta δ、theta θ、alpha α、beta β、gamma γ。读横轴时间、数完整周期来估频率；纵向高度表示电压幅度。课件列δ 1–3、θ 4–7、α 8–12、β 13–25 Hz。这些是课程采用的约定，并非唯一通用边界。其gamma“<25 Hz”标注有矛盾，不要背这个错误标签。

**EN:** Try an example: A teaching trace with ten cycles in one second is 10 Hz, fitting the slide’ s alpha range. Four cycles in one second would be slower despite taller peaks. Alpha often accompanies relaxed wakefulness and delta slow-wave sleep, but these are associations rather than individual diagnostic rules. For English multiple-choice questions, distinguish “higher amplitude” from “higher frequency” and retain the slow-to-fast band order before memorizing numerical limits.

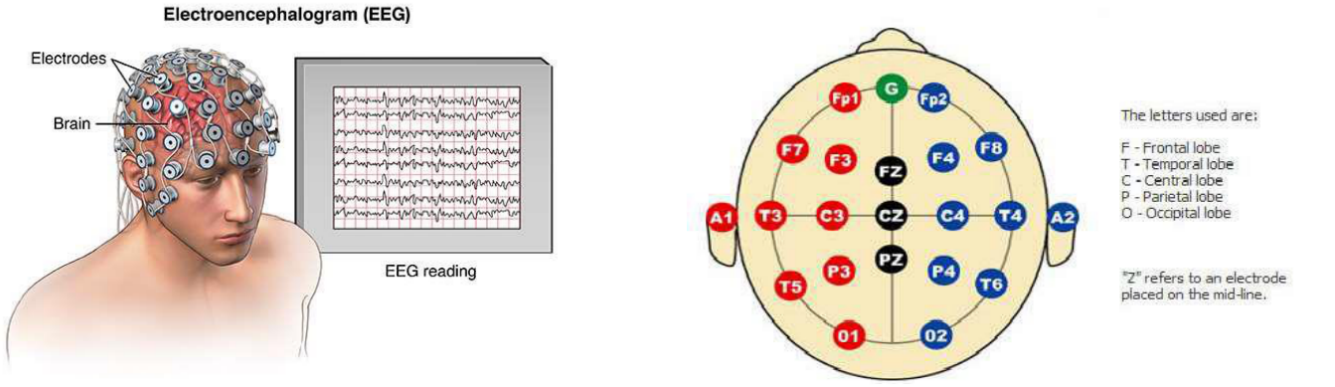
**中文：** 用例子理解：教学曲线若1秒有10个周期，就是10 Hz，落在课件的alpha范围内。另一条曲线1秒只有4个周期，即使峰更高，频率仍更慢。Alpha常伴随放松清醒，delta常见于慢波睡眠，但这只是相关表现，不是个人诊断规则。英文选择题要分清higher amplitude幅度更大与higher frequency频率更高；先记由慢到快的顺序，再记数值边界。

**EN:** Clarify the concept: In the conventional 10–20 labels shown, odd numbers identify left-hemisphere sites and even numbers right-hemisphere sites; z marks the midline. F/T/C/P/O label regions. These are spatial recording-site labels, not frequency bands, and the left/right convention refers to the subject.

**中文：** 把概念讲清：图示常规10–20编号中，奇数对应左侧、偶数对应右侧，z表示中线；F/T/C/P/O标区域。这些是空间记录位置标签，不是频率带，左右以被测者为准。

Source / 来源：Week 7 PDF p82原位置图可核对；T7-0085–0087强调位置；当前FIGV704图解；README行654

Source / 来源：W7 p79–87; Sample quiz p1 Q5 | T7-0084–0087, T7-0091–0095




Scalp EEG recording and the international 10–20 electrode layout. / 头皮EEG记录与国际10–20电极布局。

**EN:** Read the figure: The left picture links a scalp electrode cap to several recorded traces. The right top view shows named recording sites: F, T, C, P and O refer to frontal, temporal, central, parietal and occipital regions; z marks the midline. These labels identify electrode positions, not the names of brain-wave frequency bands. Multiple electrodes sample different spatial combinations of brain activity; a recording site is not an isolated single neuron.

**中文：** 读图：左图把头皮电极帽与多条记录曲线连在一起；右图俯视头部，标出各记录位置。F、T、C、P、O分别对应额、颞、中央、顶、枕区域；z表示中线。这些标签是电极位置，不是脑波频段名称。多个电极采样脑活动在不同位置的组合，不是每个位置只对应一颗神经元。

Source / 来源：Week 7 PDF p82; original slide figure / 原课件截图

## S705 • [P1] Brain-wave states and timing / 脑波状态与时间窗口

Priority / 复习标记：P1 First pass/先掌握 · Sample Q5/样卷Q5 · Teacher emphasis/课堂强调

**EN:** The slide associates beta with awake, alert activity; alpha with relaxed wakefulness; theta with deeper relaxation or drowsier states; and delta with deep sleep. Gamma is a faster band. These associations overlap and depend on recording context. A single frequency label does not uniquely prove a sleep stage, a thought or a disease.

**中文：** 课件把beta β与清醒、警觉联系，把alpha α与放松清醒联系，把theta θ与更深放松或较困倦状态联系，把delta δ与深睡联系；gamma γ是更快的频带。这些关联有重叠，也依赖记录情境。单个频率标签不能唯一证明某个睡眠阶段、想法或疾病。

**EN:** When the visible window starts at 0.2 s and ends at 0.8 s, its duration is 0.6 s, not 0.8 s and not one second. Count complete repetitions within that duration, then divide by 0.6 s. Real EEG often contains multiple frequencies; a rough cycle count estimates a representative oscillation, not the complete frequency spectrum. Vertical height and horizontal repetition rate must be read separately.

**中文：** 如果窗口从0.2 s开始、0.8 s结束，持续时间是0.6 s，不是0.8 s，也不是1秒。数这个窗口中的完整重复，再除以0.6 s。真实EEG常含多种频率，粗略数周期估计的是代表性振荡，不是完整frequency spectrum频谱。纵向高度与横向重复快慢要分开读。

Source / 来源：Week 7 PDF p84–87; T7-0091–0096; Sample quiz PDF p1 Q5

# Week 8 learning block: Cell recording, amplification and filters / 第8学习组：细胞记录、放大与滤波

## Week 8 concept map / 第8周知识框架

**EN:** Chapter logic: Choose a cell-recording method, preserve and amplify the weak signal, then select its useful frequency band; this chapter stops at Parts One and Two through PDF p29.

**中文：** 本章主线：先选择细胞记录方法，再保住并放大弱信号，最后选择有用频段；本组只到原PDF p29的第一、二章。

| English term 中文名词 | Meaning and connection 含义与知识联系 |
|---|---|
| **Patch clamp 膜片钳** | A pipette forms a tight membrane seal. Voltage clamp holds voltage/read current; current clamp applies current/read voltage. 吸管与膜紧密封接。电压钳保持电压、读电流；电流钳施加电流、读电压。 → C23 |
| **MEA / raster plot 多电极阵列 / 栅格图** | MEA records extracellular activity from many sites in parallel. A raster plot shows when each channel detects spikes. MEA并行记录多个位点的细胞外活动；栅格图显示各通道何时检测到尖峰。 → S801 |
| **Differential / common-mode 差分 / 共模** | Vout=G(V1−V2). Equal shared interference cancels ideally; unequal interference remains. Vout=G(V1−V2)。相同共有干扰理想抵消，不同干扰仍可留下。 → C24 |
| **Input impedance / output impedance / gain 输入阻抗 / 输出阻抗 / 增益** | High input impedance reduces loading; low output impedance drives a load; gain specifies amplification. They are independent roles. 高输入阻抗减少加载，低输出阻抗驱动负载，增益决定放大倍数；三者分工不同。 → S802 |
| **Preamplifier / isolation 前置放大 / 隔离** | The preamplifier handles a weak input; isolation limits unwanted electrical paths and supports patient protection. 前置放大处理弱输入；隔离限制不希望出现的电连接路径，帮助保护被测者。 → B32 |
| **Inverting / non-inverting 反相 / 同相** | In the shown ideal circuits, gains are −Rf/Ri and 1+Rf/Ri. The minus sign reverses polarity, not amplification itself. 所示理想电路增益为−Rf/Ri与1+Rf/Ri。负号表示极性翻转，不表示没有放大。 → S803 |

| English term 中文名词 | Meaning and connection 含义与知识联系 |
|---|---|
| **Low-pass / high-pass / band-pass / notch 低通 / 高通 / 带通 / 陷波** | Filters attenuate unwanted frequency components: low-pass keeps slower components, high-pass faster, band-pass an interval, notch a narrow rejected band. 滤波器削弱不需要的频率成分：低通保留较慢、高通较快、带通保留区间、陷波抑制窄频带。 → C25 |
| **Frequency-response graph 频率响应图** | Horizontal frequency and vertical gain reveal the retained region. The local RC worked example is excluded; basic filter-role recognition remains. 横轴频率、纵轴增益显示保留区域。局部RC例题被排除，基本滤波功能辨认仍保留。 → S804 |

**EN:** Formula links for this chapter: FX706 Differential voltage and shared-noise cancellation, FX707 CMRR: a rejection ratio, not voltage gain, FX710 Ideal inverting and non-inverting op-amp gains, FX711 First-order RC cutoff and low-/high-pass response, FX712 Optional active analog-filter equations.

**中文：** 本章公式定位：FX706 差分电压与共有噪声抵消；FX707 CMRR：共模抑制比，不是电压增益；FX710 理想反相与同相运放增益；FX711 一阶 RC 截止与低通/高通响应；FX712 可选：有源模拟滤波器完整式。

## C23 • [P1] Patch clamp versus multielectrode array (MEA) / 膜片钳与多电极阵列

Priority / 复习标记：P1 First pass/先掌握 · Quiz stated/Quiz明确提及 · Repeated explanation/反复讲解

**EN: Key points**

- Patch clamp forms a tight seal between a micropipette and a membrane patch and can record controlled membrane currents or potentials depending on the mode.
- A MEA records extracellular signals at many electrode sites in parallel.
- The lecturer explicitly names these two methods for Quiz/exam recognition.

**中文：必会知识点**

- Patch clamp用微吸管与膜片形成紧密封接，根据模式记录受控的膜电流或电位。
- MEA通过多个电极位点并行记录细胞外信号。
- 老师明确点名这两种方法可作为Quiz或考试的辨识内容。

**EN:** Start from zero: Patch clamp and MEA answer different measurement questions. A micropipette is a very small glass tube containing conducting liquid and an electrode. In patch clamp, its opening contacts a tiny patch of cell membrane and forms a tight seal. This close electrical connection allows detailed measurements from that cell or membrane patch. “Clamp” means controlling an electrical variable, not mechanically squeezing the whole cell.

**中文：** 从零理解：Patch clamp膜片钳与MEA解决的测量问题不同。Micropipette微吸管是非常细的玻璃管，内部装有导电液体及电极。膜片钳让管口接触一小块细胞膜，并形成紧密封接。这样的电连接让研究者能细致测量这个细胞或膜片。“Clamp钳制”指控制某个电学变量，不是用机械夹子把整个细胞夹住。

**EN:** Follow the mechanism: In voltage-clamp mode, the instrument holds a chosen membrane voltage and measures the current needed; in current-clamp mode, it applies a chosen current and records voltage. A multielectrode array, MEA, instead places many electrodes near cells and records extracellular signals in parallel. Think of one close inspection point versus many listening stations. This analogy captures measurement coverage; it does not imply that the electrical signals are sounds.

**中文：** 跟着机制走：Voltage clamp电压钳模式下，仪器把膜电压保持在选定值，测为此所需的电流；current clamp电流钳模式下，仪器施加选定电流，记录电压。Multielectrode array多电极阵列，简称MEA，则把多个电极放在细胞附近，并行记录extracellular细胞外信号。可以类比“一个近距离检查点”与“许多监听站”，用于理解覆盖范围，而不是把电信号当成声音。

**EN:** Try an example: If an English question describes a glass pipette, membrane patch and tight seal, identify patch clamp. If it describes patterned electrodes, many recording sites and simultaneous extracellular recording, identify MEA. A standard MEA does not directly report the intracellular membrane voltage at every site. Conversely, a single patch-clamp recording does not automatically provide parallel recordings from an entire network. Match the geometry and measured quantity together.

**中文：** 用例子理解：英文题出现glass pipette玻璃微吸管、membrane patch膜片、tight seal紧密封接，就认膜片钳。出现patterned electrodes图案化电极、many recording sites多个记录位点、simultaneous extracellular recording同时记录细胞外信号，就认MEA。普通MEA并不直接给出每个位点的细胞内膜电压；单个膜片钳也不自动提供整个网络的并行记录。把装置外形与测量量一起判断。

**EN:** Clarify the concept: Whole-cell patch clamp: a glass micropipette first forms a tight seal with a small membrane patch; the patch beneath the opening is then disrupted to provide electrical access to the cell interior. It is not simply a hollow needle puncturing anywhere through a cell. Configuration and clamp mode are different: voltage clamp holds voltage and reads current; current clamp applies current and reads voltage.

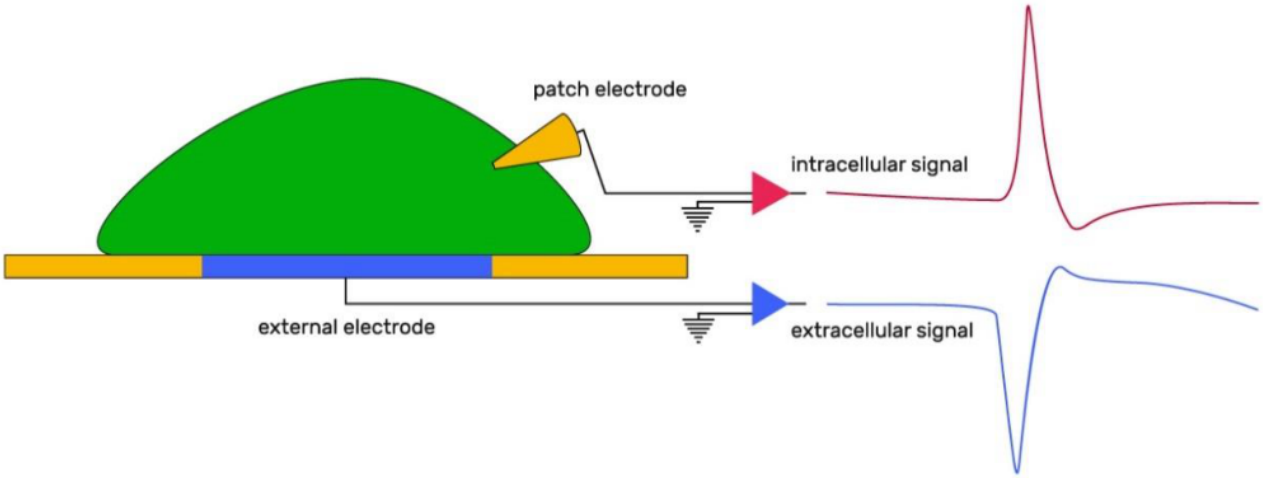
**中文：** 把概念讲清：Whole-cell patch clamp全细胞膜片钳：玻璃微吸管先与一小块膜形成紧密封接，再打开管口下方这块膜，获得细胞内电接入；不只是用中空针随便刺穿细胞。“构型”与“钳制模式”不同：电压钳保持电压、读电流；电流钳施加电流、读电压。

Source / 来源：Week 8 PDF p9–10; T8-0006–0008, T8-0014–0018; Molecular Devices: whole-cell patch clamp

**EN:** Exam wording: Recognize the pipette–membrane seal versus the patterned electrode array. Do not replace a standard electrode array with the excluded detailed conformal-electrode material/design requirements.

**中文：** 英文读题抓手：图中要分清微吸管—细胞膜封接与阵列化电极。基础MEA辨识不能被替换成已排除的贴合电极材料或技术设计细节。

Source / 来源：W8 p9–13; W8 p9–10; W8 p11–13 | T8-0005–0016, T8-0018

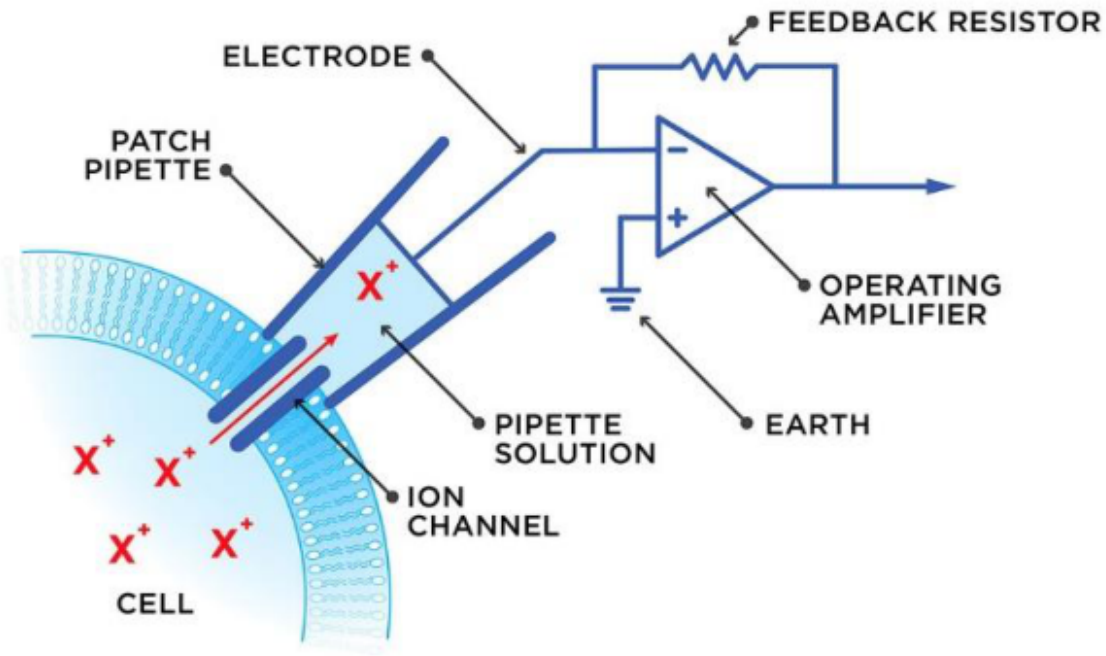



Intracellular versus extracellular recording (original lecture figure). / 细胞内与细胞外记录（原课件图）。

**EN:** Read the figure: The patch electrode accesses the cell interior; the external electrode stays outside. Compare the two illustrated waveforms: their shapes and amplitudes need not be the same. Position and recording method determine what voltage is measured.

**中文：** 读图：Patch electrode膜片电极接触细胞内；external electrode外部电极留在细胞外。比较图中两条波形：形状与幅值不必相同。位置和记录方式决定测到哪一种电压。

Source / 来源：Week 8 PDF p9; original slide figure / 原课件截图




Patch pipette connected to a feedback amplifier (original lecture figure). / 膜片吸管与反馈放大器连接（原课件图）。

**EN:** Read the figure: Find the patch pipette touching the membrane, the electrode inside its solution, and the operating amplifier with a feedback resistor. The pipette seal selects the membrane region for electrical recording; this is different from a flat MEA electrode beneath cells.

**中文：** 读图：找到接触细胞膜的patch pipette膜片吸管、吸管液中的电极，以及带反馈电阻的运算放大器。吸管与膜的封接选择电记录区域，与细胞下方平面MEA电极不同。

Source / 来源：Week 8 PDF p9; original slide figure / 原课件截图

## S801 • [P1] Recording-method trade-offs and raster plots / 记录方法的优缺点与栅格图

Priority / 复习标记：P1 First pass/先掌握 · Quiz stated/Quiz明确提及 · Repeated explanation/反复讲解 · Supporting concept/辅助理解

**EN:** Patch clamp allows detailed controlled-current or controlled-voltage measurements in a selected cell or membrane patch, but precise pipette positioning and skilled operation limit manual throughput. A standard MEA can record many extracellular sites in parallel without inserting a pipette into each cell; it is convenient for repeated network recordings. Its signal also depends on electrode contact and cell density or maturity. Many sites do not automatically mean a complete intracellular voltage map.

**中文：** 膜片钳可在选定细胞或膜片做细致的受控电流/电压测量，但微吸管需要精确定位与熟练操作，人工测量throughput通量较低。普通MEA可并行记录许多细胞外位点，不必把吸管逐个插入细胞，方便重复记录网络活动。其信号也受电极接触、细胞密度或成熟度影响。位点多不自动等于完整的细胞内电压地图。

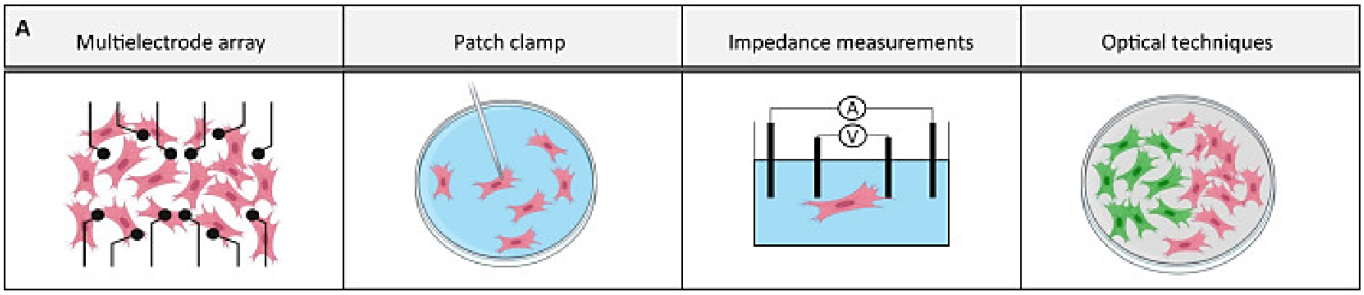
**EN:** A raster plot summarizes spike timing: the horizontal axis is time, each row represents one recording electrode, and each short mark denotes a detected spike. Marks aligned across rows suggest coordinated firing; dense bursts differ from isolated spikes. The marks primarily show when spikes occur, not their voltage amplitudes. This is how a MEA' s many channels become a readable picture of network activity.

**中文：** Raster plot栅格图概括尖峰发生时间：横轴是时间，每行代表一个记录电极，每个短刻痕代表检测到一次尖峰。多行刻痕在同一时间对齐，提示协调发放；密集burst爆发与零散尖峰不同。刻痕主要表示“何时发生”，不直接表示电压幅值。这让MEA多通道记录变成可读的网络活动图。

**EN:** The comparison slide also shows impedance and optical methods. Bulk impedance reflects an electrical response of the measured sample or culture, rather than automatically identifying each individual cell. Optical reporters can change fluorescence with cellular activity and provide many imaging sites. However, fluorescence intensity is not itself an absolute voltage in millivolts; interpreting it as electrical activity may require calibration. Patch clamp and MEA remain the two methods the lecturer explicitly asked students to recognize.

**中文：** 对比页还列出impedance阻抗与optical光学方法。整体阻抗反映被测样品或培养体系的电响应，并不自动识别每个单细胞。光学报告分子可随细胞活动改变荧光，提供许多成像位点；但荧光强度本身不是以mV表示的绝对电压，解释为电活动可能需要校准。老师明确要求辨识的两个方法仍是膜片钳与MEA。

Source / 来源：Week 8 PDF p9–13; T8-0008–0018

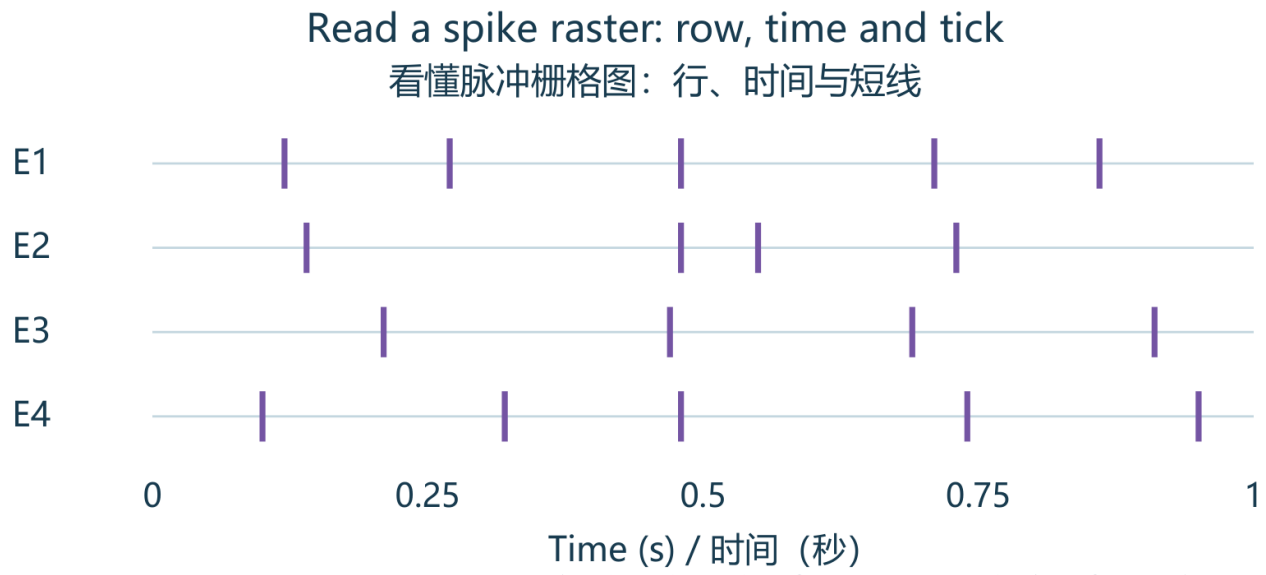



Recognize four cellular recording approaches by the probe and measured signal. / 从探头与测量信号辨认四种细胞记录方法。

**EN:** Read the figure: Read left to right: an MEA uses several electrode sites near cells; patch clamp uses a glass pipette sealed against a membrane region; impedance measurements apply electrical probing and measure the voltage–current relationship; optical techniques detect a light-based reporter response. The cartoons show different measurement interfaces, not interchangeable ways to obtain exactly the same waveform. The illustrated MEA arrangement is extracellular. For a voltage-sensitive optical reporter, quantitative voltage estimates need calibration; generic fluorescence intensity is not an absolute voltage.

**中文：** 读图：从左向右看：MEA在细胞附近设置多个电极点；patch clamp膜片钳用玻璃吸管与一块膜封接；impedance阻抗测量施加电探测并测电压—电流关系；optical techniques光学方法检测光学报告信号。图示的是不同测量界面，不表示四者能直接得到完全相同的波形。图中MEA是胞外记录；若使用电压敏感的光学报告探针，还需校准才能定量估计电压；普通荧光强度不是绝对电压。

Source / 来源：Week 8 PDF p13; original slide figure / 原课件截图




Synthetic example / 假设记录示意：短竖线的位置表示事件时间，线高不表示幅度

Read a spike raster: row, time and tick / 看懂脉冲栅格图：行、时间与短线

**EN:** Read the figure: Each row here is one electrode/channel; each tick is a detected spike event at a time. Ticks near 0.48 s across several channels illustrate roughly coordinated timing. Tick height is only a plotting symbol, not voltage amplitude. The array measures extracellular signals; interpretation depends on detection and electrode placement.

**中文：** 读图：这里每一行表示一个电极 / 通道，每条短竖线表示该时刻检测到一次Spike（脉冲事件）。多个通道在0.48 s附近出现短线，示意时间大致同步。短线高度只是画图符号，不是电压幅度。阵列记录的是细胞外信号；解释还取决于事件检测方法和电极位置。

Source / 来源：Week 8 PDF p11–13; original teaching schematic / 自绘教学示意，非实验数据

## C24 • [P1] Differential amplification and common-mode noise / 差分放大与共模噪声

Priority / 复习标记：P1 First pass/先掌握 · Teacher emphasis/课堂强调 · Worked example/例题练习

**EN: Key points**

- A differential amplifier amplifies the difference between two inputs: Vout=G($V_1-V_2$).
- A signal appearing similarly at both inputs is common-mode and is ideally rejected.
- High input impedance reduces loading of a weak biological signal.

**中文：必会知识点**

- 差分放大器放大两输入之差：Vout=G($V_1-V_2$)。
- 两端相似的信号是common-mode，理想情况下被抑制。
- 较高输入阻抗减少对微弱生物信号的负载影响。

**EN:** Start from zero: A differential amplifier receives two voltages and enlarges their difference: Vout=G($V_1-V_2$). G is voltage gain, a multiplication factor. A voltage added equally to both inputs is common-mode. If $V_1=S_1+n$ and $V_2=S_2+n$, subtraction removes the identical n, leaving $S_1-S_2$. Imagine two thermometers both shifted by the same calibration error: their difference still cancels that shared error.

**中文：** 从零理解：Differential amplifier差分放大器接收两个电压，放大它们的差：Vout=G($V_1-V_2$)。G是voltage gain电压增益，即乘上多少倍。两端同样增加的电压叫common-mode共模。如果$V_1=S_1+n$、$V_2=S_2+n$，相减后相同的n消失，只剩$S_1-S_2$。可类比两个温度计都有同样的校准偏移：相减时，这个共有误差会抵消。

**EN:** Follow the mechanism: If the noises differ, $n_1-n_2$ remains; real amplifiers also reject common-mode imperfectly. High input impedance means the measuring input draws very little current from the biological source. That reduces loading, which would otherwise alter the voltage being measured. It preserves the input signal; it does not itself set gain. Think of measuring gently without draining the source, rather than automatically making its signal bigger.

**中文：** 跟着机制走：若两端噪声不同，$n_1-n_2$仍会留下；实际放大器的共模抑制也并非完美。High input impedance高输入阻抗，表示测量端从生物信号源取走的电流很少，从而减少loading加载，否则被测电压会因测量而改变。它帮助保住输入信号，并不直接决定增益。可以理解为“轻轻测，尽量不取走源的电流”，而不是自动把源信号变大。

**EN:** Try an example: Example: $V_1$=1.4 mV, $V_2$=1.1 mV and G=1000. First subtract: 0.3 mV. Then multiply: 300 mV=0.300 V. Adding the same 5 mV interference to both inputs leaves the difference unchanged ideally. Adding 5 mV to only one input changes it greatly. Therefore “large gain removes any noise” is false: gain also enlarges noise that survives the subtraction.

**中文：** 用例子理解：例：$V_1$=1.4 mV、$V_2$=1.1 mV、G=1000。先相减得到0.3 mV，再放大得到300 mV=0.300 V。理想情况下，两端同时加同样的5 mV干扰，差值不变；只在一端加5 mV，差值就会明显改变。所以“增益大就能消除所有噪声”是错的：相减后剩下的噪声，也会被增益放大。

**EN:** Exam wording: Subtract before multiplying, and convert mV to V correctly. The reference and recording locations matter. Gain alone does not guarantee removal of noise that appears differently at the two inputs.

**中文：** 英文读题抓手：先相减再乘增益，并正确换算mV与V。记录位置和参考位置很重要。若噪声在两输入端不同，仅有增益并不保证它被消除。

Source / 来源：W7 p52–54; W8 p19–20; W8 p20; W8 p22 | T7-0043–0046, T8-0021–0026

## B32 • [P2] Amplifier chain, isolation and polarity / 放大链、隔离与极性

Priority / 复习标记：P2 Important foundations/重要基础 · Repeated explanation/反复讲解 · Lower priority/降低优先级

**EN: Key points**

- A preamplifier handles weak input;
- high-pass filtering reduces slow drift and low-pass filtering reduces fast interference.
- Isolation supports patient protection.
- High input impedance reduces loading;
- low output impedance helps drive a load.
- Neither specifies voltage gain.
- Inverting amplification reverses polarity;
- non-inverting preserves it.
- Recognize the ideal gains −Rf/Rin and 1+Rf/Rin as supporting relations, not an explicit memorization promise.

**中文：必会知识点**

- Preamplifier前置放大器处理弱输入；
- 高通减小慢漂移，低通减小快速干扰。
- Isolation隔离用于患者保护。
- 高输入阻抗减少加载，低输出阻抗便于驱动负载，二者都不直接决定电压增益。
- Inverting反相放大翻转极性；
- non-inverting同相放大保持极性。
- 理想增益−Rf/Rin和1+Rf/Rin可作辅助辨识，并非老师明确保证必背。

**EN:** Start from zero: A weak biological signal usually needs several processing stages. A preamplifier enlarges the initial small signal. High-pass filtering reduces slow drift; low-pass filtering reduces fast components outside the useful range. An isolation stage separates electrical connections to help protect the subject and limit unwanted current paths. Amplification, filtering and isolation perform different jobs: none should be described as automatically doing everything the other stages do.

**中文：** 从零理解：微弱生物信号通常需要多级处理。Preamplifier前置放大器先放大初始小信号；高通减少慢漂移，低通削弱有用频段之外的快速成分；isolation stage隔离级隔开电连接，帮助保护被测者并限制不希望出现的电流路径。放大、滤波、隔离分工不同，不能把某一级说成自动完成其他所有级的任务。

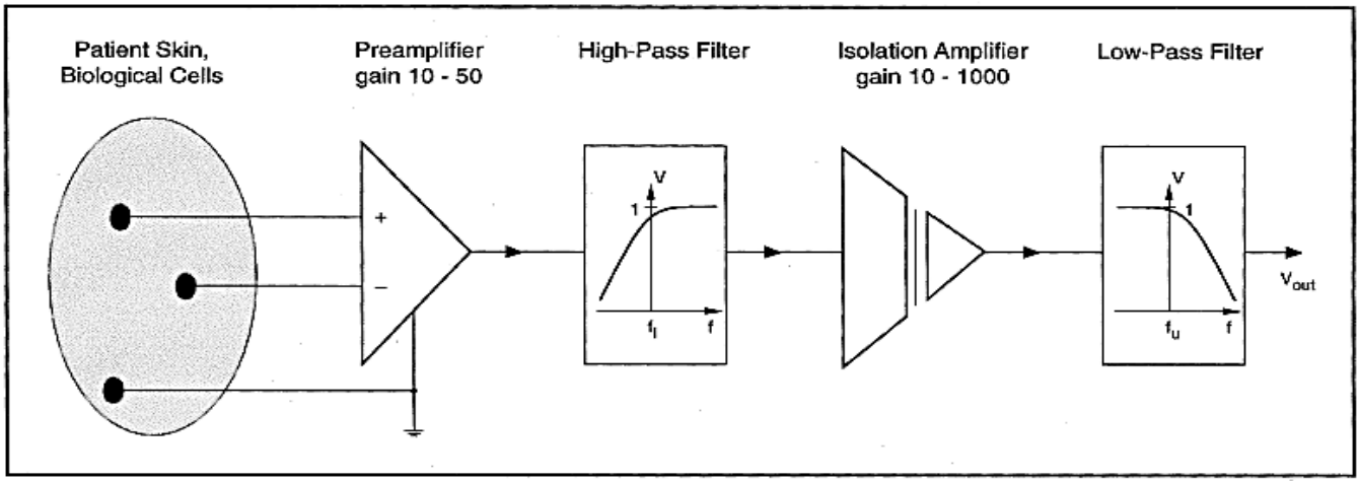
**EN:** Follow the mechanism: Input impedance describes how strongly the amplifier loads its source. A high value draws little input current and helps preserve the source voltage. Output impedance concerns driving the next device: a low value reduces voltage loss into that load. Gain is separately the ratio of output signal to input signal. Think of a gentle receiver connected to a strong sender; this illustrates loading, not a rule that high impedance creates large gain.

**中文：** 跟着机制走：Input impedance输入阻抗描述放大器对信号源施加多大负载。输入阻抗高，取走的输入电流少，有助于保住源电压。Output impedance输出阻抗涉及驱动下一级设备：输出阻抗低，可减少负载造成的电压损失。Gain增益则另行定义为输出与输入信号之比。可以理解为“轻轻接收，同时有力输出”，用来理解加载；不能据此推出阻抗高就增益大。

**EN:** Try an example: Inverting means polarity reversal: a positive input produces a negative output in the ideal circuit. Non-inverting keeps polarity. With Rf=20 kΩ and Rin=10 kΩ, the ideal inverting gain is −Rf/Rin=−2; the non-inverting gain is 1+Rf/Rin=3. Thus a +1 mV input gives −2 mV or +3 mV respectively. The minus sign describes polarity, not an absence of amplification. These simple examples explain the circuits rather than promise a specific Quiz question.

**中文：** 用例子理解：Inverting反相表示极性翻转：理想电路里，正输入产生负输出；non-inverting同相保留极性。若Rf=20 kΩ、Rin=10 kΩ，理想反相增益为−Rf/Rin=−2，同相增益为1+Rf/Rin=3。因此+1 mV输入分别得到−2 mV或+3 mV。负号描述极性，并不表示“没有放大”。这些简单例子用于理解电路，不是保证Quiz一定考这道题。

Source / 来源：W8 p13; W8 p19–20; W8 p20; W8 p21; W8 p19–29 context; W8 p22; W8 p23–24; W8 p25–26; W8 p27–28; W8 p29 | T8-0016–0018, T8-0021–0029, T8-0062–0064




Biopotential acquisition chain: preamplifier, high-pass filter, isolation and low-pass filter. / 生物电采集链：前置放大、高通、隔离与低通。

**EN:** Read the figure: Follow the arrows from the patient on the left to Vout on the right. The preamplifier raises a weak signal; the high-pass stage reduces DC and slow drift; the isolation stage provides electrical separation for safety and helps prevent ground loops; the low-pass stage attenuates excessively fast components. Each tiny filter graph has frequency on its horizontal axis. Gain, filtering and isolation serve different functions: amplification alone does not remove all noise or provide patient isolation.

**中文：** 读图：沿箭头从左侧患者读到右侧Vout输出。Preamplifier前置放大器把弱信号放大；high-pass高通减少直流与缓慢漂移；isolation隔离级提供电隔离以保护被测者，也有助于避免地环路；low-pass低通衰减过快成分。两个小滤波图的横轴都是频率。增益、滤波、隔离各司其职：单靠放大既不能消除所有噪声，也不能实现患者电隔离。

Source / 来源：Week 8 PDF p22; original slide figure / 原课件截图

## S802 • [P2] Amplifier requirements, bandwidth and safety / 放大器要求、带宽与安全

Priority / 复习标记：P2 Important foundations/重要基础 · Repeated explanation/反复讲解 · Lower priority/降低优先级

**EN:** The requirement slide gives high input impedance (>10 MΩ), low output impedance and large gain (often >1000) for weak biopotential signals. These numbers describe the stated application, not every amplifier: the preceding slide lists gains from ×10 to ×10000. Input impedance preserves the source voltage, output impedance helps drive the load, and gain sets amplification. None of these three quantities substitutes for the others.

**中文：** 要求页给出高输入阻抗（>10 MΩ）、低输出阻抗，以及微弱生物信号常需较大增益（常见>1000）。这些数值针对所述应用，不是所有放大器的统一定义：前一页也列出×10至×10000的增益。输入阻抗帮助保住源电压，输出阻抗帮助驱动负载，gain决定放大倍数，三者不能相互替代。

**EN:** Frequency response must cover the useful signal band. More gain does not restore signal removed by an unsuitable filter, and a wider bandwidth can admit more unwanted components. The amplitude-versus-frequency chart shows that ECG, EEG and EMG occupy different approximate regions; its detailed numerical boundaries were presented for reference. EOG means electrooculogram, associated with eye-related potentials; ERG means electroretinogram, recording retinal electrical responses, rather than EEG brain recordings.

**中文：** Frequency response频率响应要覆盖有用信号频段。更多增益不能恢复已被不合适滤波器削掉的信号，较宽bandwidth带宽也可放进更多不需要的成分。幅度—频率图显示ECG、EEG、EMG的大致区域不同；详细数值边界是供参考。图中EOG是electrooculogram眼电图，与眼相关电位有关；ERG是electroretinogram视网膜电图，记录视网膜电响应，不是EEG脑电。

**EN:** An isolation stage protects the subject by limiting unwanted electrical connections and current paths; it also helps prevent ground loops. A ground loop is an unintended closed path through connected grounds that can introduce interference. Safety is therefore about patient-connected current paths and protection, not merely choosing a small voltage gain. Amplification, isolation and filtering remain distinct functions.

**中文：** Isolation stage隔离级通过限制不希望出现的电连接和电流路径保护被测者，也有助于防止ground loop地环路。地环路是互连接地之间意外形成的闭合路径，可引入干扰。因此，安全涉及患者连接的电流路径与保护措施，并不只是选择较小电压增益。放大、隔离与滤波仍是不同功能。

Source / 来源：Week 8 PDF p19–22; T8-0021–0026

## S803 • [P3] Read an operational-amplifier diagram / 看懂运算放大器图

Priority / 复习标记：P3 Supporting reading/辅助阅读 · Lower priority/降低优先级 · Supporting concept/辅助理解

**EN:** The op-amp triangle has a non-inverting (+) signal input, an inverting (−) signal input and an output. The supply terminals power the circuit; they are not the two measured biological signal inputs. The +/− input labels specify which input contribution increases or decreases output polarity, not that one electrode must always have a positive measured voltage and the other a negative one.

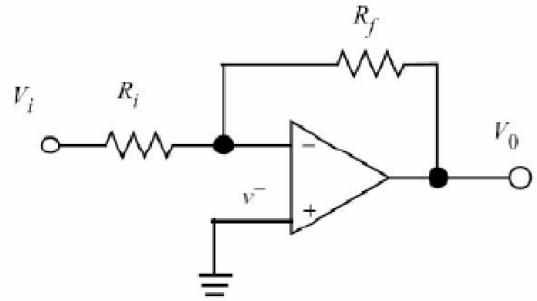
**中文：** Op-amp运算放大器三角形有non-inverting（+，同相）信号输入、inverting（−，反相）信号输入与输出。Supply terminals供电端给电路供能，不是两个被测生物信号输入。“+” “−”表示输入对输出极性的作用，不是说一个电极测得的电压必须始终为正、另一个必须始终为负。

**EN:** In the shown circuits, a feedback resistor connects output back to the inverting input. Under the ideal linear-operation assumptions, the resistor ratios set the closed-loop gains: −Rf/Ri for inverting and 1+Rf/Ri for non-inverting. A minus sign reverses polarity. These relations should not be applied as if output could grow without limit; the amplifier must remain within its available output range.

**中文：** 图中feedback resistor反馈电阻把输出接回反相输入。满足理想线性工作假设时，电阻比决定closed-loop gain闭环增益：反相为−Rf/Ri，同相为1+Rf/Ri。负号表示极性翻转。不能把公式理解成输出能无限增大；放大器必须在可用输出范围内工作。

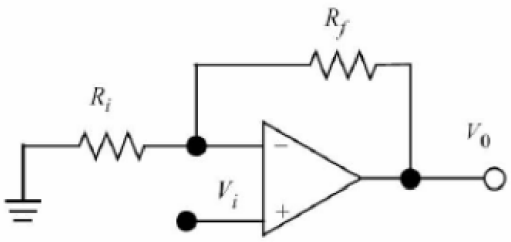
Source / 来源：Week 8 PDF p23–24; T8-0025–0026

1. Inverting Amplifier




$$V_0 = -\left(\frac{R_f}{R_i}\right)V_i$$

2. Non-inverting Amplifier




$$V_0 = \left(1 + \frac{R_f}{R_i}\right)V_i$$

Compare inverting and non-inverting op-amp feedback circuits. / 比较反相与同相运放反馈电路。

**EN:** Read the figure: On the left, Vi reaches the − input through Ri; on the right, Vi reaches the + input. In both circuits, trace Rf from the output back to the − input. Under ideal linear-operation assumptions, the gains are −Rf/Ri and 1+Rf/Ri respectively. The minus sign means output polarity is reversed; it does not mean negative signal size or negative input impedance. These relations apply while the amplifier remains within its operating and output limits.

**中文：** 读图：左图Vi经Ri到“−”输入，右图Vi直接到“+”输入。两图都可沿Rf追踪输出返回“−”输入的反馈路径。理想线性工作时，增益分别为−Rf/Ri与1+Rf/Ri。负号表示输出极性翻转，不表示信号大小为负，也不是输入阻抗为负。公式成立还要求放大器在允许的工作与输出范围内。

Source / 来源：Week 8 PDF p24; original slide figure / 原课件截图

## C25 • [P1] High-pass, low-pass, band-pass and notch filters / 高通、低通、带通与陷波滤波器

Priority / 复习标记：P1 First pass/先掌握 · Teacher emphasis/课堂强调 · Repeated explanation/反复讲解 · Lower priority/降低优先级

**EN: Key points**

- A high-pass filter attenuates low-frequency components;
- a low-pass filter attenuates high-frequency components.
- A band-pass filter preserves a frequency interval.
- A band-reject or notch filter suppresses an interval or narrow interference band, such as mains interference when appropriately selected.

**中文：必会知识点**

- High-pass抑制低频，low-pass抑制高频；
- band-pass保留一个频率区间；
- band-reject或notch抑制一个区间或窄干扰带，例如适当选择时可抑制工频干扰。

**EN:** Start from zero: A recorded signal can contain slow baseline drift, useful biological changes, and fast interference together. Frequency describes how rapidly each repeating component changes, not how tall it is. A filter treats these components differently. Pass means retain relatively well; attenuate means reduce amplitude. A low-pass filter does not turn every fast oscillation into a slower one: it mainly weakens the unwanted fast components.

**中文：** 从零理解：一条记录中，可能同时含有很慢的baseline drift基线漂移、有用的生物变化和很快的干扰。Frequency频率描述每个重复成分变化有多快，不描述它有多高。Filter滤波器区别处理这些成分。Pass通过表示较好保留；attenuate衰减表示减小幅度。低通不是把所有快速振荡“变慢”，而是主要削弱不需要的快速成分。

**EN:** Follow the mechanism: High-pass keeps higher frequencies and reduces slower ones; low-pass does the reverse. Band-pass keeps an interval between lower and upper limits. Band-reject suppresses an interval; a notch is a narrow rejection band. Think of frequency gates accepting certain repetition rates. Actual transitions are gradual, so near a cutoff the amount of attenuation matters.

**中文：** 跟着机制走：High-pass高通保留较高频率、削弱较慢成分；low-pass低通的作用相反。Band-pass带通保留上下限之间的一段频率；band-reject带阻抑制一个区间；notch陷波是窄的抑制区间。可以想成按“重复快慢”筛选的门。实际边界渐变，所以接近cutoff截止频率时，衰减多少也重要。

**EN:** Try an example: For an idealized 10–100 Hz band-pass teaching example, a 5 Hz component is reduced, a 40 Hz component is retained, and a 200 Hz component is reduced. A suitably chosen notch can reduce 50 Hz mains interference. However, useful signal at 50 Hz is also affected: the filter cannot recognize whether a component is biological or interference merely from its origin. Match the requested frequency range to the filter type.

**中文：** 用例子理解：教学例中，若采用理想化10–100 Hz带通，5 Hz成分被削弱，40 Hz被保留，200 Hz被削弱。合适的陷波可削弱50 Hz工频干扰，但有用信号若也位于50 Hz，同样会受影响：滤波器不会仅凭成分的来源，就自动区分“生物信号”与“干扰”。做题时，把要求保留或抑制的频率范围对应到滤波类型。

**EN:** Exam wording: Keep filter-role recognition in the main review. The particular RC cutoff example was excluded from this year’ s exam in its local context; this does not justify excluding every filter concept from the Quiz.

**中文：** 英文读题抓手：主复习保留滤波功能辨识。某个RC截止频率例题在局部语境中被说成今年考试不考，这不能推导为所有滤波概念都不在Quiz范围。

Source / 来源：W8 p25–26; W8 p27–28; W8 p29 | T8-0026–0029, T8-0062–0063

## S804 • [P2] Read filter responses and simple circuit trends / 读滤波响应图与简单电路趋势

Priority / 复习标记：P2 Important foundations/重要基础 · Teacher emphasis/课堂强调 · Worked example/例题练习 · Lower priority/降低优先级

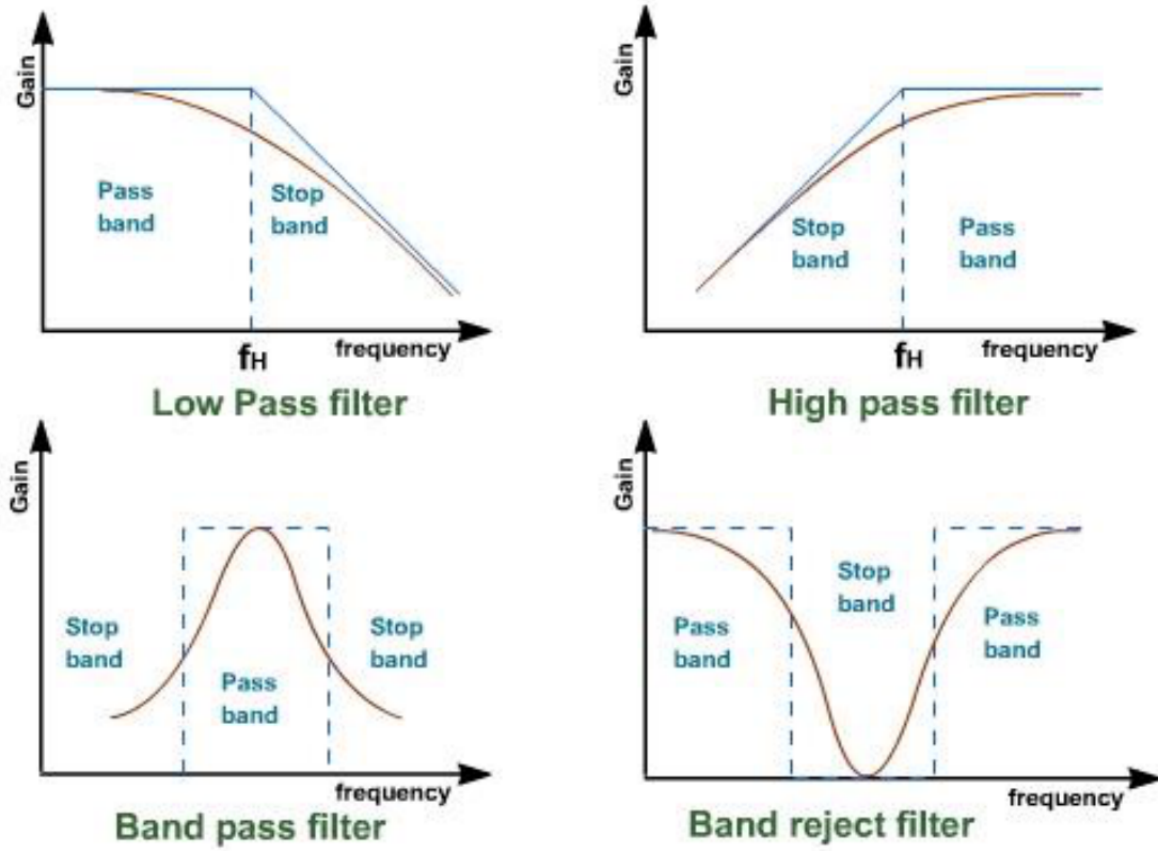
**EN:** On a frequency-response plot, the horizontal axis is frequency and the vertical axis is gain or response magnitude. A low-pass curve is relatively large at low frequency and falls at high frequency; a high-pass does the reverse. A band-pass retains a middle region and falls at both ends; a band-reject has a suppressed middle region. On a time trace, removing fast components often makes the waveform smoother; removing slow components can remove baseline drift.

**中文：** 频率响应图横轴是频率，纵轴是gain增益或响应幅度。低通在低频较大、高频下降，高通相反；带通保留中间区间、两端下降；带阻则让中间区间降低。时间曲线上，去掉快速成分往往让波形更平滑；去掉慢成分可减少基线漂移。

**EN:** The analog-filter diagrams implement these same roles using resistors, capacitors and op-amps. A high-pass stage followed by a low-pass stage can form a band-pass when the lower cutoff is below the upper cutoff. For a simple first-order RC stage, the supporting relation is fc=1/(2πRC): increasing R or C lowers fc. The lecturer excluded the particular worked RC cutoff example; retain the role and graph recognition, with this formula as supporting understanding rather than a promised Quiz calculation.

**中文：** Analog filter模拟滤波器图用电阻、电容与运放实现同样的功能。高通后接低通，在下截止频率低于上截止频率时，可组成带通。简单一阶RC级的辅助关系是fc=1/(2πRC)：增大R或C会降低fc。老师排除了那道具体RC截止频率例题；保留功能与读图辨认，公式作为辅助理解，不把它宣传成Quiz保证计算题。

Source / 来源：Week 8 PDF p25–29; T8-0026–0029




Four filter responses: low-pass, high-pass, band-pass and band-reject. / 四种滤波响应：低通、高通、带通、带阻。

**EN:** Read the figure: Read the frequency axis from left to right. A high part of a curve means relatively strong transmission, and a low part means attenuation. Blue idealized boundaries and the smooth practical curves express the same pass/stop roles.

**中文：** 读图：频率轴从左向右增大。曲线较高表示相对更好通过，较低表示衰减。蓝色理想边界与平滑的实际曲线，表达的是同样的通过/抑制功能。

Source / 来源：Week 8 PDF p25; original slide figure / 原课件截图

# Formulas explained with small examples / 公式符号与短例题（完整补全）

**EN:** Updated 2026-10-14: the original ten worked relations are expanded here by checking the classmate’ s formula sheet against the lecture sources. P1/P2/P3 set revision order, not certainty of appearing on the quiz. Long formulas remain available for recognition and substitution when quantities are supplied.

**中文：** 2026-10-14更新：原先集中讲解的10项关系式已按同学公式表及原课件扩充。P1/P2/P3决定复习先后，不代表真题保证；较长的关系式也保留，用于辨认与给定数据下的代入。

**EN:** Symbols are reused between topics: n can mean refractive index, charge-carrier density or electron count; R can mean resistance, reflectivity or resolving power. Read the definition beside each formula. Δ denotes a change or a spectral separation, $j^2 = -1$, and $\log_{10}$ is distinct from ln. Always match units before calculating.

**中文：** 不同章节会复用符号：n可能是折射率、载流子密度或电子数；R可能是电阻、反射率或分辨本领。每次按该公式旁的定义读。Δ表示变化量或谱线间隔，$j^2=-1$，$\log_{10}$与ln不同；代入前先统一单位。

**EN:** Quick route: follow a chapter’ s formula links, read the equation and symbols, then check its conditions and tiny example. The cards below retain every topic in the classmate’ s 37-row appendix and add supported short relations from the covered lectures.

**中文：** 使用顺序：从本章公式链接进入，先看式子和符号，再看适用条件与短例。下方覆盖同学附录A的全部37类主题，并补入考试范围课件中的重要短关系。

## FX101 • [P1] Calibration sensitivity and concentration LOD / 校准灵敏度与浓度检出限

**EN:** Revision priority: First: know and calculate. Worked in class and represented in Sample Q10/Q12; know the calculation and units.

**中文：** 复习优先级：优先掌握与短算。课堂讲解并对应 Sample Q10/Q12，须会算且会核对单位。

**Formula / 公式：** $y = mC + b$; $m = \Delta y / \Delta C$; $LOD_C = 3\sigma_y / |m|$

**EN:** Symbols and units: C: concentration; y: response; m: calibration slope; b: intercept; $\sigma_y$: blank/low-level response SD. m has response/concentration units.

**中文：** 符号与单位：C：浓度；y：响应；m：校准斜率；b：截距；$\sigma_y$：空白/低水平响应标准差。m 单位为响应/浓度。

**EN:** When to use: Convert a measurable response change into target amount and compare the threshold with output noise.

**中文：** 何时使用：将响应变化换算为目标量，比较检测阈值与输出噪声。

**EN:** Conditions and traps: Use a valid local linear calibration, with σ_y and m in matching response units. The 3σ rule is the course approximation. Apply an integer ceiling only when the problem requires discrete counts at or above LOD. Sensitivity is a slope; LOD is a concentration. Do not use the saturated part or mistake m for σ. W1 p62 mixes current and current-density units.

**中文：** 条件与易错点：使用有效局部线性校准，σ_y 与 m 响应单位一致。3σ 是本课程采用的近似；只有要求不低于阈值的整数计数时才向上取整。 灵敏度是斜率，检出限是浓度；不用饱和段，不把 m 当 σ。W1 p62 电流和电流密度单位混写。

**EN:** Tiny example: σ_y=2 and m=10.911 in matching units give LOD≈0.55 mM.

**中文：** 短例理解：响应单位一致时，σ_y=2、m=10.911，LOD≈0.55 mM。

Source / 来源：W1 PDF p60–62 | W2 PDF p9–10 | T1-0054 | T1-0055 | T1-0056 | T1-0057 | T1-0058 | T1-0059 | T2-0007 | T2-0008 | Sample Quiz PDF p2 Q10/Q12

## FX102 • [P3] Hall voltage / 霍尔电压

**EN:** Revision priority: Recognise; optional calculations. Printed as a background sensor example; no class-worked Hall calculation or direct Quiz question was found. Recognize the proportionality first.

**中文：** 复习优先级：辅助辨认；计算后看。课件作为传感器背景例子出现，未定位课堂霍尔计算或直接样题；先会辨认正比关系。

**Formula / 公式：** $V_H = I B_{perp} / (n_c q t)$; $|V_H| = |I B_{perp}| / (n_c |q| t)$

**EN:** Symbols and units: I: current (A); B_perp: magnetic field perpendicular to the current/sensing plate (T); n_c: carrier number density ($m^{-3}$); q: signed carrier charge (C); t: plate thickness (m).

**中文：** 符号与单位：I：电流 A；B_perp：相应垂直磁场分量 T；n_c：载流子数密度 $m^{-3}$；q：带正负号的载流子电荷 C；t：薄片厚度 m。

**EN:** When to use: Relate transverse Hall voltage to magnetic field; at fixed I, n_c, q and t, V_H is proportional to B_perp.

**中文：** 何时使用：把横向霍尔电压与磁场联系起来，其他参数固定时 V_H 与 B_perp 成正比。

**EN:** Conditions and traps: Simple uniform slab with a dominant carrier type and the stated field geometry. The sign depends on carrier type and measurement polarity. n_c here is a number density, not optical refractive index; t is thickness, not time. A voltage sign needs a defined polarity.

**中文：** 条件与易错点：均匀薄片、单一主导载流子及指定磁场几何的简化模型，符号取决于载流子与测量极性。 这里 n_c 是数密度，不是折射率；t 是厚度，不是时间。电压正负必须先定义极性。

**EN:** Tiny example: With all other factors fixed, doubling B_perp doubles |V_H|; doubling thickness halves it.

**中文：** 短例理解：其他不变时，B_perp 加倍使 |V_H| 加倍；厚度加倍使电压减半。

Source / 来源：W1 PDF p21 | T1-0022

## FX103 • [P2] Optical wave, frequency, wavelength and phase / 光波、频率、波长与相位

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|---|
| **EN:** Revision priority: Important supporting relations. W2 explicitly defines these symbols; learn the relationships and meanings, not a new wave-equation derivation. |
| **中文：** 复习优先级：重要基础关系。W2 明确定义这些符号，要懂关系与含义，不增加波动方程推导要求。 |

**Formula / 公式：** $E_{field}(z,t) = E_0 \cos(\omega t - kz)$; $\omega = 2\pi f$; $k = 2\pi/\lambda$; $v_{phase} = f\lambda$; in vacuum: $c = f\lambda_0$

| |
|---|
| **EN:** Symbols and units: $E_{field}$: electric field, not photon energy; $E_0$: field amplitude; f: Hz; ω: rad/s; k: rad/m; λ: wavelength in the propagation medium; $\lambda_0$: vacuum wavelength; $c \approx 3\times10^8$ m/s. |
| **中文：** 符号与单位：$E_{field}$：电场，不是光子能量；$E_0$：场幅；f：Hz；ω：rad/s；k：rad/m；λ：传播介质中的波长；$\lambda_0$：真空波长；$c \approx 3\times10^8$ m/s。 |
| **EN:** When to use: Recognize frequency, phase and propagation direction; shorter vacuum wavelength means higher frequency. |
| **中文：** 何时使用：辨认频率、相位、传播方向；真空波长越短，频率越高。 |
| **EN:** Conditions and traps: The cosine expression is an ideal monochromatic plane wave. Use $c=f\lambda$ only with vacuum wavelength; in a medium use its phase speed. The course writes frequency as ν. ω is not f: they differ by 2π. The electric-field symbol E and photon-energy E are different quantities. |
| **中文：** 条件与易错点：余弦式为理想单色平面波；$c=f\lambda$ 必须使用真空波长，介质中用其相速度。课件频率符号为 ν。 ω 不是 f，两者差 2π；电场 E 与光子能量 E 是不同量。 |
| **EN:** Tiny example: $\lambda_0$=600 nm gives $f \approx 5\times10^{14}$ Hz and $\omega \approx 3.14\times10^{15}$ rad/s. |
| **中文：** 短例理解：$\lambda_0$=600 nm 时 $f \approx 5\times10^{14}$ Hz，$\omega \approx 3.14\times10^{15}$ rad/s。 |

Source / 来源：W2 PDF p17 | T2-0011 | T2-0012

## FX104 • [P1] Photon energy, total energy and photon rate / 单光子能量、总能量与光子率

| |
|---|
| **EN:** Revision priority: First: know and calculate. The lecturer uses this in the 500 nm photon example and W2 p80 bacterium calculation. |
| **中文：** 复习优先级：优先掌握与短算。老师用于 500 nm 光子例子及 W2 p80 细菌光子计算。 |

**Formula / 公式：** $E_{ph} = hf = hc/\lambda_0$; $E_{ph}(eV) \approx 1240/\lambda_0(nm)$; $E_{total} = N E_{ph}$; $N_{dot} = P/E_{ph} = P\lambda_0/(hc)$

**EN:** Symbols and units: h=6.62607015×10$^{-34}$ J·s; c=2.99792458×10$^{8}$ m/s; λ0: vacuum wavelength; N: photon count; P: optical power (W=J/s); N_dot: photons/s; 1 eV=1.602176634×10$^{-19}$ J.

**中文：** 符号与单位：h=6.62607015×10$^{-34}$ J·s；c=2.99792458×10$^{8}$ m/s；λ0：真空波长；N：光子数；P：光功率 W=J/s；N_dot：光子/秒；1 eV=1.602176634×10$^{-19}$ J。

**EN:** When to use: Move between wavelength, energy per photon, total energy and photons per second.

**中文：** 何时使用：在波长、单光子能量、总能量及每秒光子数之间换算。

**EN:** Conditions and traps: Use one photon wavelength or sum over wavelengths for a broadband beam. The 1240 shortcut requires nm and eV. For photons reaching a sample, use power actually incident on that sample. P is energy per second, not photon energy. Do not multiply by the number of dye molecules, and do not confuse incident and detected photon rate.

**中文：** 条件与易错点：单色光用一个波长，宽带光需按各波长求和；1240 快捷式要求 nm 与 eV。求照到样品的光子率时用实际到达样品的功率。 P 是每秒能量，不是光子能量；不额外乘染料分子数，不混淆入射与探测光子率。

**EN:** Tiny example: 410 nm and 10 nW give E_ph≈4.845×10$^{-19}$ J and N_dot≈2.064×10$^{10}$ s$^{-1}$.

**中文：** 短例理解：410 nm、10 nW 时 E_ph≈4.845×10$^{-19}$ J，N_dot≈2.064×10$^{10}$ s$^{-1}$。

Source / 来源：W2 PDF p19–20 | W2 PDF p80 | T2-0014 | T2-0015 | T2-0016 | T2-0101 | T2-0102 | T2-0103 | T2-0104

## FX105 • [P3] Photon momentum / 光子动量

**EN:** Revision priority: Recognise; optional calculations. Shown in the particle-model slide; no dedicated momentum arithmetic was located. Recognize the concept and inverse-wavelength trend.

**中文：** 复习优先级：辅助辨认；计算后看。粒子模型页出现，未定位专门动量计算；会辨认概念与波长反比趋势。

**Formula / 公式：** $p_{ph} = h/\lambda_0 = E_{ph}/c$ (vacuum)

**EN:** Symbols and units: p_ph: photon momentum (kg·m/s); h: Planck constant; λ0: vacuum wavelength; E_ph: photon energy.

**中文：** 符号与单位：p_ph：光子动量 kg·m/s；h：普朗克常数；λ0：真空波长；E_ph：光子能量。

**EN:** When to use: Explain why absorption/reflection can transfer momentum and why light can exert pressure.

**中文：** 何时使用：解释吸收/反射为何能传递动量，以及光为何能施加压力。

**EN:** Conditions and traps: This is the vacuum photon relation. It is not an unqualified formula for momentum inside a material. Momentum, energy and photon rate have different units; P/E_ph is not a momentum.

**中文：** 条件与易错点：这是光子在真空中的关系，不可无条件拿来表示介质内动量。 动量、能量与光子率单位不同，P/E_ph 不是动量。

**EN:** Tiny example: Doubling vacuum wavelength halves each photon’ s momentum.

**中文：** 短例理解：真空波长加倍，单光子动量减半。

Source / 来源：W2 PDF p19

## FX106 • [P2] Energy fractions: reflection, absorption and transmission / 反射、吸收与透射能量比例

**EN:** Revision priority: Important supporting relations. A useful foundation for distinguishing optical interactions and absorbance notation.

**中文：** 复习优先级：重要基础关系。用于区分光学作用及吸光度记号的重要基础。

**Formula / 公式：** $R + A_{frac} + T = 1$

**EN:** Symbols and units: R: reflected fraction; A_frac: absorbed fraction; T: transmitted fraction. All are dimensionless fractions of the same incident energy.

**中文：** 符号与单位：R：反射比例；A_frac：吸收比例；T：透射比例；均为同一入射能量的无量纲比例。

**EN:** When to use: Check a simple optical energy budget and determine a missing fraction.

**中文：** 何时使用：核对简单光能量收支，并求缺失的比例。

**EN:** Conditions and traps: The measured/accounted channels must include all outgoing reflection/transmission; in a scattering sample, any separately excluded scattered light needs its own budget term. A_frac is an absorbed fraction, whereas Beer–Lambert absorbance A is −log10(T); the two A meanings are not interchangeable.

**中文：** 条件与易错点：被计入的反射/透射必须覆盖全部对应出射能量；散射样品中若有另外排除的散射光，应单独加入收支。 A_frac 是吸收比例；Beer–Lambert 的吸光度 A 是 −log10(T)，两个 A 不能混用。

**EN:** Tiny example: R=0.10 and T=0.70 imply A_frac=0.20, not an absorbance of 0.20.

**中文：** 短例理解：R=0.10、T=0.70 时 A_frac=0.20，但不能称吸光度就是 0.20。

Source / 来源：W2 PDF p30 | T2-0029 | T2-0030

## FX107 • [P2] Direct-beam attenuation and characteristic distances / 直达光衰减与特征距离

**EN:** Revision priority: Important supporting relations. W2 explains length and absorption/scattering coefficients; coefficients can be supplied, so do not memorize material tables.

**中文：** 复习优先级：重要基础关系。W2 解释光程及吸收/散射系数；材料系数可题中给，不背材料数值表。

**Formula / 公式：** $I(L) = I0 \exp(-\mu_t L)$; $\mu_t = \mu_a + \mu_s$; $\ell_a = 1/\mu_a$; $\ell_s = 1/\mu_s$; $\ell_t = 1/\mu_t$

**EN:** Symbols and units: I0, I: incident/direct-beam intensity; L: path length; $\mu_a$, $\mu_s$, $\mu_t$: absorption, scattering and total attenuation coefficients (inverse length); $\ell$: corresponding characteristic distances.

**中文：** 符号与单位：I0、I：入射/直达光强；L：光程；$\mu_a$、$\mu_s$、$\mu_t$：吸收、散射和总衰减系数，单位长度倒数；$\ell$：相应特征距离。

**EN:** When to use: Explain why a longer path or larger loss coefficient reduces direct transmitted light.

**中文：** 何时使用：解释光程更长或损耗系数更大时，直达透射光为何减少。

**EN:** Conditions and traps: Uniform medium and direct/ballistic-beam attenuation with approximately constant coefficients. Scattering removes photons from the direct beam but does not necessarily absorb them; total diffuse light is a different quantity. μL must be dimensionless. A 1/e distance is not a sharp boundary at which all light disappears.

**中文：** 条件与易错点：均匀介质、近似常系数的直达/弹道光衰减模型；散射使光离开直达束，不必然把光吸收，漫射总光是另一量。 μL 必须无量纲；1/e 距离不是光到此全部消失的硬边界。

**EN:** Tiny example: If $\mu_t = 2\ cm^{-1}$ and L=0.5 cm, $I/I0 = e^{-1} \approx 0.368$ and $\ell_t = 0.5$ cm.

**中文：** 短例理解：$\mu_t = 2\ cm^{-1}$、L=0.5 cm 时，$I/I0 = e^{-1} \approx 0.368$，$\ell_t = 0.5$ cm。

Source / 来源：W2 PDF p32–33 | T2-0034 | T2-0035 | T2-0036

## FX108 • [P1] Beer–Lambert: transmission and two logarithm conventions / Beer–Lambert：透过率与两种对数约定

**EN:** Revision priority: First: know and calculate. W2 Examples 3/4 are worked in class; know percent conversion, units and coefficient conventions.

**中文：** 复习优先级：优先掌握与短算。W2 Example 3/4 有课堂讲解，须会百分比、单位及系数约定。

**Formula / 公式：** $T = I/I0$; $A = -\log10(T) = \varepsilon CL$; $T = 10^{(-\varepsilon CL)}$; $I = I0 \exp(-\varepsilon' CL)$; $\varepsilon' = \ln(10)\ \varepsilon \approx 2.303\varepsilon$

**EN:** Symbols and units: A: dimensionless absorbance; T: transmitted fraction; C: molar concentration (M); L: cm; ε: base-10 molar absorptivity ($M^{-1}cm^{-1}$); ε′: natural-exponential coefficient in the same concentration/length units.

**中文：** 符号与单位：A：无量纲吸光度；T：透过比例；C：摩尔浓度 M；L：cm；ε：常用对数摩尔吸光系数 $M^{-1}cm^{-1}$；ε′：自然指数形式的相应系数。

**EN:** When to use: Convert transmitted fraction to absorbance and solve for concentration, path length or absorptivity.

**中文：** 何时使用：把透过比例换成吸光度，并求浓度、光程或吸光系数。

**EN:** Conditions and traps: Use a specified wavelength and a valid linear Beer–Lambert regime, after an appropriate blank/reference correction. Do not assume a buffer never absorbs at every wavelength. 15% means T=0.15; absorbance is not 1−T. Do not mix ε with ε′ or round ε before completing the calculation.

**中文：** 条件与易错点：使用指定波长、有效线性 Beer–Lambert 区间，作合适的空白/参比扣除；不能假设缓冲液在所有波长都不吸收。 15% 要写 T=0.15；吸光度不是 1−T，不混用 ε 与 ε′，计算结束前不先把 ε 粗略取整。

**EN:** Tiny example: At unchanged C and wavelength, changing L from 2 cm to 4.5 cm changes A=0.35 to 0.35×4.5/2=0.7875.

**中文：** 短例理解：C、波长不变时，L 从 2 cm 到 4.5 cm，使 A=0.35 变为 0.35×4.5/2=0.7875。

Source / 来源：W2 PDF p37–43 | T2-0041 | T2-0049 | T2-0050 | T2-0051

## FX109 • [P3] Two-wavelength hemoglobin mixture / 双波长血红蛋白混合关系

**EN:** Revision priority: Recognise; optional calculations. The lecturer says the detailed coefficients/derivation need not be memorized. Use this to recognize the two-wavelength principle or substitute supplied data.

**中文：** 复习优先级：辅助辨认；计算后看。老师说明详细系数/推导不需背；用于辨认双波长原理或题给数据代入。

**Formula / 公式：** $\mu_a(\lambda_i) = \ln(10)\,[\varepsilon_{ox}(\lambda_i)\,C_{ox} + \varepsilon_{de}(\lambda_i)\,C_{de}],\ i=1,2$

**EN:** Symbols and units: $C_{ox}$, $C_{de}$: molar oxy-/deoxyhemoglobin concentrations; $\varepsilon_{ox}$, $\varepsilon_{de}$: base-10 molar absorptivities at $\lambda_i$; $\mu_a$: natural-exponential absorption coefficient (e.g. $cm^{-1}$).

**中文：** 符号与单位：$C_{ox}$、$C_{de}$：含氧/脱氧血红蛋白摩尔浓度；$\varepsilon_{ox}$、$\varepsilon_{de}$：波长 $\lambda_i$ 下的常用对数摩尔吸光系数；$\mu_a$：自然指数吸收系数，如 $cm^{-1}$。

**EN:** When to use: Use two absorption measurements to separate two hemoglobin concentrations.

**中文：** 何时使用：用两个吸收测量分开两种血红蛋白浓度。

**EN:** Conditions and traps: Two-species additive absorption approximation with supplied coefficients and independent wavelength responses; backgrounds must be accounted for. If the two equations are proportional, two concentrations cannot be separated uniquely. ln(10) converts a base-10 ε to a natural-exponential μ_a; it is not a new concentration unit. An isosbestic wavelength alone measures a shared absorption contribution.

**中文：** 条件与易错点：两组分吸收可相加的近似，系数由题给，两个波长响应需独立，背景需处理；若两式成比例，不能唯一分出两浓度。 ln(10) 用于常用对数 ε 到自然指数 μ_a 的转换，不是新浓度单位；单独等吸收波长只有共同吸收贡献。

**EN:** Tiny example: At an isosbestic wavelength where ε_ox=ε_de=ε, μ_a=ln(10)ε(C_ox+C_de).

**中文：** 短例理解：等吸收波长 ε_ox=ε_de=ε 时，μ_a=ln(10)ε(C_ox+C_de)。

Source / 来源：W2 PDF p49 | T2-0063 | T2-0064

## FX110 • [P2] Oxygen saturation and total hemoglobin / 血氧饱和度与总血红蛋白

**EN:** Revision priority: Important supporting relations. Short conceptual relation from W2; understand the denominator and do simple ratios.

**中文：** 复习优先级：重要基础关系。W2 简短概念关系，懂分母含义并能做简单比值。

**Formula / 公式：** $SO2 = C_{ox} / (C_{ox} + C_{de})$; $CHb = C_{ox} + C_{de}$; $\text{saturation}(\%) = 100 \times SO2$

**EN:** Symbols and units: C_ox: oxyhemoglobin concentration; C_de: deoxyhemoglobin concentration; CHb: their total; SO2: dimensionless oxygenated fraction.

**中文：** 符号与单位：C_ox：含氧血红蛋白浓度；C_de：脱氧血红蛋白浓度；CHb：两者总浓度；SO2：无量纲含氧比例。

**EN:** When to use: Distinguish fraction oxygenated from the total amount of hemoglobin.

**中文：** 何时使用：分清含氧比例与血红蛋白总量。

**EN:** Conditions and traps: Both concentrations must use the same units and refer to the same sample/compartment. The denominator must be nonzero; device SpO2 is a calibrated estimate, not every abstract mixture fraction. The denominator is total hemoglobin, not deoxyhemoglobin alone; 0.8 corresponds to 80%, not 0.8%.

**中文：** 条件与易错点：两浓度单位一致、属于同一样品/区室，分母非零；仪器 SpO2 是校准估计，并非每个抽象混合比例都等同临床读数。 分母是总血红蛋白，不是只用脱氧血红蛋白；0.8 对应 80%，不是 0.8%。

**EN:** Tiny example: C_ox=8 and C_de=2 in the same units give SO2=80% and CHb=10.

**中文：** 短例理解：同单位 C_ox=8、C_de=2 时，SO2=80%，CHb=10。

Source / 来源：W2 PDF p50 | T2-0063 | T2-0064

## FX111 • [P3] Pulse-oximeter example: ratio of ratios / 血氧仪例题：比值之比

**EN:** Revision priority: Recognise; optional calculations. Worked for graph interpretation; the teacher explicitly says not to memorize the long calculation. Keep it available for supplied-formula questions.

**中文：** 复习优先级：辅助辨认；计算后看。课堂用于读图，老师明确长计算不必背；保留用于题给公式后的代入。

**Formula / 公式：** $R_{os} = \ln(I_{p,R}/I_{b,R}) / \ln(I_{p,IR}/I_{b,IR})$; $SaO_2 = [\varepsilon_{Hb,R} - \varepsilon_{Hb,IR} R_{os}] / [(\varepsilon_{Hb,R} - \varepsilon_{HbO_2,R}) + (\varepsilon_{HbO_2,IR} - \varepsilon_{Hb,IR}) R_{os}]$

**EN:** Symbols and units: Ip, Ib: trace peak and baseline intensities; R, IR subscripts: red and infrared channels; ε: the four supplied hemoglobin coefficients; R_os and SaO2 are dimensionless.

**中文：** 符号与单位：Ip、Ib：波形峰与基线光强；R、IR 下标：红光、红外通道；ε：题给四个血红蛋白系数；R_os、SaO2 都无量纲。

**EN:** When to use: Read each channel’ s peak/baseline, form the specified ratio, and substitute the provided calibration relation.

**中文：** 何时使用：读取两个通道峰/基线，按指定定义求比值，再代入给定校准关系。

**EN:** Conditions and traps: This is the relation supplied for the course example with its specified wavelengths and coefficients, not a universal uncalibrated saturation law. Intensities are positive and the IR log denominator is nonzero. Do not multiply intensities by wavelength or attach nm to the ratios. This example uses 630/940 nm, while the general device slide uses 660/940 nm.

**中文：** 条件与易错点：这是该例题指定波长/系数下给出的关系，不是无需校准的通用血氧定律；光强须为正，红外对数分母不能为零。 不要把光强再乘波长，不给比值加 nm。此例用 630/940 nm，设备概述页用 660/940 nm。

**EN:** Tiny example: 18/12 and 8/5 give R_os=0.86269. With ε_Hb,R=1.1, ε_HbO2,R=0.17, ε_Hb,IR=0.16 and ε_HbO2,IR=0.23, SaO2≈97.13%.

**中文：** 短例理解：18/12、8/5 得 R_os=0.86269。按 ε_Hb,R=1.1、ε_HbO2,R=0.17、ε_Hb,IR=0.16、ε_HbO2,IR=0.23 的明确对应关系代入，得 SaO2≈97.13%。

Source / 来源：W2 PDF p53–54 | T2-0065 | T2-0066 | T2-0067 | T2-0068

## FX112 • [P1] Raman shift and absolute wavenumber / 拉曼位移与绝对波数

**EN:** Revision priority: First: know and calculate. The lecturer works W2 p67 and stresses the forward/reverse conversion. Also supports Sample Q2 scattering concepts.

**中文：** 复习优先级：优先掌握与短算。老师讲 W2 p67 并强调正向/反向换算，也支撑 Sample Q2 散射概念。

**Formula / 公式：** $\nu_tilde = 1/\lambda(cm) = 10^7/\lambda(nm)$; $f = c_cm\ \nu_tilde$; $\Delta\nu_tilde = 10^7\ [1/\lambda0(nm) - 1/\lambda s(nm)]$; $\lambda s(nm) = 1 / [1/\lambda0(nm) - \Delta\nu_tilde/10^7]$

**EN:** Symbols and units: ν_tilde: absolute wavenumber ($cm^{-1}$); Δν_tilde: signed Raman shift ($cm^{-1}$); λ0, λs: excitation/scattered vacuum wavelengths; c_cm≈$2.998\times10^{10}$ cm/s.

**中文：** 符号与单位：ν_tilde：绝对波数 $cm^{-1}$；Δν_tilde：带符号的拉曼位移 $cm^{-1}$；λ0、λs：入射/散射真空波长；c_cm≈$2.998\times10^{10}$ cm/s。

**EN:** When to use: Convert spectral coordinates and reverse a given Raman shift to find scattered wavelength.

**中文：** 何时使用：换算光谱坐标，并由给定拉曼位移反求散射波长。

**EN:** Conditions and traps: Use the same wavelength unit in both reciprocals. With the stated signed convention, Stokes has positive shift and anti-Stokes negative shift; some plots instead use absolute magnitudes. Raman shift is a difference of reciprocal wavelengths, not λs−λ0, nor the absolute scattered wavenumber.

**中文：** 条件与易错点：两个倒数采用同一波长单位；本带符号约定下 Stokes 正、anti-Stokes 负，有些图则用绝对值。 拉曼位移是倒数波长之差，不是 λs−λ0，也不是散射光的绝对波数。

**EN:** Tiny example: 1000→1080 nm gives 740.74 $cm^{-1}$ (Stokes); a given +800 $cm^{-1}$ shift at 1000 nm gives λs≈1086.96 nm.

**中文：** 短例理解：1000→1080 nm 得 740.74 $cm^{-1}$（Stokes）；1000 nm 入射、+800 $cm^{-1}$ 位移时 λs≈1086.96 nm。

Source / 来源：W2 PDF p64–68 | T2-0084 | T2-0085 | T2-0086 | Sample Quiz PDF p1 Q2

## FX113 • [P1] Fluorescence lifetime and exponential decay / 荧光寿命与指数衰减

**EN:** Revision priority: First: know and calculate. Know lifetime, units and the 1/e meaning. The differential notation is for recognition, not a new calculus exercise.

**中文：** 复习优先级：优先掌握与短算。会寿命、单位及 1/e 含义；微分记号用于辨认，不额外要求微积分解题。

**Formula / 公式：** $k_tot = kr + knr$; $knr = k1 + k2$; $\tau = 1/k_tot$; $dN_exc/dt = -k_tot\ N_exc$; $N_exc(t) = N_exc(0)\ \exp(-t/\tau)$; $I_fl(t) \propto \exp(-t/\tau)$; $t_half = \tau\ \ln(2)$

**EN:** Symbols and units: N_exc: excited S1 population; kr, knr, k_tot: radiative, nonradiative and total exit rates ($s^{-1}$); τ: lifetime (s); I_fl: background-corrected fluorescence intensity. In the p82 diagram, k1 is nonradiative S1→S0 relaxation and k2 is S1→T1 intersystem crossing.

**中文：** 符号与单位：N_exc：S1 激发态粒子数；kr、knr、k_tot：辐射、非辐射与总离开速率，单位 $s^{-1}$；τ：寿命 s；I_fl：扣背景后的荧光强度。 原p82图中，k1为S1→S0非辐射弛豫，k2为S1→T1系间窜越。

**EN:** When to use: Connect decay rate, the 1/e lifetime and the distinct half-life.

**中文：** 何时使用：联系衰减速率、1/e 寿命及不同的半衰期。

**EN:** Conditions and traps: Single-exponential, constant first-order rates after a short excitation pulse, without continued pumping. Multi-exponential samples need a more detailed model. I_fl follows population only with unchanged detection and kr. knr=k1+k2 follows this two-channel nonradiative model; other samples can have additional nonradiative channels. At t=τ, 36.8% remains, not 50%. A lifetime is not the time until the signal becomes exactly zero; kr alone is not 1/τ unless knr=0.

**中文：** 条件与易错点：短激发脉冲后、无持续泵浦、常数一级速率的单指数模型；多指数样品需更复杂模型。检测条件与 kr 不变时 I_fl 才随粒子数衰减。 knr=k1+k2限于此图的两类非辐射通道模型；其他样品可能还有别的非辐射通道。 t=τ 时剩 36.8%，不是 50%；寿命不是恰好变零所需时间，除非 knr=0，否则 kr 不等于 1/τ。

**EN:** Tiny example: τ=10 ns gives k_tot=$10^8$ $s^{-1}$ and t_half≈6.93 ns; at 20 ns the remaining fraction is $e^{-2}$≈13.5%.

**中文：** 短例理解：τ=10 ns 时 k_tot=$10^8$ $s^{-1}$、t_half≈6.93 ns；20 ns 后剩 $e^{-2}$≈13.5%。

Source / 来源：W2 PDF p82 | T2-0105 | T2-0106 | Sample Quiz PDF p1 Q1

## FX114 • [P1] Quantum yield, radiative and nonradiative rates / 量子产率、辐射与非辐射速率

**EN:** Revision priority: First: know and calculate. W2 Examples 6/7 are explained in class; these also support the full fluorescence concepts behind Sample Q1.

**中文：** 复习优先级：优先掌握与短算。W2 Example 6/7 课堂有讲，亦支撑 Sample Q1 全部荧光概念。

**Formula / 公式：** Φ = N_emitted/N_absorbed = kr/(kr+knr) = krτ; kr = Φ/τ; knr = (1−Φ)/τ; N_emitted = Φ N_absorbed

**EN:** Symbols and units: Φ: dimensionless yield (0–1); N: emitted/absorbed photon counts; kr, knr: $s^{-1}$ if τ is in s. Use photon numbers, not unequal-wavelength optical energies.

**中文：** 符号与单位：Φ：无量纲产率 0–1；N：发射/吸收光子数；τ 用 s 时 kr、knr 为 $s^{-1}$。按光子数算，不能把不同波长的光能比直接当产率。

**EN:** When to use: Find yield or either decay rate and predict emitted photon number from absorbed photons.

**中文：** 何时使用：求产率或两种衰减速率，并由吸收光子数预测发射数量。

**EN:** Conditions and traps: The rate equality uses the single-excited-state first-order model; count-based yield needs corrected photon measurements. Incident photons equal absorbed photons only if the beam is fully absorbed. Do not use arbitrary spectral peak heights as general quantum-yield measurements. 1 $ns^{-1}=10^{9}\ s^{-1}$; the p88 25-hour unit is unusual but must be preserved if solving that source question.

**中文：** 条件与易错点：速率等式采用单激发态一级模型；计数产率须用校正光子测量。只有完全吸收时，入射光子才等于吸收光子。 不能把任意光谱峰高比当通用量子产率。1 $ns^{-1}=10^{9}\ s^{-1}$；p88 的 25 小时虽异常，做原题仍需保留。

**EN:** Tiny example: Φ=0.46, τ=8.4 ns give kr≈0.05476 $ns^{-1}=5.476\times10^{7}\ s^{-1}$ and knr≈$6.429\times10^{7}\ s^{-1}$.

**中文：** 短例理解：Φ=0.46、τ=8.4 ns 时，kr≈0.05476 $ns^{-1}=5.476\times10^{7}\ s^{-1}$，knr≈$6.429\times10^{7}\ s^{-1}$。

Source / 来源：W2 PDF p83–88 | T2-0106 | T2-0107 | Sample Quiz PDF p1 Q1

## FX115 • [P2] Molecular fluorescence brightness / 分子荧光亮度指标

**EN:** Revision priority: Important supporting relations. Printed on p85 and supports the yield/brightness explanation; retain the matched-conditions qualification.

**中文：** 复习优先级：重要基础关系。p85 明列，支撑产率/亮度解释，须保留比较条件。

**Formula / 公式：** $B_{mol} = \varepsilon(\lambda_{exc})\,\Phi$; at equal low concentration, excitation and collection: fluorescence signal $\propto C\,\varepsilon(\lambda_{exc})\,\Phi$

**EN:** Symbols and units: ε(λ_exc): molar absorptivity at the chosen excitation wavelength; Φ: quantum yield; B_mol: relative molecular brightness indicator, not a lumen/W instrument reading.

**中文：** 符号与单位：ε(λ_exc)：所选激发波长下摩尔吸光系数；Φ：量子产率；B_mol：相对分子亮度指标，不是仪器的流明/瓦读数。

**EN:** When to use: Compare a fluorophore’ s absorption strength and emission efficiency together.

**中文：** 何时使用：同时比较荧光分子的吸收能力和发射效率。

**EN:** Conditions and traps: Matched excitation/detection and dilute, unsaturated response, without major reabsorption, self-quenching or bleaching. Signal also depends on amount and instrument settings. Higher Φ alone does not always make a dye brighter if ε differs. Intensity is not necessarily linear in concentration at high concentration or saturation.

**中文：** 条件与易错点：激发/检测条件匹配，稀溶液、未饱和，重吸收、自猝灭和漂白影响不大；实际信号还取决于数量和仪器设置。 若 ε 不同，Φ 更高不一定更亮；高浓度或饱和时强度不必与浓度线性。

**EN:** Tiny example: ε=100000, Φ=0.25 give εΦ=25000; ε=20000, Φ=0.75 give 15000, so the first indicator is larger.

**中文：** 短例理解：ε=100000、Φ=0.25 得 εΦ=25000；ε=20000、Φ=0.75 得 15000，前者亮度指标更大。

Source / 来源：W2 PDF p84–86 | T2-0106 | T2-0107

## FX116 • [P1] Grating resolving power and illuminated grooves / 光栅分辨本领与受照刻线数

**EN:** Revision priority: First: know and calculate. Explicit Sample Q7 topic; the standard ideal-grating relation is added for solving that source question, not falsely attributed as printed on W2 p35.

**中文：** 复习优先级：优先掌握与短算。Sample Q7 明确涉及；补入标准理想光栅关系帮助做题，不声称 W2 p35 印了该等式。

**Formula / 公式：** $R_{grating} = \lambda/\Delta\lambda_{min} = |m| N$; $N = groove_density \times illuminated_width$; $\Delta\lambda_{min} = \lambda/(|m|N)$

**EN:** Symbols and units: $R_{grating}$: dimensionless resolving power; m: nonzero diffraction order; N: illuminated groove count; $\Delta\lambda_{min}$: smallest separable wavelength gap. Density and illuminated width must use reciprocal length units.

**中文：** 符号与单位：$R_{grating}$：无量纲分辨本领；m：非零衍射级次；N：受照刻线数；$\Delta\lambda_{min}$：最小可分辨波长间隔。刻线密度与照亮宽度长度单位须匹配。

**EN:** When to use: Find spectral resolution or compare groove density at fixed illuminated width and order.

**中文：** 何时使用：求光谱分辨率，或在固定受照宽度/级次下比较刻线密度。

**EN:** Conditions and traps: Ideal grating limit for coherently illuminated grooves; real spectrometer resolution may also be limited by slit width, optics and detector sampling. Compare at matched wavelength, order and illuminated width. Density is not N; a larger density gives more illuminated grooves only at fixed width. Resolving power R is larger-better, whereas $\Delta\lambda_{min}$ is smaller-better.

**中文：** 条件与易错点：理想相干照亮光栅的极限；实际分辨率还可受狭缝、光学系统和探测器采样限制。比较时固定波长、级次和受照宽度。 密度不是 N；只有受照宽度固定时密度越大才照亮更多刻线。R 越大越好，$\Delta\lambda_{min}$ 越小越好。

**EN:** Tiny example: At $\lambda$=600 nm, $|m|$=1 and N=50, R=50 and $\Delta\lambda_{min}$=12 nm.

**中文：** 短例理解：$\lambda$=600 nm、$|m|$=1、N=50 时，R=50、$\Delta\lambda_{min}$=12 nm。

Source / 来源：W2 PDF p35 | W1 PDF p86 | Sample Quiz PDF p1 Q7

## FX117 • [P2] Label-free spectral sensitivity and index LOD / 无标记光谱灵敏度与折射率检出限

**EN:** Revision priority: Important supporting relations. The lecturer repeatedly reads these plots and explicitly mentions sensitivity/LOD calculations for the final; this is supporting Quiz preparation, not a guaranteed Quiz item.

**中文：** 复习优先级：重要基础关系。老师反复读图并明确提到期末灵敏度/LOD 计算；用于 Quiz 辅助复习，不夸大为必出。

**Formula / 公式：** $S_\lambda = \Delta\lambda_{res}/\Delta n = (\lambda_2-\lambda_1)/(n_2-n_1)$; $\Delta\lambda_{res} \approx S_\lambda \Delta n$; $LOD_n = 3\sigma_\lambda/|S_\lambda|$

**EN:** Symbols and units: $\lambda_{res}$: resonance wavelength; $\Delta n$: index change in RIU; $S_\lambda$: nm/RIU; $\sigma_\lambda$: wavelength-noise SD in nm; $LOD_n$: RIU.

**中文：** 符号与单位：$\lambda_{res}$：共振波长；$\Delta n$：折射率变化 RIU；$S_\lambda$：nm/RIU；$\sigma_\lambda$：波长噪声标准差 nm；$LOD_n$：RIU。

**EN:** When to use: Read a peak/dip position shift, predict the response to index change, or translate wavelength noise into an index threshold.

**中文：** 何时使用：读峰/谷位置移动，预测折射率响应，或将波长噪声换成折射率阈值。

**EN:** Conditions and traps: Use the local calibrated index range, matched temperature/geometry and same resonance mode. A concentration LOD additionally needs an index–concentration calibration. Peak height and peak position are different signals. nm/RIU sensitivity is not automatically nm/mM; σ must have the same spectral unit as S.

**中文：** 条件与易错点：采用局部校准折射率范围、匹配温度/几何及同一共振模式；浓度检出限还需折射率—浓度校准。 峰高与峰位是不同信号；nm/RIU 不自动等于 nm/mM，σ 和 S 的光谱单位要一致。

**EN:** Tiny example: $S_\lambda$=30 nm/RIU and $\Delta n=10^{-4}$ RIU give a 3 pm shift. If $\sigma_\lambda$=1 pm, $LOD_n=10^{-4}$ RIU.

**中文：** 短例理解：$S_\lambda$=30 nm/RIU、$\Delta n=10^{-4}$ RIU 时位移 3 pm；若 $\sigma_\lambda$=1 pm，则 $LOD_n=10^{-4}$ RIU。

Source / 来源：W3 PDF p15 | W3 PDF p47 | W3 PDF p79 | T3-0024 | T3-0025 | T3-0026 | T3-0077 | T3-0078 | T3-0108

## FX118 • [P1] Snell refraction and critical angle / 斯涅尔折射与临界角

**EN:** Revision priority: First: know and calculate. The two TIR requirements are central to fibre questions; the numerical critical-angle calculation is a supporting basic application.

**中文：** 复习优先级：优先掌握与短算。全反射的两条件是光纤题核心；临界角数字计算属基础应用。

**Formula / 公式：** $n_1 \sin\theta_1 = n_2 \sin\theta_2$; $\theta_c = \arcsin(n_2/n_1)$ for $n_1>n_2$; TIR requires $\theta_1>\theta_c$

**EN:** Symbols and units: $n_1$,$n_2$: refractive indices; $\theta_1$,$\theta_2$: angles from the interface normal; $\theta_c$: critical angle.

**中文：** 符号与单位：$n_1$、$n_2$：折射率；$\theta_1$、$\theta_2$：相对界面法线的角；$\theta_c$：临界角。

**EN:** When to use: Decide whether a ray refracts or satisfies total internal reflection and calculate the threshold.

**中文：** 何时使用：判断折射或全反射，并求临界阈值。

**EN:** Conditions and traps: Ordinary isotropic lossless media in the course ray model. TIR needs high-index to low-index travel and incidence above the critical boundary; at θc the refracted ray is grazing. Angles are from the normal, not the surface. The p22 shorthand ≥ includes the boundary; distinguish it from strict above-critical TIR. n1<n2 has no real TIR critical angle.

**中文：** 条件与易错点：本课程射线模型中的普通各向同性无损介质。全反射需从高到低折射率且超过临界角；θc 时折射光沿界面。 角度相对法线，不是表面；p22 简写 ≥ 包括边界，应与严格超过临界值区分。n1<n2 没有实数全反射临界角。

**EN:** Tiny example: Glass n1=1.45 to water n2=1.33 gives θc≈66.53°; 70° incidence satisfies TIR.

**中文：** 短例理解：玻璃 n1=1.45 到水 n2=1.33 时 θc≈66.53°，70° 入射满足全反射。

Source / 来源：W3 PDF p21–22 | W3 PDF p25 | T3-0032 | T3-0033 | T3-0034 | T3-0035 | T3-0036

## FX119 • [P2] Fibre numerical aperture and acceptance half-angle / 光纤数值孔径与接受半角

**EN:** Revision priority: Important supporting relations. Explicit W3 formula explained in class; retain the n0 assumption rather than copying an air-only shortcut everywhere.

**中文：** 复习优先级：重要基础关系。W3 明列且课堂讲解，须保留 n0 假设，不把空气简式套到所有介质。

**Formula / 公式：** $NA_{fibre} = n_0 \sin\theta_{accept,max} = \sqrt{n_{core}^2 - n_{clad}^2}$; $\theta_{accept,max} = \arcsin(NA_{fibre}/n_0)$

**EN:** Symbols and units: n0: outside launch-medium index; n_core,n_clad: core/cladding indices; θ_accept,max: maximum launch angle from the fibre axis (acceptance cone half-angle); NA is dimensionless.

**中文：** 符号与单位：n0：光纤外入射介质折射率；n_core、n_clad：纤芯/包层折射率；θ_accept,max：相对纤轴最大入射角，即接受锥半角；NA 无量纲。

**EN:** When to use: Connect core–cladding index contrast to the external light-acceptance cone.

**中文：** 何时使用：把纤芯/包层折射率差与外部接受光锥联系起来。

**EN:** Conditions and traps: The course meridional-ray model of a step-index fibre, n_core>n_clad; arcsin requires NA/n0≤1. The slide omits n0 by assuming air. The external acceptance angle is measured from the fibre axis; the internal TIR angle is measured from an interface normal. They are different angles.

**中文：** 条件与易错点：课件阶跃折射率光纤的子午射线模型，n_core>n_clad；反正弦需 NA/n0≤1。课件省略 n0，默认空气。 外部接受角相对纤轴，内部全反射角相对界面法线，两角不同。

**EN:** Tiny example: n_core=1.50,n_clad=1.48 give NA≈0.2441; in air θmax≈14.13°. In water n0=1.33 the same NA gives θmax≈10.58°.

**中文：** 短例理解：n_core=1.50、n_clad=1.48 得 NA≈0.2441；空气中 θmax≈14.13°，水 n0=1.33 中同 NA 得 θmax≈10.58°。

Source / 来源：W3 PDF p23 | T3-0036 | T3-0037 | T3-0038

## FX120 • [P2] Evanescent intensity and field-amplitude decay / 倏逝强度与场幅衰减

**EN:** Revision priority: Important supporting relations. The surface-confinement concept is P1; the exponential expression is useful supporting algebra. Amplitude conversion is explained, not treated as an extra exam claim.

**中文：** 复习优先级：重要基础关系。近表面局限概念为 P1；指数式属辅助代数。解释场幅转换，不额外宣称是考试要求。

**Formula / 公式：** $I(d)/I0 = \exp(-d/L_I)$; $|E_{field}(d)|/|E0| = \exp(-d/L_E)$; $I \propto |E_{field}|^2$; $L_E = 2 L_I$

**EN:** Symbols and units: d: distance normal to the interface; L_I: 1/e intensity decay depth; L_E: 1/e field-amplitude decay depth. Intensity and electric field are different quantities.

**中文：** 符号与单位：d：垂直界面的距离；L_I：强度降至 1/e 的深度；L_E：场幅降至 1/e 的深度。光强与电场是不同量。

**EN:** When to use: Explain surface-confined sensing and avoid a factor-of-two error when switching amplitude/intensity conventions.

**中文：** 何时使用：解释近表面传感，并避免把场幅与强度深度混用导致二倍误差。

**EN:** Conditions and traps: A single exponential evanescent tail in a uniform lossless low-index region; keep the same depth definition. The p27 plotted normalized intensity corresponds to L_I. p27 writes I(d)=exp(−d/L0) but plots I/I0: the expression is normalized. Goos–Hänchen lateral shift is along the interface, not this normal penetration depth.

**中文：** 条件与易错点：均匀无损低折射率区的单指数倏逝尾部，需保持深度定义一致；p27 的归一化强度图对应 L_I。 p27 写 I(d)=exp(−d/L0) 而图画 I/I0，因此式子实为归一化。Goos–Hänchen 横移沿界面，不是此处垂直穿透深度。

**EN:** Tiny example: If L_I=100 nm, then at d=100 nm intensity is 36.8% and field amplitude is 60.7%; L_E=200 nm.

**中文：** 短例理解：若 L_I=100 nm，d=100 nm 时强度剩 36.8%、场幅剩 60.7%，L_E=200 nm。

Source / 来源：W3 PDF p25–27 | T3-0042 | T3-0043 | T3-0044 | T3-0045 | T3-0046

## FX121 • [P3] Evanescent penetration-depth convention (support) / 倏逝穿透深度约定（辅助）

**EN:** Revision priority: Recognise; optional calculations. README labels this as supplementary; retain it for understanding definitions, without announcing a new required calculation.

**中文：** 复习优先级：辅助辨认；计算后看。README 明确标作补充；保留帮助理解约定，不增加必做计算要求。

**Formula / 公式：** $\kappa = (2\pi/\lambda 0)\ \mathrm{sqrt}(n1^2 \sin^2\theta - n2^2)$; $L_E = 1/\kappa = \lambda 0 / [2\pi\ \mathrm{sqrt}(n1^2 \sin^2\theta - n2^2)]$; $L_I = 1/(2\kappa) = \lambda 0 / [4\pi\ \mathrm{sqrt}(n1^2 \sin^2\theta - n2^2)]$

**EN:** Symbols and units: λ0: vacuum wavelength; θ: incidence from the normal on the n1 side; n1>n2. κ is the normal field-amplitude decay constant in inverse length ($m^{-1}$ for λ0 in m); L_E and L_I are amplitude/intensity 1/e depths in the same length unit as λ0.

**中文：** 符号与单位：λ0为真空波长；θ为n1侧相对法线的入射角；n1>n2。κ为场幅垂直衰减常数，单位长度的倒数（λ0用m时κ用$m^{-1}$）；L_E、L_I为场幅 / 强度1/e深度，与λ0采用同一长度单位。

**EN:** When to use: Make the README supplemental depth formula unambiguous and connect it to the three factors listed on the slide.

**中文：** 何时使用：明确 README 补充深度式的约定，并联系课件列出的三个因素。

**EN:** Conditions and traps: Derived flat-interface TIR expression for isotropic lossless media with θ>θc. It is supplementary to p26’ s qualitative factors, not an original numeric worked formula. Very near θc the idealized model is sensitive to conditions. The README’ s 4π formula is an intensity depth. Texts using 2π may be giving field-amplitude depth; do not call either wrong without checking its definition.

**中文：** 条件与易错点：由各向同性无损平界面、θ>θc 的全反射模型推得；是对 p26 定性因素的辅助，不是原课件数字例题公式。接近 θc 时理想模型对条件很敏感。 README 的 4π 式为强度深度；资料中 2π 式可能定义场幅深度，不能不看定义就判错。

**EN:** Tiny example: At fixed n1,n2,θ, doubling λ0 doubles both depths; at fixed wavelength, approaching θc from above increases them.

**中文：** 短例理解：n1、n2、θ 固定时 λ0 加倍使两深度加倍；固定波长、从上方接近 θc 时，两深度增大。

Source / 来源：W3 PDF p26–27 | README §3.2 倏逝场（补充公式）

## FX122 • [P2] WGM resonance and small fractional changes / WGM 共振与小幅相对变化

**EN:** Revision priority: Important supporting relations. W3 shows the formula and explains index/geometry shifts; the lecturer’ s main emphasis is the reference peak and its shift.

**中文：** 复习优先级：重要基础关系。W3 明列并解释折射率/几何位移；课堂主要强调基准峰与移动。

**Formula / 公式：** $\lambda_res = (2\pi\ R\ n_eff)/m$; $\Delta\lambda_res/\lambda_res \approx \Delta R/R + \Delta n_eff/n_eff$

**EN:** Symbols and units: R: ring/sphere mode radius; n_eff: effective modal index; m: positive integer longitudinal/azimuthal mode order; λ_res: vacuum resonance wavelength.

**中文：** 符号与单位：R：环/球模式的半径；n_eff：模式有效折射率；m：正整数纵向/方位模式级次；λ_res：真空共振波长。

**EN:** When to use: Predict resonance shifts due to changes in geometry or effective index.

**中文：** 何时使用：预测几何或有效折射率变化引起的共振位移。

**EN:** Conditions and traps: Same mode m, small perturbations and negligible/localized dispersion in the course optical-path model. The fractional expression is first order; it neglects the product ΔRΔn and does not automatically use bulk index in place of n_eff. The slide uses r/R and n for the same role; use one consistent notation. m is not generally 1 just because a lecture gives that verbal simplification. A radius change and an index change may both shift the peak.

**中文：** 条件与易错点：同一模式 m、小扰动及课件光程模型中的可忽略/局部色散。相对变化式为一阶近似，略去 ΔRΔn 乘积，不能自动用体相折射率替换 n_eff。 课件 r/R、n 记号混用，要统一含义。不能因课堂口头简化就总令 m=1；半径与折射率变化都可使峰移动。

**EN:** Tiny example: If $\Delta R/R=2\times10^{-5}$ and $\Delta n_{eff}/n_{eff}=4\times10^{-5}$, a 1550 nm resonance shifts by ≈0.093 nm.

**中文：** 短例理解：$\Delta R/R=2\times10^{-5}$、$\Delta n_{eff}/n_{eff}=4\times10^{-5}$ 时，1550 nm 共振约移动 0.093 nm。

Source / 来源：W3 PDF p46–47 | T3-0073 | T3-0074 | T3-0075 | T3-0076

## FX123 • [P3] Two-mirror cavity resonance (support) / 双镜腔共振（辅助）

**EN:** Revision priority: Recognise; optional calculations. The Quiz-relevant lesson is cavity recognition; this supporting equation is not added as a teacher-confirmed Quiz calculation.

**中文：** 复习优先级：辅助辨认；计算后看。Quiz 相关核心为腔结构辨识，此辅助等式不标成老师确认的 Quiz 计算。

**Formula / 公式：** $2 n_{cav} L = m \lambda_0;\ \lambda_0 = 2 n_{cav} L/m$

**EN:** Symbols and units: L: physical mirror separation; n_cav: cavity medium index; m: positive integer mode order; λ0: vacuum resonance wavelength. The wavelength in the cavity is λ0/n_cav.

**中文：** 符号与单位：L：两镜物理间距；n_cav：腔介质折射率；m：正整数模式级次；λ0：真空共振波长，腔内波长为 λ0/n_cav。

**EN:** When to use: Explain the standing-wave optical-path condition for a simple Fabry–Pérot cavity.

**中文：** 何时使用：解释简单 Fabry–Pérot 腔的驻波光程条件。

**EN:** Conditions and traps: Normal incidence, a simple two-mirror path and negligible additional reflection phase. This is the generalized interpretation of the slide’ s half-wavelength illustration; real coatings and angled paths modify it. The integer mode label in a standing-wave drawing is not the same as the refractive-index symbol n. Do not confuse physical length L with optical path n_cav L.

**中文：** 条件与易错点：正入射、简单双镜光程且额外反射相位可忽略；是课件半波长示意的规范一般化解释。真实镀膜与倾斜光路会修改条件。 驻波图中的整数模式编号不是折射率 n；物理长度 L 与光程 n_cav L 不同。

**EN:** Tiny example: In air, L=1 μm and m=4 give λ0=500 nm.

**中文：** 短例理解：空气中 L=1 μm、m=4 时 λ0=500 nm。

Source / 来源：W3 PDF p44 | W3 PDF p48

## FX124 • [P2] FWHM, Q and unit-consistent FOM / 半高宽、Q 与单位一致的 FOM

**EN:** Revision priority: Important supporting relations. The teacher explicitly mentions these for the final; retain them as important support for broad Quiz concepts, without a guarantee.

**中文：** 复习优先级：重要基础关系。老师明确提到期末，此处作为宽泛 Quiz 概念的重要辅助，不保证必出。

**Formula / 公式：** I_half,peak = I_base + (I_max−I_base)/2; I_half,dip = I_base − (I_base−I_min)/2 = (I_base+I_min)/2; width_x = x_right−x_left at the appropriate half level; Q = f0/Δf_FWHM ≈ λ0/Δλ_FWHM (narrow line); FOM_λ = |S_λ|/Δλ_FWHM; FOM_f = |S_f|/Δf_FWHM

**EN:** Symbols and units: x: chosen spectral coordinate; Δf: linewidth in Hz; Δλ: linewidth in nm; S_λ: nm/RIU; S_f: Hz/RIU; Q: dimensionless; both consistent FOMs: $RIU^{-1}$.

**中文：** 符号与单位：x：所选光谱坐标；Δf：Hz 线宽；Δλ：nm 线宽；S_λ：nm/RIU；S_f：Hz/RIU；Q 无量纲，两种一致 FOM 都为 $RIU^{-1}$。

**EN:** When to use: Read the full half-height width of an upward peak or the full half-depth width of a downward dip, then compare sharpness and index-shift performance.

**中文：** 何时使用：上凸峰读半高全宽，向下谷读半深全宽，再比较共振尖锐程度与折射率位移表现。

**EN:** Conditions and traps: Account for baseline and distinguish a peak from a dip; SPR reflectance often has a dip. Subtract baseline before identifying half-height. Q≈λ0/Δλ assumes a narrow resonance; FOM must use the same spectral coordinate for sensitivity and width. Neither alone proves concentration LOD without noise/calibration. The README calls wavelength Q exactly equivalent and tells readers to follow Δf in a mixed-unit FOM; correct this: wavelength Q is a narrow-line approximation, and nm/RIU cannot be divided directly by Hz. Δf is width, not a before/after shift.

**中文：** 条件与易错点：先看基线并区分上凸峰与向下谷；SPR反射谱常是向下谷。先扣基线再找半高；Q≈λ0/Δλ 要求窄共振。FOM 的灵敏度和宽度须用同一光谱坐标；若缺噪声/浓度校准，均不能单独证明浓度检出限。 README 把波长 Q 说成完全等价、混单位 FOM 说应照 Δf，须修正：波长式是窄线近似，nm/RIU 不能直接除 Hz。Δf 是宽度，不是前后峰位移。

**EN:** Tiny example: λ0=600 nm, Δλ=10 nm and S_λ=200 nm/RIU give Q≈60 and FOM_λ=20 $RIU^{-1}$. Halving Δλ doubles both if the other quantities stay fixed.

**中文：** 短例理解：λ0=600 nm、Δλ=10 nm、S_λ=200 nm/RIU 时，Q≈60、FOM_λ=20 $RIU^{-1}$；其他不变、线宽减半，两者加倍。

Source / 来源：W3 PDF p80 | T3-0109 | T3-0110

## FX125 • [P3] Weak Raman signals and reported SERS enhancement / 微弱拉曼信号与所报 SERS 增强量级

**EN:** Revision priority: Recognise; optional calculations. Keep the weakness/enhancement concept; no need to memorize every reported exponent as a guaranteed Quiz number.

**中文：** 复习优先级：辅助辨认；计算后看。保留弱信号/增强概念，不把所报每个指数标成必背 Quiz 数字。

**Formula / 公式：** $I_{Rayleigh}/I_0 \sim 10^{-3}$; $I_{Raman}/I_0 \sim 10^{-6}$–$10^{-9}$; reported SERS enhancement can be ~$10^{9}$–$10^{12}$

**EN:** Symbols and units: I0: incident intensity; I_Rayleigh and I_Raman: illustrative scattered intensities; enhancement: a stated comparison under defined measurement conditions.

**中文：** 符号与单位：I0：入射强度；I_Rayleigh、I_Raman：示例散射强度；增强：在指定测量条件下的比较。

**EN:** When to use: Recognize why ordinary Raman is weak and why local-field enhancement is useful.

**中文：** 何时使用：理解普通拉曼为何微弱，以及局域场增强为何有用。

**EN:** Conditions and traps: These are rough reported orders from the slides, not universal conversion fractions or guaranteed enhancement for every substrate/sample. Comparisons depend on geometry, molecules and normalization. Do not multiply arbitrary input power by $10^{12}$ and treat it as physically emitted power; enhancement is relative to a reference Raman measurement.

**中文：** 条件与易错点：这些是课件所报粗略量级，不是统一转换比例，也不保证每个基底/样品都达到；比较取决于几何、分子及归一化方式。 不能把任意入射功率乘 $10^{12}$ 当实际出射功率；增强是相对参照拉曼测量的比值。

**EN:** Tiny example: Qualitative example: a weak molecular fingerprint can become easier to measure at a metal nanoparticle hot spot; the amount still needs calibration.

**中文：** 短例理解：趋势例：金属纳米颗粒热点可让弱分子指纹更容易测到，定量仍需校准。

Source / 来源：W3 PDF p89–91

## FX401 • [P2] Thin lens and magnification / 薄透镜与放大率

**EN:** Revision priority: Important supporting relations. Slide relation and basic image-formation reasoning; no explicit quiz promise for lens algebra.

**中文：** 复习优先级：重要基础关系。课件短关系和成像基础，未找到薄透镜代数的明确Quiz承诺。

**Formula / 公式：** $1/S_1 + 1/S_2 = 1/f$; $|M| = |S_2/S_1|$; signed $M = -S_2/S_1$

**EN:** Symbols and units: $S_1$: object distance; $S_2$: image distance; f: focal length, all in the same length unit; M: dimensionless magnification.

**中文：** 符号与单位：$S_1$物距、$S_2$像距、f焦距，统一长度单位；M为无量纲放大率。

**EN:** When to use: Find where the image forms and how large it is.

**中文：** 何时使用：求像的位置与图像大小。

**EN:** Conditions and traps: Paraxial thin-lens model. The slide uses positive real-object/real-image distance magnitudes; other sign conventions must be stated. The minus sign describes inversion. More magnification alone does not improve diffraction resolution.

**中文：** 条件与易错点：近轴薄透镜模型；课件图用实物、实像距离的正值大小，其他符号约定需另说明。 负号表示倒像；放大率增大本身不提高衍射分辨率。

**EN:** Tiny example: $S_1$=20 cm, f=10 cm → $S_2$=20 cm, |M|=1; the real image is inverted.

**中文：** 短例理解：$S_1$=20 cm、f=10 cm，得$S_2$=20 cm、|M|=1，实像倒立。

Source / 来源：W4 PDF p23–26 | T4-0026–0029

## FX402 • [P1] Objective numerical aperture / 物镜数值孔径

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| **EN:** Revision priority: First: know and calculate. Repeated explanation; the W5 classroom objective-comparison question was explicitly linked to quiz. |
| **中文：** 复习优先级：优先掌握与短算。反复讲解；W5课堂物镜比较题明确与Quiz联系。 |

**Formula / 公式：** $NA = n \sin\theta$

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|---|
| **EN:** Symbols and units: n: refractive index between specimen and objective; θ: acceptance-cone half-angle; NA: dimensionless. |
| **中文：** 符号与单位：n为样品与物镜之间介质折射率，θ为接收光锥半角，NA无量纲。 |
| **EN:** When to use: Compare light collection and calculate the NA used in resolution formulas. |
| **中文：** 何时使用：比较收光能力，并为分辨率公式计算NA。 |
| **EN:** Conditions and traps: Use the surrounding-medium index, not the lens-glass index; in air $n \approx 1$ and $NA \le 1$. The full cone angle is 2θ. An immersion objective can have $NA > 1$. |
| **中文：** 条件与易错点：用周围介质折射率，不用镜片玻璃折射率；空气$n \approx 1$时$NA \le 1$。 完整光锥角是2θ；浸没物镜的NA可大于1。 |
| **EN:** Tiny example: $n = 1.00$, $\theta = 30° \rightarrow NA = 0.50$. |
| **中文：** 短例理解：$n = 1.00$、$\theta = 30°$，$NA = 0.50$。 |

Source / 来源：W4 PDF p28–31 | T4-0031–0038 | T5-0059–0063

## FX403 • [P1] Rayleigh separation and Airy radius/diameter / 瑞利间距与艾里斑半径 / 直径

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| **EN:** Revision priority: First: know and calculate. Core classroom comparison; distinguish physical quantities rather than memorize an unlabeled coefficient. |
| **中文：** 复习优先级：优先掌握与短算。课堂核心比较，重点区分物理量，不只背一个无标签系数。 |

**Formula / 公式：** $\Delta d = 0.61\lambda/NA$; $r_{Airy} \approx 0.61\lambda/NA$; $D_{Airy} \approx 1.22\lambda/NA$ (slide p29 prints $1.21\lambda/NA$)

| |
|---|
| **EN:** Symbols and units: λ: vacuum/free-space wavelength in the model; Δd: minimum lateral separation; $r_{Airy}$: first-zero radius; $D_{Airy}$: diameter, all lengths. |
| **中文：** 符号与单位：λ为该模型使用的真空 / 空气波长；Δd最小横向间距，$r_{Airy}$第一零点半径，$D_{Airy}$直径，均为长度。 |
| **EN:** When to use: Resolve two points and distinguish radius/diameter from the Rayleigh separation. |
| **中文：** 何时使用：判断两点能否分开，区分斑半径、直径与瑞利间距。 |

**EN:** Conditions and traps: Ideal circular pupil, incoherent point-object Rayleigh criterion and diffraction-limited scalar model; comparable wavelength and optics when ranking NA. High resolution means smaller Δd. Do not double Δd merely because two points are present. The 1.21/1.22 difference is a source coefficient discrepancy. If a question supplies the source diameter coefficient1.21, use that stated coefficient.

**中文：** 条件与易错点：理想圆形孔径、非相干点物的瑞利判据及衍射限制标量模型；比较NA须有可比波长及光学条件。 高分辨率指更小Δd；不能因有两点就把Δd翻倍。1.21/1.22是源材料系数差异。若题目给课件直径系数1.21，计算按题给值。

**EN:** Tiny example: λ=550 nm, NA=0.80 → Δd≈0.419 μm; conventional D_Airy≈0.839 μm.

**中文：** 短例理解：λ=550 nm、NA=0.80，Δd≈0.419 μm；通常艾里斑直径约0.839 μm。

Source / 来源：W4 PDF p27–31, p81 | T4-0034–0044 | T5-0059–0061

## FX404 • [P2] Detector sampling and magnification / 探测器采样与倍率

**EN:** Revision priority: Important supporting relations. Teacher explained the bright–dark–bright sampling example; useful short calculation.

**中文：** 复习优先级：重要基础关系。老师讲过亮—暗—亮的采样例子，适合短计算。

**Formula / 公式：** $\Delta x_{object} = \Delta x_{detector}/|M|$; $|M|\Delta d/\Delta x_{detector} \approx 2.3–3$; $|M| \geq q\Delta x_{detector}/\Delta d$

**EN:** Symbols and units: $\Delta x_{detector}$: detector pixel pitch; $\Delta x_{object}$: object-plane equivalent pitch; q: chosen samples per resolved separation; lengths must match units.

**中文：** 符号与单位：$\Delta x_{detector}$相机像素间距；$\Delta x_{object}$物面等效间距；q为选定的每分辨距离采样数；长度单位一致。

**EN:** When to use: Check whether pixels sample an optically resolvable feature finely enough.

**中文：** 何时使用：检查像素是否足够细地采到光学上可分辨的细节。

**EN:** Conditions and traps: The slide recommends q≈2.3–3 for its two-peak illustration; this is a practical sampling criterion, not a universal three-pixel law. Do not compare detector pitch with object-plane Δd without M; finer pixels do not remove the optical diffraction limit.

**中文：** 条件与易错点：课件双峰示例建议q≈2.3–3；这是实用采样准则，不是所有信号统一三像素定律。 不要漏掉M直接比较像面像素与物面Δd；细像素不能消除光学衍射限制。

**EN:** Tiny example: Δd=0.50 μm, detector pitch=6 μm, q=3 → |M|≥36.

**中文：** 短例理解：Δd=0.50 μm、像素6 μm、q=3，需|M|≥36。

Source / 来源：W4 PDF p32–34 | T4-0045–0047

## FX405 • [P1] Two-photon intensity law / 双光子强度关系

**EN:** Revision priority: First: know and calculate. Sample Q1 and classroom single/two-photon demonstration; remember conditions for the square law.

**中文：** 复习优先级：优先掌握与短算。Sample Q1及课堂单双光子演示；平方规律须带条件。

**Formula / 公式：** $F \propto I_exc^2$; $F_2/F_1 = (I_2/I_1)^2$; $2hc/\lambda_2 ph \approx hc/\lambda_1 ph \Rightarrow \lambda_2 ph \approx 2\lambda_1 ph$

**EN:** Symbols and units: F: fluorescence signal/rate; I_exc: local excitation intensity; $\lambda_2 ph$: wavelength of each equal-energy two-photon excitation photon.

**中文：** 符号与单位：F为荧光信号 / 速率，I_exc局部激发强度，$\lambda_2 ph$为每个等能双光子激发光子的波长。

**EN:** When to use: Predict signal ratios and understand low-energy photons exciting the same state.

**中文：** 何时使用：求信号比，并理解两个低能光子激发同一能态。

**EN:** Conditions and traps: Unsaturated two-photon absorption, fixed dye, pulse shape and collection conditions; the wavelength relation matches the same excitation energy gap. Longer wavelength means lower energy per photon; λ_em can be shorter than each two-photon excitation wavelength.

**中文：** 条件与易错点：未饱和双光子吸收，染料、脉冲形状及收集条件固定；波长关系针对同一激发能隙。 每个光子的长波长对应低能量；发射波长可短于每个双光子激发波长。

**EN:** Tiny example: Excitation intensity doubles → F grows fourfold under these assumptions.

**中文：** 短例理解：上述条件下激发强度翻倍，F变四倍。

Source / 来源：W4 PDF p55–61 | T4-0067–0069 | T4-0077–0079 | Sample Q1

## FX406 • [P2] Michelson path, phase and intensity / 迈克尔逊光程、相位与强度

**EN:** Revision priority: Important supporting relations. Explicit OCT working-principle relation; phase/intensity algebra is supporting rather than a quiz promise.

**中文：** 复习优先级：重要基础关系。课件OCT原理关系；相位 / 强度代数为辅助，不标Quiz承诺。

**Formula / 公式：** $OPD = 2n\Delta L$; $\Delta\varphi = 2\pi \cdot OPD/\lambda_0$; $I = I_1 + I_2 + 2\sqrt{(I_1 I_2)}\cos\Delta\varphi$; in air $OPD=2\Delta L$, $\Delta\varphi=4\pi\Delta L/\lambda_0$

**EN:** Symbols and units: ΔL: geometric one-way arm-length difference; OPD: optical-path difference; n: uniform index; $\lambda_0$: vacuum wavelength; phase in radians.

**中文：** 符号与单位：ΔL为几何单程臂长差，OPD光程差，n均匀折射率，$\lambda_0$真空波长；相位用弧度。

**EN:** When to use: Explain why mirror motion changes the interference pattern and why reflection gives a factor two.

**中文：** 何时使用：解释移镜如何改干涉条纹，以及反射光路为何乘二。

**EN:** Conditions and traps: Ideal coherent, matching-polarization beams at the detector; broadband OCT adds a coherence envelope and phase depends on spectral component. L is an arm difference in this equation, not automatically the whole arm length; do not treat OCT as ordinary electronic light-echo timing.

**中文：** 条件与易错点：探测处为理想相干、偏振匹配光束；宽带OCT另有相干包络，相位随光谱分量而变。 此式L是臂长差，不是随便取整条臂长度；不能把OCT当普通电子直接计光回波时间。

**EN:** Tiny example: In air a mirror displacement $\Delta L=\lambda_0/4$ changes phase byπ and swaps bright/dark for equal beams.

**中文：** 短例理解：空气中移镜$\lambda_0/4$，相位变π；等强束由亮变暗。

Source / 来源：W4 PDF p75–79 | T4-0091–0098

## FX407 • [P2] OCT coherence-limited axial resolution / OCT相干长度限制的轴向分辨率

**EN:** Revision priority: Important supporting relations. Teacher emphasized which parameter controls axial versus lateral resolution; full constants are secondary to the bandwidth trend.

**中文：** 复习优先级：重要基础关系。老师强调轴向与横向控制参数不同；先掌握带宽趋势，再掌握系数。

**Formula / 公式：** $\Delta z_{air} \approx (2\ln 2/\pi)\lambda_0^2/\Delta\lambda \approx 0.441\lambda_0^2/\Delta\lambda$; $\Delta z_{medium} \approx \Delta z_{air}/n_g$

**EN:** Symbols and units: $\lambda_0$: center wavelength; Δλ: source spectral FWHM; $n_g$: group refractive index converting optical delay to geometric depth; Δz: axial length.

**中文：** 符号与单位：$\lambda_0$中心波长；Δλ光源光谱半高全宽；$n_g$用于将光学延迟换算几何深度的群折射率；Δz轴向长度。

**EN:** When to use: Calculate layer separation and compare source bandwidths.

**中文：** 何时使用：计算能分开的层间距，并比较光源带宽。

**EN:** Conditions and traps: Gaussian source spectrum, the stated FWHM convention and narrow fractional bandwidth approximation; weak dispersion can use supplied n≈$n_g$. Use consistent units for $\lambda_0$ and Δλ. Axial coherence resolution is not lateral Rayleigh resolution or the focused depth range.

**中文：** 条件与易错点：高斯光源光谱、给定FWHM定义及相对窄带近似；弱色散时可用题给n≈$n_g$。$\lambda_0$与Δλ统一单位；相干轴向分辨率不等于横向瑞利分辨率，也不等于景深。

**EN:** Tiny example: 900 nm/40 nm → $\Delta z_{air}$≈8.94 μm; in a medium with $n_g$=1.4 → ≈6.38 μm.

**中文：** 短例理解：900 nm / 40 nm得空气Δz≈8.94 μm；$n_g$=1.4时组织中约6.38 μm。

Source / 来源：W4 PDF p81–84 | T4-0099–0104

## FX408 • [P3] Object-side depth of field with detector term / 含探测器项的物方景深

**EN:** Revision priority: Recognise; optional calculations. Keep the distinction and source correction; detailed detector-term arithmetic was not explicitly promised for quiz.

**中文：** 复习优先级：辅助辨认；计算后看。保留概念区分与原式补正；复杂探测器项计算未明确点名Quiz。

**Formula / 公式：** $d_{tot} \approx \lambda_0 n/NA^2 + n \cdot e/(|M| \cdot NA)$

**EN:** Symbols and units: d_tot: object-side in-focus depth range; e: smallest image-plane spacing resolved by detector; $\lambda_0$, e, d_tot in matching length units; M, NA, n dimensionless.

**中文：** 符号与单位：d_tot物方清晰深度范围；e探测器在像面可分辨的最小间距；$\lambda_0$、e、d_tot长度单位一致；M、NA、n无量纲。

**EN:** When to use: Recognize the NA trade-off and use the corrected geometric detector term when all data are supplied.

**中文：** 何时使用：辨认NA与景深的权衡；给全参数时使用补正的探测器几何项。

**EN:** Conditions and traps: A stated microscopy focus criterion combining diffraction and detector/geometric limits; not a universal OCT axial-resolution formula. p83 omits e, leaving a dimensionless term that cannot be added to a length. Its object-side “Depth of Focus” should be called depth of field; standard depth of focus is image-side.

**中文：** 条件与易错点：该显微焦清晰判据合并衍射与探测器 / 几何限制，并非通用OCT轴向分辨率公式。 p83漏e，使第二项无量纲而不能与长度相加；原页物方“Depth of Focus”应称景深depth of field，标准depth of focus指像方焦深。

**EN:** Tiny example: $\lambda_0$=0.55 μm, n=1, NA=0.5, M=20, e=6 μm → 2.20+0.60=2.80 μm.

**中文：** 短例理解：$\lambda_0$=0.55 μm、n=1、NA=0.5、M=20、e=6 μm，得2.20+0.60=2.80 μm。

Source / 来源：W4 PDF p83 | T4-0101–0104 | Nikon MicroscopyU: https://www.microscopyu.com/microscopy-basics/depth-of-field-and-depth-of-focus

## FX409 • [P3] Hyperspectral cube data counts / 高光谱立方的数据数目

**EN:** Revision priority: Recognise; optional calculations. Short counting aid to the teacher’ s 5×5/25-spectrum illustration, not an added exam requirement.

**中文：** 复习优先级：辅助辨认；计算后看。辅助老师5×5 / 25条谱的解释，不新增考试要求。

**Formula / 公式：** $N_{spectra} = N_x N_y$; $N_{values} = N_x N_y N_\lambda$

**EN:** Symbols and units: N_x, N_y: spatial sample counts; N_λ: spectral-band count; counts are dimensionless integers.

**中文：** 符号与单位：N_x、N_y为空间采样数；N_λ为光谱波段数；均为无量纲整数。

| |
|---|
| **EN:** When to use: Distinguish spectra per spatial pixel from all scalar intensity samples. |
| **中文：** 何时使用：区分每像素光谱条数与全部强度标量数。 |
| **EN:** Conditions and traps: A complete x, y,λ cube with one intensity per spatial pixel and band; not every scanning step acquires the whole cube. The λ axis is spectral, not anatomical depth z; spatial sampling per mm is not a frequency in Hz. |
| **中文：** 条件与易错点：完整x,y,λ立方，每个空间像素每个波段一个强度；不能说每次扫描步骤都取得整块立方。 λ轴是光谱，不是解剖深度z；每毫米空间采样数不是Hz。 |
| **EN:** Tiny example: 5×5 pixels, 120 bands →25 spectra and3000 intensity samples. |
| **中文：** 短例理解：5×5像素、120波段，得25条谱、3000个强度值。 |

Source / 来源：W4 PDF p99–103 | T4-0111–0113

## FX410 • [P2] Acoustic wave, wavelength and frequency / 声波、波长与频率

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| **EN:** Revision priority: Important supporting relations. Foundational source-wave relations; not part of the locally excluded scanning details. |
| **中文：** 复习优先级：重要基础关系。基本声源关系，不属于老师局部列参考的扫描细节。 |

**Formula / 公式：** $p(x,t)=p_0\cos(\omega t-kx)$; $v=\lambda f$; $\omega=2\pi f$; $k=2\pi/\lambda$; $T=1/f$

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|---|
| **EN:** Symbols and units: $p_0$: peak acoustic pressure Pa; v: m/s; λ: m; f: Hz; ω: rad/s; k: rad/m; T: s. |
| **中文：** 符号与单位：$p_0$为压力峰值Pa；v用m/s，λ用m，f用Hz，ω用rad/s，k用rad/m，T用s。 |
| **EN:** When to use: Convert frequency, period and wavelength and read a traveling pressure wave. |
| **中文：** 何时使用：换算频率、周期、波长并辨认传播压力波。 |
| **EN:** Conditions and traps: Harmonic longitudinal plane-wave model with the stated sound speed; tissue speed is not the speed of light. Frequency f and angular frequency ω differ by2π; a full pressure cycle contains compression and rarefaction. |
| **中文：** 条件与易错点：给定声速的简谐纵向平面波模型；组织声速不是光速。 f与ω相差2π；一个完整压力周期含压缩和稀疏。 |
| **EN:** Tiny example: v=1540 m/s, f=2 MHz → λ=0.770 mm, T=0.50 μs. |
| **中文：** 短例理解：v=1540 m/s、f=2 MHz，λ=0.770 mm、T=0.50 μs。 |

Source / 来源：W5 PDF p9–14 | T5-0008–0010

## FX411 • [P1] Echo depth versus one-way photoacoustic path / 回波深度与光声单程声路

**EN:** Revision priority: First: know and calculate. Teacher directly explained divide-by-two echo timing; core short calculation.

**中文：** 复习优先级：优先掌握与短算。老师直接解释回波除二，属于基础短计算。

**Formula / 公式：** $d_{echo} = vt_{echo}/2$; $d_{oneway} = vt_{oneway}$

**EN:** Symbols and units: d: geometric path/depth m; v: sound speed m/s; t: propagation time s.

**中文：** 符号与单位：d为几何路径 / 深度m；v声速m/s；t传播时间s。

**EN:** When to use: Choose the correct time-of-flight factor from the actual path.

**中文：** 何时使用：根据实际路径选择飞行时间公式中的因子。

**EN:** Conditions and traps: Echo uses the same source/receiver path out and back; one-way PA assumes sound starts at the absorber and optical/system delay is negligible or removed. The names alone do not determine the factor: read the stated geometry and time reference.

**中文：** 条件与易错点：回波沿同路径去返；单程光声假设声源在吸收体，光 / 系统延迟可忽略或已扣。不能只看成像名字就定因子，须看几何与计时起点。

**EN:** Tiny example: t=40 μs, v=1540 m/s → echo depth3.08 cm; the same one-way time gives6.16 cm.

**中文：** 短例理解：t=40 μs、v=1540 m/s，回波深度3.08 cm；同时间单程为6.16 cm。

Source / 来源：W5 PDF p10–11,p21,p45–46 | T5-0007–0008 | T5-0066–0070

## FX412 • [P2] Acoustic impedance and pressure/intensity / 声阻抗与压力 / 强度

**EN:** Revision priority: Important supporting relations. Teacher introduced acoustic material properties; essential for interpreting reflection and attenuation.

**中文：** 复习优先级：重要基础关系。老师介绍声学材料性质，是反射与衰减的理解基础。

**Formula / 公式：** $Z_a = \rho v$; $I(t) = p(t)^2/Z_a$; for $p(t) = p_0 \cos\omega t$: $\bar{I} = p_0^2/(2Z_a) = p_{rms}^2/Z_a$

**EN:** Symbols and units: ρ: density kg/m$^3$; $Z_a$: acoustic impedance rayl=kg/(m$^2$·s); p: acoustic excess pressure Pa; I: W/m$^2$.

**中文：** 符号与单位：ρ密度kg/m$^3$；$Z_a$声阻抗rayl=kg/(m$^2$·s)；p声学过压Pa；I强度W/m$^2$。

**EN:** When to use: Relate pressure, material properties and carried acoustic intensity.

**中文：** 何时使用：联系压力、材料性质与声强。

**EN:** Conditions and traps: Progressive plane wave in a uniform lossless medium; distinguish instantaneous p(t), peak $p_0$ and RMS pressure. The factor1/2 applies when using peak pressure to compute time-average intensity; it is not inserted into the slide’ s instantaneous formula.

**中文：** 条件与易错点：均匀无损介质中的行进平面波；区分瞬时p(t)、峰值$p_0$与有效值压力。 用压力峰值求时间平均强度才出现1/2，不能硬加进课件瞬时公式。

**EN:** Tiny example: $p_0$=2000 Pa, Z_a=1.5×$10^6$ rayl → Ī≈1.33 W/$m^2$.

**中文：** 短例理解：$p_0$=2000 Pa、Z_a=1.5×$10^6$ rayl，平均声强约1.33 W/$m^2$。

Source / 来源：W5 PDF p13–14,p19–22 | T5-0010–0012

## FX413 • [P1] Acoustic reflection and normal-incidence optical analogue / 声反射与正入射光学类比

**EN:** Revision priority: First: know and calculate. Repeated classroom emphasis on impedance mismatch and echo contrast; Fresnel is a secondary comparison.

**中文：** 复习优先级：优先掌握与短算。声阻抗失配与回波对比课堂反复强调；Fresnel仅辅助类比。

**Formula / 公式：** $R_I = [(Z_2-Z_1)/(Z_2+Z_1)]^2$; $T_I=1-R_I$; optical $R=[(n_2-n_1)/(n_2+n_1)]^2$

**EN:** Symbols and units: $Z_1$, $Z_2$: acoustic impedances in identical units; $n_1$, $n_2$: refractive indices; R_I, T_I: dimensionless intensity fractions.

**中文：** 符号与单位：$Z_1$、$Z_2$为同单位声阻抗；$n_1$、$n_2$折射率；R_I、T_I为无量纲强度比例。

**EN:** When to use: Calculate reflected/transmitted energy fractions and understand coupling gel.

**中文：** 何时使用：求反射 / 透射能量比例并理解耦合凝胶。

**EN:** Conditions and traps: Normal incidence at an ideal lossless interface; optical expression additionally assumes nonmagnetic, transparent media and no interface absorption. These are intensity coefficients, not signed pressure-amplitude reflection coefficients; no universal T=1−R in a lossy whole tissue path.

**中文：** 条件与易错点：理想无损界面正入射；光学式另要求非磁性透明介质且界面不吸收。 这是强度系数，不是带正负号的压力反射系数；有损整条组织路径不通用T=1−R。

**EN:** Tiny example: $Z_2=2Z_1$ → R=1/9, T=8/9; air/glass n=1/1.5 → optical R≈4%.

**中文：** 短例理解：$Z_2=2Z_1$时R=1/9、T=8/9；空气 / 玻璃n=1 / 1.5时光反射约4%。

Source / 来源：W5 PDF p19–22,p30 | T5-0019–0026

## FX414 • [P1] Exponential pressure and intensity attenuation / 压力与强度的指数衰减

**EN:** Revision priority: First: know and calculate. Core attenuation concept; teacher highlighted total path and frequency dependence.

**中文：** 复习优先级：优先掌握与短算。衰减基础，老师强调总路径及频率依赖。

**Formula / 公式：** $p_{amp}(L)=p_{amp,0}\,e^{(-\alpha_p L)}$; $I(L)=I_0\,e^{(-\alpha_I L)}$; $\alpha_I=2\alpha_p$ (same medium)

**EN:** Symbols and units: L: actual traveled length; α_p: pressure-amplitude coefficient in inverse length; α_I: intensity coefficient in inverse length.

**中文：** 符号与单位：L为实际传播长度；α_p压力幅度衰减系数、α_I强度衰减系数，均用长度倒数。

**EN:** When to use: Apply exponential attenuation without mixing pressure and intensity definitions.

**中文：** 何时使用：应用指数衰减，同时区分压力和强度定义。

**EN:** Conditions and traps: Uniform medium with unchanged acoustic impedance and stated constant coefficients; α values can depend on frequency. p23 uses an intensity α and p24 uses a pressure α. Equal symbols do not mean equal numerical definitions. For echoes L can be2d.

**中文：** 条件与易错点：介质均匀、声阻抗不变、系数给定且恒定；α可随频率改变。 p23强度α与p24压力α不是同一定义；同符号不代表同数值。回波L可为2d。

**EN:** Tiny example: If $\alpha_p L=1$, pressure ratio=$e^{-1}\approx0.368$ while intensity ratio=$e^{-2}\approx0.135$.

**中文：** 短例理解：$\alpha_p L=1$时压力比$e^{-1}\approx0.368$，强度比$e^{-2}\approx0.135$。

Source / 来源：W5 PDF p23–25 | T5-0027–0032

## FX415 • [P2] dB loss and exponential-coefficient conversion / dB损耗与指数系数换算

**EN:** Revision priority: Important supporting relations. Supports the classroom attenuation calculation and prevents unit-definition mistakes.

**中文：** 复习优先级：重要基础关系。支持课堂衰减计算，避免单位与定义错误。

**Formula / 公式：** $A_{dB}=10\log_{10}(I_0/I)=20\log_{10}(p_0/p)$; $I/I_0=10^{(-A_{dB}/10)}$; $p/p_0=10^{(-A_{dB}/20)}$; $a_{dB}=4.343\alpha_I=8.686\alpha_p$

**EN:** Symbols and units: A_dB: positive attenuation loss dB; a_dB: loss per unit length dB/length; α_I,α_p: inverse-length coefficients. p0 and p are positive pressure amplitudes or RMS values; I0 and I are corresponding time-averaged intensities.

**中文：** 符号与单位：A_dB为正的衰减损耗dB；a_dB每单位长度损耗；α_I、α_p自然指数系数，单位长度倒数。 p0、p为正的声压幅度或均方根值；I0、I为对应时间平均声强。

**EN:** When to use: Convert the p24–25 dB data and distinguish intensity from amplitude ratios.

**中文：** 何时使用：换算p24–25的dB数据，区分强度比和幅度比。

**EN:** Conditions and traps: Pressure and intensity conversions assume the same medium/impedance. Positive loss uses input/output; gain convention uses output/input instead. 8.686 converts the pressure exponential coefficient, not an intensity α. Never put a dB/cm value directly into e^(−αL). Do not take the logarithm of signed instantaneous p(t).

**中文：** 条件与易错点：压力与强度换算需同介质 / 声阻抗；正损耗用输入 / 输出，增益定义则相反。8.686对应压力指数系数，不是强度α；不能直接把dB/cm放进e^(−αL)。 不对带正负号的瞬时p(t)直接取对数。

**EN:** Tiny example: A_dB=20 dB → intensity1%, pressure10%; α_I=1 $cm^{-1}$ → a_dB≈4.343 dB/cm.

**中文：** 短例理解：20 dB损耗对应强度1%、压力10%；α_I=1 $cm^{-1}$对应约4.343 dB/cm。

Source / 来源：W5 PDF p23–25 | T5-0030–0032

## FX416 • [P1] Frequency-dependent ultrasound path loss / 频率相关的超声路径损耗

**EN:** Revision priority: First: know and calculate. p25 worked example and explicit teacher correction of the two-way route.

**中文：** 复习优先级：优先掌握与短算。p25例题及老师对双程路径的明确纠正。

**Formula / 公式：** A_dB = a·f·L_total; uniform pulse-echo L_total=2d; layered path $A_{dB}=\Sigma\, a_i f L_i$

**EN:** Symbols and units: a: dB/(MHz·cm) if f uses MHz and L uses cm; d: reflector depth; L_total: full travel length.

**中文：** 符号与单位：若f用MHz、L用cm，a须用dB/(MHz·cm)；d反射点深度；L_total全路径长度。

**EN:** When to use: Use the muscle example and include outward and return losses.

**中文：** 何时使用：使用肌肉例子，并计入去返程衰减。

**EN:** Conditions and traps: Approximately linear frequency dependence over the stated range; ignores interface reflection loss unless separately included. Depth and traveled length differ in echo measurements; do not use MHz values as Hz in this empirical-unit formula.

**中文：** 条件与易错点：给定范围内近似随频率线性变化；除非另计，忽略界面反射损失。 回波深度不等于传播总长度；这条经验单位式不能把MHz数值当Hz。

**EN:** Tiny example: a=1.3 dB/(MHz·cm), f=5 MHz, d=3 cm → A=1.3×5×6=39 dB.

**中文：** 短例理解：a=1.3 dB/(MHz·cm)、f=5 MHz、d=3 cm，A=1.3×5×6=39 dB。

Source / 来源：W5 PDF p24–25 | T5-0030–0032

## FX417 • [P3] Ultrasound pulse length and slide half-depth heuristic / 超声脉冲长度与课件半深度经验式

**EN:** Revision priority: Recognise; optional calculations. Recognition/reference only: T5-0040 lowers detailed resolution calculation priority; it does not exclude basic echo/attenuation formulas.

**中文：** 复习优先级：辅助辨认；计算后看。仅辨认 / 参考：T5-0040降低详细分辨率计算优先级，不排除基础测距和衰减式。

**Formula / 公式：** SPL=N_cyclesλ; axial separation≈SPL/2; for $I=I_0e^{(-\alpha_IL)}$, $L_half,I=ln2/\alpha_I$; p31 also prints $L_half\approx6/f_\nu$

**EN:** Symbols and units: N_cycles: cycles in pulse; SPL: spatial pulse length. The README assumes f_ν in MHz and 6/f_ν in cm, but source p31 does not print these units.

**中文：** 符号与单位：N_cycles为脉冲周期数，SPL空间脉冲长度。README假设f_ν用MHz、6/f_ν得cm，但p31原页未印出该单位约定。

**EN:** When to use: Recognize why shorter pulses and higher frequency can improve axial separation; identify the half-depth definition.

**中文：** 何时使用：辨认短脉冲 / 高频如何改善轴向间距，并认清半深度定义。

**EN:** Conditions and traps: SPL/2 is pulse-echo axial geometry. The6/f expression is a tissue/frequency heuristic with unstated coefficient assumptions, not a universal law. Half intensity and half pressure require different depths. The teacher called detailed ultrasound resolution calculation reference material.

**中文：** 条件与易错点：SPL/2是脉冲回波轴向几何；6/f是未充分说明组织 / 系数条件的经验式，不是普遍定律。 强度减半与压力减半深度不同；老师把详细超声分辨率计算列为参考。

**EN:** Tiny example: At v=1540 m/s, f=2 MHz, N=2: λ=0.77 mm, SPL/2=0.77 mm.

**中文：** 短例理解：v=1540 m/s、f=2 MHz、N=2时，λ=0.77 mm，SPL/2=0.77 mm。

Source / 来源：W5 PDF p27–31 | T5-0037–0041

## FX418 • [P1] X-ray single- and multilayer attenuation / X射线单层与分层衰减

**EN:** Revision priority: First: know and calculate. p60–61 worked in class; important exponential layering and unit check.

**中文：** 复习优先级：优先掌握与短算。p60–61上课讲解，重点是分层指数与单位。

**Formula / 公式：** $I/I_0=e^{(-\mu x)}$; $I/I_0=e^{(-\Sigma\mu_i x_i)}$; $x_1/e=1/\mu$; $x_half=ln2/\mu$

**EN:** Symbols and units: μ: linear attenuation coefficient $cm^{-1}$ or $m^{-1}$; x: matching path length; $I/I_0$: dimensionless transmitted intensity.

**中文：** 符号与单位：μ线性衰减系数$cm^{-1}$或$m^{-1}$；x使用对应长度单位；$I/I_0$为无量纲透射强度。

**EN:** When to use: Calculate a film/detector intensity and accumulate different tissue layers.

**中文：** 何时使用：计算胶片 / 探测器收到的强度，并累加不同组织层。

**EN:** Conditions and traps: Monoenergetic beam, stated fixed layer coefficients, primary transmitted beam; neglect scatter and beam hardening. Add $\mu_i x_i$ in the exponent, not intensities; projection X-ray is one-way in the source problem.

**中文：** 条件与易错点：单能光束、各层系数给定且固定、直接透射束；忽略散射与束硬化。 累加指数中的$\mu_i x_i$，不累加强度；原题X射线为单程透射。

**EN:** Tiny example: 1 cm bone $\mu=10\ cm^{-1}$ plus3 cm tissue $\mu=1\ cm^{-1} \rightarrow I/I_0=e^{-13}\approx 2.26\times10^{-6}$.

**中文：** 短例理解：1 cm骨$\mu=10\ cm^{-1}$加3 cm组织$\mu=1\ cm^{-1}$，$I/I_0=e^{-13}\approx 2.26\times10^{-6}$。

Source / 来源：W5 PDF p58–61 | T5-0082–0085

## FX419 • [P2] Signed transmission-image contrast / 透射图像的带符号对比度

**EN:** Revision priority: Important supporting relations. Source formula and imaging concept; no specific quiz promise for this numerical ratio.

**中文：** 复习优先级：重要基础关系。原课件公式及成像概念，未找到此比值的明确Quiz承诺。

**Formula / 公式：** $C_{local}=(I_B-I_O)/I_B = 1-I_O/I_B$

**EN:** Symbols and units: $I_B$: background transmitted intensity; $I_O$: object-path transmitted intensity; $C_{local}$: dimensionless, often expressed as%.

**中文：** 符号与单位：$I_B$背景透射强度，$I_O$目标路径透射强度；$C_{local}$无量纲，可表示百分比。

**EN:** When to use: Quantify the relative intensity difference between object and background.

**中文：** 何时使用：量化目标与背景的相对强度差。

**EN:** Conditions and traps: Use the p62 sign convention and the same acquisition/reference exposure. Other contrast definitions may use absolute values or reversed signs. Greater attenuation gives smaller detector intensity and positive C here. Displayed black/white can be inverted; do not infer sign from screen color alone.

**中文：** 条件与易错点：用p62的符号定义，且两强度采集参照相同；其他定义可能取绝对值或反号。衰减更强使探测强度更小，在本定义下C为正；显示灰度可反相，不能只看黑白判断符号。

**EN:** Tiny example: $I_B=100$, $I_O=80 \rightarrow C=0.20$; $I_O=120 \rightarrow C=-0.20$.

**中文：** 短例理解：$I_B=100$、$I_O=80$时$C=0.20$；$I_O=120$时$C=-0.20$。

Source / 来源：W5 PDF p62 | T5-0085–0087

## FX420 • [P3] Photoelectric threshold and absorption trend / 光电阈值与吸收趋势

**EN:** Revision priority: Recognise; optional calculations. Supporting mechanism/trend recognition; source technical details were not promised for quiz.

**中文：** 复习优先级：辅助辨认；计算后看。辅助机制 / 趋势辨认；源技术细节未明确点名Quiz。

**Formula / 公式：** $K_{max}=E_{photon}-W$ (if $E_{photon}\geq W$); photoelectric probability $\propto \rho Z_{atom}^3/E_{photon}^3$ (source approximation)

**EN:** Symbols and units: W: work function/binding threshold energy; K_max: electron kinetic energy; Z_atom: atomic number, not acoustic impedance; ρ: density.

**中文：** 符号与单位：W逸出功 / 束缚阈值；K_max电子最大动能；Z_atom为原子序数，不是声阻抗；ρ密度。

**EN:** When to use: Recognize threshold emission and why high-Z contrast agents can absorb strongly.

**中文：** 何时使用：辨认阈值发射及高原子序数造影剂强吸收趋势。

**EN:** Conditions and traps: K_max is the simple one-photon photoelectric illustration. The absorption power law is a rough energy/material trend away from absorption edges. The p57 sodium example uses visible photons, not X-ray wavelengths. Shell absorption edges break a smooth $E^{-3}$ extrapolation.

**中文：** 条件与易错点：K_max用于简单单光子光电示例；吸收幂律是在吸收边之外的粗略能量 / 材料趋势。 p57钠例使用可见光子，不是X射线波长；壳层吸收边使平滑$E^{-3}$外推失效。

**EN:** Tiny example: p57: $E_{photon}=2.48$ eV, $W=2.28$ eV → $K_{max}=0.20$ eV.

**中文：** 短例理解：p57示例2.48 eV光子、阈值2.28 eV，$K_{max}=0.20$ eV。

Source / 来源：W5 PDF p57–58 | T5-0077–0081

## FX421 • [P3] Bit depth, storage and spatial sampling / 位深、存储与空间采样

**EN:** Revision priority: Recognise; optional calculations. Useful image-data interpretation; storage counting is an auxiliary derived relation.

**中文：** 复习优先级：辅助辨认；计算后看。辅助图像数据理解；存储计数为派生短关系。

**Formula / 公式：** $N_{gray}=2^b$; highest unsigned code $=2^b-1$; $N_{bits}=N_{pixels}\cdot b$; raw bytes $=N_{bits}/8$; $f_{spatial}=1/\Delta x$

**EN:** Symbols and units: b: bits/pixel for grayscale; N_pixels: pixel count; Δx: spatial pitch in mm or m; f_spatial: samples/mm or samples/m, not Hz.

**中文：** 符号与单位：b为灰度每像素位数，N_pixels像素数；Δx空间间距mm或m，f_spatial为空间采样数/mm或/m，不是Hz。

**EN:** When to use: Read quantization tables and distinguish gray levels from spatial resolution.

**中文：** 何时使用：读量化表，区分灰度等级与空间分辨率。

**EN:** Conditions and traps: Storage formula assumes packed raw grayscale without header/compression; implementation can store12-bit values in16-bit containers. 256 levels means codes0–255. More gray levels do not create new optically resolved spatial detail.

**中文：** 条件与易错点：存储式假设紧凑原始灰度且无文件头 / 压缩；12bit可实际装入16bit容器。256级编码是0–255；增加灰度级不会生成新的光学空间细节。

**EN:** Tiny example: 8 bits→256 levels; 256×256×12 bits/8=98, 304 bytes when packed.

**中文：** 短例理解：8bit→256级；256×256×12bit紧凑存储为98,304字节。

Source / 来源：W5 PDF p73–76 | T5-0098–0101

## FX422 • [P3] Convolution and averaging kernels / 卷积与均值核

**EN:** Revision priority: Recognise; optional calculations. Teacher described convolution; full indexing algebra is recognition/reference rather than first-pass quiz memorization.

**中文：** 复习优先级：辅助辨认；计算后看。老师讲卷积作用；完整下标代数仅辨认 / 参考，不列第一轮Quiz背诵。

**Formula / 公式：** $g(i,j)=\Sigma_a\Sigma_b\ h(a,b)f(i-a,j-b)$; uniform K×K mean: $h=1/K^2$, $\Sigma h=1$

**EN:** Symbols and units: f: input pixel intensity; g: filtered intensity; h: dimensionless kernel weights; K: integer kernel side length.

**中文：** 符号与单位：f输入像素强度，g滤波后强度，h无量纲核权重，K核边长整数。

**EN:** When to use: Predict smoothing and calculate one explicitly defined neighborhood output.

**中文：** 何时使用：判断平滑作用，计算明确定义邻域的输出。

**EN:** Conditions and traps: Use the specified convolution/correlation convention and border handling; a symmetric mean kernel gives the same local mean in either convention. p82 mixes a5×5 label with a3×3 red-center illustration. Follow the stated kernel, not a copied output.

**中文：** 条件与易错点：须说明卷积 / 相关约定与边界处理；对称均值核在两种约定下局部均值相同。p82混用5×5标题与3×3红中心示例；按题给核，不盲背输出。

**EN:** Tiny example: A5×5 mean gives weight1/25 per sample; the illustrated3×3 red-center mean is489/9≈54.33.

**中文：** 短例理解：5×5均值每权重1/25；图示3×3红中心均值为489/9≈54.33。

Source / 来源：W5 PDF p82 | T5-0102–0103

## FX423 • [P3] Diagnostic sensitivity, specificity and ROC / 诊断敏感度、特异度与ROC

**EN:** Revision priority: Recognise; optional calculations. Teacher discussed the definitions but called the extension reference/not very important; formulas support understanding.

**中文：** 复习优先级：辅助辨认；计算后看。老师讲定义但把扩展列为参考、不很重要；此式辅助理解。

**Formula / 公式：** TPR=TP/(TP+FN); specificity=TN/(TN+FP); FPR=FP/(FP+TN)=1−specificity; accuracy=(TP+TN)/N; precision=TP/(TP+FP)

**EN:** Symbols and units: TP, TN: correct positive/negative counts; FP, FN: false positive/negative counts; N=TP+TN+FP+FN; all ratios dimensionless.

**中文：** 符号与单位：TP、TN为正确阳 / 阴计数；FP、FN为假阳 / 阴；N为四格总和；比例无量纲。

**EN:** When to use: Interpret the confusion matrix and ROC axes (FPR horizontal, TPR vertical).

**中文：** 何时使用：读混淆矩阵与ROC轴：横FPR，纵TPR。

**EN:** Conditions and traps: Binary classifier at a given threshold; denominators must be nonzero. Threshold changes generate ROC points. Diagnostic sensitivity is not calibration slope. Precision uses predicted positives, while TPR uses actual positives.

**中文：** 条件与易错点：给定阈值下的二分类器，分母需非零；改变阈值形成ROC点。 诊断敏感度不是校准斜率；precision分母是预测阳性，TPR分母是真实阳性。

**EN:** Tiny example: TP=80, FN=20, TN=90, FP=10 → TPR=80%, specificity=90%, accuracy=85%.

**中文：** 短例理解：TP=80、FN=20、TN=90、FP=10时，敏感度80%、特异度90%、准确率85%。

Source / 来源：W5 PDF p84–86 | T5-0103–0105

## FX424 • [P2] Cell potential and electrochemical Nernst equation / 电池电势与电化学能斯特式

**EN:** Revision priority: Important supporting relations. Source equation and relative-potential concept; teacher called chemistry tables supporting, so long reaction bookkeeping is not a quiz promise.

**中文：** 复习优先级：重要基础关系。原课件式与相对电势概念；老师把化学表列支持资料，不把长反应配平宣传Quiz必考。

**Formula / 公式：** $E_{cell}=E_{cathode}-E_{anode}$ (both reduction potentials); $E=E^\circ-\frac{RT}{nF}\ln Q_{rxn}$; at 25°C (298.15 K): $E=E^\circ-\frac{0.05916\ V}{n}\log_{10}Q_{rxn}$

**EN:** Symbols and units: R=8.314 J/(mol·K); T: K; n: integer stoichiometric electron number per balanced reaction; F≈96485 C/mol; $Q_{rxn}$: dimensionless activity quotient; E: V.

**中文：** 符号与单位：R=8.314 J/(mol·K)，T开尔文，n为配平反应中转移电子数（整数，不是电子摩尔数），F≈96485 C/mol，$Q_{rxn}$无量纲活度反应商，E用V。

**EN:** When to use: Relate concentration/activity and temperature to equilibrium electrochemical potential.

**中文：** 何时使用：联系活度 / 浓度、温度与平衡电化学电势。

**EN:** Conditions and traps: Equilibrium/reversible electrode response; 0.05916 applies at 25°C (298.15 K) and base10 log. Pure solids/liquids usually have unit activity in Q. Q_rxn is not electric charge Q; do not flip anode reduction potential twice or mix ln with the0.05916 coefficient.

**中文：** 条件与易错点：平衡 / 可逆电极响应；0.05916针对25°C (298.15 K)及常用对数；反应商中纯固液活度通常取1。 Q_rxn不是电荷Q；不要对阳极还原电势反号两次，也不要把ln和0.05916混用。

**EN:** Tiny example: n=1, Q_rxn=10 at 25°C (298.15 K) → E is59.16 mV below E°.

**中文：** 短例理解：25°C (298.15 K)、n=1、Q_rxn=10时，E比E°低59.16 mV。

Source / 来源：W6 PDF p34–36 | T6-0035–0038

## FX425 • [P2] Ideal ion-selective response andpH / 理想离子选择响应与pH

**EN:** Revision priority: Important supporting relations. pH is explicitly explained; the ideal Nernst slope is a useful conditioned supporting formula.

**中文：** 复习优先级：重要基础关系。pH课堂明确解释；理想Nernst斜率是带条件的辅助短公式。

**Formula / 公式：** $E=K+[2.303RT/(zF)]\log_{10}a_ion$; $pH=-\log_{10}a_H^{+}$; ideal $H^{+}$ response at 25°C (298.15 K): $E=K-0.05916\ V\cdot pH$

**EN:** Symbols and units: a_ion: dimensionless ion activity; z: signed ion valence; K: reference/constant terms V; pH: dimensionless.

**中文：** 符号与单位：a_ion离子活度无量纲，z带符号价数，K为参比 / 常数电势项V，pH无量纲。

**EN:** When to use: Recognize logarithmic sensing and a59 mV-per-pH-unit ideal slope.

**中文：** 何时使用：认清对数感测及理想每pH单位约59 mV斜率。

**EN:** Conditions and traps: Ideal selective equilibrium response at stated temperature and voltage polarity; dilute solutions can approximate activity with normalized concentration. The source phrase “proportional to concentration” must not become E∝c. Reversing measured electrode polarity reverses the slope sign.

**中文：** 条件与易错点：给定温度与电压极性的理想选择性平衡响应；稀溶液可用归一浓度近似活度。源文“与浓度成比例”不能解读成E∝c；反转测量电极极性，斜率符号也反转。

**EN:** Tiny example: For $E=K-59.16\ mV\cdot pH$, pH6→8 gives $\Delta E=-118.32$ mV.

**中文：** 短例理解：按$E=K-59.16\ mV\cdot pH$，pH6→8得$\Delta E=-118.32$ mV。

Source / 来源：W6 PDF p36,p40–43 | T6-0042–0045

## FX426 • [P2] Reference-electrode conversion / 参比电极换算

**EN:** Revision priority: Important supporting relations. Explicit slide worked question; short reference subtraction, not chemical table memorization.

**中文：** 复习优先级：重要基础关系。原课件明确例题；会短参比减法，不背化学表。

**Formula / 公式：** $E(vsRef)=E_{ind}(vsSHE)-E_{Ref}(vsSHE)$; $E(vsRefB)=E(vsRefA)+E_{RefA}-E_{RefB}$

**EN:** Symbols and units: $E_{ind}$: indicator half-cell potential; $E_{Ref}$: specified reference potential; all V relative to a commonSHE zero.

**中文：** 符号与单位：$E_{ind}$指示半电池电势，$E_{Ref}$指定参比电势；均用V并相对共同SHE零点。

**EN:** When to use: Change the reference zero without changing the underlying indicator electrode.

**中文：** 何时使用：变换参照零点而不改变指示电极本身。

**EN:** Conditions and traps: Same polarity and reference conditions; ignore or separately include liquid-junction potential, which the source diagram labels $E_j$. Subtract the new reference. The p38 source actually prints SCE+0.2444 V, not+0.2440 V.

**中文：** 条件与易错点：极性、参照条件相同；液接电势需忽略或另计，源图标为$E_j$。 减去新参照；p38原图实际印SCE+0.2444 V，不是+0.2440 V。

**EN:** Tiny example: $Fe^{3+}/Fe^{2+}$ +0.750 V vsSHE → +0.5056 V vsSCE; +0.553 V vsAg/AgCl(+0.197 V).

**中文：** 短例理解：$Fe^{3+}/Fe^{2+}$相对SHE+0.750 V，换成SCE为+0.5056 V，换Ag/AgCl(+0.197 V)为+0.553 V。

Source / 来源：W6 PDF p32,p37–38 | T6-0034–0039

## FX427 • [P1] Faraday charge, electron moles and counts / 法拉第电荷、电子摩尔与个数

**EN:** Revision priority: First: know and calculate. Three classroom worked examples and repeated unit/stoichiometry emphasis.

**中文：** 复习优先级：优先掌握与短算。三个课堂例题及反复单位 / 计量强调。

**Formula / 公式：** $Q=\int i(t)dt$; constant current $Q=It$; $n_e=|Q|/F$; $N_e=n_eN_A=|Q|/e_{charge}$

**EN:** Symbols and units: I: A; t: s; Q: C; F≈96500 C/mol electrons; $N_A\approx6.022\times10^{23}$ $mol^{-1}$; $e_{charge}\approx1.602\times10^{-19}$ C; $N_e$: electron count.

**中文：** 符号与单位：I用A，t用s，Q用C；F≈96500 C/mol电子，$N_A\approx6.022\times10^{23}$ $mol^{-1}$，$e_{charge}\approx1.602\times10^{-19}$ C，$N_e$电子个数。

**EN:** When to use: Convert measured current/time into electron moles or electron count.

**中文：** 何时使用：把电流与时间换成电子摩尔数或个数。

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| **EN:** Conditions and traps: Use integrated faradaic reaction charge; the requested count uses magnitude, whereas signed Q follows the current convention. 1 faraday of charge is not1 farad of capacitance. Electron moles and individual electrons are different answers. |
| **中文：** 条件与易错点：使用反应的法拉第电荷积分；电子数量用大小，带符号Q遵循电流方向约定。一法拉第电荷不是一法拉电容；电子摩尔数与单个电子数量不同。 |
| **EN:** Tiny example: 0.10 A×600 s=60 C →$6.22\times10^{-4}$ mol electrons. |
| **中文：** 短例理解：0.10 A×600 s=60 C，对应$6.22\times10^{-4}$ mol电子。 |

Source / 来源：W6 PDF p50–52,p60 | T6-0054–0063

## FX428 • [P1] Analyte stoichiometry, mass and current efficiency / 分析物计量、质量与电流效率

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| **EN:** Revision priority: First: know and calculate. Teacher said the lactate question resembles past exam questions; it supports this method, not a guaranteed quiz replica. |
| **中文：** 复习优先级：优先掌握与短算。老师说乳酸题类似历年题，支持这一方法，不代表本次Quiz原样必出。 |

**Formula / 公式：** $n_{analyte}=m/M_{molar}$; $n_e=z_e\cdot n_{analyte}$; $|Q_{reaction}|=z_eFm/M_{molar}$; $m=\eta|It|M_{molar}/(z_eF)$

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| **EN:** Symbols and units: $M_{molar}$: g/mol with m: g; $z_e$: electron coefficient per analyte; η: current efficiency0–1; I: A and t: s. |
| **中文：** 符号与单位：$M_{molar}$用g/mol时m用g；$z_e$每份待测物电子系数；η电流效率0–1；I用A、t用s。 |
| **EN:** When to use: Convert a given reaction coefficient to charge, analyte mass or average current. |
| **中文：** 何时使用：用给定电子系数求电荷、分析物质量或平均电流。 |
| **EN:** Conditions and traps: The stated reaction model, complete/known reaction yield and collected charge. η includes the specified useful-current fraction. Convert mg to g before dividing by g/mol. Do not import the real lactate molar mass or reaction into p60 without noting its simplified100 g/mol, 1: 12 assumptions. |
| **中文：** 条件与易错点：按给定反应模型、已知 / 完全反应产率与收集电荷；η是指定有效电流比例。除以g/mol前把mg换g；不能静默替换p60简化100 g/mol、1:12假设。 |
| **EN:** Tiny example: 10 mL×1 mg/mL=0.010 g; M=100 g/mol, $z_e=12$ → $n_e=1.2\times10^{-3}$ mol. |
| **中文：** 短例理解：10 mL×1 mg/mL=0.010 g；M=100 g/mol、$z_e=12$，得$n_e=1.2\times10^{-3}$ mol。 |

Source / 来源：W6 PDF p51–52,p60 | T6-0055–0063

## FX429 • [P2] Glucose-strip background subtraction / 葡萄糖试纸背景扣除

**EN:** Revision priority: Important supporting relations. Explicit source arithmetic relationship and practical glucose readout.

**中文：** 复习优先级：重要基础关系。原图明确的减法关系与葡萄糖实际读出。

**Formula / 公式：** $I_{glucose} \approx I_{WE1} - I_{WE2}$; if $I_{corr} = S\,c + b$, $c = (I_{corr} - b)/S$

**EN:** Symbols and units: WE1: GOx-containing working electrode signal+background; WE2: matched enzyme-free background; S: calibration current/concentration.

**中文：** 符号与单位：WE1含GOx，测目标加背景；WE2不含酶，测匹配背景；S校准斜率，单位电流/浓度。

**EN:** When to use: Subtract matched interference and use the calibration for the corrected signal.

**中文：** 何时使用：扣除匹配干扰，再用扣背景信号的校准线。

**EN:** Conditions and traps: Background currents are suitably matched and calibration is valid for the corrected channel. p58 is two working electrodes plusAg/AgCl reference; do not rename the secondWE as a counter electrode.

**中文：** 条件与易错点：两路背景电流适当匹配，且校准线对扣背景通道有效。 p58是双WE加Ag/AgCl参比，不能把第二WE改叫对电极。

**EN:** Tiny example: 18−5=13 μA; S=2 μA/mM, b=0 →c=6.5 mM.

**中文：** 短例理解：18−5=13 μA；S=2 μA/mM、b=0，c=6.5 mM。

Source / 来源：W6 PDF p58 | T6-0066–0069

## FX430 • [P3] Conductance, conductivity and geometry / 电导、电导率与几何

**EN:** Revision priority: Recognise; optional calculations. Supporting uniform-path relation derived for interpreting conductometric diagrams; not an explicit new quiz formula.

**中文：** 复习优先级：辅助辨认；计算后看。用于理解电导图的均匀路径辅助关系，不新增明确Quiz公式。

**Formula / 公式：** $G = 1/R$; $I = GV$; $R = \rho_e L/A$; $G = \kappa A/L$; $\kappa = 1/\rho_e$

**EN:** Symbols and units: G: siemens S; R:Ω; κ: conductivity S/m; $\rho_e$: resistivity Ω·m; L: m; A: $m^2$.

**中文：** 符号与单位：G电导用S，R电阻Ω，κ电导率S/m，$\rho_e$电阻率Ω·m，L用m，A用$m^2$。

**EN:** When to use: Distinguish a measured conductance from material conductivity and account for geometry.

**中文：** 何时使用：区分测得的电导与材料电导率，并考虑几何形状。

**EN:** Conditions and traps: Uniform ohmic path with specified length/cross-section; electrode interfaces and complex geometries need a fuller model. G depends on geometry as well as material. Acoustic densityρ is unrelated to electrical resistivityρ_e.

**中文：** 条件与易错点：指定长度、截面的均匀欧姆导体；电极界面与复杂几何需更完整模型。 G随几何与材料共同变；声学密度ρ与电阻率ρ_e不是同一量。

**EN:** Tiny example: κ=1 S/m, A=1 $mm^2$, L=1 mm →G=1 mS, R=1 kΩ.

**中文：** 短例理解：κ=1 S/m、A=1 $mm^2$、L=1 mm，G=1 mS、R=1 kΩ。

Source / 来源：W6 PDF p19,p67–69 | T6-0020 | T6-0077–0079

## FX431 • [P2] Parallel-plate capacitance model / 平行板电容模型

**EN:** Revision priority: Important supporting relations. README core formula supports the course capacitance mechanism; treat it as a stated model, not every electrode’ s universal law.

**中文：** 复习优先级：重要基础关系。README核心式支持课程电容机制，但须声明模型，不当成所有电极通用定律。

**Formula / 公式：** $C=\varepsilon A/d=\varepsilon_0\varepsilon_r A/d$; for fixedA: $C_2/C_1=(\varepsilon_2/\varepsilon_1)(d_1/d_2)$

**EN:** Symbols and units: C: F; $\varepsilon_0\approx8.854\times10^{-12}$ F/m; ε_r: relative permittivity; A: $m^2$; d: m; ε: absolute permittivityF/m.

**中文：** 符号与单位：C用F；$\varepsilon_0\approx8.854\times10^{-12}$ F/m，ε_r相对介电常数，A用$m^2$、d用m；ε为绝对介电率F/m。

**EN:** When to use: Predict capacitance change with dielectric, area and separation.

**中文：** 何时使用：判断介电、面积及间距变化怎样改变电容。

**EN:** Conditions and traps: Ideal parallel plates, uniform dielectric, negligible fringing and fixed measurement frequency; biosensor interfaces need an appropriate equivalent model. C=εA/d already uses absoluteε; do not multiply $\varepsilon_0$ twice. Binding does not universally make C rise—thickness and dielectric can change together.

**中文：** 条件与易错点：理想平行板、均匀介电、忽略边缘效应且测量频率固定；真实传感界面需适当等效模型。 C=εA/d中的ε已是绝对介电率，不乘两次$\varepsilon_0$；结合可同时改厚度与介电，不保证C一定上升。

**EN:** Tiny example: Fixedε, A and doubledd →C halves.

**中文：** 短例理解：ε、A固定，d翻倍，C减半。

Source / 来源：W6 PDF p80–81,p86 | T6-0088–0092 | T6-0097–0099 | Sample Q10

## FX432 • [P2] R,C,L complex impedances / 电阻、电容、电感复阻抗

**EN:** Revision priority: Important supporting relations. Basic impedance recognition is covered; full multi-element algebra/Bode drawing is secondary/final emphasis.

**中文：** 复习优先级：重要基础关系。基础阻抗辨认在课程内；完整多元件代数 / Bode作图为次要 / 期末强调。

**Formula / 公式：** $\omega=2\pi f$; $Z_R=R$; $Z_C=1/(j\omega C)=-j/(\omega C)$; $Z_L=j\omega L$; $|X_C|=1/(2\pi fC)$; $|X_L|=2\pi fL$

**EN:** Symbols and units: $j^2=-1$; f: Hz; ω: rad/s; C: F; L: H; allZ andX:Ω; phases R0°, C−90°, L+90° for the stated convention.

**中文：** 符号与单位：$j^2=-1$；f用Hz、ω用rad/s、C用F、L用H；Z、X均用Ω；按此约定相位R0°、C−90°、L+90°。

**EN:** When to use: Read frequency trends and calculate reactive impedance.

**中文：** 何时使用：辨认频率趋势并计算电抗阻抗。

**EN:** Conditions and traps: Ideal linear lumped elements and e^(jωt) phasor convention; actual components can have parasitic losses. p83’ s graphic/ASR incorrectly puts f in the denominator of inductive reactance: correct $X_L=2\pi fL$. Capacitive magnitude falls as f rises.

**中文：** 条件与易错点：理想线性集中参数元件，采用e^(jωt)相量约定；真实器件有寄生损耗。 p83图 / 字幕把电感电抗的f放分母有误；正确$X_L=2\pi fL$。电容幅值随频率升高而降低。

**EN:** Tiny example: f=1 MHz, C=500 pF →$Z_C=-j318.3\ \Omega$.

**中文：** 短例理解：f=1 MHz、C=500 pF，$Z_C=-j318.3\ \Omega$。

Source / 来源：W6 PDF p82–86 | T6-0093–0101

## FX433 • [P3] Combining AC impedances / 交流阻抗串并联

**EN:** Revision priority: Recognise; optional calculations. Necessary support for source Example4; lengthy network calculations are not a quiz guarantee.

**中文：** 复习优先级：辅助辨认；计算后看。支持原例4所需关系；长网络计算不宣传Quiz保证。

**Formula / 公式：** $Z_{series}=\Sigma Z_i$; $1/Z_{parallel}=\Sigma(1/Z_i)$; two branches $Z_{parallel}=Z_1Z_2/(Z_1+Z_2)$; $|a+jb|=\sqrt{(a^2+b^2)}$

**EN:** Symbols and units: $Z_i$: complex impedancesΩ; a, b: real and imaginary partsΩ; |Z|: magnitudeΩ.

**中文：** 符号与单位：$Z_i$复阻抗Ω；a、b为实部、虚部Ω；|Z|幅值Ω。

**EN:** When to use: Simplify equivalent circuits before computing current or voltage division.

**中文：** 何时使用：先化简等效电路，再算电流或分压。

**EN:** Conditions and traps: Linear sinusoidal steady state at a common frequency; use the actual connection diagram. Add complexZ first, then take magnitude. In general $|Z_1+Z_2| \neq |Z_1|+|Z_2|$.

**中文：** 条件与易错点：同一频率下的线性正弦稳态；连接关系按实际电路图。 先加复数Z，再求模；一般$|Z_1+Z_2|$不等于$|Z_1|+|Z_2|$。

**EN:** Tiny example: 100 Ω in series with−j318.3 Ω →|Z|≈333.65 Ω, not418.3 Ω.

**中文：** 短例理解：100 Ω串联−j318.3 Ω，|Z|≈333.65 Ω，不是418.3 Ω。

Source / 来源：W6 PDF p85–86 | T6-0096–0101

## FX434 • [P3] SeriesRC capacitor voltage divider / 串联RC的电容分压

**EN:** Revision priority: Recognise; optional calculations. Worked in class; teacher explicitly emphasized final impedance/Bode questions, not an assured quiz complex calculation.

**中文：** 复习优先级：辅助辨认；计算后看。课堂讲过；老师明确强调期末阻抗 / Bode，不等于保证Quiz复数计算。

**Formula / 公式：** $V_C/V_s=Z_C/(R+Z_C)=1/(1+j\omega RC)$; $|V_C|=|V_s|/\sqrt{[1+(\omega RC)^2]}$; $\varphi_C=-\operatorname{atan}(\omega RC)$

**EN:** Symbols and units: R:Ω; C: F; ω: rad/s; ωRC dimensionless; V_s, V_C use the same peak/RMS convention.

**中文：** 符号与单位：R用Ω、C用F、ω用rad/s；ωRC无量纲；V_s与V_C统一峰值 / 有效值约定。

**EN:** When to use: Solve the p86 voltage across the series sensor capacitor.

**中文：** 何时使用：求p86串联传感电容两端电压。

**EN:** Conditions and traps: Ideal seriesR–C driven by a voltage source, output acrossC, no loading; not the same as a parallelRC total impedance. Do not divide by R+|Z_C| as scalars. Phase is relative to the source; peak5 V gives a peak answer.

**中文：** 条件与易错点：理想电压源驱动串联R–C，输出取电容、无负载；不同于并联RC总阻抗。 分母不能用R+|Z_C|的标量和；相位相对源，5 V峰值得到峰值结果。

**EN:** Tiny example: R=100 Ω, C=500 pF, f=1 MHz, V_s=5 V peak →$V_C \approx 4.77\angle -17.44°$ V peak.

**中文：** 短例理解：R=100 Ω、C=500 pF、f=1 MHz、源5 V峰值，$V_C \approx 4.77\angle -17.44°$ V峰值。

Source / 来源：W6 PDF p86 | T6-0097–0101

## FX435 • [P3] Pure parallelRC and Bode conventions / 纯并联RC与Bode约定

**EN:** Revision priority: Recognise; optional calculations. Reference/final-algebra support; included chiefly to correct the copied phase-peak claim.

**中文：** 复习优先级：辅助辨认；计算后看。参考 / 期末代数辅助，主要用于纠正照抄相位峰的说法。

**Formula / 公式：** $Z_{(R\|C)}=R/(1+j\omega RC)$; $|Z|=R/\sqrt{[1+(\omega RC)^2]}$; $\varphi=-\operatorname{atan}(\omega RC)$; $f_c=1/(2\pi RC)$; normalized magnitude_dB$=20\log_{10}(|Z|/Z_{ref})$

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| **EN:** Symbols and units: $Z_{ref}$: stated positive reference impedance; R:Ω, C: F, $f_c$: Hz; magnitude ratio for dB must be dimensionless. |
| **中文：** 符号与单位：$Z_{ref}$为声明的正参考阻抗；R用Ω、C用F、$f_c$用Hz；dB的幅值比必须无量纲。 |
| **EN:** When to use: Recognize a corner frequency, limiting impedance and phase behavior. |
| **中文：** 何时使用：辨认转折频率、阻抗极限与相位变化。 |
| **EN:** Conditions and traps: Pure parallelR‖C. The source may plotlog\|Z\| rather than normalizeddB: read the axis label; at$f_c$,$|Z|=R/\sqrt{2}$ and$\varphi=-45°$. A pure parallelRC phase changes monotonically0→−90° and has no interior phase peak. The p85 generic peak requires additional circuit structure. |
| **中文：** 条件与易错点：纯并联R‖C；课件可画log\|Z\|而非归一dB，须读轴；$f_c$处$|Z|=R/\sqrt{2}$、$\varphi=-45°$。纯并联RC相位从0到−90°单调变化，无内部相位峰；p85一般峰形需额外电路结构。 |
| **EN:** Tiny example: R=1 kΩ, C=1 μF →$f_c\approx159$ Hz; at$f_c$,$|Z|\approx707$ Ω. |
| **中文：** 短例理解：R=1 kΩ、C=1 μF，$f_c\approx159$ Hz，此处$|Z|\approx707$ Ω。 |

Source / 来源：W6 PDF p85–86 | T6-0095–0101

## FX436 • [P3] Series resistance with a parallelRC/Nyquist semicircle / 串联电阻加并联RC / Nyquist半圆

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| **EN:** Revision priority: Recognise; optional calculations. Auxiliary equivalent-circuit recognition; longer derivation belongs to final/advanced practice, not quiz first-pass memory. |
| **中文：** 复习优先级：辅助辨认；计算后看。等效电路辅助辨认；长推导放期末 / 进阶，不列Quiz第一轮背诵。 |

**Formula / 公式：** $Z=R_s+R_p/(1+j\omega R_pC)$; $Z'=R_s+R_p/(1+x^2)$; $Z''=-R_px/(1+x^2)$, $x=\omega R_pC$; $(Z'-R_s-R_p/2)^2+(Z'')^2=(R_p/2)^2$

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| **EN:** Symbols and units: $R_s$: series resistanceΩ; $R_p$: parallel resistanceΩ; C: F; x: dimensionless; Nyquist convention plots−Z″ againstZ′. |
| **中文：** 符号与单位：$R_s$串联电阻Ω，$R_p$并联电阻Ω，C用F，x无量纲；Nyquist常画−Z″对Z′。 |
| **EN:** When to use: Recognize the semicircle and why adding$R_s$ changes Bode phase behavior. |
| **中文：** 何时使用：辨认半圆，并理解加$R_s$如何改变Bode相位。 |

**EN:** Conditions and traps: Ideal $R_s+(R_p \| C)$ with positive parameters; no diffusion/constant-phase elements. Different electrode models give other shapes. The circuit, not every biosensor, predicts a semicircle. This network can have a phase feature; do not apply it to pure$R \| C$ indiscriminately.

**中文：** 条件与易错点：理想正参数$R_s+(R_p \| C)$，不含扩散 / 常相位元件；不同电极模型可有其他曲线。 是该电路预测半圆，并非所有传感器都半圆；本网络可有相位特征，不能无条件套纯$R \| C$。

**EN:** Tiny example: Nyquist real-axis intercepts are$R_s$ and$R_s+R_p$; with$R_s=300\ \Omega$, $R_p=20\ k\Omega$ they are$300\ \Omega$ and$20.3\ k\Omega$.

**中文：** 短例理解：实轴截距为$R_s$和$R_s+R_p$；$R_s=300\ \Omega$、$R_p=20\ k\Omega$时为$300\ \Omega$和$20.3\ k\Omega$。

Source / 来源：W6 PDF p66,p70,p84–85 | T6-0095–0101

## FX701 • [P2] Cell Nernst potential: valence, sign and temperature / 细胞 Nernst 电位：价态、符号与温度

**EN:** Revision priority: Important supporting relations. Understand the one-ion meaning and signs; the teacher did not require memorizing long cellular expressions. Numerical use is supporting practice.

**中文：** 复习优先级：重要基础关系。理解单离子含义与符号；老师说细胞长表达式不需死记。完整数值代入作辅助练习。

**Formula / 公式：** $E_{ion} = (RT/(zF))\ \ln(a_{out}/a_{in}) = (2.303RT/(zF))\ \log_{10}(a_{out}/a_{in})$; 25°C: $E_{ion} = (59.16\ mV/z)\ \log_{10}(a_{out}/a_{in})$; 37°C: $E_{ion} = (61.54\ mV/z)\ \log_{10}(a_{out}/a_{in})$

**EN:** Symbols and units: $E_{ion}$: inside-minus-outside equilibrium voltage; z: signed ionic charge (+1 for $K^+$, −1 for $Cl^-$); a: activity, approximated by concentration when appropriate. $R=8.314\ J/(mol\cdot K)$, T in K, $F\approx96485\ C/mol$.

**中文：** 符号与单位：$E_{ion}$：胞内减胞外的单离子平衡电位；z 是带符号价态（$K^+$为 +1，$Cl^-$为 −1）；a 为活度，适当条件下用浓度近似。$R=8.314\ J/(mol\cdot K)$，T 用 K，$F\approx96485\ C/mol$。

**EN:** When to use: Find the voltage balancing one ion’ s concentration drive; compare how a gradient or temperature changes that voltage.

**中文：** 何时使用：求使某一种离子的浓度驱动与电作用平衡的电压；判断梯度或温度变化的影响。

**EN:** Conditions and traps: One specified ion at electrochemical equilibrium; inside and outside use the same concentration units. 25°C=298.15 K; 37°C=310.15 K. The slide’ s −61 coefficient is a rounded 37°C coefficient for z=+1 with the ratio reversed. Do not put ln into a base-10 coefficient, omit the sign of z, or use −61 for every ion. This single-ion voltage is not automatically the multi-ion resting potential.

**中文：** 条件与易错点：针对一种特定离子的电化学平衡；内外浓度单位一致。25°C=298.15 K；37°C=310.15 K。课件 −61 是 z=+1、浓度比倒置后使用的 37°C 粗略系数。 不能把 ln 配上常用对数系数、漏掉 z 的正负号，或所有离子都套 −61。单离子平衡电位不自动等于多离子细胞的实际静息电位。

**EN:** Tiny example: $K^+$: aout/ain≈4/140 at 37°C gives −95.0 mV using 61.54; the source’ s rounded 61 gives −94.2 mV. For z=−1, the coefficient changes sign.

**中文：** 短例理解：$K^+$在 37°C，aout/ain≈4/140：用 61.54 得 −95.0 mV，课件用粗略 61 得 −94.2 mV。z=−1 时系数反号。

Source / 来源：W7 PDF p27 | T7-0017–0018

## FX702 • [P2] GHK: permeability weights and reversed chloride placement / GHK：通透性权重与 $Cl^-$ 的反向位置

**EN:** Revision priority: Important supporting relations. Learn what is weighted and the $Cl^-$ placement; retain the full expression for reference rather than treating it as a promised Quiz derivation.

**中文：** 复习优先级：重要基础关系。先会权重与 $Cl^-$ 位置的概念辨认；完整式保留供查阅，不宣传为 Quiz 保证推导题。

**Formula / 公式：** $Vm = (RT/F)\ \ln[(PK[K^+]out + PNa[Na^+]out + PCl[Cl^-]in)/(PK[K^+]in + PNa[Na^+]in + PCl[Cl^-]out)]$; At a steady zero-net-current state: $I_{Na} + I_K + I_{Cl} = 0$

**EN:** Symbols and units: Vm: inside−outside voltage (V); P: relative permeability for each ion; brackets: ion concentration, in consistent units; R,T,F as in Nernst.

**中文：** 符号与单位：Vm：胞内减胞外电位（V）；P：各离子的相对通透性；方括号表示统一单位的离子浓度；R、T、F 同 Nernst。

**EN:** When to use: Recognize how several monovalent ions contribute to membrane voltage; identify why changing a permeability can change Vm without changing the concentrations.

**中文：** 何时使用：识别多个单价离子如何共同决定膜电位；理解浓度不变时通透性改变也能改变 Vm。

**EN:** Conditions and traps: The shown form is the constant-field, monovalent $K^+/Na^+/Cl^-$ model. Use equal concentration units and a consistent permeability convention. Zero summed ionic current does not require zero current in every branch. $Cl^-$ has “in” in the numerator and “out” in the denominator, opposite to the cations. GHK is not an arithmetic average of Nernst voltages, and −85 mV is the source’ s example, not a universal constant.

**中文：** 条件与易错点：此式针对恒电场、单价 $K^+/Na^+/Cl^-$ 模型。浓度单位与通透性约定要一致。总离子电流为零，不要求每个支路电流都为零。 $Cl^-$ 的胞内项在分子、胞外项在分母，与阳离子相反。GHK 不是各 Nernst 电位的算术平均；−85 mV 是课件示例，不是普适常数。

**EN:** Tiny example: If PNa and PCl are negligible, the ratio reduces to $[K^+]out/[K^+]in$ and the $K^+$ Nernst result. Currents +2, −1.5 and −0.5 nA can sum to zero while each is nonzero.

**中文：** 短例理解：若 PNa、PCl 可忽略，式子退化为 $K^+$ 的胞外/胞内浓度比，即 $K^+$ Nernst 结果。+2、−1.5、−0.5 nA 总和为零，但各项仍不为零。

Source / 来源：W7 PDF p28–31 | T7-0017–0018

## FX703 • [P3] Ionic branch: conductance and driving force / 离子支路：电导与驱动力

**EN:** Revision priority: Recognise; optional calculations. Supporting circuit vocabulary from a slide the teacher described as reference; not a teacher-confirmed Quiz formula to memorize.

**中文：** 复习优先级：辅助辨认；计算后看。课堂称相关细胞数学式供参考；这是支路模型补充，不是老师明确要求 Quiz 必背的公式。

**Formula / 公式：** $I_{ion} = g_{ion}(V_m - E_{ion}) = (V_m - E_{ion})/R_{ion}$; $g_{ion} = 1/R_{ion}$

**EN:** Symbols and units: Iion: branch current (A), positive outward under the chosen convention; gion: conductance (S); Rion: resistance (Ω); Vm and Eion: voltages in the same unit.

**中文：** 符号与单位：Iion：支路电流（A），按所选约定取向外为正；gion：电导（S）；Rion：电阻（Ω）；Vm、Eion 用同一电压单位。

**EN:** When to use: Separate ion concentration equilibrium voltage from the current at the actual membrane voltage. The difference Vm−Eion is the electrical driving force in this model.

**中文：** 何时使用：把该离子的平衡电位与实际膜电位下的电流分开；这个模型的驱动力是 Vm−Eion。

**EN:** Conditions and traps: A passive, locally ohmic ion-channel branch with specified conductance and sign convention. Channel conductance can change with time or voltage; the formula is not a complete cell model. Zero current in this branch occurs at Vm=Eion, but zero total membrane current can arise from nonzero cancelling branches. Do not equate conductance with permeability.

**中文：** 条件与易错点：针对给定电导与符号约定的被动、局部欧姆性离子通道支路。通道电导可随时间/电压改变；此式不是完整细胞模型。 此支路在 Vm=Eion 时电流为零；总膜电流也可由多个非零支路相抵而为零。电导不应直接等同通透性。

**EN:** Tiny example: g=10 nS, Vm=−70 mV and Eion=−90 mV give I=10 nS×20 mV=200 pA outward.

**中文：** 短例理解：g=10 nS、Vm=−70 mV、Eion=−90 mV，得到 I=10 nS×20 mV=200 pA，按约定向外。

Source / 来源：W7 PDF p30 | T7-0017

## FX704 • [P2] Einthoven limb leads: write the polarity first / Einthoven 肢体导联：先写清极性

**EN:** Revision priority: Important supporting relations. A short polarity identity printed in the ECG slide; understand it without elevating deep cardiac physiology.

**中文：** 复习优先级：重要基础关系。原 ECG 课件的短极性关系，应能理解；不因此提高深层心脏生理的复习权重。

**Formula / 公式：** Lead I = V_LA − V_RA; Lead II = V_LL − V_RA; Lead III = V_LL − V_LA; Lead II = Lead I + Lead III

**EN:** Symbols and units: RA: right arm, LA: left arm, LL: left leg of the person; each V is an electrode potential referred to the same reference, usually in mV.

**中文：** 符号与单位：RA 是人体右臂，LA 是左臂，LL 是左腿；每个 V 是相对同一参考点的电极电位，通常用 mV。

**EN:** When to use: Recover the third standard limb-lead voltage from two leads and avoid confusing electrode locations with P/QRS/T waveform features.

**中文：** 何时使用：由两个标准肢体导联求第三个电压；区分电极位置与 P/QRS/T 波形特征。

**EN:** Conditions and traps: The standard simultaneous limb-lead definitions must be used consistently. On W7 p69, electrode 1=RA, 2=LA and 3=LL; left/right is the subject’ s anatomy, not the viewer’ s image side. Reversing a subtraction reverses the sign. Lead II is not the QRS complex, and the identity does not add three different stages of the cardiac cycle.

**中文：** 条件与易错点：必须统一采用同时测量的标准肢体导联定义。W7 p69 的 1=RA、2=LA、3=LL；左右指人体解剖学左右，不是观看图纸的左右。 颠倒相减顺序就会反号。Lead II 不是 QRS 波群，此关系不是把三个不同心动阶段相加。

**EN:** Tiny example: Lead I=+0.2 mV and Lead III=+0.3 mV imply Lead II=+0.5 mV. If the lead-I terminals are swapped, that measured trace changes sign.

**中文：** 短例理解：Lead I=+0.2 mV、Lead III=+0.3 mV，则 Lead II=+0.5 mV。若交换 I 导联的测量端，这条记录反号。

Source / 来源：W7 PDF p68–69 | T7-0072–0073

## FX705 • [P1] Frequency, peak intervals and heart rate / 频率、峰间隔与心率

**EN:** Revision priority: First: know and calculate. Time-window frequency reading is a repeated classroom and Sample Q5 priority. RR calculation is an additional short ECG application; the source’ s blurry Q5 letter remains provisional.

**中文：** 复习优先级：优先掌握与短算。时间窗读频率为课堂反复强调且 Sample Q5 涉及的重点；RR 心率是短应用。模糊 Q5 的参考字母仍不能当已确认精确答案。

**Formula / 公式：** $f = N/(t_2 - t_1) = 1/T$; For n successive matching peaks: $f = (n-1)/(t_{last} - t_{first})$; Heart rate (beats/min) = 60/RR(s)

**EN:** Symbols and units: N: complete cycles in the selected time window; n: number of successive matching peaks; T: one period (s); RR: time between successive ECG R peaks (s); f in Hz.

**中文：** 符号与单位：N：时间窗内的完整周期数；n：连续同相位峰的个数；T：单周期（s）；RR：连续心电 R 峰的间隔（s）；f 单位 Hz。

**EN:** When to use: Read a time-axis waveform rather than guessing frequency from amplitude. Use RR specifically when estimating heart rate.

**中文：** 何时使用：根据时间轴读频率，不用幅度猜频率；计算心率时要选 RR 间隔。

**EN:** Conditions and traps: A reasonably periodic signal with a defined cycle. Convert ms to s; use the actual difference between the time endpoints. Irregular EEG permits only a representative-window estimate. n peak positions delimit n−1 intervals. A window starting at 0.2 s does not have duration equal to its final timestamp. HR arithmetic must not be applied to arbitrary EMG spikes.

**中文：** 条件与易错点：信号近似周期性，且已定义一个周期。ms 要换成 s；时长是两端相减。对不规则 EEG，只能用代表性时间窗估读。 n 个峰位置划出 n−1 个间隔。时间从 0.2 s 开始时，不能把末时刻当总时长。不能把心率算法随便套到任意 EMG 尖峰。

**EN:** Tiny example: Eight complete cycles in 0.20 s give 40 Hz. Nine successive peak positions spanning 0.20 s also give 40 Hz. RR=0.8 s gives 75 beats/min.

**中文：** 短例理解：0.20 s 内 8 个完整周期得 40 Hz；9 个连续峰从首到末跨 0.20 s 也得 40 Hz。RR=0.8 s 得 75 次/分。

Source / 来源：W7 PDF p64–67, p83–85 | T7-0070–0071 | T7-0092–0095 | Sample Quiz Q5

## FX708 • [P3] Loading and contact mismatch: voltage-divider model / 加载与接触不匹配：分压模型

**EN:** Revision priority: Recognise; optional calculations. Supporting model from a worked historical final-exam example; the Quiz priority is loading/matched-contact understanding, not the long network calculation.

**中文：** 复习优先级：辅助辨认；计算后看。来自课堂讲过的往年期末模型，作为辅助；Quiz 应先理解加载与接触匹配，不必把长网络计算提为重点。

**Formula / 公式：** $V_{input} = V_{source} \cdot Z_{in}/(Z_{source}+Z_{in})$; $V_A - V_B = V_{cm}[Z_{in}/(Z_1+Z_{in}) - Z_{in}/(Z_2+Z_{in})]$

**EN:** Symbols and units: Zin: amplifier input impedance; Zsource: source impedance; $Z_1$,$Z_2$: the two electrode-contact impedances; Vcm: shared body voltage; A is on the $Z_1$ branch and B on $Z_2$ here.

**中文：** 符号与单位：Zin：放大器输入阻抗；Zsource：信号源阻抗；$Z_1$、$Z_2$：两端电极接触阻抗；Vcm：身体共有电压；这里 A 接 $Z_1$ 支路、B 接 $Z_2$。

**EN:** When to use: Explain reduced loading when Zin is large and why unequal electrode impedances can convert common-mode interference into a differential input.

**中文：** 何时使用：说明大 Zin 如何减少加载，以及两电极阻抗不等如何把共模干扰转成差分输入。

**EN:** Conditions and traps: The second relation is the simplified equal-input-impedance circuit on W7 p74–75. For AC, use complex impedances and phasors consistently; the slide’ s supplied real values are one special case. Do not divide using impedance magnitudes alone when phase matters. In p74 A has $Z_1$=20 kΩ and B has $Z_2$=10 kΩ; the source’ s positive 137 μV is an interference amplitude, not signed VA−VB.

**中文：** 条件与易错点：第二式对应 W7 p74–75 的等输入阻抗简化电路。交流要统一用复阻抗与相量；原题给的实数阻抗是一个特例。 相位重要时，不能只用阻抗的模来分压。p74 中 A 对 $Z_1$=20 kΩ、B 对 $Z_2$=10 kΩ；源解 +137 μV 是干扰幅值，不是严格的有符号 VA−VB。

**EN:** Tiny example: Zsource=100 kΩ gives Vinput/Vsource=0.50 for Zin=100 kΩ, but ≈0.990 for Zin=10 MΩ. If $Z_1$=$Z_2$, the ideal common-mode differential term is zero.

**中文：** 短例理解：Zsource=100 kΩ：Zin=100 kΩ 时保留 0.50 倍电压，Zin=10 MΩ 时保留约 0.990 倍。$Z_1$=$Z_2$ 时，理想共模转差分项为零。

Source / 来源：W7 PDF p74–75 | W8 PDF p20 | T7-0074–0077 | T8-0023

## FX709 • [P3] Electrode-interface RC impedance and limits / 电极界面 RC 阻抗与频率极限

**EN:** Revision priority: Recognise; optional calculations. Full impedance algebra is retained for source completeness and final-exam support. Matching-contact and frequency trends are more important for the current concept-heavy Quiz.

**中文：** 复习优先级：辅助辨认；计算后看。完整阻抗式供源式查阅与期末辅助；当前概念型 Quiz 更应先会接触匹配与频率趋势。

**Formula / 公式：** $Z(\omega) = Rs + Rd/(1+j\omega RdCd)$; $|Z| = \sqrt{[(Rs+Rd)^2+(\omega RsRdCd)^2]}/\sqrt{[1+(\omega RdCd)^2]}$; Low f: Z→Rs+Rd; high f: Z→Rs; ω=2πf

**EN:** Symbols and units: Rs: solution/series resistance; Rd: interfacial resistance; Cd: interfacial capacitance; Z in Ω, C in F, ω in rad/s, f in Hz; $j^2=-1$. The p76 example uses Rs=$R_1$, Rd=$R_2$, Cd=C.

**中文：** 符号与单位：Rs：溶液/串联电阻；Rd：界面电阻；Cd：界面电容；Z 用 Ω、C 用 F、ω 用 rad/s、f 用 Hz；$j^2=-1$。p76 例题中 Rs=$R_1$、Rd=$R_2$、Cd=C。

**EN:** When to use: Turn the illustrated Rs+(Rd∥Cd) interface into a usable formula and predict low-/high-frequency behavior.

**中文：** 何时使用：把 Rs 串联 Rd∥Cd 的界面示意转为可使用的式子，并判断低频/高频行为。

**EN:** Conditions and traps: The simple linear lumped interface model omits diffusion and additional skin layers. A dc half-cell voltage offset is separate from this small-signal impedance. Substitute ω=2πf before calculation. W7 p78’ s table is not consistent with its stated components when read as Hz. An EIS Nyquist plot has Z′ and usually −Z″ axes; frequency labels belong to points, not its horizontal axis.

**中文：** 条件与易错点：此线性集中参数简化模型未包含扩散与额外皮肤层。直流半电池电位偏移与这里的小信号阻抗不同。 计算前代入 ω=2πf。W7 p78 的原表按 Hz 读时与给定元件值不一致。EIS Nyquist 图横轴为 Z′、纵轴常为 −Z″；频率属于各测量点的标签，不是横坐标。

**EN:** Tiny example: Rs=300 Ω, Rd=20 kΩ and Cd=100 nF give limits 20.3 kΩ and 300 Ω; at f=100 Hz, |Z|≈12.64 kΩ.

**中文：** 短例理解：Rs=300 Ω、Rd=20 kΩ、Cd=100 nF，低/高频极限为 20.3 kΩ 与 300 Ω；f=100 Hz 时 |Z|≈12.64 kΩ。

Source / 来源：W7 PDF p41–42, p76–78 | W6 PDF p66, p84–85 (Nyquist/Bode comparison) | T7-0035–0038 | T7-0077–0081

## FX713 • [P3] Sampling Nyquist frequency is not an impedance-plot axis / 采样 Nyquist 频率不是阻抗图的横轴

**EN:** Revision priority: Recognise; optional calculations. The temporal sampling formula was not found as a required W7–8 formula. It is a bridge to W4 spatial sampling and the guide’ s S505; do not label it teacher-confirmed Quiz material.

**中文：** 复习优先级：辅助辨认；计算后看。没有在 W7–8 找到该时间采样式的必考要求。它用于联系 W4 空间采样及原讲义 S505，不标成老师确认的 Quiz 考点。

**Formula / 公式：** $f_s = 1/\Delta t_s$; $f_{Nyquist} = f_s/2$; For a baseband band-limited signal: $f_s > 2f_{max}$

**EN:** Symbols and units: fs: samples per second (Hz); Δts: sample interval (s); fNyquist: half the sample rate; fmax: highest retained signal frequency after suitable anti-alias filtering.

**中文：** 符号与单位：fs：每秒采样数（Hz）；Δts：采样间隔（s）；fNyquist：采样率的一半；fmax：经过适当抗混叠滤波后需保留的最高信号频率。

**EN:** When to use: Separate a waveform’ s oscillation frequency from its digital sampling rate, and separate the sampling theorem from an EIS Nyquist impedance plot.

**中文：** 何时使用：区分波形振荡频率与数字采样率，并区分采样定理和 EIS Nyquist 阻抗图。

**EN:** Conditions and traps: This is a supplementary baseband sampling relation. The signal must be band-limited and the acquisition path appropriate; exactly two samples per cycle can lose information at unfavorable phase, so allow margin. Neither a taller EEG wave nor more spatial electrodes determines fs. An EIS Nyquist plot uses Z′ versus −Z″ and its points have frequency labels; fNyquist=fs/2 is a different concept.

**中文：** 条件与易错点：这是补充的基带采样关系。信号需限带且采集通路合适；恰好每周期两次在不利相位下可能丢失信息，应留余量。 EEG 更高的波幅或更多空间电极都不直接决定 fs。EIS Nyquist 图画 Z′ 与 −Z″，每点带有测量频率；fNyquist=fs/2 是另一概念。

**EN:** Tiny example: fs=2000 samples/s gives fNyquist=1000 Hz. A retained 400 Hz baseband component is below that limit, provided higher-frequency contamination is adequately filtered.

**中文：** 短例理解：fs=2000 次/秒，则 fNyquist=1000 Hz。要保留的 400 Hz 基带成分低于此限，但还需适当滤掉更高频干扰。

Source / 来源：W4 PDF p33–34 (spatial-sampling context) | W7 PDF p83–85 (time-axis context) | W6 PDF p66, p84–85 (impedance-plot comparison) | T7-0092–0095 | NI primary digitization guide (fNyquist=fs/2): https://download.ni.com/evaluation/daq/NI_Engineers_Guide_to_Digitization.pdf

## FX706 • [P1] Differential voltage and shared-noise cancellation / 差分电压与共有噪声抵消

**EN:** Revision priority: First: know and calculate. The source repeatedly explains differential recording and gain; this consolidates the existing short formula with its cancellation conditions.

**中文：** 复习优先级：优先掌握与短算。课堂反复讲差分与增益；此卡把已有短式与抵消条件合在一起，不只是孤立背公式。

**Formula / 公式：** $Vd = V_1 - V_2$; $Vout = G \cdot Vd$; If $V_1 = s_1 + n$ and $V_2 = s_2 + n$: $Vd = s_1 - s_2$; If noises differ: $Vd = (s_1 - s_2) + (n_1 - n_2)$

**EN:** Symbols and units: Vd: differential input voltage; $V_1$,$V_2$: input electrode voltages; s: wanted components; n: interference; G: dimensionless voltage gain. All voltage terms use matching units.

**中文：** 符号与单位：Vd：差分输入电压；$V_1$、$V_2$：两输入电极电位；s 为有用成分，n 为干扰；G 是无单位电压增益。所有电压项用一致单位。

**EN:** When to use: Explain why a large shared background can cancel while a small biological difference remains; then calculate the output voltage.

**中文：** 何时使用：解释为什么大共同背景可以抵消，小的生物电位差仍可保留，再计算输出。

**EN:** Conditions and traps: Linear amplifier operation within input/output limits. Ideal cancellation requires equal interference at both inputs; real finite common-mode rejection and unequal contact cause residual error. High input impedance reduces loading; it does not set G. Gain amplifies remaining differential noise as well as signal. It is not a filter or a guarantee of better SNR.

**中文：** 条件与易错点：要求放大器在输入/输出限制内线性工作。理想抵消要求两端干扰相同；有限共模抑制与接触差异仍会留下误差。 高输入阻抗减少加载，不决定 G。增益也会放大残留差分噪声；放大不是滤波，也不保证信噪比提高。

**EN:** Tiny example: $V_1 = 2.02$ mV, $V_2 = 2.00$ mV and G=1000 give Vd=20 µV and Vout=20 mV.

**中文：** 短例理解：$V_1 = 2.02$ mV、$V_2 = 2.00$ mV、G=1000，得 Vd=20 µV、Vout=20 mV。

Source / 来源：W7 PDF p52–54 | W8 PDF p19–22 | T7-0043–0046 | T8-0021–0025

## FX707 • [P3] CMRR: a rejection ratio, not voltage gain / CMRR：共模抑制比，不是电压增益

**EN:** Revision priority: Recognise; optional calculations. Definition-level support for common-mode rejection, verified against TI primary documentation. CMRR was not found as a printed formula or required term in the supplied W7–8 slides/transcripts.

**中文：** 复习优先级：辅助辨认；计算后看。这是共模抑制的定义型补充，已据 TI 原厂资料核验；提供的 W7–8 课件/字幕中未找到 CMRR 印刷公式或必背要求。

**Formula / 公式：** $V_{cm} = (V_1+V_2)/2$; $V_{out} = A_d \cdot V_d + A_{cm} \cdot V_{cm}$; $CMRR_{linear} = |A_d/A_{cm}|$; $CMRR_{dB} = 20 \log_{10}(CMRR_{linear})$; $CMRR_{linear} = 10^{(CMRR_{dB}/20)}$; Voltage-gain dB = $20 \log_{10}|A_d|$

**EN:** Symbols and units: Vcm: common-mode input; Ad: differential voltage gain; Acm: gain from common-mode input to output; both gains are dimensionless and taken at the same frequency.

**中文：** 符号与单位：Vcm：共模输入；Ad：差模电压增益；Acm：共模输入到输出的增益。两个增益均无单位，必须在同一频率比较。

**EN:** When to use: Quantify how strongly an amplifier rejects a shared input compared with amplifying a difference; distinguish a linear ratio from its dB representation.

**中文：** 何时使用：量化放大器在放大差值的同时抑制共有输入的能力，区分线性比值与 dB 表示。

**EN:** Conditions and traps: A linear small-signal model with the stated Vd/Vcm definitions. CMRR can depend on frequency and operating conditions. Ideal Acm=0 gives infinite CMRR; a real device is finite. Ad alone is not CMRR. Use 20 log for a voltage-gain ratio, not 10 log, and do not insert a dB number directly as G in Vout=G Vin.

**中文：** 条件与易错点：采用给定 Vd/Vcm 定义的线性小信号模型。CMRR 可随频率和工作条件改变。理想 Acm=0 时 CMRR 无穷大，真实器件为有限值。 只知道 Ad 不等于知道 CMRR。电压增益比用 20 log，不是 10 log；不能把 dB 数直接当 G 代入 Vout=G Vin。

**EN:** Tiny example: Ad=1000 and Acm=0.1 give CMRRlinear=10000 and CMRRdB=80 dB. The differential voltage gain alone is 60 dB, a different quantity.

**中文：** 短例理解：Ad=1000、Acm=0.1 时，CMRR线性比=10000，CMRR=80 dB；差模电压增益本身为 60 dB，是不同的量。

Source / 来源：W7 PDF p53–54 (underlying subtraction concept, not a printed CMRR formula) | W8 PDF p20–22 (amplifier context) | T7-0044–0045 | TI Precision Labs instrumentation-amplifier introduction: https://www.ti.com/video/series/precision-labs/ti-precision-labs-instrumentation-amplifiers.html

## FX710 • [P3] Ideal inverting and non-inverting op-amp gains / 理想反相与同相运放增益

**EN:** Revision priority: Recognise; optional calculations. The circuits are printed on p24, but the lecturer’ s nearby “reference” wording does not make their memorization a confirmed Quiz requirement.

**中文：** 复习优先级：辅助辨认；计算后看。p24 印有这两式；附近“仅供参考”的课堂措辞不足以确认它们属于 Quiz 必背公式，保留作辅助。

**Formula / 公式：** Inverting: $G = -R_f/R_i$; $V_{out} = -(R_f/R_i)V_{in}$; Non-inverting: $G = 1+R_f/R_i$; $V_{out} = (1+R_f/R_i)V_{in}$; Ideal negative-feedback approximation: $V_+ \approx V_-$; $I_+ \approx I_- \approx 0$

**EN:** Symbols and units: Rf: feedback resistance from output to the − input; Ri: input resistance in the inverting circuit or lower feedback-divider resistance in the non-inverting circuit; G is dimensionless.

**中文：** 符号与单位：Rf：输出返回“−”端的反馈电阻；Ri：反相连接的输入电阻，或同相连接反馈分压网络的下臂电阻；G 无单位。

**EN:** When to use: Identify polarity and compute a short output-voltage example from the resistor ratio.

**中文：** 何时使用：根据电阻比判断极性，并作简短输出电压计算。

**EN:** Conditions and traps: Ideal op-amp, stable negative feedback and linear unsaturated operation, with sufficient bandwidth and input common-mode range. Virtual equality of inputs is an approximation, not a physical short circuit. The non-inverting gain includes +1. A negative gain reverses polarity; it does not mean zero amplification or a negative resistance. Do not apply the formulas after clipping.

**中文：** 条件与易错点：理想运放、稳定负反馈、未饱和线性工作，还需合适带宽与输入共模范围。两输入虚短是近似，不是实体导线短接。 同相增益不能漏掉 +1。负增益表示反号，不是没有放大或电阻为负；削顶后不能继续套线性式。

**EN:** Tiny example: Rf=20 kΩ, Ri=10 kΩ and Vin=+1 mV give −2 mV (inverting) or +3 mV (non-inverting).

**中文：** 短例理解：Rf=20 kΩ、Ri=10 kΩ、Vin=+1 mV，反相输出 −2 mV，同相输出 +3 mV。

Source / 来源：W8 PDF p23–24 | T8-0025–0026

## FX711 • [P3] First-order RC cutoff and low-/high-pass response / 一阶 RC 截止与低通/高通响应

**EN:** Revision priority: Recognise; optional calculations. Filter-role recognition is P1 elsewhere. These equations are supporting reference; the source’ s old RC example was locally excluded from this year’ s exam.

**中文：** 复习优先级：辅助辨认；计算后看。滤波功能辨认在正文仍为 P1；这些关系式作辅助查阅。原旧 RC 例题在局部语境中被说成今年 exam 不出。

**Formula / 公式：** $\tau = RC$; $\omega c = 1/(RC)$; $fc = 1/(2\pi RC)$; $x = \omega RC = f/fc$; $HLP = 1/(1+jx)$; $HHP = jx/(1+jx)$; $|HLP| = 1/\sqrt{(1+x^2)}$; $|HHP| = x/\sqrt{(1+x^2)}$; At fc: $|H| = 1/\sqrt{2} \approx 0.707$ (−3.01 dB relative to the passband)

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|---|
| **EN:** Symbols and units: R in Ω, C in F, τ in s, fc in Hz, ωc in rad/s; H=Vout/Vin is dimensionless; LP/HP mean low-/high-pass; $j^2=-1$. |
| **中文：** 符号与单位：R 用 Ω、C 用 F、τ 用 s、fc 用 Hz、ωc 用 rad/s；H=Vout/Vin 无单位；LP/HP 为低通/高通；$j^2=-1$。 |
| **EN:** When to use: Locate the transition and connect a formula to the filter graph: low-pass keeps the low-frequency side; high-pass keeps the high-frequency side. Increasing R or C lowers the cutoff. |
| **中文：** 何时使用：定位过渡区并把公式和曲线联系起来：低通保留低频侧，高通保留高频侧；R 或 C 增大都会降低截止频率。 |
| **EN:** Conditions and traps: These are standard passive, first-order RC stages with negligible source/loading effects. Output is across C for LP and across R for HP. Active-filter passband gain need not be unity. Convert $pF=10^{-12}$ F and $k\Omega=10^3$ Ω. ωc and fc differ by 2π. The teacher excluded the particular repeated 82 pF/240 kΩ exam example, not every filter formula or every simple Quiz calculation. |
| **中文：** 条件与易错点：针对信号源/负载影响可忽略的标准无源一阶 RC 级。低通输出取 C 两端，高通输出取 R 两端。有源滤波器的通带增益不必为 1。 先换 $pF=10^{-12}$ F、$k\Omega=10^3$ Ω。ωc 与 fc 相差 2π。老师局部排除的是重复的 82 pF/240 kΩ exam 例题，不是所有滤波公式或所有 Quiz 短计算。 |
| **EN:** Tiny example: R=10 kΩ and C=100 nF give τ=1 ms, ωc=1000 rad/s and fc≈159 Hz. At 10fc, LP magnitude≈0.100 and HP≈0.995. |
| **中文：** 短例理解：R=10 kΩ、C=100 nF，得 τ=1 ms、ωc=1000 rad/s、fc≈159 Hz。10fc 时，低通幅度约 0.100，高通约 0.995。 |

Source / 来源：W8 PDF p25–29 | T8-0026–0029

## FX712 • [P3] Optional active analog-filter equations / 可选：有源模拟滤波器完整式

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| **EN:** Revision priority: Recognise; optional calculations. Source completeness only. The teacher prioritized meanings and short relations, and did not ask students to memorize long filter algebra. |
| **中文：** 复习优先级：辅助辨认；计算后看。仅为源式完整性保留。老师强调含义与短式，没有要求把长滤波代数式当必背题。 |

**Formula / 公式：** Define $H(s)=Vout/Vin$; $s=j\omega$ for a sinusoidal steady state; High-pass: $H(s) = -sCR_2/(1+sCR_1)$; $f_0 = 1/(2\pi R_1 C)$; Band-pass: $H(s) = -sC_1R_2/[(1+sC_1R_1)(1+sC_2R_2)]$; $f_1 = 1/(2\pi R_1 C_1)$; $f_2 = 1/(2\pi R_2 C_2)$

**EN:** Symbols and units: $R_1$: input-path resistance; $R_2$: feedback resistance; C or $C_1$: input-path capacitance; $C_2$: feedback capacitance in the p29 circuit; s in $s^{-1}$, f in Hz.

**中文：** 符号与单位：$R_1$：输入通路电阻；$R_2$：反馈电阻；C 或 $C_1$：输入通路电容；$C_2$：p29 电路的反馈电容；s 单位 $s^{-1}$，f 单位 Hz。

**EN:** When to use: Keep the actual p27/p29 source equations available, while understanding that a high-pass stage and a low-pass factor can make a band-pass response.

**中文：** 何时使用：保留 p27/p29 的实际源式供查看，同时理解高通项配合低通项可形成带通响应。

**EN:** Conditions and traps: The illustrated ideal inverting op-amp circuits must remain linear. The two characteristic stage frequencies are not automatically the overall −3 dB boundaries; a clear passband requires suitably separated $f_1 < f_2$. p27 prints an extra minus sign before Vout/Vin that conflicts with its inverting diagram and negative transfer function. Here H is consistently Vout/Vin. Do not memorize the long transfer function as a teacher-confirmed Quiz requirement.

**中文：** 条件与易错点：针对图示理想反相运放，须保持线性。两个级的特征频率不自动等于总响应的精确 −3 dB 边界；清楚的通带要求 $f_1 < f_2$ 且有适当间隔。 p27 把 Vout/Vin 前多印了一个负号，与反相图和后面的负传递函数不一致；此处统一定义 H=Vout/Vin。长传递函数不能宣传成老师确认的 Quiz 必背式。

**EN:** Tiny example: For the high-pass stage with $R_2/R_1=10$, low-frequency gain tends to zero; at sufficiently high frequency its ideal gain tends to −10, so the magnitude tends to 10.

**中文：** 短例理解：有源高通若 $R_2/R_1=10$，低频增益趋于零；足够高频的理想增益趋于 −10，幅度趋于 10。

Source / 来源：W8 PDF p27, p29 | T8-0026–0029 | T8-0062–0064

# Read English exam questions / 看懂英文题目

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|---|
| **EN:** Which statement is incorrect / wrong? |
| **中文：** 哪项陈述错误？优先找错误方向或条件。 |
| **EN:** Which is NOT related to...? |
| **中文：** 哪项不属于……？不要漏看NOT。 |
| **EN:** Minimum detectable number |
| **中文：** 最小可检测数量；若是整数个体，需达到连续阈值。 |
| **EN:** Linear region / calibration slope |
| **中文：** 线性区 / 校准斜率；排除饱和点。 |
| **EN:** Assume all other conditions are identical |
| **中文：** 假设其他条件相同；才能只比较一个因素。 |
| **EN:** Higher resolving power / smaller resolvable separation |
| **中文：** 更高分辨本领 / 更小可分辨间距：都表示分辨更好。 |
| **EN:** Incident / absorbed / emitted photons |
| **中文：** 入射 / 吸收 / 发射光子：量子产率分母用吸收光子。 |
| **EN:** Reference electrode / reference arm |
| **中文：** 参比电极 / 参考光臂：一个用于电位基准，一个用于干涉比较。 |
| **EN:** Intracellular / extracellular |
| **中文：** 细胞内 / 细胞外：膜电位定义的相减方向不能反。 |
| **EN:** Round trip / one way |
| **中文：** 往返 / 单程：决定传播时间算深度是否除以2。 |
| **EN:** Common-mode / differential-mode |
| **中文：** 共模 / 差模：两输入共有分量 / 两输入差值。 |
| **EN:** High-pass / low-pass / band-pass / notch |
| **中文：** 高通 / 低通 / 带通 / 陷波：按保留或抑制的频率判断。 |

# Source corrections and sample cautions / 源材料纠错与样题注意点

**EN:** PMT correction: the lecture description at T4-0049–0050 wrongly describes tiny mirrors and optical reflection. The actual diagram (W4 p35) labels dynodes: multiplication is by secondary electron emission. Amplification does not remove all noise. See Hamamatsu’ s PMT explanation.

**中文：** PMT纠错：T4-0049–0050把结构说成小镜子与光反射，讲解不准确。原图W4 p35标有dynodes倍增极，实际为二次电子发射倍增；放大不等于消除全部噪声。已核对 Hamamatsu官方PMT说明。

**EN:** The X-ray “non-ionizing” error also occurs in the original W4 p7, not only a student sheet. Correct it to ionizing; see FDA medical X-ray imaging. W7 p84 also misprints gamma as <25 Hz; learn the band ordering here rather than memorizing this inconsistent boundary.

**中文：** X射线“non-ionizing”的错误也出现在原课件W4 p7，不只是学生一页纸；应改为电离性，见FDA医学X射线说明。W7 p84还把gamma误写为<25 Hz；这里保留频带顺序，不让你死背不一致的边界。

**EN:** Further wording safeguards: the glucose/peroxide example is not universal to amperometry; high input impedance is not voltage gain; below-threshold neurons still have a resting potential; ECG leads are not the P/QRS/T waves. These conceptual corrections take precedence over loose oral wording.

**中文：** 另已防止四种过度概括：葡萄糖/过氧化氢例子不代表全部安培传感；高输入阻抗不等于高电压增益；未达阈值仍有静息电位；心电导联不等于P/QRS/T波。复习采用纠正后的概念，不照抄宽泛口头表述。

**EN:** Sample Q10 follows the lecturer’ s integer-threshold answer B=5. Q5 remains a provisional waveform estimate; Q9 is a detector-principle reference answer rather than an unambiguous spoken final letter; Q12 retains a current/current-density unit inconsistency. Use the review report for details.

**中文：** 样题Q10采用老师整数阈值解释，答案B=5。Q5仍为暂定波形估读；Q9是按探测器原理推导的参考答案，课堂没有明确最后字母；Q12保留电流/电流密度单位冲突说明。细节见复核报告。

# How to use this learning edition / 怎样使用讲解版

**EN:** For the first pass, read F01–F08 and the Chinese explanation of each lesson, including the adjacent S supplements and figure guidance. Before moving on, explain what is measured and why the output changes. For the second pass, match the English terms and judge every option. The old 32-page summary recalls the original main line but does not contain this coverage update; use this updated guide for the added distinctions.

**中文：** 第一遍读F01–F08及各项中文讲解，包括相邻S补充与读图说明；进入下一项前，自己说出测什么、为什么输出会变。第二遍对应英文术语，并解释每个选项。旧32页速记版可回顾原主线，但未含本次补漏；新增区别应以这份更新后的讲解版为准。